A TEXT BOOK OF

B.Sc.

MATHEMATICS

Third Year, A.P.

LINEAR ALGEBRA

SEMESTER V

(3rd year, 5th semester-Paper 6)

(THEORY & PRACTICAL)

&

Key Book included

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Third Year, A.P.

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(SEMESTER V)

(*3rd year 5th semester-Paper 6*)

(As per New Common Core Syllabus 2015-16 (revised in 2016) based on CBCS for B. Sc. Third Year Fifth Semester Students of the Universities in Andhra Pradesh)

V. VENKATESWARA RAO
M. Sc.
Former Vice Principal and
Head of the Department of Mathematics
Hindu College, GUNTUR

N. KRISHNAMURTHY
M. Sc.
Former Head of the Department of Mathematics
V.V. College, HYDERABAD

B.V.S.S. SARMA
M. Sc.
Former Senior Lecturer in Mathematics
V.S.R. & N.V.R. College, TENALI

S. ANJANEYA SASTRY
M.A.
Former Head of the Deptt. of Mathematics
V.S.R. & N.V.R. College, TENALI.

S. RANGANATHAM
M. Sc. M. Phil.
Former Head of the Deptt. of Mathematics
Jawahar Bharathi Degree College, KAVALI.

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Branches

Chennai : Ph: 23632120; chennai@schandpublishing.com

Guwahati : Ph: 2738811, 2735640; guwahati@schandpublishing.com

Hyderabad : Ph: 40186018; hyderabad@schandpublishing.com

Jalandhar : Ph: 4645630; jalandhar@schandpublishing.com

Kolkata : Ph: 23357458, 23353914; kolkata@schandpublishing.com

Lucknow : Ph: 4003633; lucknow@schandpublishing.com

Mumbai : Ph: 25000297; mumbai@schandpublishing.com

Patna : Ph: 2260011; patna@schandpublishing.com

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PREFACE

This *Text book of B.Sc Mathematics* for the students studying Third year in all universities of Andhra Pradesh was first published in the year 1988 and has undergone several editions and many reprints. The authors are very happy that the earlier editions have been very well used by the students.

The **revised syllabus** is being adopted by all the universities in Andhra Pradesh, following **Common Core model curriculam** from the academic year **2015 - 2016 (revised in 2016)** based on **CBCS (Choice Based Credit System)**. This book strictly covers the new curriculum for **Semester V** (*3rd year, 5th semester-Paper 6*).

Objective type questions covering multiple choice type and fill in the blank type questions which are useful to M.Sc. entrance tests are given with answers at the end of each section.

Detailed solutions for all the problems in the various exercises of different chapters are given at the end.

All suggestions for the further improvement of the book are welcome.

We are thankful to the Management Team and the Editorial Department of S Chand And Company Limited, New Delhi for all help and support in the publication of this book.

AUTHORS

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SYLLABUS

AS PER COMMON CORE MODEL CURRICULUM FROM THE ACADEMIC YEAR 2015 - 2016
(revised in 2016)

SEMESTER - V, PAPER - 6
LINEAR ALGEBRA

UNIT - I : (12 hours)

Vector Spaces - I :

Vector Spaces, General properties of vector spaces, n-dimensional vectors, addition and scalar multiplication of Vectors, internal and external composition, Null space, Vector subspaces, Algebra of subspaces, Linear Sum of two subspaces, linear combination of Vectors, Linear span Linear independence and Linear dependence of Vectors.

UNIT - II : (12 hours)

Vector Spaces - II :

Basis of Vector space, Finite dimensional Vector spaces, basis extension, co-ordinates, Dimension of a Vector space, Dimension of a subspace, Quotient space and Dimension of Quotient space.

UNIT - III : (12 hours)

Linear Transformations :

Linear transformations, linear operators, Properties of L.T., sum and product of LTs, Algebra of Linear Operators, Range and null space of linear transformation, Rank and Nullity of linear transformations - Rank - Nullity Theorem.

UNIT - IV: (12 hours)

Matrix

Matrices, Elementary properties of Matrices, Inverse Matrices, Rank of Matrix, Linear Equations, Characteristic Roots, Characteristic Values & Vectors of square Matrix, Cayley-Hamilton Theorem.

UNIT - V: (12 hours)

Inner Product Space :

Inner product spaces, Euclidean and unitary spaces, Norm or length of a vector, Schwartz inequality, Triangle in inequality, Parallelogram law, Orthogonality, Orthonormal set, complete orthonormal set, Gram-Schmidt orthogonalisation process. Bessel's inequality and Parseval's Identitty.

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LINEAR ALGEBRA

UNIT - I

Vector Spaces I

Vector Spaces, General properties of vector spaces, n-dimensional Vectors, addition and scalar multiplication of Vectors, internal and external composition, Null space, Vector subspaces, Algebra of subspaces, Linear Sum of two subspaces, linear combination of Vectors, Linear span Linear independence and Linear dependence of Vectors.

1. Vector Spaces

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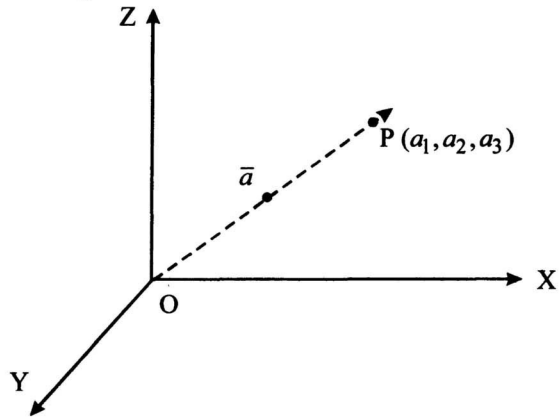
1
Vector Spaces

1.1. Let P be a point (a_1, a_2, a_3) with respect to the frame of reference OXYZ. Let $\overline{OP}$ be denoted by the vector $\bar{a}$. Now the vector $\bar{a}$ is associated with the point P given by the ordered trial (a_1, a_2, a_3). Conversely the ordered triad (a_1, a_2, a_3) defines a point P associated with the vector $\bar{a}$.



This shows that the set of all points in 3-D space has a one to one correspondence with the set of all vectors starting from the origin. Thus each vector is representable as an ordered triad of three real numbers. This enables us to write $\bar{a} = (a_1, a_2, a_3)$.

From this we can visualise the 3.dimensional space as an ordered set of triads (a_1, a_2, a_3) where a_1, a_2, a_3 are real numbers.

This space is denoted by R^3.

1.2. ALGEBRAIC STRUCTURE OF 3 - D VECTORS

Let V be the set of 3.D vectors and F be the field of scalars.

Now it is easy to verify the following algebraic structure with the vector addition (+) and scalar multiplication (•) of a vector.

G :(1) Closure : $\alpha + \beta \in V \ \forall \ \alpha, \beta \in V$

(2) Associativity : $\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma \ \forall \ \alpha, \beta, \gamma \in V$

(3) Identity : $\alpha + \bar{O} = \bar{O} + \alpha = \alpha \ \forall \ \alpha \in V$ null vector $\bar{O}$ is the additive identity in V.

(4) Inverse : $\alpha + (-\alpha) = \bar{O} = (-\alpha) + \alpha$. Every vector in V has the additive inverse.

(5) Commutativity : $\alpha + \beta = \beta + \alpha \ \forall \ \alpha, \beta \in V$

$\therefore$ (V, +) is an abelian group.

A : *Admission of scalar multiplication in V*

(1) $a\alpha \in V \ \forall \ a \in F$ and $\alpha \in V$

(2) $1.\alpha = \alpha \ \forall \ \alpha \in V$, where 1 is the unity of F.

S : *Scalar multiplication*

(1) $a(b\alpha) = (ab)\alpha \qquad \forall \ a, b \in F$ and $\alpha \in V$

Scalar multiplication of vectors is associative

(2) $a(\alpha + \beta) = a\alpha + a\beta \quad \forall \ a \in F$ and $\alpha, \beta \in V$

Multiplication by scalars is distributive over vector addition.

(3) $(a+b)\alpha = a\alpha + b\alpha \quad \forall \ a, b \in F$ and $\alpha \in V$

Multiplication by vectors is distributive over scalar addition.

The above verified **GAS** structure leads us to define abstract vector spaces or simply vector spaces or linear spaces.

1.3. THE *n* - DIMENSIONAL VECTORS

Now we can generalise the 3-D vector concepts to *n*-dimensional vectors. Thereby, a point in the *n*-dimensional vector space has *n* coordinates of the form $(a_1, a_2, ..., a_n)$. Hence we define an *n*-dimensional vector $\bar{a}$ as an ordered *n*-tuple, such as $\bar{a} = (a_1, a_2, a_3, ..., a_n)$

where each a_i is a number, real or complex. This space is denoted by $\mathbf{R}^n$ or $\mathbf{C}^n$.

1.4. SCALAR MULTIPLICATION OF A VECTOR

Let $\alpha = (a_1, a_2, ..., a_n)$ be a vector in R^n and k be a scalar.

Then we define $k\alpha = (ka_1, ka_2, ..., ka_n)$.

1.5. ADDITION OF VECTORS

Let $\alpha = (a_1, a_2, ..., a_n)$, $\beta = (b_1, b_2, ..., b_n)$ be two vectors of R^n.

Then we define $\alpha + \beta = (a_1 + b_1, a_2 + b_2, ..., a_n + b_n)$

1.6. INTERNAL COMPOSITION

Let F be a set. If $aob \in F$ for all $a, b \in F$. Then 'o' is said to be an internal composition in the set F.

1.7. EXTERNAL COMPOSITION

Let V and F be two sets. If $a\,o\,\alpha \in V$ for all $a \in F$ and for all $\alpha \in V$, then 'o' is said the be an external composition in V over F. Here the resulting element $a\,o\,\alpha$ is an element of the set V.

1.8. VECTOR SPACES

Definition. *Let V be a non-empty set whose elements are called vectors. Let F be* any set whose elements are called scalars where $(F, +, \bullet)$ *is a field.*

The set V is said to be a vector space if **(S.K. U. M 2015)**

(1) *there is defined an internal composition in V called addition of vectors denoted by +, for which (V, +) is an abelian group.*

(2) *there is defined an external composition in V over F, called the scalar multiplication in which* $a\alpha \in V$ *for all* $a \in F$ *and* $\alpha \in V$.

(3) *the above two compositions satisfy the following postulates*

(*i*) $a(\alpha+\beta) = a\alpha + a\beta$ (*ii*) $(a+b)\alpha = a\alpha + b\alpha$ (*iii*) $(ab)\alpha = a(b\alpha)$ (*iv*) $1\alpha = \alpha$

$\forall\ a, b \in F$ and $\alpha, \beta \in V$ and 1 is the unity element of F. Instead of saying that V is a vector space over the field F, we simply say V(F) is a vector space. Sometimes if the field F is understood, then we simply say that V is a *'vector space'.*

1.9. VECTOR SPACE

(O.U. 2011)

If R is a field of real numbers, then V(R) is called the real vector space.

If C is the field of complex numbers, then V(C) is called the complex vector space.

Note.1. In the above definition (V, +) is an abelian group implies that for all $\alpha, \beta, \gamma \in V$

(*a*) $\alpha+\beta \in V$ (*b*) $\alpha + (\beta+\gamma) = (\alpha+\beta)+\gamma$

(*c*) there exists an element $\bar{O} \in V$ such that $\alpha+\bar{O}=\alpha$

(*d*) To every vector $\alpha \in V$ there exists $-\alpha \in V$ such that $\alpha + (-\alpha) = 0$

(*e*) $\alpha+\beta=\beta+\alpha$

2. The use of symbol + for two different compositions (for the addition in the field F and for the internal composition in V) should not cause any confusion. It should be understood according to the context in which it is used.

3. In vector space, two types of zero elements come into operation. One is the zero vector $\bar{O}$ of V and the other is the zero scalar 0 of the field F.

1.10. NULL SPACE OR ZERO VECTOR SPACE

The vector space having only one zero vector $\bar{O}$ is called the zero vector space or null space. It clearly satisfies all the postulates on any field of scalars.

Theorem : ***Let V(F) be a vector space and*** $O, \bar{O}$ ***be the zero scalar and zero vector respectively. Then***

(1) $a\bar{O}=\bar{O}\ \forall\ a \in F$ **(2)** $0\alpha=\bar{O}\ \forall\ \alpha \in V$

(3) $a(-\alpha) = -(a\alpha)\ \forall\ a \in F,\ \forall \alpha \in V$ **(4)** $(-a)\alpha = (-a\alpha)\ \forall\ a \in F,\ \forall \alpha \in V$

(5) $a\alpha=\bar{O} \Rightarrow a=0$ or $\alpha=\bar{O}$ **(6)** $a(\alpha-\beta) = a\alpha - a\beta\ \forall\ a \in F,\ \forall \alpha, \beta \in V$

(7) $(-a)(-\alpha) = a\alpha\ \forall\ a \in F,\ \forall \alpha \in V$

Proof. (1) $a\bar{O}=a(\bar{O}+\bar{O})=a\bar{O}+a\bar{O}$

$\therefore\ a\bar{O}+\bar{O}=a\bar{O}+a\bar{O}$ (Distributive Law)

$\bar{O}=a\bar{O}$ (Left Cancellation Law)

(2) $0\alpha = (0+0)\alpha = 0\alpha + 0\alpha$

$\therefore \bar{O} + 0\alpha = 0\alpha + 0\alpha \Rightarrow \bar{O} = 0\alpha$ (Cancellation Law)

(3) $a[\alpha + (-\alpha)] = a\bar{O} \Rightarrow a\alpha + a(-\alpha) = \bar{O}$

$\Rightarrow a(-\alpha)$ is the additive inverse of $a\alpha \Rightarrow a(-\alpha) = -(a\alpha)$

(4) $[a + (-a)]\alpha = 0\alpha \Rightarrow a\alpha + (-a)\alpha = \bar{O}$

$\Rightarrow (-a)\alpha$ is the additive inverse of $a\alpha$

$\Rightarrow (-a)\alpha = -(a\alpha)$

(5) $a\alpha = \bar{O}$ and $a = 0$. Then there is nothing to prove. $a\alpha = \bar{O}$ and $a \neq 0$

Now $a \neq 0, a \in F \Rightarrow$ there exists. $a^{-1} \in F$ such that $a\,a^{-1} = a^{-1}a = 1$

Now $a\alpha = \bar{O} \Rightarrow a^{-1}(a\alpha) = a^{-1}\bar{O} \Rightarrow 1\alpha = \bar{O} \Rightarrow \alpha = \bar{O}$

$\therefore a\alpha = \bar{O} \Rightarrow a = 0$ or $\alpha = \bar{O}$

(6) $a(\alpha - \beta) = a[\alpha + (-\beta)] = a\alpha + a(-\beta) = a\alpha - a\beta$

(7) $(-a)(-\alpha) = -[a(-\alpha)] = -[-(a\alpha)] = a\alpha$

1.11. Theorem. *Let $V(F)$ be a vector space*

(1) If $a, b \in F$ and $\alpha \in V$ where $\alpha \neq \bar{O}$ then $a\alpha = b\alpha \Rightarrow a = b$

(2) If $a \in F$ where $a \neq 0$ and $\alpha, \beta \in V$ then $a\alpha = a\beta \Rightarrow \alpha = \beta$.

Proof : (1) $a\alpha = b\alpha \Rightarrow a\alpha + (-b\alpha) = b\alpha + (-b\alpha) \Rightarrow (a-b)\alpha = \bar{O}$

$\Rightarrow a - b = 0$ as $\alpha \neq \bar{O} \Rightarrow a = b$

(2) $a\alpha = a\beta \Rightarrow a\alpha - a\beta = \bar{O} \Rightarrow a(\alpha - \beta) = \bar{O} \Rightarrow \alpha - \beta = \bar{O}$ as $a \neq 0 \Rightarrow \alpha = \beta$.

SOLVED PROBLEMS

Ex. 1. *The set C_n of all n-tuples of complex numbers with addition as the external composition and scalar multiplication of complex numbers by complex numbers is a vector space over the field of complex numbers with the following definitions.*

(i) If $\alpha, \beta \in C_n$ and $\alpha = (a_1, a_2, ..., a_n)$ and $\beta = (b_1, b_2, ..., b_n)$ for all $a_k, b_k \in C$

$\alpha + \beta = (a_1 + b_1, a_2 + b_2, ..., a_n + b_n) \in C_n$

(ii) $x\alpha = (x\,a_1, x\,a_2, ..., x\,a_n)\ \forall\ x \in C$

Sol. *(i)* By definition $x + y$ is an *n*-tuple of complex numbers. Hence C_n is closed.

(ii) $(\alpha + \beta) + \gamma = (a_1 + b_1, a_2 + b_2, ..., a_n + b_n) + (c_1, ..., c_n)$

where $\gamma = (c_1, c_2, ..., c_n)$ with c_i's $\in C$

$= \{(a_1 + b_1) + c_1, (a_2 + b_2) + c_2, ..., (a_n + b_n) + c_n\}$

$= \{a_1 + (b_1 + c_1), a_2 + (b_2 + c_2), ..., a_n + (b_n + c_n)\}$ since + is associative in C

$= (a_1, a_2, ..., a_n) + (b_1 + c_1, b_2 + c_2, ..., b_n + c_n) = \alpha + (\beta + \gamma)$

$\therefore$ Vector addition is associative in C_n.

(iii) $\alpha+\bar{O}=(a_1, a_2, \ldots, a_n)+(0, 0, \ldots, 0) = (a_1+0, a_2+0, \ldots, a_n+0) = (a_1, a_2, \ldots, a_n) = \alpha$

Similarly $\bar{O}+\alpha=\alpha \Rightarrow \alpha+\bar{O}=\bar{O}+\alpha=\alpha$

$\therefore$ The n-tuple $\bar{O}=(0, 0, \ldots, 0)$ is the identity in C_n.

(iv) $\alpha+(-\alpha)=(a_1, a_2, \ldots, a_n)+(-a_1, -a_2, \ldots, -a_n)$

$=(a_1-a_1, a_2-a_2, \ldots, a_n-a_n) = (0, 0, 0, \ldots, 0) = \bar{O}$. $\quad \therefore \alpha+(-\alpha)=\bar{O}$

$\Rightarrow (-\alpha) \in C$, is the additive inverse of α in C

(v) $\alpha+\beta=(a_1+b_1, a_2+b_2, \ldots, a_n+b_n)$

$=(b_1+a_1, a_2+b_2, \ldots, b_n+a_n)$ ($\because$ + is commutative in C)

$=\beta+\alpha$.

(vi) By definition $x \in C, \alpha \in C_n \Rightarrow x\alpha \in C_n$

(vii) Let $x, y \in C$. $\quad x(y\alpha)=x(ya_1, ya_2, \ldots, ya_n) = \{x(ya_1), x(ya_2), \ldots, x(ya_n)\}$

$=\{(xy)a_1, (xy)a_2, \ldots, (xy)a_n\} = xy(a_1, a_2, \ldots, a_n) = (xy)\alpha$

$(viii)$ $x \in C$ and $\alpha, \beta \in C_n$

$\Rightarrow x(\alpha+\beta)=x(a_1+b_1, a_2+b_2, \ldots, a_n+b_n)$

$=\{x(a_1+b_1), x(a_2+b_2), \ldots, x(a_n+b_n)\} = (xa_1+xb_1, xa_2+xb_2, \ldots, xa_n+xb_n)$

$=(xa_1, xa_2, \ldots, xa_n)+(xb_1, xb_2, \ldots, xb_n) = x\alpha+x\beta$

(ix) $x, y \in C$ and $\alpha \in C_n$

$\Rightarrow (x+y)\alpha=(x+y)(a_1, a_2, \ldots, a_n) = \{(x+y)a_1, (x+y)a_2, \ldots, (x+y)a_n\}$

$=\{(xa_1+ya_1), (xa_2+ya_2), \ldots, (xa_n+ya_n)\}$ (Since distributive law is true in C)

$=(xa_1, xa_2, xa_n)+(ya_1, ya_2, \ldots, ya_n) = x\alpha+y\alpha$

(x) $1.\alpha=1(a_1, a_2, \ldots, a_n)=(a_1, a_2, \ldots, a_n)=\alpha$

Since all the postulates are verified C_n (C) is a vector space.

Ex.2. *Prove that the set of all real valued continuous functions defined in the open interval (0,1) is a vector space over the field of real numbers, with respect to the operations of addition and scalar multiplication defined as*

$(f+g)(x)=f(x)+g(x)$ $\quad \ldots (1)$

$(af)(x)=af(x)$. a is real $\quad \ldots (2)$ where $0<x<1$

Sol: Let V be the set of all such real values continuous functions and R be the field of real numbers.

(i) The sum of two continuous functions is again a continuous function

$(f+g) \in V \ \forall \ f, g \in V$.

(ii) $\{(f+g)+h\}(x)=(f+g)x+h(x)$ $\quad$ (by def. (1))

$= f(x)+g(x)+h(x) = f(x)+(g+h)(x) = \{f+(g+h)\}(x)$

$\Rightarrow (f+g)+h = f+(g+h)$. $\therefore$ V is associative

(*iii*) Let the function O be defined as $\bar{O}(x)=0$

$\therefore (0+b)(x)=\bar{O}(x)=f(x)$ (by def (1))

$=0+f(x)=f(x)$ as the real number 0 is the additive identity in R.

$\therefore \bar{O}+f=f,\ \forall f\in V$

(*iv*) $\{f+(-f)\}(x)=f(x)+(-f(x))\} = f(x)-f(x)=0=\bar{O}(x)$

The function $(-f)$ is the additive inverse of f

$\therefore f+(-f)=\bar{O}$ (the identity function)

(*v*) $(f+g)(x)=f(x)+g(x)$

$=g(x)+f(x)$ (real numbers are commutative under addition)

$=(g+f)(x)$ by definition (1)

Thus $f+g=g+f,\ \forall\ g,f\in V$. $\therefore$ (V, +) is an abelian group

(*vi*) For all $a\in R$ and $f,g\in V$, $a(f+g)=af+ag$

Now $[a(f+g)](x)=a[(f+g)(x)]=a[f(x)+g(x)] = af(x)+ag(x)$

$=(af)(x)+(ag)(x)$ $=(af+ag)(x)$ $\therefore a(f+g)=af+ag$

(*vii*) If $a,b\in R$ and $f\in V$ then, $\{(a+b)f\}(x)=(a+b)f(x)$ (by (2))

$=af(x)+bf(x)$ (as $f(x)$ is real)

$=(af)(x)+(bf)(x)=(af+bf)(x)$ (by (1))

$\Rightarrow (a+b)f=af+bf$

(*viii*) If $a,b\in R$ and $f\in V$ then, $\{a(bf)\}(x)$ (by (2))

$=a\{(bf)(x)\}=a\{bf(x)\}$

$=(ab)f(x)$ as $f(x)$ is real $=\{(ab)f\}(x)\Rightarrow a(bf)=(ab)f$

(*ix*) Since 1 is the identity of the field R and $f\in V$,

we have $(1f)(x)=1f(x)=f(x)$ $\therefore 1f=f$

All the postulates of vector space are verified. Hence V (R) is a vector space.

Ex. 3. *V is the set of all $m\times n$ matrices with real entries and R is the field of real numbers. "Addition of matrices' is the internal composition and 'multiplication of a matrix by a real number' an external composition in V. Show that V (R) is a vector space.*

Sol: Let $\alpha,\beta,\gamma\in V$ and $x, y\in R$ where $\alpha=[a_{ij}]$, $\beta=[b_{ij}]$, $\gamma=[c_{ij}]$ for $a's, b's, c's\in R$

(*i*) Addition of two matrices is a matrix. $\therefore \alpha+\beta\in V\ \forall\ \alpha,\beta\in V$

(*ii*) $\alpha+(\beta+\gamma)=[a_{ij}]+[b_{ij}+c_{ij}]=[a_{ij}+(b_{ij}+c_{ij})] \quad =[(a_{ij}+b_{ij})+c_{ij}]$

(Since real numbers are associative in R)

$=[a_{ij}+b_{ij}]+[c_{ij}] \quad =(\alpha+\beta)+\gamma \qquad \therefore$ V is associative.

(*iii*) If $O=[o_{ij}]$ is the null matrix then $\alpha+O=[a_{ij}]+[o_{ij}]=[a_{ij}+o_{ij}] \quad =[a_{ij}]=\alpha$

Thus $\alpha+O=O+\alpha=\alpha$. $\Rightarrow$ The null matrix O is the additive identity in V.

(*iv*) $\alpha+(-\alpha)=[a_{ij}]+[-a_{ij}] \quad =[a_{ij}-a_{ij}]=[-o_{ij}]=0$

$\therefore$ $(-\alpha)$ is the additive inverse of every α in V

(*v*) Clearly $\alpha+\beta=\beta+\alpha$ $\therefore$ (V, +) is an abelian group.

(*vi*) For $x \in R$ and $\alpha \in V$. $x\alpha=x[a_{ij}]=[xa_{ij}] \Rightarrow x\alpha \in V$

(*vii*) For $x, y \in R$ and $\alpha \in V$

$x(y\alpha)=x(y[a_{ij}])=x[ya_{ij}]=[xya_{ij}]=(xy)[a_{ij}]=(xy)\alpha$

(*viii*) $x(\alpha+\beta)=[a_{ij}+b_{ij}]=[x(a_{ij}+b_{ij})]=[xa_{ij}+xb_{ij}] \quad =[xa_{ij}]+[xb_{ij}]$

$=x[a_{ij}]+x[b_{ij}]=x\alpha+x\beta$

(*ix*) $(x+y)\,\alpha=(x+y)\,[a_{ij}] \quad =[(x+y)\,a_{ij}]=[x\,a_{ij}+y\,a_{ij}]$

$=[x\,a_{ij}]+[y\,a_{ij}] \quad =x[a_{ij}]+y[a_{ij}]=x\,\alpha+y\,\alpha$

(*x*) $1.\alpha=1[a_{ij}]=[a_{ij}]=\alpha$ Hence V (R) is a vector space.

Ex.4. *Let V be the set of all pairs (a, b) of real numbers and R be the field of real numbers. Show that with the operation* $(a_1,b_1)+(a_2,b_2)=(a_1+a_2,o)$, $c(a_1,b_1)=(ca_1,b_1)$ *V (R) is not a vector space.*

Sol: If any one of the postulates of a vector space is not satisfied, then V (R) cannot be a vector space. Let (x, y) be the identity in V.

Now $(a,b)+(x,y)=(a,b) \ \forall \ a,b \in V \Rightarrow (a+x,0)=(a,b)$ ... (1)

If $b \neq 0$, then we cannot have the equality (1).

Thus there exists no element $(x,y) \in V$ such that $(a,b)+(x,y)=(a,b) \ \forall \ a,b \in V$

$\therefore$ Additive identity does not exist in V and hence V (R) is not a vector space.

EXERCISE 1 (*a*)

1. Show that the set of all triads (x_1,x_2,x_3) where x_1,x_2,x_3 are real numbers forms a vector space over the field of real numbers with respect to the operations of addition and scalar multiplication defined as

(*i*) $(x_1,x_2,x_3)+(y_1,y_2,y_3)=(x_1+y_1,x_2+y_2,x_3+y_3)$ and

(*ii*) $c\,(x_1,x_2,x_3)=(c\,x_1,c\,x_2,c\,x_3)\,c$ is a real number.

2. Let P be the set of all polynomials in one indeterminate with real coefficients. Show that P (R) is vector space, if in P the addition of polynomials is taken as the internal composition and the multiplication of polynomial by a constant polynomial (*i.e.* by an element k) as scalar multiplication.
3. If F is a field, the F (F) is a vector space, if in F the addition of field F is taken as the internal composition and the multiplication of the field F is taken as the external composition in F over F.
4. Let R^n be the set of all ordered n-tuples of real numbers given by

   $R^n = [(x_1, x_2, \ldots, x_n) \mid x_i \in R]$

   The internal and external composition in R^n are defined by

   (*i*) $(x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$

   (*ii*) $a(x_1, x_2, \ldots, x_n) = (ax_1, ax_2, \ldots, ax_n)$, $a \in R$. Show that R^n (R) is a vector space.
5. Let F be any field and K any subfield of F. Then F (K) is a vector space, if the addition of the field F is taken as the internal composition in F, and the field multiplication is taken as the external composition in F over K.
6. Let V (R) be a vector space and $W = \{(x, y) \mid x, y \in V\}$. For (x_1, y_1) and (x_2, y_2) in W we have $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$ and if $\alpha = a_1 + ia_2 \in C$, we define $d(x_1, y_1) = (a_1 x_1 - a_2 y_1, a_2 x_1 + a_1 y_1)$. Prove that W (C) is a vector space.
7. The set of all real valued differentiable or integrable functions defined in some interval [0,1] is a vector space.
8. Let V be the set of all pairs of real numbers and let F be the field of real numbers with the definition $(x_1, y_1) + (x_2, y_2) = (3y_1 + 3y_2, -x_1 - x_2)$, $c(x_1, y_1) = (3c\, y_1, -c\, x_1)$

   Show that V (F) is not a vector space.
9. V is the set of all polynomials over real numbers of degree atmost one and F = R.

   If $f(t) = a_0 + a_1 t$ and $g(t) = b_0 + b_1(t)$ in V ; define

   $f(t) + g(t) = (a_0 + b_0) + (a_0 b_1 + a_1 b_0)\, t$ and $kf(t) = (ka_0) + (ka_1)\, t$, $k \in F$

   Show that V (F) is not a vector space. (**Hint:** Additive identity does not exist)
10. C is the field of complex numbers and R is the field of real numbers. Show that C (R) is a vector space and R (C) is not a vector space.
11. Let V be the set of all pairs (x, y) of real numbers, and let F be the field of real numbers. Define $(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$, $c(x, y) = (cx, y)$

    Show that V (F) is not a vector space.

1.12. VECTOR SUBSPACES

Definition. *Let V (F) be a vector space and* $W \subseteq V$. *Then W is said to be a subspace of V if W itself is a vector space over F with the same operations of vector addition and scalar multiplication in V.*

Note.1. If W (F) is a subspace of V (F) then W is a sub-group of V.

Note.2. Let V (F) be a vector space. The zero vector space $\{\bar{O}\} \subseteq V$ and $V \subseteq V$.

$\therefore$ $\{\bar{O}\}$ and V are the trivial subspaces of V.

1.13. Theorem. *Let V (F) be a vector space and let* $W \subseteq V$. *The necessary and sufficient conditions for W to be a subspace of V are*

(i) $\alpha \in W, \beta \in W \Rightarrow \alpha - \beta \in W$ *(ii)* $a \in F, \alpha \in W \Rightarrow a\alpha \in W$. *(S.V. U 2011)*

Proof. *Conditions are necessary*

(*i*) W is a vector subspace of V

$\Rightarrow$ W is a subgroup of $(V, +)$ $\Rightarrow$ $(W, +)$ is a group

$\Rightarrow$ if $a, \beta \in W$ then $a - \beta \in W$.

(*ii*) W is a subspace of V

$\Rightarrow$ W is closed under scalar multiplication $\Rightarrow$ for $a \in F$, $\alpha \in W$; $a\alpha \in W$

Conditions are sufficient.

Let W be a nonempty subset of V satisfying the two given conditions

$\alpha \in W$, $\alpha \in W \Rightarrow \alpha - \alpha \in W \Rightarrow \bar{O} \in W$ (by (*i*))

$\therefore$ The zero vector of V is also the zero vector of W

$\bar{O} \in W$, $\alpha \in W \Rightarrow \bar{O} - \alpha \in W \Rightarrow (-\alpha) \in W$ (by (*i*))

$\Rightarrow$ additive inverse of each element of W is also in W

Again $\alpha \in W$, $\beta \in W \Rightarrow \alpha \in W$, $(-\beta) \in W \Rightarrow \alpha - (-\beta) \in W$ (by (*i*))

$\Rightarrow \alpha + \beta \in W$

i.e., W is closed under vector addition

As $W \subseteq V$, all the elements of W are also the elements of V. Thereby vector addition in W will be associative and commutative. This implies that (W, +) is an abelian group.

Further by (*ii*), W is closed under scalar multiplication and the other postulates of vector space hold in w as $W \subseteq V$.

$\therefore$ W itself is a vector space under the operations of V.

Hence W (F) is a vector subspace of V (F)

A more useful and abridged form of the above theorem is the following.

1.14. Theorem. *Let V (F) be a vector space. A non-empty set* $W \subseteq V$. *The necessary and sufficient condition for W to be a subsapce of V is*

$a, b \in F$ *and* $\alpha, \beta \in V \Rightarrow a\alpha + b\beta \in W$ *... (I)*

(A. U. M 2014, S.V. U. M 14, S97, O. U. M 13, A.N.U., K.U. 11, M 13, S. K. U. M13, M14)

Proof. *Condition is necessary.*

W (F) is a subspace of V (F) $\Rightarrow$ W (F) is a vector space

$\therefore\ a \in F,\ \alpha \in W \Rightarrow a\alpha \in W$ and $b \in F,\ \beta \in W \Rightarrow b\beta \in W$

Now $a\alpha \in W,\ b\beta \in W \Rightarrow a\alpha + b\beta \in W$

Condition is sufficient.

Let W be the non-empty subset of V satisfying the given condition

i.e., $a, b \in F$ and $\alpha, \beta \in W \Rightarrow a\alpha + b\beta \in W$ ... (I)

Taking $a = 1, b = -1$ and $\alpha, \beta \in W \Rightarrow 1\alpha + (-1)\beta \in W$

$\Rightarrow \alpha - \beta \in W$ [$\because\ \alpha \in W \Rightarrow \alpha \in V$ and $1\alpha = \alpha$ in V]

($H \subseteq G$ and $a, b \in H \Rightarrow a\,ob^{-1} \in H$ then (H, o) is subgroup of (G, 0)).

$\therefore$ (W, +) is a subgroup of the abelian group (V, +).

$\Rightarrow$ (W, +) is an abelian group. Again taking $b = 0$

$a, 0 \in F$ and $\alpha, \beta \in W \Rightarrow a\alpha + 0\beta \in W \Rightarrow a\alpha \in W \Rightarrow a \in F$ and $\alpha \in W \Rightarrow a\alpha \in W$

$\therefore$ W is closed under scalar multiplication.

The remaining postulates of vector space hold in W as $W \subseteq V$.

$\therefore$ W (F) is a vector subspace of V (F).

1.15. Theorem. *A non-empty set W is a subset of vector space V (F). W is a subspace of W if and only if* $a \in F$ *and* $\alpha, \beta \in V \Rightarrow a\alpha + \beta \in W$. *(K.U. M13, S. V. U. M15)*

Proof. *Condition is necessary.* W is a subspace of V (F)

$\Rightarrow$ W (F) is a vector space. $\therefore\ a \in F, \alpha \in W \Rightarrow a\alpha \in W$

Further $a\alpha \in W, \beta \in W \Rightarrow a\alpha + \beta \in W$

Condition is sufficient. W is a non-empty subset of V satisfying the condition.

$a \in F$ and $\alpha, \beta \in W \Rightarrow a\alpha + \beta \in W$

(*i*) Now taking $a = -1$, for $\alpha \in W$ we have $(-1)\alpha + \alpha \in W \Rightarrow \bar{O} \in W$

(*ii*) Again $a \in F, \alpha, \bar{O} \in W \Rightarrow a\alpha + \bar{O} \in W \Rightarrow a\alpha \in W$

$\therefore$ W is closed with respect to scalar multiplication.

(*iii*) $-1 \in F$ and $\alpha, \bar{O} \in W \Rightarrow (-1)\alpha + \bar{O} \in W \Rightarrow -\alpha \in W$. $\therefore$ Inverse exists in W.

The remaining postulates of vector space hold good in W since they hold in V of which W is a subset.

Hence W is a subspace of V (F).

SOLVED PROBLEMS

Ex.1. *The set W of ordered triads (x, y, 0) where* $x, y \in F$ *is a subspace of* V_3 *(F).*

Sol: Let $\alpha, \beta \in W$ where $\alpha = (x_1, y_1, 0)$ and $\beta = (x_2, y_2, 0)$ for some $x_1, y_1, x_2, y_2 \in F$.

Let $a, b \in F$. $\therefore\ a\alpha + b\beta = a(x_1, y_1, 0) + b(x_2, y_2, 0)$

$= (ax_1, ay_1, 0) + (bx_2, by_2, 0) = (ax_1 + bx_2, ay_1 + by_2, 0)$

Clearly $ax_1 + bx_2, ay_1 + by_2 \in F \Rightarrow a\alpha + b\beta \in W$ for all $a, b \in F$ and $\alpha, \beta \in W$

Hence W is subspace of V_3 (F).

Ex.2. *Let p, q, r be the fixed elements of a field F. Show that the set W of all triads (x, y, z) of elements of F, such that* $px+qy+rz=0$ *is a vector subspace of* $V_3(F)$. ***(S. V. U. S97)***

Sol: By definition $W \neq \phi$. Let $\alpha, \beta \in W$ where $\alpha = (x_1, y_1, z_1)$ and $\beta = (x_2, y_2, z_2)$ for some $x_1, x_2, y_1, y_2, z_1, z_2 \in F$.

Then by definition $px_1 + qy_1 + rz_1 = 0$ ... (1) $px_2 + qy_2 + rz_2 = 0$ ... (2)

If $a, b \in F$, then we have $a\alpha + b\beta = a(x_1, y_1, z_1) + b(x_2, y_2, z_2)$

$= (ax_1, ay_1, az_1) + (bx_2, by_2, bz_2) = (ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2)$

Now $p(ax_1 + bx_2) + q(ay_1 + by_2) + r(az_1 + bz_2)$

$= a(px_1 + qy_1 + rz_1) + b(px_2 + qy_2 + rz_2) = a.0 + b.0 = 0$ (by (1) and (2))

$\therefore a\alpha + b\beta = (ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2) \in W$. Hence W is a subspace of $V_3(F)$.

Ex. 3. *Let R be the field of real numbers and W = { (x, y, z) / x, y, z are rational numbers }. Is W a subspace of* $V_3(R)$.

Sol: Let $\alpha = (2,3,4)$ be an element of W ; $a = \sqrt{7}$ is an element of R.

Now $a\alpha = \sqrt{7}(2,3,4) = (2\sqrt{7}, 3\sqrt{7}, 4\sqrt{7}) \notin W$.

($\therefore 2\sqrt{7}, 3\sqrt{7}, 4\sqrt{7}$ not rational numbers).

$\therefore$ W is not closed under scalar multiplication.

Hence W is not a subspace of $V_3(R)$.

Ex. 4. *Let V be the vector space of all polynomials in an indeterminate x over the field F. Let W be a subset of V consisting of all polynomials of degree* $\leq n$. *Then W is a subspace of V.*

Sol: Let $\alpha, \beta \in W$. Then α, β are polynomials over F of degree $\leq n$.

If $a, b \in F$ then $a\alpha + b\beta$ will also be a polynomial of degree $\leq n$.

$\therefore a\alpha + b\beta \in W \Rightarrow$ W is a subspace of V.

Ex. 5. *Let V be the set of all* $n \times n$ *matrices and F be the field. If W is the subset of* $n \times n$ *symmetric matrices in V, show that W is a subspace of V (F).*

Sol: $O_{n\times n} = \begin{vmatrix} 0 & 0 & 0 & .... & 0 \\ 0 & 0 & .... & .... & 0 \\ .... & .... & .... & .... & .... \\ 0 & 0 & .... & .... & 0 \end{vmatrix}$ is a symmetric matrix

since $i-j$ th element of $O = 0 = j-i$ th element of 0

$\therefore W \neq \phi$. Let $a, b \in F$ and $P, Q \in W$ where $P = [p_{ij}], Q = [q_{ij}]$

$\because$ P, Q are symmetric, $p_{ij} = p_{ji}$ and $q_{ij} = q_{ji}$ ... (1)

$\therefore aP + bQ = a[p_{ij}] + b[q_{ij}] = [ap_{ij}] + [bq_{ij}] = [ap_{ij} + bq_{ij}]_{n\times n}$

The $i-j$ th element in $aP+bQ = ap_{ij} + bq_{ij} = ap_{ji} + bq_{ji}$ (by(1))

$= j-i$ th element of $aP+bQ$

$\Rightarrow aP+bQ$ is a symmetric $n \times n$ matrix $\Rightarrow aP+bQ \in W$

$\therefore$ W is a vector subspace of V (F).

Ex. 6. *Let F be a field and A be a $m \times n$ matrix over F.*

$F_{1\times m}$ *is the set of all* $1 \times m$ *matrices defined over f forming the vector space* $F_{1\times m}(F)$.

Define $W = \{ X = [x_1, x_2, \ldots x_n] \in F_{1\times m} \; 1 \times A = O_{1\times n} \}$.

Prove that W is a subspace of $F_{1\times m}(F)$.

Sol: Let $X, Y \in W$ and $a, b \in F$. By def. of W, $XA = O_{1\times n}$ and $YA = O_{1\times n}$

Now $(aX+bY)\,A = (aX)\,A + (bY)\,A = a\,(XA) + b\,(YA) = a\,O_{1\times n} + b\,O_{1\times n} = O_{1\times n}$

$\therefore aX+bY \in W$ $\therefore$ W is a subspace of $F_{1\times m}$ (F).

Note. W is called a solution space of XA = O.

EXERCISE 1 (*b*)

1. Let R be the field of real numbers. Show the set of triads that
   (1) $\{(x, 2y, 3z) / x, y, z \in R\}$ ***(S. K. U. 2013)***
   (2) $\{(x, x, x) / x \in R\}$ form the subspaces of $R^3(R)$.
2. Let $V = R^3 = \{ (x,y,z) / x,y,z \in R \}$ and W be the set of triads (x,y,z) such that $x-3y+4z = 0$. Show that W is a subspace of V (R).
3. Show that the set W of the elements of the vector space $V_3(R)$ of the form $(x+2y, y, -x+3y)$ where $x, y \in R$ is a subspace of V_3 (R).
4. Prove that the set of solutions (x,y,z) of the equation $x+y+2z = 0$ is a subspace of the space R^3 (R).
5. Show that the solutions of the differential equation $(D^2 - 5D + 6)\,y = 0$ is a subspace of the vector space of all real-valued continuous functions over R.
   [**Hint.** Let $y = f(x)$ and $y = g(x)$ be two solutions.
   $\therefore D^2 f(x) - 5Df(x) + 6f(x) = 0$ ... (1), $D^2 g(x) - 5Dg(x) + 6g(x) = 0$ ... (2)
   (1) a + (2) b gives : $D^2[af(x)+bg(x)] - 5D[af(x)+bg(x)] + 6[af(x)+bg(x)] = 0$
   $\therefore af(x)+bg(x)$ is also a solution of D.E. etc.]
6. Show that the subset W defined below is not a subspace of R^3 (R).
   $W = \{ (a,b,c) \mid a^2 + b^2 + c^2 \le 1 \}$
7. Let V be the vector of $n \times n$ matrices over a field F. Show that W the set of matrices which commute with a given matrix T is a subspace of V.
   $W = \{ A = [a_{ij}] \in V : AT = TA \}$

8. V is the vector space of 2×2 matrices over a field F. Show that W is not a subspace of V where (*i*) $W = \{ A \in V \mid \det. A = 0 \}$ (*ii*) $W = \{ A \in V \mid A^2 = A \}$

[**Hint.** (*i*) Take $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ in W.

$A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ with det. $(A + B) \neq 0$ etc.

[**Hint.** (*ii*) Clearly $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in W$. As $(2\,I)^2 \neq 2\,I$ scalar multiplication in W fails].

9. Let V be the set of all real valued functions defined on $[-1,1]$. Which of the following are subspaces of V (R).

(*i*) $W_1 = \{ f \in V \mid f(0) = 0 \}$ (*ii*) $W_2 = \{ f \in V \mid f(1) = f(-1) \}$

(*iii*) $W_3 = \{ f \in V \mid f(x) = 0 \text{ if } x < 0 \}$ (*iv*) $W_4 = \{ f \in V \mid f(x) = f(-x) \}$

10. W is the subset of C^3. Which of the following are the subsapces of C^3 (C).

(*i*) $W = \{ (\alpha_1, \alpha_2, \alpha_3) \in C^3 \mid \alpha_1 \text{ is real} \}$

(*ii*) $W = \{ (\alpha_1, \alpha_2, \alpha_3) \in C^3 \mid \text{either } \alpha_1 = 0 \text{ or } \alpha_2 = 0 \}$

11. Let x be an indeterminate over the field F. Which of the following are subspaces of F $[x]$ over F ?

(*i*) All monic polynomials of degree atmost 10.

(*ii*) All polynomials having α and β in F as roots. (*iii*) All polynomials divisible by x.

(*iv*) All polynomials $f(x)$ such that $2 f(0) = f(1)$.

12. Prove that $W = \{ \lambda (1, 1, 1) \mid \lambda \in R \}$ is a subspace of R^3.

ALGEBRA OF SUBSPACES

1.16. Theorem. *The intersection of any two subspaces W_1 and W_2 of vector space V (F) is also a subspace.* *(O.U. A 2014)*

Proof. W_1 and W_2 are subspaces V (F)

$\Rightarrow \bar{O} \in W_1$ and $\bar{O} \in W_2 \Rightarrow \bar{O} \in W_1 \cap W_2$ $\quad\therefore\ W_1 \cap W_2 \neq \phi$

Let $a, b \in F$ and $\alpha, \beta \in W_1 \cap W_2$ $\quad\therefore\ \alpha, \beta \in W_1$ and $\alpha, \beta \in W_2$

Now $a, b \in F$ and $\alpha, \beta \in W_1 \Rightarrow a\alpha + b\beta \in W_1$

$a, b \in F$ and $\alpha, \beta \in W_2 \Rightarrow a\alpha + b\beta \in W_2$

$\therefore\ a\alpha + b\beta \in W_1 \cap W_2$ $\quad\therefore\ W_1 \cap W_2$ is a subspace of V (F)

1.17. Theorem. *The intersection of any family of subspaces of a vector space is also a subspace.*

Proof. Let V (F) be a vector space.

Let $W_1, W_2, W_3 \ldots W_n$ be the n subspaces of V (F).

Let $W = \bigcap_{i=1}^{n} W_i = \{\alpha \in V : \alpha \in W_n \ \forall\ n\}$

$\bar{O} \in W_i$ for $i = 1, 2, \ldots, n \Rightarrow \bar{O} \bigcap_{1}^{n} W_i$ and $\bigcap_{i=1}^{n} W_i \neq \phi$

Let $\alpha, \beta \in \bigcap_{i=1}^{n} W_i \Rightarrow \alpha, \beta \in W_n \ \forall\ n$

Since each W_n is a subspace we have $a, b \in F$ and $\alpha, \beta \in W_n \Rightarrow a\alpha + b\beta \in W_n \ \forall\ n$

$\Rightarrow a\alpha + b\beta \in \bigcap_{1}^{n} W_i$. Hence $\bigcap_{i=1}^{n} W_i$ is a subspace of V (F).

Note : The union of two subspaces of V (F) may not be a subspace of V (F).

e.g. Let W_1 and W_2 be two subspaces of V_3 (R) given by

$$W_1 = \{(0, y, 0) \mid y \in R\},\ W_2 = \{(0, 0, z) \mid z \in R\}$$

$\therefore\ W_1 \cup W_2 \{(0, y, 0) \cup (0, 0, z) \mid y, z \in R\}$

Now $(0, y, 0) + (0, 0, z) = (0, y, z) \notin W_1 \cup W_2$

Since neither $(0, y, z) \in W_1$ nor $(0, y, z) \in W_2$

Thus $W_1 \cup W_2$ is not closed under vector addition.

$\therefore\ W_1 \cup W_2$ is not a subspace of V (F).

1.18. Theorem. ***The union of two subspaces is a subspace if and only if one is contained in the other.***

Proof. Let W_1 and W_2 be two subspaces of V (F).

Condition is necessary. Let $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$

$\therefore\ W_1 \cup W_2 = W_2$ or $W_1 \Rightarrow W_1 \cup W_2$ is a subspace of V (F)

Condition is sufficient. Let $W_1 \cup W_2$ be a subspace.

Let us suppose that $W_1 \not\subset W_2$ and $W_2 \not\subset W_1$

Now $W_1 \not\subset W_2 \Rightarrow$ there exists $x \in W_1$ and $x \notin | W_2$ ... (1)

$W_2 \not\subset W_1 \Rightarrow$ there exists $y \in W_2$ and $y \notin W_1$ ... (2)

$\therefore\ x \in W_1 \cup W_2$ and $y \in W_1 \cup W_2$

$\Rightarrow x + y \in W_1 \cup W_2$ ($\because\ W_1 \cup W_2$ is a subspace)

$\Rightarrow x + y \in W_1$ or $x + y \in W_2$.

Now $x + y, x \in W_1$ (subspace) $\Rightarrow 1(x + y) + (-1)\ x \in W_1 \Rightarrow y \in W_1$ ... (3)

Similarly $x + y, y \in W_2$ (subspace) $\Rightarrow 1(x + y) + (-1)\ y \in W_2 \Rightarrow x \in W_2$ ... (4)

Thus (3) and (4) contradict (2) and (1). $\therefore$ Either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

1.19. LINEAR SUM OF TWO SUBSPACES

(S. V. U. 2011)

Definition. *Let* W_1 *and* W_2 *be two subspaces of the vector space V (F). Then the linear sum of the subspaces* W_1 *and* W_2, *denoted by* $W_1 + W_2$, *is the set of all sums* $\alpha_1 + \alpha_2$ *such that* $\alpha_1 \in W_1$, $\alpha_2 \in W_2$ *i.e.,* $W_1 + W_2 = \{\alpha_1 + \alpha_2 / \alpha_1 \in W_1, \alpha_2 \in W_2\}$

1.20. Theorem. *If* W_1 *and* W_2 *are any two subspaces of a vector space (F) then*

(i) $W_1 + W_2$ *is a subspace of V (F).*

(ii) $W_1 \subseteq W_1 + W_2$ *and* $W_2 \subseteq W_1 + W_2$. *(S. V. U. 11, A. N. U. 94 M, M 13, S. K. U. M2015)*

Proof. *(i)* Let $\alpha, \beta \in W_1 + W_2$. Then

$\alpha = \alpha_1 + \alpha_2$ and $\beta = \beta_1 + \beta_2$ where $\alpha_1, \beta_1 \in W_1$ and $\alpha_2, \beta_2 \in W_2$.

If $a, b \in F$ then $a\alpha_1, b\beta_1 \in W_1$ ($\because$ W_1 subspace)

and $a\alpha_2, b\beta_2 \in W_2$ ($\because$ W_2 subspace)

Now $a\alpha + b\beta = a(\alpha_1 + \alpha_2) + b(\beta_1 + \beta_2) = (a\alpha_1, b\beta_1) + (a\alpha_2, b\beta_2) \in W_1 | W_2$

$\therefore$ $a, b \in F$ and $\alpha, \beta \in W_1 + W_2 \Rightarrow a\alpha + b\beta \in W_1 + W_2$

Hence $W_1 + W_2$ is a subspace of V (F).

(ii) $\alpha_1 \in W_1$ and $\bar{O} \in W_2 \Rightarrow \alpha_1 + \bar{O} \in W_1 + W_2$. $\therefore$ $\alpha_1 \in W_1 \Rightarrow \alpha_1 \in W_1 + W_2$

$\Rightarrow$ $W_1 \subseteq W_1 + W_2$. Similarly, $W_2 \subseteq W_1 + W_2$. Hence $W_1 \cup W_2 \subseteq W_1 + W_2$.

SOLVED PROBLEMS

Ex. 1. *Let V be the vector space of all functions from R into R. Let* S_e *be the subset of even functions,* $f(-x) = f(x)$ S_o *be the subset of odd functions,* $f(-x) = -f(x)$. *Prove that* (1) S_e *and* S_o *are subspaces of V* (2) $S_e + S_o = V$ (3) $S_e \cap S_o = \{\bar{o}\}$

Sol: (1) Let $f_e, g_e \in S_e$ and $a, b \in R$

$\therefore$ $(af_e + bg_e)(-x) = af_e(-x) + bg_e(-x) = af_e(x) + bg_e(x) = (af_e + bg_e)(x)$

$\Rightarrow$ $af_e + bg_e$ is an even function.

$\therefore$ $f_e, g_e \in S_e, a$ and $b \in R \Rightarrow af_e + bg_e \in S_e$

Hence S_e is a subspace of V. Similarly we can prove that S_o is a subspace of V.

(2) Since S_e and S_o are subspaces of V. $S_e + S_o$ is also a subspace of V

$\therefore$ $S_e + S_o \subseteq V$. Let $g_e(x) = \frac{1}{2}[f(x) + f(-x)]$, $h_o(x) = \frac{1}{2}[f(x) - f(-x)]$

Clearly g_e is an even function and h_o is an odd function

Now $f(x) = \frac{1}{2}[f(x) + f(-x)] + \frac{1}{2}[f(x) - f(-x)] = g_e(x) + h_o(x) = (g_e + h_o)(x)$

$\Rightarrow$ $f = g_e + h_o$ where $g_e \in S_e, h_o \in S_o$

Thus $f \in V \Rightarrow f \in S_e + S_o$ $\therefore$ $V \subseteq S_e + S_o$. Hence $S_e + S_o = V$.

(3) Let $\bar{O}$ denote the zero function *i.e.*, $\bar{O}(x)=0\,\forall x\in R$

Let $f\in S_e\cap S_o$ then $f(-x)=f(x)$ and $f(-x)=-f(x)$

$\therefore\ f(x)=-f(x)\Rightarrow\ 2f(x)=0$

$\Rightarrow f(x)=0=\bar{O}(x)\Rightarrow f=\bar{O}$. Hence $S_e\cap S_o=\{\bar{O}\}$.

Ex.2. *Let* W_1 *and* W_2 *be subspaces of a vector space V such that* $W_1+W_2=V$ *and* $W_1\cap W_2=\{\bar{O}\}$. *Prove that for each vector* α *in V there are unique vectors* $\alpha_1\in W_1,\alpha_2\in W_2$ *such that* $\alpha=\alpha_1+\alpha_2$. ***(A.U. 2011)***

Sol: $\alpha\in V\Rightarrow\alpha\in W_1+W_2$ $(\because\ V=W_1+W_2)$

$\Rightarrow\ \alpha=\alpha_1+\alpha_2$ where $\alpha_1\in W_1,\alpha_2\in W_2$

If possible let $\alpha=\beta_1+\beta_2$ where $\beta_1\in W_1,\beta_2\in W_2$

$\therefore\ \alpha_1+\alpha_2=\beta_1+\beta_2$. $\alpha_1-\beta_1=\beta_2-\alpha_2$

Now $\alpha_1-\beta_1\in W_1$ and $\beta_2-\alpha_2\in W_2$

$\Rightarrow\ \alpha_1-\beta_1\in W_1$ and $\alpha_1-\beta_1\in W_2$

$\Rightarrow\ \alpha_1-\beta_1\in W_1\cap W_2\ \Rightarrow\ \alpha_1-\beta_1=\bar{O}$ $(\because\ W_1\cap W_2=\{\bar{O}\})$

$\Rightarrow\ \beta_2-\alpha_2=\bar{O}$

$\Rightarrow\ \alpha_1=\beta_1$ and $\alpha_2=\beta_2$. Hence $\alpha=\alpha_1+\alpha_2$ is unique.

1.21. LINEAR COMBINATION OF VECTORS

Definition. *Let V (F) be a vector space. If* $\alpha_1,\alpha_2,...,\alpha_n\in V$ *then any vector* $\gamma=a_1\alpha_1+a_2\alpha_2+...+a_n\alpha_n$ *where* $a_1,a_2,...,a_n\in F$ *is called a linear combination of the vectors* $\alpha_1,\alpha_2,...,\alpha_n$.

Note.1. γ is a vector belonging to V (F).

1.22. LINEAR SPAN OF A SET

Definition. *Let S be a non-empty subset of a vector space V (F). The linear span of S is the set of all possible linear combinations of all possible finite subsets of S.*

The linear span of S is denoted by L (S)

$\therefore\ L(S)=\{\gamma:\gamma=\sum a_i\alpha_i,a_i\in F,\alpha_i\in S\}$

Note.1. S may be a finite set but L (S) is infinite set.

L (S) is said to be generated or spanned by S

2. If S is an empty subset of V then we define L (S) = {0}

3. $S\subseteq L\{S\}$

1.23. Theorem. *The linear span L(S) of any subset S of a vector space V (F) is a subspace of v (F).* ***(A. U. M 14, A. N. U. M 13, S. V. U. 94 S)***

Proof. Let $\alpha,\beta\in L(S)$ and $a,b\in F$

$\therefore\ \alpha = a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m, \qquad \beta = b_1\beta_1 + b_2\beta_2 + ... + b_n\beta_n$

where $a'_i s, b'_i s \in F$ and $\alpha'_i s, \beta'_i s \in S$

$\therefore\ a\alpha + b\beta = a(a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m) + b(b_1\beta_1 + b_2\beta_2 + ... + b_n\beta_n)$

$= (a\,a_1)\,\alpha_1 + (a\,a_2)\,\alpha_2 ... + (aa_m)\alpha_m + (b\,b_1)\beta_1 + (b\,b_2)\beta_2 + ... + (bb_n)\beta_n$

$\Rightarrow a\alpha + b\beta$ is a linear combination of finite set

$\alpha_1, \alpha_2 ... \alpha_m, \beta_1, \beta_2 ... \beta_n$ of the elements of S. $\Rightarrow (a\alpha + b\beta) \in L(S)$

Thus $a, b \in F$ and $\alpha, \beta \in L(S) \Rightarrow (a\alpha + b\beta) \in L(S)$ $\therefore$ L (S) is a subspace of V (F).

1.24. Theorem. ***Let S be a non-empty subset of the vector space V (F). The linear span L (S) is the intersection of all subspaces of V which contain S.***

Proof. Let W be the subspace of V (F) containing S. Every linear combination of finite set of elements of S is an element of W as W is closed.

But the set of every linear combination of finite set of elements of S is L (S).

$\therefore\ L(S) \subseteq W$

$\therefore\ L(S) \subseteq$ Intersection of all subspaces of V containing S $\quad (\because\ S \subseteq L(S))$

The intersection of all subspaces of V containing $S \subseteq L(S)$

Hence L (S) is the intersection of all subspaces of V containing S.

SOLVED PROBLEMS

Ex.1. *Express the vector* $\alpha = (1, -2, 5)$ *as a linear combination of the vectors* $e_1 = (1,1,1), e_2 = (1,2,3)$ *and* $e_3 = (2, -1, 1)$ ***(A.U. M 2014, A. N. U. M 2014)***

Sol: Let $\alpha = (1, -2, 5) = x(1,1,1) + y(1,2,3) + z(2, -1, 1)$

$= (x + y + 2z, x + 2y - z, x + 3y + z)$

Hence $x + y + 2z = 1$, $x + 2y - z = -2$, $x + 3y + z = 5$

Reducing to echelon form we get $\alpha + \beta + 2\gamma = 1$, $\beta - 3\gamma = -3$, $5\gamma = 10$

These equations are consistant and have a solution given by $\alpha = -6, \beta = 3, \gamma = 2$

Hence $\alpha = -6\,e_1 + 3e_2 + 2e_3$

Ex.2. *Show that the vector* $\alpha = (2, -5, 3)$ *in* R^3 *cannot be expressed as a linear combination of the vectors* $e_1 = (1, -3, 2), e_2 = (2, -4, -1)$ *and* $e_3 = (1, -5, 7)$

Sol: Let $\alpha = ae_1 + be_2 + ce_3$

i.e., $(2, -5, 3) = a(1, -3, 2) + b(2, -4, -1) + c(1, -5, 7)$

$= (a + 2b + c, -3a - 4b - 5c, 2a - b + 7c)$

$\Rightarrow \quad a + 2b + c = 2, \quad -3a - 4b - 5c = -5, \quad 2a - b + 7c = 3$

Now reducing to echelan form we get $a + 2b + c = 2$, $2b - 2c = 1$, $0 = 3$

As the system is inconsistant, the equations have no solution. Hence α cannot be expressed as a l . c of e_1, e_2, e_3 .

Ex.3. *The subset* $S = \{(1,0,0), (0,1,0), (0,0,1)\}$ *of* $V_3(R)$ *generates the entire vector space* $V_3(R)$.

Sol: Let $(a, b, c) \in V$ then we can write $(a, b, c) = a(1,0,0) + b(0,1,0) + c(0,0,1)$

$\Rightarrow (a, b, c) \in L(S) \Rightarrow V \subseteq L(S)$. But $L(S) \subseteq V$ Hence $L(S) = V$.

Ex.4. *In the vector space* $R^3(R)$. *Let* $\alpha = (1,2,1), \beta = (3,1,5), \gamma = (3,-4,5)$. *Show that subspace spanned by* $S = \{\alpha, \beta\}$ *and* $T = \{\alpha, \beta, \gamma\}$ *are the same.*

Sol: $L(T) = l$. c . of the vectors of $T = \{a\alpha + b\beta + c\gamma / a, b, c \in R\}$

Let $\gamma = x\alpha + y\beta \Rightarrow (3, -4, 5) = x(1, 2, 1) + y(3, 1, 5)$

$\therefore \quad x + 3y = 3,\ 2x + y = -4,\ x + 5y = 5$

$x = -3,\ y = 2$ satisfy all the three equations. $\therefore (3, -4, 5) = -3(1, 2, 1) + 2(3, 1, 5)$

$\therefore c(3, -4, 5) = -3c(1, 2, 1) + 2c(3, 1, 5) \Rightarrow c\gamma = -3c\alpha + 2c\beta$

$\therefore a\alpha + b\beta + c\gamma = a\alpha + b\beta - 3c\alpha + 2c\beta$

$= (a - 3c)\alpha + (b + 2c)\beta = p\alpha + q\beta$ when $p, q \in R$

$\therefore$ But $p\alpha + q\beta \in L(S)$. $\therefore \quad L(T) = L(S)$.

Ex.5. *In the vector space* $R^3(R)$ *let* $S = \{(1, 0, 0), (0, 1, 0)\}$. *Find* $L(S)$.

Sol: $L(S) = \{\alpha / \alpha = a(1,0,0) + b(0,1,0), a, b \in R\} = \{\alpha / \alpha = (a, b, 0); a, b \in R\}$

Geometrically the linear span is the plane $Z = 0$.

1.25. Theorem. *If* W_1 *and* W_2 *are two subspaces of a vector space* $V(F)$ *then* $L(W_1 \cup W_2) = W_1 + W_2$. *(A. N. U. 2013)*

Proof. Let $\alpha_1 \in W_1$ and $\bar{O} \in W_2$ $\therefore \alpha_1 \in W_1$ and $\bar{O} \in W_2 \Rightarrow \alpha_1 + \bar{O} \in W_1 + W_2$

$\therefore$ Every $\alpha_1 \in W_1 \Rightarrow \alpha_1 \in W_1 + W_2$

$\therefore W_1 \subseteq W_1 + W_2$. Similarly $W_2 \subseteq W_1 + W_2$. Hence $W_1 \cup W_2 \subseteq W_1 + W_2$

Also W_1 and W_2 subspaces of $V \Rightarrow W_1 \cup W_2$ subspace of $W_1 + W_2$.

$\therefore L(W_1 \cup W_2)$ is a subspace of $W_1 + W_2$. Again let $\alpha \in W_1 + W_2$

$\therefore$ By def. $\alpha = \alpha_1 + \alpha_2$ where $\alpha_1 \in W_1$ and $\alpha_2 \in W_2$ $\therefore \alpha_1, \alpha_2 \in W_1 \cup W_2$

Now, $\alpha = \alpha_1 + \alpha_2 = 1.\alpha_1 + 1.\alpha_2 = l$. c. of elements of $W_1 \cup W_2$

$\therefore \alpha \in L(W_1 \cup W_2)$ $\therefore W_1 + W_2 \subseteq L(W_1 \cup W_2)$ ... (i)

Also we know that $L(W_1 \cup W_2)$ is the smallest subsapce containing $W_1 \cup W_2$ and $W_1 \cup W_2 \subseteq W_1 + W_2$. $\therefore L(W_1 \cup W_2) \subseteq W_1 + W_2$ ... (ii)

From (i) and (ii) $L(W_1 \cup W_2) = W_1 + W_2$

1.26. Theorem. *If S is a subset of a vector space V (F) then prove that*
(I) *S is a subspace of V* $\Leftrightarrow$ *L (S) = S* *(S. V. U. 2001/O)* **(II)** *L (L (S)) = L (S).*

(A. N. U. M 2013, S. K. U. 2014, M 2015)

Proof. I. (*i*) Let S be a subspace of V,

Let $\alpha \in L(S)$. Then $\alpha = a_1 \alpha_1 + a_2 \alpha_2 + ... + a_m \alpha_m$ where $a_1, a_2, ... a_m \in F$

and $\alpha_1, \alpha_2 ... \alpha_m \in S$

$\because$ S is a subspace of V, it is closed w.r. to scalar multiplication and vector addition.

$\therefore a_1\alpha_1 + a_2\alpha_2 ... + a_m\alpha_m = \alpha \in S$

Thus $\alpha \in L(S) \Rightarrow \alpha \in S$ $\therefore L(S) \subseteq S$

Let $\beta \in S$. Now $\beta = 1.\beta =$ *l.c.* of infinite elements of S

$\Rightarrow \beta \in L(S)$ $\therefore S \subseteq L(S)$. Hence L (S) = S.

(*ii*) Suppose L (S) = S.

We know that L (S) is a subspace of *v*. $\Rightarrow$ S is a subspace of V.

II. We know that L (S) is a subspace of V. By **I**, L (L (S)) = L (S).

1.27. Theorem. *If S, T are the subsets of a vector space V (F), then*
(I) $S \subseteq T \Rightarrow L(S) \subseteq L(T)$ **(II)** $L(S \cup T) = L(S) + L(T)$

(S. V. U. M14, S. K. U. M 15, 11, A.N. U. M 2014, 95)

Proof. (I) Let $\alpha \in L(S)$

$\therefore \alpha = a_1\alpha_1 + a_2\alpha_2 + ... + a_n\alpha_n$ where $a_i's \in F$ and $\alpha_i's \in S$

$\because S \subseteq T$, then $\{\alpha_1, \alpha_2 ... \alpha_n\} \subseteq T$

$\therefore \alpha = l.c.$ of finite subset of T $\Rightarrow \alpha \in L(T)$

Hence $L(S) \subseteq L(T)$.

(II) Let $\alpha \in L(S \cup T)$

$\therefore \alpha = a_1\alpha_1 + a_2\alpha_2 ... + a_m\alpha_m + b_1\beta_1 + b_2\beta_2 ... + b_n\beta_n$ where $a_1, a_2 ... a_m, b_1, b_2 ... b_n \in F$

$\alpha_1, \alpha_2 ... \alpha_m \in S$ and $\beta_1, \beta_2 ... \beta_n \in T$.

(*i*) But $a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m \in L(S)$, $b_1\beta_1, b_2\beta_2 + ... + \beta_n \in L(T)$

$\therefore \alpha =$ an element of L (S) + an element of L (T)

$\therefore \alpha \in L(S) + L(T) \Rightarrow L(S \cup T) \subseteq L(S) + L(T)$ ... (1)

(*ii*) Let $\alpha \in L(S) + L(T)$

$\therefore \alpha = \gamma + \delta$ where $\gamma \in L(S)$, $\delta \in L(T)$ $\therefore \gamma =$ *l.c.* of a finite no. of elements of S

$\delta =$ *l.c.* of a finite no. of elements of T

$\therefore \alpha = \gamma + \delta =$ *l.c.* of a finite no. of elements of $S \cup T \Rightarrow \alpha = \gamma + \delta \in L(S \cup T)$

$\therefore L(S) + L(T) \subseteq L(S \cup T)$ ... (2)

Hence from (1) and (2) : $L(S \cup T) = L(S) + L(T)$.

EXERCISE 1 (c)

1. Show that each of the following set of vectors generates $R^3(R)$.

(*i*) $\{(1,0,0),(1,1,0),(1,1,1)\}$ (*ii*) $\{(1,2,3),(0,1,2),(0,0,1)\}$ (*iii*) $\{(1,2,1),(2,1,0),(1,-1,2)\}$

2. If $\alpha=(1,2,-1),\beta=(2,-3,2),\gamma=(4,1,3)$ and $\delta=(-3,1,2)$ are the vectors of $V_3(R)$ show that $L(\{\alpha,\beta\})\neq L(\{\gamma,\delta\})$.

3. Prove that the subspace spanned by a non-empty subset S of a vector space V is the set of all linear combinations of vectors in S.

1.28. LINEAR DEPENDENCE OF VECTORS

Definition. *Let V (F) be a vector space. A finite subset* $\{\alpha_1,\alpha_2,...,\alpha_n\}$ *of vectors of V is said to be a linearly dependent (L.D.) set if there exist scalars* $\alpha_1,\alpha_2,...,\alpha_n\in F$, *not all zero, such that* $a_1\alpha_1+a_2\alpha_2+...+a_n\alpha_n=\bar{O}$.

1.29. LINEAR INDEPENDENCE OF VECTORS

Definition. Let V (F) be a vector space. A finite subset $\{\alpha_1,\alpha_2,...,\alpha_n\}$ of vectors of V is said to be linearly independent (L.I,) if every relation of the form

$$a_1\alpha_1+a_2\alpha_2+...+a_n\alpha_n=\bar{O},\ a_i's\in F$$

$$\Rightarrow a_1=0,a_2=0,...,a_n=0.$$

1.30. Theorem. ***Every superset of a linearly dependent (L.D) set of vectors is linearly dependent (L.D.).*** *(S. V. U. M 2015)*

Proof. Let $S=\{\alpha_1,\alpha_2,...,\alpha_n\}$ be a linearly dependent set of vectors.

$\therefore$ There exist scalars $a_1,a_2,...,a_n\in F$, not all zero such that

$$a_1\alpha_1+a_2\alpha_2+...+a_n\alpha_n=\bar{O} \qquad ...(1)$$

Let $S'=\{\alpha_1,\alpha_2...\alpha_n,\beta_1,\beta_2...\beta_n\}$ be a super set of S. then (1) can be re-written as

$$a_1\alpha_1+a_2\alpha_2+...+a_n\alpha_n+0\beta_1,0\beta_2+...+0\beta_n=\bar{O} \qquad ...(2)$$

In (2) all the scalars are not zero $\Rightarrow$ S' is linearly dependent (L.D.)

Hence any super set of an L.D. set is L.D.

1.31. Theorem. ***Every non-empty subset of a linearly independent (L.I.) set of vectors is linearly independent (L.I.)*** *(O. U. M 15, N. U. 94 M)*

Proof. Let $S=\{\alpha_1,\alpha_2,...,\alpha_m\}$ be a L.I. set of vectors

Let us consider the subset $\{\alpha_1,\alpha_2,...,\alpha_k\}$ where $1\le k\le m$.

Now $a_1\alpha_1+a_2\alpha_2+...+a_k\alpha_k=\bar{O}$

$\Rightarrow a_1\alpha_1 + a_2\alpha_2 + ... + a_k\alpha_k + 0\alpha_{k+1} + ... + 0\alpha_m = \bar{O}$

$\Rightarrow a_1 = 0, a_2 = 0, a_k = 0$ ($\because$ S is L.I. set.)

Hence the subset $\{\alpha_1, \alpha_2, ..., \alpha_k\}$ is L.I.

1.32. Theorem. ***A set of vectors which contains zero vector is linearly dependent.***

Proof. Let $\alpha_1 = \bar{O}$ and $\alpha_2, \alpha_3 ... \alpha_m \neq \bar{O}$

Then $1\alpha_1 + 0\alpha_2 + 0\alpha_3 ... + \bar{O}\alpha_m = \bar{O}$, $a_1 = 1 \in F$ with atleast one scalar $a_1 \neq 0$

$\Rightarrow \{\bar{O}, \alpha_2, ..., \alpha_m\}$ L.D. set.

Hence the set of vectors containing zero vector is linearly dependent (L.D.).

Note. A L.I. subset of a vector space V (F) does not contain zero vector.

1.33. Theorem. ***A single non - zero vector forms a linearly independent (L.I.) set.***

Proof. Let L = $\{\alpha\}$ be a subset of V (F) where $a \neq \bar{O}$

If $a \in F$ then $a\alpha = \bar{O} \Rightarrow a = 0$. $\therefore$ The set S is linearly independent.

Note. $\{\alpha\}$ is a L.D. set.

SOLVED PROBLEMS

Ex.1. *Show that the system of vectors (1,3,2),* $(1,-7,-8), (2,1,-1)$ *of* $V_3(R)$ *is linearly dependent.* ***(O. U. A14, A. U. M14, S. V. U. M 12, A. N. U. M 2013, S. K. .U. 2011)***

Sol. Let $a, b, c \in R$, then $a(1,3,2) + b(1,-7,-8) + c(2,1,-1) = \bar{O}$

$\Rightarrow (a + b + 2c, 3a - 7b + c, 2a - 8b - c) = (0,0,0)$

$\Rightarrow a + b + 2c = 0, 3a - 7b + c = 0, 2a - 8b - c = 0$

$\Rightarrow a = 3, b = 1, c = -2$ $\therefore$ The given vectors are linearly dependent.

Ex.2. *Show that the system of vectors* $(1,2,0), (0,3,1), (-1,0,1)$ *of* $V_3(Q)$ *is L.I. where Q is the field of rational numbers.*

Sol. Let $x, y, z, \in Q$ then $x(1,2,0) + y(0,3,1) + z(-1,0,1) = \bar{O}$

$\Rightarrow (x - z, 2x + 3y, y + z) = (0,0,0)$

$\Rightarrow x - z = 0, 2x + 3y = 0, y + z = 0 \Rightarrow x = 0, y = 0, z = 0$

Hence the system is L.I.

Ex.3. *If two vectors are linearly dependent, prove that one of them is a scalar multiple of the other.*

Sol: Let α, β be two L.D. vectors of V (F)

Then there exist $a, b \in F$, not both zero such that $a\alpha + b\beta = \bar{O}$

Let $a \neq 0$, then $\alpha = \left(-\frac{b}{a}\right)\beta$ $\Rightarrow$ α is a scallar multiple of β

If $b \neq 0$, similarly we get $\beta = a$ scalar multiple of α.

Ex.4. *Prove that the vectors* (x_1, y_1) *and* (x_2, y_2) *of* $V_2(F)$ *are L.D. if* $x_1y_2 - x_2y_1 = 0$.

Sol. Let $a, b \in F$ then $a(x_1y_1) + b(x_2y_2) = (0,0)$

$\Rightarrow (a x_1 + b x_2, a y_1 + b y_2) = (0,0) \Rightarrow a x_1 + b x_2 = 0, a y_1 + b y_2 = 0$

These equations are consistant.

If $\det. \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} = 0 \Rightarrow x_1y_2 - x_2y_1 = 0$

Ex.5. *In the vector space* $V_n(F)$, *the system of n vectors* $e_1 = (1,0,0 \ldots 0)$, $e_2 = (0,1,0 \ldots 0)$, ..., $e_n = (0,0, \ldots 1)$ *is linearly independent where 1 is unity of f.*

Sol. Let $a_1, a_2, \ldots, a_n \in F$, then $a_1 e_1 + a_2e_2 + \ldots + a_ne_n = \bar{O}$

$\Rightarrow a_1(1,0, \ldots, 0) + a_2(0,1,0 \ldots) \ldots + a_n(0,0, \ldots, 0,1) = \bar{O}$

$\Rightarrow (a_1, a_2, \ldots, a_n) = (0,0, \ldots, 0) \Rightarrow a_1 = 0, a_2 = 0, \ldots a_n = 0$

$\Rightarrow$ The given set of vectors is linearly independent.

Ex.6. *Prove that the four vector* $\alpha = (1,0,0), \beta = (0,1,0), \gamma = (0,0,1), \delta = (1,1,1)$ *in* $V_3(C)$ *form L.D. set, but any three of them are L.I.* **(K.U. M 2014)**

Sol. Let $a, b, c, d \in C$ $\quad \therefore a\alpha + b\beta + c\gamma + d\delta = \bar{O}$

$\Rightarrow a(1,0,0) + b(0,1,0) + c(0,0,1) + d(1,1,1) = \bar{O}$

$\Rightarrow (a+d, b+d, c+d) = (0,0,0)$

$\Rightarrow a+d = 0, b+d = 0, c+d = 0 \quad \therefore a = -d, b = -d, c = -d$

Thus if $d = -k$, then $a = k, b = k, c = k$, showing that $\alpha + \beta + \gamma - \delta = \bar{O}$

$\therefore$ The four vectors $\alpha, \beta, \gamma, \delta$ are L.D.

(*ii*) But $a\alpha + b\beta + c\delta = \bar{O} \Rightarrow a(1,0,0) + b(0,1,0) + c(1,1,1) = \bar{O}$

$\Rightarrow (a+c, b+c, c) = (0,0,0) \Rightarrow a+c = 0, b+c = 0, c = 0,$

$\Rightarrow a = 0, b = 0, c = 0 \Rightarrow$ the vectors α, β, δ are L.I.

Similarly we can show that any other three vectors are L.I.

Ex.7. *If* α, β, γ *are linearly independent vectors of V (R) show that* $\alpha + \beta, \beta + \gamma, \gamma + \alpha$ *are also L.I.* **(O. U. 2001/O)**

Sol. Let $a, b, c \in R$.

$a(\alpha + \beta) + b(\beta + \gamma) + c(\gamma + \alpha) = \bar{O} \Rightarrow (a+c)\alpha + (a+b)\beta + (b+c)\gamma = \bar{O}$

Given α, β, γ are linearly independent.

$\Rightarrow a + 0.b + c = 0, \quad a + b + 0.c = 0, \quad 0.a + b + c = 0.$

Now the coefficient matrix A of these equations is $A=\begin{bmatrix}1 & 0 & 1\\ 1 & 1 & 0\\ 0 & 1 & 1\end{bmatrix}$,

where rank of A = 3 *i.e.,* equal to the no. of unknowns.

$\Rightarrow$ $a=0, b=0, c=0$ is the only solution of the given equations.

$\therefore$ $\alpha+\beta, \beta+\gamma, \gamma+\alpha$ are also L.I.

Ex.8. *Let* $F(x)$ *be the vector space of all polynomials over the field F. Show that the infinite set* $S = \{1, x, x^2, ...\}$ *is linearly independent.*

Sol. Let $S' = \{x^{m_1}, x^{m_2}, ..., x^{m_n}\}$ be any finite subset of having a vectors, where $m_1, m_2, ... m_n$ are non-negative integers.

Let $a_1, a_2 ... a_n \in F$

$\therefore$ $a_1x^{m^1} + a_2x^{m^2} + ... + a_nx^{m^n} = \bar{O}$ (*i.e.*zero polynomial)

Then by definition of equality of two vectors we have $a_1 = 0, a_2 = 0, ..., a_n = 0$

$\Rightarrow$ Every finite subset of S is L.I. and hence s is linearly independent.

Ex.9. *Let V be the vector space of* 2×3 *matrices over R. Show that the vectors A, B, C form L.I set where*

$$A=\begin{bmatrix}2 & 1 & -1\\ 3 & -2 & 4\end{bmatrix} \quad B=\begin{bmatrix}1 & 1 & -3\\ -2 & 0 & 5\end{bmatrix} \quad \textit{and} \quad C=\begin{bmatrix}4 & -1 & 2\\ 1 & -2 & -3\end{bmatrix}$$

Sol. Let $a, b, c \in R$ then for L.I.

$a A + b B = c C = 0 \quad \Rightarrow \quad a=0, b=0, c=0$

Now $a A + b B = c C = 0$

$$\Rightarrow a\begin{bmatrix}2 & 1 & -1\\ 3 & -2 & 4\end{bmatrix}+b\begin{bmatrix}1 & 1 & -3\\ -2 & 0 & 5\end{bmatrix}+c\begin{bmatrix}4 & -1 & 2\\ 1 & -2 & -3\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}2a+b+4c & a+b-c & -a-3b+2c\\ 3a-2b+c & -2a-2c & 4a+5b-3c\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$\Rightarrow \quad 2a+b+4c=0, \quad a+b-c=0, \quad -a-3b+2c=0$$
$$3a-2b+c=0, \quad -2a-2c=0, \quad 4a+5b-3c=0$$

These equations have only one solution $a=0, b=0, c=0$.

Ex.10. *Prove that the set* $\{1, i\}$ *is L.D. in the vector space C (C) but is L.I. in the vector space C (R).*

Sol. Since $(-i)\, i = 1$ with $(-i) \in C$, (one vector is a multiple of other)

$\Rightarrow$ the set $\{1, i\}$ in C (C) is L.D.

There exists no real number a such that $a\, i = 1$.

Hence $\{1, i\}$ is not L.D and so L.I in C (R).

Ex. 11. *Prove that the set* $\{(1,0,0,-1),(0,1,0,-1),(0,0,1,-1),(0,0,0,1)\}$ *is linearly independent in* $V_4(R)$ ***(O. U. 2011)***

Sol. Let $\alpha_1 = (1,0,0,-1), \alpha_2 = (0,1,0,-1), \alpha_3 = (0,0,1,-1), \alpha_4 = (0,0,0,1)$

Let $a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3 + a_4\alpha_4 = \bar{O}, \quad \bar{O} \in V_4(R)$

$\Rightarrow a_1(1,0,0,-1) + a_2(0,1,0,-1) + a_3(0,0,1,-1) + a_4(0,0,0,1) = (0,0,0,0)$

$\Rightarrow (a_1, a_2, a_3, -a_1 - a_2 - a_3 + a_4) = (0,0,0,0)$

$\Rightarrow a_1 = 0, a_2 = 0, a_3 = 0, a_4 = 0$

Since all the co-ordinates are equal to zer0, the set is linearly independent.

1.34. Theorem. ***Let V (F) be a vector space and*** $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ ***is a finite subset of non-zero vectors of V (F). Then S is linearly dependent if and only if some vector*** $\alpha_k \in S$, $2 \le k \le n$, ***can be expressed as a linear combination of its preceding vectors.*** ***(S. K. U. M13, N. U. 93S, O. U. 2001/O, S. V. U. 2001/O)***

Proof. Given $S = \{\alpha_1, \alpha_2, ..., \alpha_k, ..., \alpha_n\}$ is L.D. Then there exist $\alpha_1, \alpha_2, ..., \alpha_n \in F$, not all zero, such that $a_1\alpha_1 + a_2\alpha_2 + ... + a_k\alpha_k + .. a_n\alpha_n = \bar{O}$

Let k be the greatest suffix of a for which $\alpha_k \neq 0$.

Then $a_1\alpha_1 + a_2\alpha_2 + ... + a_k\alpha_k + 0\alpha_{k+1} + ... + 0\alpha_n = \bar{O} \Rightarrow a_1\alpha_1 + a_2\alpha_2 + ... + a_k\alpha_k = \bar{O}$

Now suppose $k = 1$, (1) $\Rightarrow a_1\alpha_1 = \bar{O}$

But $a_1 \neq 0$, so $\alpha_1 = 0$, which contradicts that each element of S is a non-zero vector.

Hence $k > 1$, *i.e.*, $2 \le k \le n$.

Again from (1) we have $a_k\alpha_k = -a_1\alpha_1 - a_2\alpha_2 .... - a_{k-1}\alpha_{k-1}$

$\Rightarrow a_k^{-1}(a_k\alpha_k) = a_k^{-1}(-a_1\alpha_1 - a_2\alpha_2 .... - a_{k-1}\alpha_{k-1})$

$\Rightarrow \alpha_k = (-a_k^{-1}a_1)\alpha_1 + (-a_k^{-1}a_2)\alpha_2 + ... + (-a_k^{-1}\alpha_{k-1})\alpha_{k-1}$

= (L.C. of its preceding vectors).

Conversely. Let some $\alpha_p \in S$ be expressible as a linear combination of its preceding vectors *i.e.*, for $b_1, b_2, ..., b_{p-1} \in F$.

$\alpha_p = b_1\alpha_1 + b_2\alpha_2 + ... + b_{p-1}\alpha_{p-1} \Rightarrow b_1\alpha_1 + b_2\alpha_2 + ... + b_{p-1}\alpha_{p-1} + (-1)\alpha_p = \bar{O}$

$\Rightarrow$ the set $\{\alpha_1, \alpha_2, ..., \alpha_n\}$ is L.D. ... [(-1) is a non-zero coefft.]

Hence the superset $S = \{\alpha_1, \alpha_2, ..., \alpha_p, ..., \alpha_n\}$ is L.D.

Note. If β is a linear combination of the set of vectors $\alpha_1, \alpha_2, ..., \alpha_n$, then the set of vectors $\{\beta, \alpha_1, \alpha_2, ..., \alpha_n\}$ is L.D.

1.35. Theorem. *Let* $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ *be a subset of the vector space* $V(F)$. *If* $\alpha_i \in S$ *is a linear combination of its preceding vectors then* $L(S) = (S')$ *where*

$S' = \{\alpha_1, \alpha_2, ..., \alpha_{i-1}, \alpha_{i+1}, ..., \alpha_n\}$

Proof. Clearly $S' \subset S \Rightarrow L(S') \subset L(S)$. Let $\beta \in L(S)$ then for $a_1, a_2 ... a_n \in F$.

$\therefore \beta = a_1 \alpha_1 + a_2 \alpha_2 + ... + a_i \alpha_i + ... + a_n \alpha_n$

But given that $\alpha_i = b_1 \alpha_1 + b_2 \alpha_2 ... + b_{i-1} \alpha_{i-1} \quad \forall \ b's \in F$

$\therefore \beta = a_1\alpha_1 + a_2\alpha_2 + ... + a_i (b_1\alpha_1 + b_2\alpha_2 ... + b_{i-1} \alpha_{i-1}) + a_{i+1} \alpha_{i+1} + ... + a_n \alpha_n$

$= (a_1 + a_i b_1) \alpha_1 + (a_2 + a_i b_2) \alpha_2 + ... + (a_{i-1} + a_i b_{i-1}) \alpha_{i-1} + a_{i+1}\alpha_{i+1} ... + a_n\alpha_n$

= L.C. of the elements of S'

$\therefore \beta \in L(S') \Rightarrow L(S) \subset L(S')$. Hence $L(S) = L(S')$.

EXERCISE 1 (*d*)

1. Determine whether the following sets of vectors are L.D. or L.I.

(*a*) $\{(2,-3), (6,-9)\}$ in $R^2(R)$ (*b*) $\{(4,3,-2), (2,-6,7)\}$ in $V_3(R)$.

(*c*) $\{A, B\}$ where $A = \begin{bmatrix} 1 & -2 & 4 \\ 3 & 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -6 & 12 \\ 9 & 3 & -3 \end{bmatrix}$ in the vector space of 2×3 matrices over R.

(*d*) $\{p_1(x), p_2(x)\}$ where $p_1(x) = 1 - 2x + 3x^2$ and $p_2(x) = 2 - 3x$ in the vector space of all polynomials over R.

2. Determine whether the following sets of vectors in $R^3(R)$ are L.D or L.I.

(*a*) $\{(1,-2,1), (2,1,-1), (7,-4,1)\}$ (*b*) $\{(2,0,5), (3,-5,8), (4,-2,1), (0,0,1)\}$

(*c*) $\{(-1,2,1), (3,0,-1), (-5,4,3)\}$ ***(O. U., M 06)***

3. Determine whether each of the following sets of vectors $V_4(Q)$ is or L.D. or L.I., Q is the field of rational numbers.

(*a*) $\{(2,1,1,1), (1,3,1,-2), (1,2,-1,3)\}$ (*b*) $\{(0,1,0,1), (1,2,3,-1), (1,0,1,0), (0,3,2,0)\}$

(*c*) $\{(1,2,-1,1), (0,1,-1,2), (2,1,0,3), (1,1,0,0)\}$

(*d*) $\{(1,1,2,4), (2,-1,-5,2), (1,-1,-4,0), (2,1,1,6)\}$ ***(K.U. 2011)***

4. Show that the set $\{1, x, x - x^2\}$ is L.I., set of vectors in F (x) over the field of real numbers.

5. If α, β, γ are the vectors of V (F), and $a, b \in F$ show that the set $\{\alpha, \beta, \gamma\}$ is L.D, if the set $\{\alpha + a\beta + b\gamma, \beta, \gamma\}$ is linearly dependent.

6. If $S = \{\alpha_1, \alpha_2 \ldots \alpha_n\}$ is any linearly independent subset of a vector space V and $\beta \notin L(S)$, then prove that $\alpha_1, \alpha_2 \ldots \alpha_n, \beta\}$ is linearly independent.

7. Under what conditions on the scalar β is the subset $\{(\beta,1,0),(1,\beta,1),(0,1,\beta)\}$ of R^3, linearly dependent ?

8. If $f(x)$ is a polynomial of degree n with real coefficients prove that

$$\left\{f(x), \frac{d}{dx} f(x), \frac{d^2}{dx^2} f(x), \ldots \frac{d^n}{dx^n} f(x)\right\}$$ is a linearly independent set.

LINEAR ALGEBRA

UNIT - II

Vector Spaces II

Basis of Vector space, Finite dimensional Vector spaces, basis extension, co-ordinates, Dimension of a Vector space, Dimension of a subspace, Quotient space and Dimension of Quotientspace.

2. Basis and Dimension

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2

Basis and Dimension

2.1. BASIS OF VECTOR SPACE

Definition. *A subset S of a vector space V (F) is said to be the basis of V, if*

(*i*) S is *linearly independent.* (*ii*) *the linear span of* S is V *i.e.,* $L(S) = V$

Note: A vector space may have more than one basis.

2.2. FINITE DIMENSIONAL SPACE.

Definition. *A vector space V (F) is said to be finite dimensional if it has a finite basis.* (OR)

A vector space V (F) is said to be finite dimensional if there is a finite subset S in V such that L (S) = V.

SOLVED PROBLEMS

Ex. 1. *Show that the set of n vectors* $e_1 = (1, 0, \ldots, 0), e_2 = (0, 1, 0, \ldots, 0), e_n = (0, 0, \ldots, 1)$ *is a basis of* $V_n(F)$.

Sol. Let $S = \{e_1, e_2, \ldots, e_n\}$. (*i*) It can be easily verified that the set S is L.I,

(*ii*) Now any vector $\alpha = (a_1, a_2 \ldots a_n) \in V_n(F)$ can be put in the form

$(a_1, a_2, \ldots, a_n) = a_1(1, 0, \ldots, 0) + a_2(0, 1, \ldots, 0) \ldots + a_n(0, \ldots, 1) = a_1e_1 + a_2e_2 + \ldots + a_ne_n$

= *l. c.* of elements of the set S $\Rightarrow$ $\alpha \in L(S)$.

$\therefore$ $\alpha \in V \Rightarrow$ $\alpha \in L(S)$ $\therefore$ $V = L(S)$. Hence S is a basis of $V_n(F)$

Note 1. The set $S = \{e_1, e_2, \ldots, e_n\}$ is called the standard basis of $V_n(F)$ or F^n

Note 2. The set $\{(1,0,0), (0,1,0), (0,0,1)\}$ is the standard basis of $V_3(R)$ or R^3.

Ex. 2. *Show that the infinite set* $S = \{1, x, x^2, \ldots, x^n, \ldots\}$ *is a basis of the vector space* $F[x]$ *over the field F.*

Sol. (*i*) The set S is clearly L. I. (*ii*) Let $f(x) = a_0 + a_1x + \ldots + a_nx^n \in F[x]$

$\therefore$ where $a_0, a_1, \ldots, a_n \in F$

$\therefore$ $f(x) \in L(S)$ $\Rightarrow$ $F[x] = L(S)$ $\therefore$ S is a basis of $F[x]$

Let $S' = \{1, x, x^2, \ldots, x^m\}$ be a finite subset of S. If $g(x) = c_0 + c_1 x + c_2 x^2 \ldots + c_nx^n$ and $n > m$ it is not possible to write $g(x)$ as a *l.c.* of the element of S'.

Hence the vector space $F[x]$ is not finite dimensional.

2.3 Theorem. ***IF V(F) is a finite dimensional vector space, then there exists a basis set of V.*** ***(K. U. M 2013, N. U. 95, S. V. U. S97, S. K. U. O2001, M14)***

Proof. Since V (F) is finite dimensional, there exists a finite set S such that, L (S) = V

Let $S=\{\alpha_1, \alpha_2, ..., \alpha_n\}$

We may assume that S does not contain the $\bar{0}$ vector.

If S is L. I., then S is a basis set of V. If S is L. D. set, then there exists a vector $\alpha_i \in S$ which can be expressed as a linear combination of the preceding vectors,.

Omitting α_i from S, let $S_1 = \{\alpha_1, \alpha_2, ..., \alpha_{i-1}, \alpha_{i+1}, ..., \alpha_n\} \Rightarrow S_1 \subset S$

By previous theorem $L(S_1) = L(S)$. Now $L(S) = V \Rightarrow L(S_1) = V$

If S_1 is L.I set then S_1 will be a basis of V.

If S_1 is linearly dependent, then proceeding as above for a finite number of steps, we will be left with a L. I. set S_k and $L(S_k) = V$. Hence S_k will be the basis of V (F)

Thus there exists a basis set for V (F)

2.4. BASIS EXTENSION

Theorem. ***Let V (F) be a finite dimensional vector space and*** $S=\{\alpha_1, \alpha_2, ..., \alpha_m\}$ ***a linearly independent subset of V. Then either S itself a basis of V or S can be extended to form a basis of V.*** ***(S. K. U. 2011)***

Proof. Given $S=\{\alpha_1, \alpha_2, ..., \alpha_m\}$ is L. I. subset of V.

Since V (F) is finite dimensional it has a finite basis B. Let $B=\{\beta_1, \beta_2 ... \beta_n\}$

Now consider the set $S_1 = \{\alpha_1, \alpha_2, ..., \alpha_m, \beta_1, \beta_2, ..., \beta_n\}$

written in this order. Clearly L (S) = V.

Each α can be expressed as a l.c. of β's as B is the basis of V $\Rightarrow$ S_1 is L. D.

Hence some vector in S_1 can be expressed as a *l.c.* of its proceeding vectors. This vector cannot be any of α's, since S is L. I. So this vector must be some β_i.

Consider now the set.

$S_2 = \{\alpha_1, \alpha_2, ..., \alpha_m, \beta_1, \beta_{i-1}, \beta_{i+1}, ..., \beta_n\} = S_1 - \{\beta_i\}$ obviously $L(S_n) = L(S_1) = V$.

If S_2 is L. I., then S_2 forms a basis of V and it is the extended set.

If S_2 is L. D. then continue this procedure till we get a set $S_k \subset S$ such that S_k is L. I.

$\therefore L(S_k) = L(S) = V$

S_k will be extended set of S forming a basis of V.

Note. 1. Every basis is a spanning set but every spanning set is not a basis.

2. If a basis of V(F) contains n elements, then $m \leq n$.

2.5. Theorem. *Let* $S=\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ *be a basis set of a finite dimensional vector space* $V(F)$. *Then for every* $\alpha \in V$ *there exists a unique set of scalars* $a_1, a_2, \ldots, a_n \in F$ *such that* $\alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n$

Proof. If possible let there exist another set of scalars $b_1, b_2 \ldots b_n \in F$ such that

$\alpha = b_1\alpha_1 + b_2\alpha_2 + \ldots + b_n\alpha_n$

$\therefore\ a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n = b_1\alpha_1 + b_2\alpha_2 + \ldots + b_n\alpha_n$

$\Rightarrow (a_1 - b_1)\alpha_1 + (a_2 - b_2)\alpha_2 \ldots + (a_n - b_n)\alpha_n = \bar{O}$

Since the set $S=\{\alpha_1, \alpha_2 \ldots \alpha_n\}$ is L. I.

we have $a_1 - b_1 = 0, a_2 - b_2 = 0 \ldots a_n - b_n = 0 \Rightarrow a_1 = b_1, a_2 = b_2, \ldots, a_n = b_n$

2.6. COORDINATES

Definition. *Let* $S=\{\alpha_1, \alpha_2 \ldots \alpha_n\}$ *be the basis set of a finite dimensional vector space* $V(F)$. *Let* $\beta \in V$ *be given by* $\beta = a_1\alpha_1 + a_2\alpha_2 \ldots + a_n\alpha_n$ *for* $a_1, a_2, \ldots, a_n \in F$ *then the scalars* $(a_1, a_2, \ldots, a_n)$ *are called the coordinates.*

Note. Coordinates change with the change of basis.

SOLVED PROBLEMS

Ex. 1. *Show that the vector (1,1,2), (1,2,5), (5,3,4) of* $R^3(R)$ *do not form a basis set of* $R^3(R)$. **(S. V. U. M 2015)**

Sol. Writing these vectors as rows of a matrix we have $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 5 \\ 5 & 3 & 4 \end{bmatrix}$

Reducing to echelon form, we get $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & -2 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$

But in the last matrix there are only two non-zero rows, hence the given vectors are L.D. Therefore they ean not form a basis of $R^3(R)$.

Ex. 2. *Show that the set* $\{(1,0,0),(1,1,0),(1,1,1)\}$ *is a basis of* $C^3(C)$. *Hence find the coordinates of the vector* $(3+4i, 6i, 3+7i)$ *in* $C^3(C)$. **(S. V. U. 2011)**

Sol. Let $S=\{(1,0,0), (1,1,0), (1,1,1)\}$

Keeping these vectors as rows of a matrix we get $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

Reducing to echelen form, $A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ *i.e.*, $R_2 - R_1$ and $R_3 - R_2$

$\therefore$ rank A = no. of unknowns. $\Rightarrow$ The given set is L. I.

Let $z \in C^3$ be $z = (a,b,c)$ where $a,b,c \in C$

Now $(a,b,c) = p(1,0,0) + q(1,1,0) + r(1,1,1)$ for $p,q,r \in C$ $= (p+q+r, q+r, r)$

$\Rightarrow$ $a = p+q+r$, $b = q+r, c = r$. $\Rightarrow$ $r = c, q = b-c, p = a-b$

$\therefore$ $z = (a-b)(1,0,0) + (b-c)(1,1,0) + c(1,1,0)$

= *l.c.* of the given vectors of S $\Rightarrow$ $z \in L(S)$. $\therefore$ S is a basis of $C^3(C)$

Now if $(a,b,c) = (3+4i, 6i, 3+7i)$ then $p = 3-2i, q = -3-i, r = 3+7i$ which are the coordinates of the given vector.

Ex. 3. *The set* $S_4 = \{\alpha, \beta, \gamma, \delta\}$ *where* $\alpha = (1,0,0), \beta = (1,1,0), \gamma = (1,1,1)$ *and* $\delta = (0,1,0)$ *is a spaning set of* $R^3(R)$ *but not a basis of set.*

Sol. Keeping the vectors in matrix form $\begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

δ - row reduced to zero row.

$\Rightarrow$ δ can be expressed as *l.c* of α, β, γ $\Rightarrow$ S is L. D.

Consider $S_3 = \{\alpha, \beta, \gamma\}$ $\therefore$ $\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Whenever $[\alpha\, \beta\, \gamma]$ can be reduced to the standard basis set, S_3 will be the basis of $V_3(F)$ *i.e.*, $L(S_3) = V$. As $S_3 \subset S_4, L(S_3) = L(S_4) = V$

Thus S_4 is a spaning set of $V_3(F)$ but not a basis set.

Ex. 4. *If* $\alpha = (1,-1,0), \beta = (2,1,3)$ *find a basis for* R^3 *containing* α *and* β.

(S. K. U. 2011)

Sol. Let $e_1 = (1,0,0), e_2 = (0,1,0), e_3 = (0,0,1)$ and $S = \{e_1, e_2, e_3\}$ we know that S is a standard basis for R^3

We have $\beta \in R^3$. Let $S_1 = \{\beta, e_1, e_2, e_3\}$ then $L(S_1) = L(S) = R^3$

Now $\beta = 2e_1 + e_2 + 3e_3$

$\Rightarrow e_3 = \frac{1}{3}\beta - \frac{2}{3}e_1 - \frac{1}{3}e_2$ which shows that e_3 is a L. C. of β, e_1, e_2

Let $S_1 = \{\beta, e_1, e_2\}$ then we have $L(S_1) = L(S) = R^3$. Again $\alpha \in R^3$

Let $S_2 = \{\alpha, \beta, e_1, e_2\}$ then $L(S_2) = L(S_1) = R^3$

We have $\alpha = e_1 - e_2 \Rightarrow e_2 = -\alpha + 0\beta + e_1$ which show that e_2 is a L. C.of α, β, e_1

Remove e_2 from S_2 and let $S_3 = \{\alpha, \beta, e_1\}$ then $L(S_3) = L(S) = R^3$

Since the dimension of R^3 is 3. We have S_3 is a basis of R^3.

$\therefore$ $\{(1,-1,0)(2,1,3)(1,0,0)\}$ is a basis of R^3.

Note: We can prove that $\{\alpha, \beta, e_2\}$ and $\{\alpha, \beta, e_3\}$ are also basis of R^3.

Ex. 5. *Show that* $(2,2,1)(2,1,1)(2,1,0)$ *from a basis of* $(1,2,1)$ *in* $V_3(F)$

(A. N. U. M 14)

Sol. Let $(1,2,1) = \alpha(2,2,1) + \beta(2,1,1) + \gamma(2,1,0)$

$= (2\alpha + 2\beta + 2\gamma, 2\alpha + \beta + \gamma, \alpha + \beta)$

Comparing, $2\alpha + 2\beta + 2\gamma = 1$ ... (1)

$2\alpha + 4\gamma = 2$ ... (2)

$\alpha + \beta = 1$ ... (3)

Solving we get $\alpha = 3/2, \beta = -1/2, \gamma = -1/2$

$\therefore (1,2,1) = \frac{3}{2}(2,2,1) - \frac{1}{2}(2,1,1) - \frac{1}{2}(2,1,0)$

Ex. 6. *Find a basis for* R^4 *write* $(3,2,5,4), (6,3,5,2)$ *as numbers.*

Sol. Let $\alpha_1 = (3,2,5,4), \alpha_2 = (6,3,5,2)$

$e_1 = (1,0,0,0), e_2 = (0,1,0,0), e_3 = (0,0,1,0), e_4 = (0,0,0,1)$

We know that $\{e_1, e_2, e_3, e_4\}$ is a basis of R^4

we have $\alpha_2 = 6e_1 + 3e_2 + 5e_3 + 2e_4$

$\therefore$ $\alpha_2, e_1, e_2, e_3, e_4$ are L.D. *i.e.* e_4 is a L.C. of α_2, e_1, e_2, e_3

Suppose we remove e_4 from the set $\{\alpha_2, e_1, e_2, e_3, e_4\}$ then $\{\alpha_2, e_1, e_2, e_3\}$ is L.I. set

Consider $\{\alpha_1, \alpha_2, e_1, e_2. e_3\}$. Since the dimension of R^4 is 4 this must be a L.D. set.

Consider $x_1\alpha_1 + x_2\alpha_2 + x_3e_1 + x_4e_2 + x_5e_3 = 0$

then $3x_1 + 6x_2 + x_3 = 0$ ... (1) $2x_1 + 3x_2 + x_4 = 0$ ... (2)

$5x_1 - 5x_2 + x_5 = 0$ ... (3) $4x_1 + 2x_2 = 0$ ... (4)

(4) $\Rightarrow x_2 = -2x_1$ substituting (I), (2), (3) we get

$$-9x_1 + x_3 = 0 \Rightarrow x_1 = \frac{1}{9}x_3\,; \quad -4x_1 + x_4 = 0 \Rightarrow x_1 = \frac{1}{4}x_4\,; \quad -5x_1 + x_5 = 0 \Rightarrow x_1 = \frac{1}{5}x_5$$

$\therefore\ x_1 = 1, x_2 = -2, x_3 = 9, x_4 = 4, x_5 = 5$

$$\therefore\ \alpha_1 - 2\alpha_2 + 9e_1 + 4e_2 + 5e_3 = 0 \ \textit{i.e.,}\ \ e_3 = \frac{-1}{5}\alpha_1 + \frac{2}{5}\alpha_2 - \frac{9}{5}e_1 - \frac{4}{5}e_2$$

which proves that e_3 is a L. C. of $\alpha_1, \alpha_2, e_1, e_2$

We can prove that $\{\alpha_1, \alpha_2, e_1, e_2\}$ is L.I. set, which gives a basis for R^4.

Ex. 7. *If* $\alpha_1 = (1,2,-1), \alpha_2 = (-3,-6,3), \alpha_3 = (2,1,3)$ *and* $\alpha_4 = (8,7,7)$ *and if* $S = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ *is such that L (S) = W, find a basis by reducing S.*

Sol. We observe $\alpha_2 = -3\alpha_1 = -3\alpha_1 + 0\alpha_3 + 0\alpha_4$

$\therefore\ \alpha_2$ is a L. C. of $\alpha_1, \alpha_3, \alpha_4$

Let $S_1 = \{\alpha_1, \alpha_3, \alpha_4\}$. We have $L(S_1) = L(S) = W$

Again $\alpha_4 = 2\alpha_1 + 3\alpha_3$. Thus α_4 is L. C. of α_1 and α_3

Let $S_2 = \{\alpha_1, \alpha_3\}$ then $L(S_2) = L(S_1) = W$

Now $a\alpha_1 + b\alpha_2 = 0 \Rightarrow a(1,2,-1) + b(2,1,3) = 0$

$\Rightarrow (a+2b, 2a+b, -a+3b) = (0,0,0) \Rightarrow a + 2b = 0$

$2a + b = 0,\ -a + 3b = 0$

on solving : $a = 0 = b$ is the only solution which proves that S_2 is L.I. set

Thus S_2 is a basis for W

Note : In every step we can remove one vector which is a L. C. of other vectors.

In the above problem any subset of S containing 2 vectors will become a basis of W.

Ex. 8. *Find the co-ordinates of* $(2,3,4,-1)$ *with respect to the basis of* $V_4(R)$, $B = \{(1,1,1,2),(1,-1,0,0),(0,0,1,1),(0,1,0,0)\}$ ***(A.N. U. 2011)***

Sol. Let $(2,3,4,-1) = a(1,1,1,2) + b(1,-1,0,0) + c(0,0,1,1) + d(0,1,0,0)$

$= (a+b, a-b+d, a+c, 2a+c)$

Equating the components, we get $a+b=2, a-b+d=3, a+c=4, 2a+c=-1$

Solving we get $a=-5, b=7, c=9, d=15$

$\therefore$ The co-ordinates are $(-5,7,9,15)$

EXERCISE 2 (*a*)

1. Show that the set $\{(1,0,0),(1,1,0),(1,1,1)\}$ is a basis of $C^3(C)$ but not a basis of $C^3(R)$.

2. Find the coordinates of α with respect to the basis set $\{x, y, z\}$ where

(*i*) $\alpha=(4,5,6)$, $x=(1,1,1)$, $y=(-1,1,1)$, $z=(1,0,-1)$

(*ii*) $\alpha=(1,0,-1)$, $x=(0,1,-1)$, $y=(1,1,0)$, $z=(1,0,2)$

(*iii*) $\alpha=(2,1,3)$, $x=(1,1,1)$, $y=(-1,1,0)$, $z=(1,0,-1)$

3. Show that the four vectors $\{(2,1,0),(0,1,2),(-7,2,5),(8,0,0)\}$ is not a basis of $R^3(R)$

4. Show that if $\{\alpha_1,\alpha_2,\alpha_3\}$ is a basis of $R^3(R)$ then the set $\{\alpha_1+\alpha_2,\alpha_2+\alpha_3,\alpha_3+\alpha_1\}$ is also a basis of $R^3(R)$

5. (*i*) Show that the set $\{(1,2,1),(2,1,0),(1,-1,2)\}$ forms a basis of $V_3(F)$

(A. N. U. M13, S. V. U. S97, S. K. U. M14)

(*ii*) Show that the set of vectors $\{(2,1,4),(1,-1,2),(3,1,-2)\}$ form a basis for R^3.

(S. V. U. M 14, S97)

6. Show that the set $\{(1,i,0),(2i,1,1),(0,1+i,1-i)\}$ is a basis of $V_3(C)$ and find the coordinates of the vectors $(1,0,0),(0,1,0),(0,0,1)$ with respect to this basis.

7. Show that the set $\{(3-i,2+2i,4),(2,2+4i,3),(1-i,-2i,1)\}$ a basis of $V_3(C)$ and find the coordinates of the vectors (I, 0, 0) and (0, 0, I) with respect to this basis.

8. Show that the vectors (1,1,2), (1,2,5), (5,3,4) in $R^3(R)$ do not fonn a basis set of R^3.

9. Find the coordinates of (*i*) $(2i,3+4i,5)$ (*ii*) $(6i,7,8i)$ (*iii*) $(3+4i,6i,3+7i)$ with respect to the basis set $\{(1,0,0),(1,1,0),(1,1,1)\}$ of $C^3(C)$.

10. Under what conditions on the scalar $a\in R$ is the set $\{(0,1,a),(a,0,1),(a,1,1+a)\}$ a basis of $R^3(R)$.

2.7 Invariance Theorem . ***Let V(F) be a finite dimensional vector space. Then any two bases of V have the same number of elements.***

(A.N.U. 11, M13, S. V. U. O2001, 11,M12, M 15, S. K. U. M13, O. U. M 06, M13, A. U. M14)

Proof. Let S_m and S_n be the two bases of V(F) where $S_m=\{\alpha_1,\alpha_2,\ldots,\alpha_m\}$, $S_n=\{\beta_1,\beta_2,\ldots,\beta_n\}$

Obviously both S_m and S_n are L.I. subsets of V.

(*i*) Consider S_m as the basis of V and S_n and L. I. set.

$\Rightarrow$ $L(S_m)=V$ and $n(S_m)=m$

$\therefore$ S_n can be extended to be a basis of V $\Rightarrow$ $n \leq m$

(*ii*) Consider S_n as the basis of V and S_m as L.I. set $\Rightarrow$ $L(S_n) = V$ and $n(S_n) = n$

$\therefore$ S_m can be extended to form a basis of V $\Rightarrow$ $m \leq n$

But both S_m and S_n are bases of V. $\therefore$ $n \leq m$ and $m \leq n$ $\Rightarrow$ $m = n$

Thus any two bases of V have the same number of elements.

2.8. DIMENSION OF A VECTOR SPACE

Definition. *Let V (F) be a finite dimensional vector space. The number of elements in any basis of V is called the dimension of V and is denoted by dim V.*

Note: The dimension of zero vector space $\{\bar{O}\}$ is said be zero.

e.g. The set S = {(1,0,0), (0,1,0), (0,0,1)} is a basis of $V_3(R)$

$\therefore$ dim V = no. of elements of S = 3.

2.9. Theorem. *Every set of $(n+1)$ or more vectors in an n dimensional vector space is linearly dependent.*

Proof. Let V (F) be the vector space with V = n

Let $S = \{\alpha_1, \alpha_2, ..., \alpha_{n+1}\}$ be any subset consisting of $(n+1)$ vectors of V (F)

If S is L I. then S itself is the basis or can be extended to be a basis of V (F).

In any of these two cases S will have more than or equal to $(n+1)$ vectors. But every basis of V must contain exactly n vectors. Therefore S can not be L.I.

Hence S (*i.e.* every $n+1$ or more vectors in V) is L.D.

Note. The largest L.I. subset of a finite dimensional vector space of dimension n is a basis.

2. 10. Theorem. *Let V (F) be a finite dimensional vector space of dimension n. Then any set of n linearly independent vectors in V forms a basis of V.*

Proof. Given dim V = n.

Let $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ be L. I. set of n vectors in V.

If S is not a basis of V, then it can be extended to form a basis of V. In such a case the basis will contain more than n vectors.

But every basis of V must contain exactly n vectors. Therefore our presumption is wrong and S must be a basis of V.

2. 11. Theorem. *Let V (S) be a finite dimensional vector space of dimension n. Let S be a set of n vectors of V such that L (S) = V. Then S is a basis of V (F).*

Proof. Let $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ be the subset of vector space V (F).

If S is L.I. and because L (S) = V, the set S becomes the basis of V.

If S is L.D. then there exists a proper subset of S forming a basis of V. In such case we get a basis of V consisting less than a vectors.

As every basis of V must contain exactly n vectors, S cannot be L.D. Hence S is a basis of V.

Note. If V is a finite dimensional vector space of dimension n, then V cannot be generated by a set of vectors whose number of elements is less than n.

2.12.Theorem. ***V is a vector space which is spanned by a finite set of vectors*** $\beta_1, \beta_2, \ldots, \beta_n$ ***then any independent set of vectors in V is finite and contains no more than n elements.***

Proof. Let $\{\alpha_1, \alpha_2, \ldots, \alpha_m\}$ be a linearly independent subset of vector space V (F)

Let $S = \{\beta_1, \beta_2, \ldots, \beta_n\}$ and L (S) = V.

Then any vector of V is *l.c.* of the elements of S.

Let $\alpha_m \in V$ be a linear combination of the elements of S.

$\Rightarrow S' = \{\alpha_m, \beta_1, \beta_2, \ldots, \beta_n\}$ is L. D.

$\therefore$ There exists a vector in S' which is a linear combination of its preceeding vectors. This vector must be one among $\beta's$.

Let it be β_i. $\therefore$ β_i = *l.c.* of $\alpha_m, \beta_1, \beta_2, \ldots, \beta_{i-1}$

If $S_1 = \{\alpha_m, \beta_1, \beta_2, \ldots, \beta_{i-1}, \beta_{i+1}, \ldots, \beta_n\}$. Then $L(S_1) = L(S) = V$

Again the vector $\alpha_{m-1} \in V$ is a *l.c.* of the elements of S_1

$\therefore$ $\alpha_{m-1}, \alpha_m, \beta_1, \beta_2, \ldots, \beta_{i-1}, \beta_{i+1}, \ldots, \beta_n$ are L.D.

$\therefore$ There exists a vector in this set which is a *l.c.* of its predecessors.

Such vector is one of $\beta's$ as α_{m-1}, α_m are L. I. Let it be β_k.

If $S_2 = \{\alpha_{m-1}, \alpha_m, \beta_1, \ldots, \beta_{k-1}, \beta_{k+1}, \ldots, \beta_{i-1}, \beta_{i+1}, \ldots, \beta_n\}$ then

β_k = *l.c* of the elements of S_3. $\therefore$ $L(S) = L(S_1) = L(S) = V$

Continuing this process further for $m-3$ times

We get $S_{m-1} = \{\alpha_2, \alpha_3, \ldots, \alpha_m, \beta_{i1}, \beta_{i2}, \beta_{in-m+1}\}$. So that $L(S_{m-1}) = V$.

This set consists of atleast one β_i. Otherwise $S_{m-1} = \{\alpha_2, \alpha_3 \ldots \alpha_m\}$

So that α_1 is a *l.c.* of $\alpha_2, \alpha_3 \ldots \alpha_m$

This can not happen as the set $\alpha_2, \alpha_3 \ldots, \alpha_m$ is L. I.

Hence S_{m-1} consists of atleast one $\beta_i \Rightarrow n-m+1 \geq 1 \Rightarrow n \geq m$

$\therefore$ The no. of elements of the L. I. set in $V \leq n$.

DIMENSION OF A SUBSPACE

2.13. Theorem. ***Let V(F) be a finite dimensional vector space of dimension n and W be the subspace of V. Then W is a finite dimensional vector space with dim*** $W \leq n$.

Proof. Dim V = $n \Rightarrow$ each $(n+1)$ or more vectors of V form an L.D. set.

Given W is a subspace of V(F)

$\Rightarrow$ each set of $(n+1)$ vectors in W is a subset of V and hence L.D.

Thus any LI. set of vectors in W can contain at the most n vectors .

Let $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ be the largest L.I. subset of W, where $m \leq n$.

Now we shall prove that S is the basis of W.

For any $\beta \in W$ consider, $S_1 = \{\alpha_1, \alpha_2, ..., \alpha_m, \beta\}$

Since S is the largest set of L.I. vectors, S_1 is L.D.

$\therefore$ Therefore there exists $a_1, a_2, ..., a_m$, $b \in F$ not all zero such that

$a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m + b\beta = 0$

Let $b = 0$, then we have $a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m = 0$

$\Rightarrow a_1 = 0, a_2 = 0, ..., a_m = 0$ as S is L. I.

This proves that S_1 is L.I which is a contradiction.

$\therefore$ $b \neq 0$. Therefore there exists $b^{-1} \in F$ such that $bb^{-1} = 1$

$\therefore$ $a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m + b\beta = 0 \Rightarrow b\beta = -a_1\alpha_1 - a_2\alpha_2 ... a_m\alpha_m$

$\Rightarrow \beta = (-b^{-1}a_1) + (-b^{-1}a_2)\,\alpha_2 + ... + (-b^{-1}a_m)\,\alpha_m$

$\Rightarrow \beta = a$ linear combination of elements of S. $\Rightarrow \beta \in L(S)$

Also S is L.I. Hence S is the basis of W.

$\therefore$ W is a finite dimensional vector space with dim $W \leq n$.

2.14. Theorem. ***Let*** W_1 ***and*** W_2 ***be two subspaces of a finite dimensional vector space V (F). Then*** $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$

(K.U. 11, M 14, A.N.U. 11, M 14, O. U. M13, S . V. U. M 15, M14, M12, M13, S. K. U. M14)

Proof. Since W_1 and W_2 are subspace of V, $W_1 + W_2$ and $W_1 \cap W_2$ are also subspace of V.

Let $\dim(W_1 \cap W_2) = k$ and $S = \{\gamma_1, \gamma_2, ..., \gamma_k\}$ be a basis of $W_1 \cap W_2$.

Clearly $S \subseteq W_1$ and $S \subseteq W_2$ and S is L.I.

Since S is L.I. and $S \subseteq W_1$, the set S can be extended to form a basis of W_1 .

Let $B_1 = \{\gamma_1, \gamma_2, ..., \gamma_k, \alpha_1, \alpha_2, ..., \alpha_m\}$ be a basis of W_1 . $\therefore$ $\dim W_1 = k + m$.

Again since S is L.I. and $S \subseteq W_2$, the set S can be extended to form a basis of W_2 .

Let $B_2 = \{\gamma_1, \gamma_2, ..., \gamma_k, \beta_1, \beta_2, ..., \beta_t\}$ be a basis of W_2. $\therefore$ $\dim W_2 = k + t$

$\therefore$ $\dim W_1 + \dim W_2 - \dim (W_1 \cap W_2) = (k+m)+(k+t)-k = k+m+t$

Now we shall prove that the set

$S' = \{\gamma_1, \gamma_2, ..., \gamma_k, \alpha_1, \alpha_2, ..., \alpha_m, \beta_1, \beta_2, ..., \beta_t\}$ is a basis of $W_1 + W_2$ and hence $\dim (W_1 + W_2) = k + m + t$.

(*i*) *To prove that S' is L.I.*

Now $c_1\gamma_1 + c_2\gamma_2 + ... + c_k\gamma_k + a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m + b_1\beta_1 + b_2\beta_2 + ... + b_t\beta_t = 0$ ... (1)

$\Rightarrow$ $b_1\beta_1 + b_2\beta_2 + ... + b_t\beta_t = -c_1\gamma_1 - ... - c_k\gamma_k - a_1\alpha_1 - ... - a_m\alpha_m$

= *l.c.* of elements of B_1 and $\therefore$ $\in W_1$

$\Rightarrow$ $b_1\beta_1 + b_2\beta_2 + ... + b_t\beta_t \in W_1$

Again $0\gamma_1 + 0\gamma_2 + ... 0\gamma_k + b_1\beta_1 + b_2\beta_2 + ... + b_t\beta_t$

= *l.c.* of elements of B_2 and $\therefore \in W_2$ $\Rightarrow$ $b_1\beta_1 + b_2\beta_2 + ... + b_t\beta_t \in W_2$

Therefore by (1) and (2) $\Rightarrow$ $b_1\beta_1 + b_2\beta_2 + ... + b_t\beta_t \in W_1 \cap W_2$

Hence it can be expressed as a *l.c.*of the elements of the basis S of $W_1 \cap W_2$

Let $b_1\beta_1 + b_2\beta_2 + ... + b_t\beta_t = d_1\gamma_1 + d_2\gamma_2 + ... + d_k\gamma_k$

$\therefore$ $b_1\beta_1 + b_2\beta_2 ... + b_t\beta_t - d_1\gamma_1 - d_2\gamma_2 ... - d_k\gamma_k = \bar{O} \Rightarrow$ (*l.c.* of elements of basis B_2) = $\bar{O}$

$b_1 = 0, b_2 = 0 ... b_t = 0, d_1 = 0, ... k_k = 0$

Substituting in I we have $c_1\gamma_1 + c_2\gamma_2 + ... + c_k\gamma_k + a_1\alpha_1 + a_2\alpha_2 ... + a_m\alpha_m = \bar{O}$

$\Rightarrow$ (*l.c.* of elements of basis B_1) = $\bar{O}$ $\Rightarrow$ $c_1 = 0, c_2 = 0 ... c_k = 0, a_1 = 0, a_2 = 0 ... a_m = 0$

Thus the relation I implies that $c_1 = c_2 = ... = c_k = a_1 = a_2 = ... \alpha_m = o, b_1 = b_2 = ... = b_t = o$

$\therefore$ S' is L.I. set.

(*ii*) *To prove* $L(S') = W_1 + W_2$

Every vector of S' is a vector of $W_1 + W_2$. $\therefore$ $L(S') \subseteq W_2 + W_2$

Let $\delta \in W_1 + W_2$. $\therefore$ $\delta = \alpha + \beta$ where $\alpha \in W_1, \beta \in W_2$

$\delta =$ (*l.c.* of elements of B_1) + (*l.c.* of elements of B_2)

= (*l.c.* of γ's and α's) + (*l.c.* of γ's and β's) = *l.c.* of γ's , α's and β's.

= *l.c.* of elements of S'.

$\therefore$ $\delta \in L(S')$ $\therefore$ $W_1 + W_2 \subseteq L(S')$

$\therefore$ $L(S') = W_1 + W_2$. Hence S' is the basis of $W_1 + W_2$.

$\therefore$ $\dim (W_1 + W_2) = k + m + t$. Hence the theorem.

SOLVED PROBLEMS

Ex. 1. *Let* W_1 *and* W_2 *be two subspaces of* R^4 *given by* $W_1 = \{(a,b,c,d) : b-2c+d=0\}$, $W_2 = \{(a,b,c,d) : a=d, b=2c\}$. *Find the basis and dimension of* (*i*) W_1 (*ii*) W_2 (*iii*) $W_1 \cap W_2$ *and hence find* $\dim(W_1 + W_2)$.

(S. K. U. M15, A.N.U, O.U., 2011)

Sol. (i) Given $V = \{(a,b,c,d) : b-2c-d\}$

Let $(a,b,c,d) \in V$ then $(a,b,c,d) = (a, 2c-d, c, d)$

$= a(1,0,0,0) + c(0,2,1,0) + d(0,-1,0,1)$, $\therefore (a,b,c,d)$ = *l.c.* of L.I. set

$\{(1,0,0,0), (0,2,1,0), (0,-1,0,1)\}$ which forms the basis of W_1 $\therefore \dim W_1 = 3$.

(*ii*) Given $W_2 = \{(a,b,c,d) : a=d, b=2c\}$.

Let $\alpha \in W_2 \Rightarrow \alpha = (a,b,c,d)$ where $a=d, b=2c$

$\therefore \alpha = (d, 2c, c, d) = d(1,0,0,1), +c(0,2,1,0) \Rightarrow \alpha = l.c$ of L.I. set $\{(1,0,0,1), (0,2,1,0)\}$

$\therefore$ It forms a basis $\therefore \dim W_2 = 2$.

(*iii*) $W_1 \cap W_2 = \{(a,b,c,d) / b-2c+d=0, a=d, b=2c\}$

Now $b-2c+d=0, a=d, b=2c$ gives $b=2c, a=0, d=0$

$\therefore (a,b,c,d) = (0,2c,c,0) = c(0,2,1,0) \Rightarrow (a,b,c,d) =$ multiple of the vector $(0,2,1,0)$

$\therefore$ Basis of $W_1 \cap W_2 = (0,2,1,0) \Rightarrow \dim W_1 \cap W_2 = 1$

(*iii*) $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim W_1 \cap W_2 = 3+2-1=4$

Ex. 2. *If W is the subspace of* $V_4(R)$ *generated by the vectors* $(1,-2,5,-3), (2,3,1,-4)$ *and* $(3,8,-3,-5)$ *find a basis of W and its dimension.*

Sol. Arranging the given vectors as rows of a matrix

Let $A = \begin{bmatrix} 1 & -2 & 5 & -3 \\ 2 & 3 & 1 & -4 \\ 3 & 8 & -3 & -5 \end{bmatrix}$. Reducing to echelon form,

$R_2 - 2R_1, R_2 - 3R_1; A \sim \begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 14 & -18 & 4 \end{bmatrix}$. Again by $R_3 - 2R_2, A \sim \begin{bmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Hence the two non-zero rows *viz* ; $(1,-2,5,-3), (0,7,-9,2)$ form the least L. I. set and hence a basis of W. $\therefore \dim W = 2$

Ex. 3. *V is the space generated by the polynomials* $\alpha = x^3 + 2x^2 - 2x + 1$, $\beta = x^3 + 3x^2 - x + 4$, $\gamma = 2x^3 + x^2 - 7x - 7$. *Find a basis of V and its dimension.*

Sol. Now V is the polynomial space generated by $\{\alpha, \beta, \gamma\}$ given above

$\alpha = x^3 + 2x^2 - 2x + 1$

$\Rightarrow$ the coordinates of are $(1,2,-2,1)$ w. r. to the base $\{x^3, x^2, x, 1\}$

Similarly the coordinates of β and γ w. r. to the same base are $(1,3,-1,4)$ and $(2,1,-7,-7)$.

Forming the matrix A with these co-ordinate as rows

$$A = \begin{bmatrix} 1 & 2 & -2 & 1 \\ 1 & 3 & -1 & 4 \\ 2 & 1 & -7 & -7 \end{bmatrix}$$ Reducing to echilon form,

$$\text{by } R_2 - R_1, R_3 - 2R_1, \sim \begin{bmatrix} 1 & 2 & -2 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & -3 & -3 & -9 \end{bmatrix}$$

$$\text{again by } R_3 + 3R_1, A \sim \begin{bmatrix} 1 & 2 & -2 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Hence the basis of V is formed by the coordinate vectors $(1,2,-2,1)$ and $(0,1,1,3)$

Now the polynomials are $\alpha = x^3 - 2x^2 - 2x + 1$ and $\delta = x^2 + x + 3$

Thus the base of V is $\{\alpha, \delta\}$ $\quad \therefore$ dim W = 2

Ex. 4. *V is the vector space of polynomials over R.* W_1 *and* W_2 *are the subspaces generated by* $\{x^3 + x^2 - 1, x^3 + 2x^2 + 3x, 2x^3 + 3x^2 + 3x - 1\}$ *and*

$\{x^3 + 2x^2 + 2x - 2, 2x^3 + 3x^2 + 2x - 3, x^3 + 3x^2 + 4x - 3\}$ *respectively.*

Find (*i*) $\dim(W_1 + W_2)$ (*ii*) $\dim(W_1 \cap W_2)$

Sol. The coordinates of the polynomials w.r.to the basis $\{x^3, x^2, x, 1\}$ are respectively

$\{(1,1,0,-1), (1,2,3,0), (2,3,3,-1)\}$ and $\{(1,2,2,-2), (2,3,2,-3), (1,3,4,-3)\}$.

(*i*) Arranging the vectors generating W_1 as rows of a matrix and reducing to echelon

$$\text{form } A = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 1 & 2 & 3 & 0 \\ 2 & 3 & 3 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\Rightarrow$ the basis of W_1 contains 2 vectors $\Rightarrow$ dim $W_1 = 2$ similarly we get dim $W_2 = 2$.

(*ii*) Now the subspace $W_1 + W_2$ is by all the six vectors. Hence arranging them in rows of a matrix and reducing to echelon form.

$$P = \begin{Bmatrix} 1 & 1 & 0 & -1 \\ 1 & 2 & 3 & 0 \\ 2 & 3 & 3 & -1 \\ 1 & 2 & 2 & -2 \\ 2 & 3 & 2 & -3 \\ 1 & 3 & 4 & -3 \end{Bmatrix} \sim \begin{Bmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{Bmatrix} = Q$$

$\Rightarrow$ The basis of $W_1 + W_2$ consists of the three non zero row vectors in Q.

$\Rightarrow \dim(W_1 + W_2) = 3$.

Now we have the formula $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$

$\Rightarrow \dim(W_1 \cap W_2) = \dim W_1 + \dim W_2 - \dim(W_1 + W_2) = 2 + 2 - 3 = 1$

Ex. 5. *If* $U = \{(1,2,1),(0,1,2)\}, W = \{(1,0,0),(0,1,0)\}$ *determine the dimension of* $U + W$ ***(A. U. 2011)***

Sol. Given $U = \{(1,2,1),(0,1,2)\}$, $W = \{(1,0,0),(0,1,0)\}$ are vector spaces

We can observe that, $\dim U = 2, \dim W = 2$

To find a basis for $U \cap W$

Let $x \in U \cap W \quad \Rightarrow x \in U$ and $x \in W$

$\Rightarrow x = a(1,2,1) + b(0,1,2) = (a, 2a+b, a+2b)$ and $x = c(1,0,0) + d(0,1,0) = (c,d,0)$

Equating the values $a = c, 2a + b = d$ and $a + 2b = 0$

Solving we get, $a = (2/3)d, b = (-1/3)d, c = (2/3)d$

$\therefore x = (c,d,0) = \left(\frac{2}{3}d, d, 0\right) = \frac{1}{3}d(2,3,0) \quad \therefore U \cap W$ is generated by $(2,3,0)$

$\dim U \cap W = 1$. $\quad \dim(U + W) = \dim U + \dim W - \dim(U \cap W) = 2 + 2 - 1 = 3$

EXERCISE 2 (*b*)

1. P and Q are the two subspaccs of R^4 defined by $P = \{(a, b, c, d) : b + c + d = 0\}$;

$Q = \{(a, b, c, d) : a + b = 0, c = 2d\}$. Find the dimension and basis of P, Q and $P \cap Q$.

2. Let W_1 and W_2 be the subspaces of R^4 generated by

$\{1,1,0,-1), (1,2,3,0), (2,3,3,-1)\}$ and $\{1,2,2,-2), (2,3,2,-3), (1,3,4,-3)\}$ respectively.

Find the dimension of (*i*) W_1 (*ii*) W_2 (*iii*) $W_1 + W_2$ (*iv*) $W_1 \cap W_2$.

QUOTIENT SPACE

2.15. Coset : *Let W be a subspace of a vector space V (F) then for any element* $\alpha \in V$, *the set* $W + \alpha = \{x + \alpha / x \in W\}$ *is called the right coset of W in V,generated by* α.

Similarly the set $\alpha + W = \{\alpha + x / x \in W\}$ *is called the left coset of W in V, generated by* α.

Since (W, +) is a subgroup of the abelian group (V. +), by commutative property

$x + \alpha = \alpha + x \quad \forall \alpha \in V,\ x \in W$

Hence $W + \alpha$ is called simply the coset of W in V, generated by α.

Note 1. For $\bar{O} \in V,\ \bar{O} + W = W$. $\therefore$ W is itself a coset in V, generated by $\bar{O}$.

Note. 2. For $x \in W,\ x + W = W$. $\therefore$ Coset $x + W =$ coset W.

Note. 3. Any two cosets of W in V are either identical or disjoint.

i.e. Either $\alpha + W = \beta + W$ or $(\alpha + W) \cap (\beta + W) = \phi$

Note 4. If $\alpha + W$ and $\beta + W$ are two cosets of W in V then

$\alpha + W = \beta + W \Leftrightarrow \alpha - \beta \in W$

2.16. QUOTIENT SET

Let W be a subspace of V (F). Then the set of all cosets of W in V denoted by

$V/W = \{W + \alpha \ \forall \alpha \in V\}$ is called Quotient set.

2.17. Theorem. ***Let W be a subspace of vector V (F). Then the set V /W is a vector space over F for the vector addition and scalar multiplication defined by***

$$(W + \alpha) + (W + \beta) = W + (\alpha + \beta) \ \forall \alpha, \beta \in V \quad \ldots (1)$$

$$a(W + \alpha) = W + a\alpha \ \forall a \in F,\ \alpha \in V \quad \ldots (2)$$

Proof. First of all let us prove that the above two compositions are well defined.

(*i*) Let $W + \alpha = W + \alpha'$ then it $\Rightarrow \alpha - \alpha' \in W$

$W + \beta = W + \beta'$ then it $\Rightarrow \beta - \beta' \in W$. Hence $(\alpha - \alpha') + (\beta - \beta') \in W$

$\Rightarrow (\alpha + \beta) - (\alpha' + \beta') \in W \Rightarrow W + (\alpha + \beta) = W + (\alpha' + \beta')$

$\therefore$ Addition of cosets is well defined

(*ii*) $W + \alpha = W + \alpha' \Rightarrow \alpha - \alpha' \in W$

$\alpha - \alpha' \in W, a \in F \Rightarrow a(\alpha - \alpha') \in W \Rightarrow a\alpha - a\alpha' \in W \Rightarrow W + a\alpha = W + a\alpha'$

$\therefore$ Scalar multiplication is well defined.

Now we shall show that V/W is a vector space.

1. Closure axiom. $\alpha, \beta \in V \Rightarrow \alpha + \beta \in V$ and $W + \alpha, W + \beta \in V/W$

$\therefore (W + \alpha) + (W + \beta) = W + (\alpha + \beta)$ by (1)

As $W+(\alpha+\beta) \in V/W$, it is closed under addition.

2. Associativity. Let $W+\alpha, W+\beta, W+\gamma \in V/W$ for $\alpha, \beta, \gamma \in V$.

$\therefore\ [(W+\alpha)+(W+\beta)]+(W+\gamma)=[W+(\alpha+\beta)]+(W+\gamma)$

$W+(\alpha+\beta)+\gamma = W+\alpha+(\beta+\gamma)$

$[W+\alpha]+[W+(\beta+\gamma)] = (W+\alpha)+[(W+\beta)+(W+\gamma)]$

$\therefore$ V/W is associative.

3. Identity. $W+\bar{O}, W+\alpha \in V/W$ for $\alpha \in V$

$\therefore\ (W+\bar{O})+(W+\alpha)=W+(\bar{O}+\alpha)=W+\alpha$ ($\bar{O}$ being the identity of V)

and $(W+\alpha)+(W+\bar{O})=W+\alpha \Rightarrow W+\bar{O}$ is additive identity in V /W

4. Inverse. $\alpha \in V \Rightarrow -\alpha \in V$

$\therefore\ (W+\alpha)+[W+(-\alpha)]=W+(\alpha-\alpha)=W+\bar{O}$ = additive identity

$\Rightarrow$ additive inverse exists in V /W.

5. Commutative. $(W+\alpha)+(W+\beta)=W+(\alpha+\beta)$ for $\alpha, \beta \in V$

$=W+(\beta+\alpha)=(W+\beta)+(W+\alpha) \Rightarrow$ addition is commutative

$\therefore$ V/W is an abelian group for addition defined in (1).

6. $a \in F$ and $W+\alpha, W+\beta \in V/W \Rightarrow\ a[(W+\alpha)+(W+\beta)]=a[W+\alpha+\beta)]$

$=W+a(\alpha+\beta)=W+(a\alpha+a\beta)=(W+a\alpha)+W+a\beta=a(W+\alpha)+a(W+\beta)$

7. $a,b \in F$ and $W+\alpha \in V/W \Rightarrow\ (a+b)(W+\alpha)=W+(a+b)\alpha$

$=W+(a\alpha+b\alpha)=(W+a\alpha)+(W+b\alpha)=a(W+\alpha)+b(W+\alpha)$

8. $a,b \in F$ and $W+\alpha \in V/W$

$(a\,b)(W+\alpha)=W+(a\,b)\alpha=W+a(b\alpha)=a[W+b\alpha]=a[b(W+\alpha)]$

9. $W+\alpha \in V/W$ and the unity $1 \in F, \Rightarrow\ 1(W+\alpha)=W+1.\alpha=W+\alpha$

This shows that all the postulates of vector space are satisfied. Hence V/W is a vector space.

The vector space V /W is called the Quotient space of V relative to W.

DIMENSION OF QUOTIENT SPACE

2.18. Theorem. *Let W be a sub space of a finite dimensional vector space V(F), then* **dim V/W = dim V − dim W.** *(S. K. U. 2001O, 11, A. N. U. M 2013, 94M, 95)*

Proof. Since V is finite dimensional, W is also finite dimensional.

Let the set $B=\{\alpha_1, \alpha_2, \ldots \alpha_m\}$ be the basis of W. $\therefore$ dim $W = m$.

Since the set B is L.I. it can be extended to form a basis of V.

Let the set $S = \{\alpha_1, \alpha_2, \ldots \alpha_m, \beta_1, \beta_2 \ldots \beta_l\}$ be the basis of V. $\therefore$ dim $V = m + l$

$\therefore$ dim V − dim W $= (m + l) - m = l$

Now we shall prove that the set $S' = \{W + \beta_1, W + \beta_2 \ldots W + \beta_l\}$ is a basis of V/W and hence dim $V/W = l$.

(i) To prove S' is L.I.

The zero vector of V/W is W

Now $b_1 (W + \beta_1) + b_2 (W + \beta_2) + \ldots + b_l (W + \beta_l) = W$ .... (1)

$\Rightarrow (W + b_1\beta_1) + (W + b_2\beta_2) + \ldots + (W + b_l\beta_l) = W$

$\Rightarrow W + (b_1\beta_1 + b_2\beta_2 + \ldots + b_l\beta_l) = W \Rightarrow (b_1\beta_1 + b_2\beta_2 + \ldots + b_l\beta_l) \in W$

But any vector of W is *l.c.* of elements of B.

Let $b_1\beta_1 + b_2\beta_2 + \ldots + b_l\beta_l = a_1\alpha_1 + a_2\alpha_2 \ldots + a_m\alpha_m$

$\Rightarrow b_1\beta_1 + b_2\beta_2 + \ldots + b_l\beta_l - a_1\alpha_1 - a_2\alpha_2 \ldots - a_m\alpha_m = \bar{O}$

$\Rightarrow$ (l.c. of elements of L.I set) = $\bar{O}$

$\Rightarrow b_1 = b_2 = \ldots = b_l = 0, a_1 = a_2 = \ldots = a_m = 0 \Rightarrow$ The set S' is L.I.

(ii) To Prove L (S') = V/W.

Since S is the basis of V, $\alpha \in V$ can be expressed as

$\alpha = c_1\alpha_1 + c_2\alpha_2 + \ldots + c_m\alpha_m + d_1\beta_1 + d_2\beta_2 \ldots + d_l\beta_l$

$= \gamma + d_1\beta_1 + d_2\beta_2 \ldots + d_l\beta_l$ where $c's, d's \in F$ where $\gamma = c_1\alpha_1 + c_2\alpha_2 \ldots c_m\alpha_m$

= (*l.c.* of the elements of B) $\Rightarrow \gamma \in W$

For $\alpha \in V$, $W + \alpha \in V/W$

Now $W + \alpha = W + \gamma + d_1\beta_1 + d_2\beta_2 \ldots + d_l\beta_l$

$= W + d_1\beta_1 + d_2\beta_2 \ldots + d_l\beta_l$ [$\gamma \in W \Rightarrow W + \gamma = W$]

$= (W + d_1\beta_1) + (W + d_2\beta_2) \ldots + (W + d_l\beta_l) = d_1(W + \beta_1) + d_2(W + \beta_2) \ldots + d_l(W + \beta_l)$

(*l.c.* of elements of S')

$\therefore W + \alpha \in L(S')$ $\therefore L(S') = V/W$

Therefore S' is the basis of V/W and hence dim $V/W = l =$ dim V − dim W.

Ex. *If W is a subspace of V (F) then for $a, b \in V$ show that*

(i) $\alpha \in (\beta + W)$ *(ii)* $\beta \in (\alpha + W)$ *are equivalent.*

Sol. $\alpha \in \beta + W \Rightarrow \alpha = \beta + \gamma$ for some $\gamma \in W \Rightarrow \alpha - \beta \in W$

For $(-1) \in F, (-1)(\alpha - \beta) \in W \Rightarrow \beta - \alpha \in W \Rightarrow (\beta - \alpha + \alpha) \in W + \alpha \Rightarrow \beta \in W + \alpha$.

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LINEAR ALGEBRA

UNIT - III

Linear Transformations

Linear transformations, linear operators, Properties of L.T, sum and product of LTs, Algebra of Linear Operators, Range and null space of linear transformation, Rank and Nullity of linear transformations – Rank – Nullity Theorem.

3. **Linear Transformations**
4. **Vector Space Isomorphism**

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3

Linear Transformations

3.1. VECTOR SPACE HOMOMORPHISM

Definition. *Let U and V be two vector spaces over the same field F. Thus the mapping* $f: U \to V$ *is called a homomorphism from U into V if*

(*i*) $f(\alpha+\beta) = f(\alpha)+f(\beta)\ \forall\ \alpha,\beta \in U$ (*ii*) $f(a\alpha) = af(\alpha)\ \forall a \in F; \forall\ \alpha \in U$

Note. 1. If f is onto function then V is called the homomorphic image of f.

2. If f is one - one onto function then f is called an isomorphism. Thus it is said that U is isomorphic to V denoted by $U \cong V$.

3. The two conditions of homomorphism are combined into a single condition, called the linear property to define the linear transformation as below.

3.2. LINEAR TRANSFORMATION

Definition. *Let U(F) and V(F) be two vector spaces. Then the function.* $T: U \to V$ *is called a linear transformation of U into V if* $T(a\alpha+b\beta) = aT(\alpha)+bT(\beta)\ \forall a,b \in F; \alpha,\beta \in U$.

Clearly the vector space homomorphism is equivalent to linear transformation.

Linear Operator : Definition. *If* $T: U \to U$ *(i.e T transforms U into itself) then T is called a linear operator on U.*

Linear Functional : Definition. *If* $T: U \to F$ *(i.e. T transforms U into the field F) then T is called a linear functional on U.*

3.3. ZERO TRANSFORMATION

Theorem. ***Let U(F) and V(F) be two vector spaces. Let the mapping*** $T: U \to V$ ***be defined by*** $T(\alpha) = \hat{0}\ \forall\ \alpha \in U$ ***where*** $\hat{0}$ ***(zero crown) is the zero vector of V. Then T is a linear transformation.***

Proof. For $a,b \in F$ and $\alpha,\beta \in U \Rightarrow a\alpha+b\beta \in U$ ($\because$ U is V.S.)

By definition we have $T(a\alpha+b\beta) = \hat{0} = a\hat{0}+b\hat{0} = aT(\alpha)+bT(\beta)$

$\therefore$ By the definition of linearity T is a linear transformation.

Such a L.T. is called the zero transformation and is denoted by O.

3.4. IDENTITY OPERATOR

Theorem. ***Let V(F) be a vector spaee and the mapping*** $I: V \to V$ ***be defined by*** $I(\alpha) = \alpha\ \ \forall\ \alpha \in V$. ***Then, I is a linear operator from V into itself.***

Proof. $a, b \in F$ and $\alpha, \beta \in V \Rightarrow a\alpha + b\beta \in V$ ($\because$ V is L.S.)

By definition we have $I(a\alpha + b\beta) = a\alpha + b\beta = a\,I(\alpha) + b\,I(\beta)$ (by def.)

$\therefore$ I is a L. T from V into itself and I is called the **Identity Operator.**

3.5. NEGATIVE OF TRANSFORMATION

Theorem. *Let U (F) and V (F) be two vector spaces and* $T: U \to V$ *be a linear transformation. Then the mapping* $(-T)$ *defined by* $(-T)(\alpha) = -T(\alpha)\ \forall\ \alpha \in U$ *is a linear transformation.*

Proof. $a, b \in F$ and $\alpha, \beta \in U \Rightarrow a\alpha + b\beta \in U$ ($\because$ U is V.S.)

Now by definition $(-T)(a\alpha + b\beta) = -[T(a\alpha + b\beta)]$

$= -[a\,T(\alpha) + b\,T(\beta)] = -a\,T(\alpha) - b\,T(\beta)$

$= a[-T(\alpha)] + b[-T(\beta)] = a[(-T)(\alpha)] + b[(-T)(\beta)]$

$\Rightarrow -T$ is a linear transformation.

PROPERTIES OF LINEAR TRANSFORMATIONS

3.6. Theorem. *Let* $T: U \to V$ *is a linear transformation from the vector space U(F) to the vector space V(F). Then* (*i*) $T(\bar{O}) = \hat{\bar{O}}$, *where* $\hat{\bar{O}} \in U$ *and* $\hat{\bar{O}} \in V$

(*ii*) $T(-\alpha) = -T(\alpha)\ \forall\ \alpha \in U$ (*iii*) $T(\alpha - \beta) = T(\alpha) - T(\beta)\ \forall\ \alpha, \beta \in U$

(*iv*) $T(a_1\alpha_1 + a_2\alpha_2 \ldots + a_n\alpha_n) = a_1T(\alpha_1) + a_2T(\alpha_2) \ldots + a_nT(\alpha_n)\ \forall\ a_i \in F$ *and* $\alpha's \in U$.

(S. V.U. 2011)

Proof. (*i*) $\alpha, \bar{O} \in U \Rightarrow T(\alpha), T(\bar{O}) \in V$

Now $T(\alpha) + T(\bar{O}) = T(\alpha + \bar{O})$ (T is L.T.) $= T(\alpha) = T(\alpha) + \hat{\bar{O}}$ ($\bar{O} \in V$)

By cancellation law $T(\bar{O}) = \hat{\bar{O}}$

(*ii*) $T(-\alpha) = T(-1.\alpha) = (-1)\,T(\alpha) = -T(\alpha)$

(*iii*) $T(\alpha - \beta) = T[\alpha + (-1)\beta] = T(\alpha) + (-1)T(\beta) = T(\alpha) - T(\beta)$ ($\because$ T is L.T.)

(*iv*) For $n = 1$, $T(a_1\alpha_1) = a_1\,T(\alpha_1)$ ($\because$ T is L.T.)

$n = 2$, $T(a_1\alpha_1 + a_2\alpha_2) = a_1T(\alpha_1) + a_2T(\alpha_2)$

Let this be true for $n = m$

$\therefore T(a_1\alpha_1 + a_2\alpha_2 + a_m\alpha_m) = a_1T(\alpha_1) + a_2T(\alpha_2) \ldots + a_mT(\alpha_m)$ ... (1)

Now $T[a_1\alpha_1 + \ldots + a_m\alpha_m) + a_{m+1}\alpha_{m+1}] = T(a_1\alpha_1 + \ldots + a_m\alpha_m) + T(a_{m+1}\alpha_{m+1})$

$= a\,T(\alpha_1) + \ldots + a_mT(\alpha_m) + a_{m+1}T(\alpha_{m+1})$ $\therefore$ The relation is true for $n = m+1$

Hence it is true for all integral values of n.

DETERMINATION OF LINEAR TRANSFORMATION

3.7. Theorem. *Let U(F) and V(F) be two vector spaces and* $S = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ *be a basis of U. Let* $\{\delta_1, \delta_2, \ldots, \delta_n\}$ *be a set of n vectors in V. Then there exists a unique linear transformation* $T : U \to V$ *such that* $T(\alpha_i) = \delta_i$ *for i = 1, 2, ... n.*

Proof. Let $\alpha \in U$. Since $S = \{\alpha_1, \alpha_2 .. \alpha_n\}$ is a basis of U, there exist unique scalars $a_1, a_2 \ldots a_n \in F$ such that $\alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n$

(*i*) Existence of T, $\delta_1, \delta_2, \ldots, \delta_n \in V \Rightarrow (a_1\delta_1 + a_2\delta_2 + \ldots + a_n\delta_n) \in V$

We define $T : U \to V$, such that $T(\alpha) = a_1\delta_1 + a_2\delta_2 + \ldots + a_n\delta_n$

$\therefore$ T is a mapping from U into V.

Now $\alpha_i = 0.\alpha_1 + 0.\alpha_1 + 0.\alpha_2 .. + 1.\alpha_i + 0.\alpha_{i+1} + \ldots + 0.\alpha_n$

$\therefore$ by the definition of T mapping

$T(\alpha_i) = 0.\delta_1 + 0.\delta_1 + 0.\delta_2 .. + \delta_i + 0.\delta_{i+1} \ldots + 0.\delta_n \Rightarrow T(\alpha_i) = \delta_i, \ \forall\, i = 1, 2, \ldots n$

(*ii*) To show that T is L.T:

Let $a, b \in F$ and $\alpha, \beta \in U$

$\therefore \alpha = a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n ; \qquad \beta = b_1\alpha_1 + b_2\alpha_2 + \ldots + b_n\alpha_n$

$\therefore T(\alpha) = a_1\delta_1 + a_2\delta_2 + \ldots + a_n\delta_n ; \qquad T(\beta) = b_1\delta_1 + b_2\delta_2 + \ldots + b_n\delta_n$

$\therefore a\alpha + b\beta = a(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n) + b(b_1\alpha_1 + b_2\alpha_2 + \ldots + b_n\alpha_n)$ (by def.)

$= (aa_1 + bb_1)\alpha_1 + (aa_2 + bb_2)\alpha_2 + \ldots + (aa_n + bb_n)\alpha_n$

$\therefore T(a\alpha + b\beta) = T[(aa_1 + bb_1)\alpha_1 + (aa_2 + bb_2)\alpha_2 + \ldots + (aa_n + bb_n)\alpha_n]$

$= (aa_1 + bb_1)\delta_1 + (aa_2 + bb_2)\delta_2 + \ldots + (aa_n + bb_n)\delta_n$

$= a(a_1\delta_1 + a_2\delta_2 + \ldots + a_n\delta_n) + b(b_1\delta_1 + \ldots + b_n\delta_n) = a\,T(\alpha) + b\,T(\beta)$

$\therefore$ T is a L.T.

(*iii*) To show that T is unique.

Let $T' : U \to V$ be another L.T. so that $T(\alpha_i) = \delta_i$ for $i = 1, 2, \ldots n$

If $\alpha = a_1\alpha_1 + a_2\alpha_2 \ldots + a_n\alpha_n$

then $T'(\alpha) = T'(a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n) = a_1T'(\alpha_1) + a_2T'(\alpha_2) + \ldots + a_nT'(\alpha_n)$

$= a_1\delta_1 + a_2\delta_2 + \ldots + a_n\delta_n = T(\alpha)$

$\therefore$ T' = T and hence T is unique.

Note. In determining the L.T. the assumption that $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ is a basis of U is essential.

3.8. Theorem. *Let $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ and $S' = \{\beta_1, \beta_2, ..., \beta_n\}$ be two ordered bases of n, dimensional vector space V(F). Let $\{a_1, a_2, ..., a_n\}$ be an ordered set of n scalars such that $\alpha = a_1\alpha_1 + a_2\alpha_2, ..., a_n\alpha_n$ and $\beta = a_1\beta_1 + a_2\beta_2, ..., a_n\beta_n$. Show that $T(\alpha) = \beta$ where T is the linear operator on V defined by $T(\alpha_i) = \beta_i$, $i = 1, 2, ..., n$.*

Proof. Now $T(\alpha) = T(a_1\alpha_1 + a_2\alpha_2 + ... + a_n\alpha_n)$

$= a_1T(\alpha_1) + a_2T(\alpha_2) + ... + a_nT(\alpha_n)$ (T is L.T.)

$= a_1\beta_1 + a_2\beta_2 + ... + a_n\beta_n = \beta$

SOLVED PROBLEMS

Ex. 1. *The mapping $T : V_3(R) \to V_2(R)$ is defined by T (x, y, z) = $(x - y, x - z)$. Show that T is a linear transformation.* ***(A. N. U. M 2013, S.V.U. 2011, M 12)***

Sol. Let $\alpha = (x_1, y_1, z_1)$ and $\beta = (x_2, y_2, z_2)$ be two vectors of $V_3(R)$

For $a, b \in R$

$T[a\alpha + b\beta] = T[a(x_1, y_1, z_1) + b(x_2, y_2, z_2)] = T(ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2)$

$= ((ax_1 + bx_2) - (ay_1 + by_2),\ ax_1 + bx_2 - (az_1 + bz_2))$

$= (a(x_1 - y_1) + b(x_2 - y_2),\ a(x_1 - z_1) + b(x_2 - z_2))$

$= (a(x_1 - y_1), a(x_1 - z_1) + (b(x_2 - y_2), b(x_2 - z_2))$

$= a(x_1 - y_1, x_1 - z_1) + b(x_2 - y_2, x_2 - z_2)$

$= aT(x_1, y_1, z_1) + bT(x_2, y_2, z_2) = aT(\alpha) + bT(\beta)$

$\Rightarrow$ T is a linear transformation from $V_3(R)$ to $V_2(R)$.

Ex. 2. *If $T : V_3(R) \to V_2(R)$ is defined as $T(x_1, x_2, x_3) = (x_1 - x_2, x_1 + x_3)$*

Prove that T is a linear transformation. ***(S.K. U. M 14)***

Sol. Let $\alpha = (p_1, q_1, r_1), \beta = (p_2, q_2, r_2)$ be two vectors of $V_3(R)$. Let $a, b \in R$.

Let $T(a\alpha + b\beta) = T[a(p_1, q_1, r_1) + b(p_2, q_2, r_2)]$

$= T[ap_1 + bp_2 . \overline{aq_1 + bq_2}, ar_1 + br_2]$

$= (ap_1 + bp_2 - \overline{aq_1 + bq_2}, ap_1 + bp_2 + ar_1 + br_2)$

$= (a(p_1 - q_1) + b(p_2 - q_2), a(p_1 + r_1) + b(p_2 + r_2))$

$= a(p_1 - q_1, p_1 + r_1) + b(p_2 - q_2, p_2 + r_2)$

$= aT(p_1, q_1, r_1) + bT(p_2, q_2, r_2) = aT(\alpha) + bT(\beta)$

Similarly we can prove $T(c\alpha) = cT(\alpha)$ for $c \in R$

Thus, $T : V_3(R) \to V_2(R)$ is a linear transformation.

Ex. 3. *The mapping* $T : V_3(R) \to V_1(R)$ *is defined by* $T(a,b,c) = a^2 + b^2 + c^2$*; Can T be a linear transformation ?* ***(O. U. 2001/O)***

Sol. Let $\alpha = (a,b,c)$ and $\beta = (x,y,z)$ be two vectors of $V_3(R)$.

For $p, q \in R$, $T(pa + qx,\ pb + qy,\ pc + qz) = (pa+qx)^2 + (pb+qy)^2 + (pc+qz)^2$

Now $= p\,T(\alpha) + q\,T(\beta) = p\,T(a,b,c) + q\,T(x,y,z)$

$= p(a^2 + b^2 + c^2) + q(x^2 + y^2 + z^2) = T(p\alpha + q\beta) \neq p\,T(\alpha) + q\,T(\beta)$

$\therefore$ T is not a L.T from $V_3(R)$ to $V_1(R)$.

Ex. 4. *Let V be the vector space of polynomials in the variable x over R. Let* $f(x) \in V(R)$*; show that*

(*i*) $D : V \to V$ *defined by* $Df(x) = \dfrac{df(x)}{dx}$

(*ii*) $I : V \to V$ *defined by* $If(x) = \int_0^x f(x)\,dx$ *are linear transformations.*

Sol. Let $f(x), g(x) \in V(R)$ and $a, b \in R$

(*i*) $D[af(x) + bg(x)] = \dfrac{d}{dx}[af(x) + bg(x)] = \dfrac{d}{dx}[af(x)] + \dfrac{d}{dx}[bg(x)]$

$= a\dfrac{d}{dx}[f(x)] + b\dfrac{d}{dx}[g(x)] = a\,D\,f(x) + b\,D\,g(x)$

$\therefore$ D is a linear transformation and D is called a differential operator.

(ii) $I[af(x) + bg(x)] = \int_0^x [af(x) + bg(x)]\,dx = a\int_0^x f(x)\,dx + b\int_0^x g(x)\,dx = a\,I\,f(x) + b\,I\,g(x)$

$\therefore$ I is a linear transformation and is called Integral transformation.

Ex. 5. *Let* $P_n(R)$ *be the vector space of all polynomials of degree n over a field R. If a linear operator T on* $P_n(R)$ *is such that* $Tf(x) = f(x+1)$, $f(x) \in P_n(R)$.

Show that $T = 1 + \dfrac{D}{1!} + \dfrac{D^2}{2!} + \dfrac{D^3}{3!} + \ldots + \dfrac{D^n}{n!}$

Sol. Let $f(x) = a_0 + a_1x + a_2x^2 \ldots + a_nx^n \quad \forall\ a_i \in R$

$$\left[1 + \frac{D}{1!} + \frac{D^2}{2!} + \frac{D^3}{3!} + \ldots + \frac{D^n}{n!}\right] f(x) = a_0 + a_1x + a_2x^2 \ldots + a_nx^n$$

$$+ \frac{1}{1!}(0 + a_1 + 2a_2x + 3a_3x^2 + \ldots + na_nx^{n-1}) + \frac{1}{2!}(0 + 0 + 2a_2 + 6a_3 + \ldots + n(n-1)a_nx^{n-2})$$

$$+ \ldots\ldots \ldots\ldots \quad \ldots\ldots \quad \ldots\ldots \quad + \frac{1}{n!}(0+0+C \ldots + a_n\, n!)$$

$$= a_0 + a_1 (x+1) + a_2 (x+1)^2 + \ldots + a_n (x+1)^n$$

$$= f(x+1) = \mathrm{T}\, f(x) \qquad \text{(by def.)}$$

$$\therefore \ \mathrm{T} = \left(1 + \frac{D}{1!} + \frac{D^2}{2!} + \ldots + \frac{D^n}{n!}\right).$$

Thus T is a linear operator from $\mathrm{P}_n(\mathrm{R})$ into $\mathrm{P}_n(\mathrm{R})$.

Ex. 6. *Is the mapping* $\mathrm{T}: \mathrm{R}^3 \to \mathrm{R}^2$ *defined by T (x, y, z) = (| x |, 0)a linear transformation.*

Sol. We have $\mathrm{T}: \mathrm{R}^3 \to \mathrm{R}^2$ defined by $\mathrm{T}(x, y, z) = (|x|, 0)$

Let $\alpha, \beta \in \mathrm{R}^3$ where $\alpha = (x_1, y_1, z_1)$ and $\beta = (x_2, y_2, z_2)$

For $a, b \in \mathrm{R}$, $a\alpha + b\beta = a(x_1, y_1, z_1) + b(x_2, y_2, z_2) = (ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2)$

$\therefore \ \mathrm{T}(a\alpha + b\beta) = \mathrm{T}(ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2) = (|ax_1 + bx_2|, 0)$

And $a\,\mathrm{T}(\alpha) + b\,\mathrm{T}(\beta) = a\,\mathrm{T}(x_1, y_1, z_1) + b\,\mathrm{T}(x_2, y_2, z_2)$

$$= a(|x_1|, 0) + b(|x_2|, 0) = (a|x_1| + b\,|x_2|, 0)$$

Clearly $\mathrm{T}(a\alpha + b\beta) \neq a\mathrm{T}(\alpha) + b\,\mathrm{T}(\beta)$. Hence T is not a linear transformation.

Ex. 7. *Let T be a linear transformation on a vector space U into V. Prove that the vectors* $x_1, x_2, \ldots x_n \in \mathrm{U}$ *are linearly independent if* $\mathrm{T}(x_1), \mathrm{T}(x_2) \ldots \mathrm{T}(x_n)$ *are L.I.*

Sol. Given $\mathrm{T}: \mathrm{U(F)} \to \mathrm{V(F)}$ is a L.T. and $x_1, x_2, \ldots x_n \in \mathrm{U}$.

Let there exist $a_1, a_2 \ldots a_n \in \mathrm{F}$ such that $a_1x_1 + a_2x_2 + \ldots + a_nx_n = \bar{\mathrm{O}}$ ... (1) $(\bar{\mathrm{O}} \in \mathrm{U})$

$\therefore \ \mathrm{T}(a_1x_1 + a_2x_2 + \ldots + a_nx_n = \mathrm{T}(\bar{\mathrm{O}}) \Rightarrow a_1\,\mathrm{T}(x_1) + a_2\,\mathrm{T}(x_2) \ldots + a_n\,\mathrm{T}(x_n) = \hat{\bar{\mathrm{O}}}$ $(\bar{\mathrm{O}} \in \mathrm{V})$

But $\mathrm{T}(x_1), \mathrm{T}(x_2), \ldots, \mathrm{T}(x_n)$ are L. I. $\therefore \ a_1 = a_2 = \ldots = a_n = 0$.

$\therefore$ From (1) $x_1, x_2 \ldots x_n$ are L. I.

Ex. 8. *Let V be a vector space of* $n \times n$ *matrices over the field F. M is a fixed matrix in V. The mapping* $\mathrm{T}: \mathrm{V} \to \mathrm{V}$ *is defined by T (A) = AM + MA where* $\mathrm{A} \in \mathrm{V}$. *Show that T is linear.*

Sol. Let $a, b \in \mathrm{F}$ and $\mathrm{A}, \mathrm{B} \in \mathrm{V}$. Then $\mathrm{T}(\mathrm{A}) = \mathrm{AM} + \mathrm{MA}$ and $\mathrm{T}(\mathrm{B}) = \mathrm{BM} + \mathrm{MB}$

$\therefore \ \mathrm{T}(a\mathrm{A} + b\mathrm{B}) = (a\mathrm{A} + b\mathrm{B})\,\mathrm{M} + \mathrm{M}(a\mathrm{A} + b\mathrm{B}) = a(\mathrm{AM} + \mathrm{MA}) + b(\mathrm{BM} + \mathrm{MB})$

$= a\,\mathrm{T}(\mathrm{A}) + b\,\mathrm{T}(\mathrm{B})$. T is a linear transformation.

Ex. 9. *Describe explicitly the linear transformation* $T: R^2 \to R^2$ *such that* $T(2,3) = (4,5)$ *and* $T(1,0) = (0,0)$. ***(S. V. U. M14, 11, S97, S.K.D. 11)***

Sol. First of all we have to show that the vectors (2,3) and (1, 0) are L.I.

Let $= a(2,3) + b(1,0) = \bar{O}$

$\Rightarrow (2a+b, 3a+0) = (0,0) \Rightarrow 2a+b=0, 3a=0 \Rightarrow 2a=0, b=0$

$\therefore$ $S = \{(2,3), (1,0)\}$ is L.I. Let us prove that L (S) = R^2

Let $(x,y) \in R^2$ and $(x,y) = a(2,3) + b(1,0) = (2a+b, 3a)$

$\Rightarrow 2a+b=x,\ 3a=y \Rightarrow a=\frac{y}{3}; b=\frac{3x-2y}{3}$. Hence S spans R^2

Now $T(x,y) = T\left[\frac{y}{3}(2,3) + \frac{3x-2y}{3}(1,0)\right]$

$$= \frac{y}{3}T(2,3) + \frac{3x-2y}{3}T(1,0) = \frac{y}{3}(4,5) + \frac{3x-2y}{3}(0,0) = \left(\frac{4y}{3}, \frac{5y}{3}\right)$$

$\therefore$ This is the required transformation.

Ex. 10. *Find* $T(x, y, z)$ *where* $T: R^3 \to R$ *is defined by* $T(1,1,1) = 3$, $T(0,1,-2) = 1$, $T(0,0,1) = -2$. ***(S. V. U. M 15, M 2012)***

Sol. Let $S = \{(1,1,1), (0,1,-2), (0,0,1)\}$

(*i*) Let $a(1,1,1) + b(0,1,-2) + c(0,0,1) = \bar{O}$

$\Rightarrow (a, a+b, a-2b+c) = (0,0,0)$ $(\because \bar{O} \in R^3)$

$\Rightarrow a=0, a+b=0, a-2b+c=0 \Rightarrow a=0, b=0, c=0$ $(\therefore$ S is L.I. set)

(*ii*) Let $(x,y,z) \in R^3$

$(x,y,z) = a(1,1,1) + b(0,1,-2) + c(0,0,1) = (a, a+b, a-2b+c)$

$\Rightarrow a=x, a+b=y, a-2b+c=z \Rightarrow a=x, b=y-x, c=z+2y-3x$

$\therefore$ S spans R^3

Hence $T(x,y,z) = T[x(1,1,1) + (y-x)(0,1,-2) + (z+2y-3x)(0,0,1)]$

$= x\,T(1,1,1) + (y-x)\,T(0,1,-2) + (z+2y-3x)\,T(0,0,1)$

$= x(3) + (y-x)(1) + (z+2y-3x)(-2)$

$= 8x - 3y - 2z$ which is the required linear functional.

Ex. 11. *Let* $T:U \to V$ *be a linear transformation* $\{(1,2,1),(2,1,0),(1,-1,-2)\}$ *and* $\{(1,0,0),(0,1,0),(1,1,1)\}$ *are the basis of U and V. Find T for the transformation of the basis of U to the basis of V.* **(A. U. 2011)**

Sol. Let $\alpha_1 = (1,2,1), \alpha_2 = (2,1,0), \alpha_3 = (1,-1,-2)$ and let $\{\alpha_1, \alpha_2, \alpha_3\}$ form a basis for U.

Let $\beta_1 = (1,0,0), \beta_2 = (0,1,0), \beta_3 = (1,1,1)$ and $\{\beta_1, \beta_2, \beta_3\}$ form a basis for V.

We define $T: U \to V$ as follows : $T(\alpha_i) = \beta_i, i = 1,2,3$

Let $\alpha \in U$. Let $\alpha = a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3$

$= a_1(1,2,1) + a_2(2,1,0) + a_3(1,-1,-2)$

$= (a_1 + 2a_2 + a_3, 2a_1 + a_2 - a_3, a_1 - 2a_3) = (x,y,z)$ (say)

Then $x = a_1 + 2a_2 + a_3; y = 2a_1 + a_2 - a_3; z = a_1 - 2a_3$

Equating the components, $x = a_1 + 2a_2 + a_3, y = 2a_1 + a_2 - a_3, z = a_1 - 2a_3$

Solving we get, $a_1 = \frac{1}{3}(-2x + 4y - 3z); a_2 = x - y + z; a_3 = \frac{1}{3}(-x + 2y - 3z)$

$T(\alpha) = T(a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3)$

$= a_1T(\alpha_1) + a_2T(\alpha_2) + a_3T(\alpha_3)$ ($\because$ T is a L. T.)

$= a_1\beta_1 + a_2\beta_2 + a_3\beta_3$ (by the dfinition of T)

$= a_1(1,0,0) + a_2(0,1,0) + a_3(1,1,1)$

$= (a_1 + a_3, a_2 + a_3, a_3)$

$\therefore\ T(\alpha) = T(x,y,z) = (a_1 + a_3, a_2 + a_3, a_3)$

$$= \left(-x + 2y - 2z, \frac{2x - y}{3}, \frac{-x - 2y + 3z}{3}\right)$$

This gives the required linear transformation.

Ex. 12. *Show that the transformation* $T: R^3 \to R^3$ *defined by* $T(x,y,z) = (x - y, 0, y + z)$ *is a linear transformation* **(A. U. 2011)**

Sol. $T: R^3 \to R^3$ is defined as $T(x,y,z) = (x - y, 0, y + z)$

$\alpha = (x_1, y_1, z_1)$ and $\beta = (x_2, y_2, z_2) \in R^3$ (domain)

then $\alpha + \beta = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$ and $c\alpha = (cx_1, cy_1, cz_1)$

$T(\alpha + \beta) = T(x_1 + x_2, y_1 + y_2, z_1 + z_2)$

$$= (x_1 + x_2 - \overline{y_1 + y_2}, 0, y_1 + y_2 + \overline{z_1 + z_2})$$

$$= (x_1 - y_1 + x_2 - y_2, 0, y_1 + z_1 + y_2 + z_2)$$

$$= (x_1 - y_1, 0, y_1 + z_1) + (x_2 - y_2, 0, y_2 + z_2)$$

$$= T(\alpha) + T(\beta)$$

$\therefore$ $T(\alpha + \beta) = T(\alpha) + T(\beta)$ ... (1)

$T(c\alpha) = T(cx_1, cy_1, cz_1)$

$= (cx_1 - cy_1, 0, cy_1 + cz_1) = c(x_1 - y_1, 0, y_1 + z_1) = cT(\alpha)$

$T(c\alpha) = cT(\alpha)$ ... (2)

by (1) & (2) T is a linear transformation.

EXERCISE 3 (*a*)

1. Which of the following maps are linear transformations ,

(*a*) $T : V_1 \to V_3$ defined by $T(x) = (x, 2x, 3x)$.

(*b*) $T : R^2 \to R^2$ defined by $T(a,b) = (2a + 3b, 3a - 4b)$.

(*c*) $T : P \to P$ defined by $T(x) = x^2 + x$.

(*d*) $T : P \to P$ defined by $T\, p(x) = p(0) + xp'(0) + \frac{x^2}{2!} p''(0)$.

(*e*) $T : R^3 \to R^3$ defined by $T(x, y, z) = (x + 1, y, z)$.

(*f*) $T : R^2 \to R^2$ defined by $T(x, y) = (x^3, y^3)$.

2. Let V be the space of $m \times n$ matrices over the field F. Let P be a fixed $m \times m$ matrix and Q is a fixed $n \times n$ matrix over F.

$T : V \to V$ is defined by T (A) = PAQ. Then show that T is a linear transformation.

{ **Hint :** $T(aA + bB) = P(aA + bB)Q = (a\,PA + b\,PB)Q$

$= a\,PAQ + b\,PBQ = a\,T(A) + b\,T(B)$).

Find a linear transformation.

3. $T : R^2 \to R^2$ such that $T(1,0) = (1,1)$ and $T(0,1) = (-1,2)$ ***(A.N. U. M 13)***

4. $T : V_2 \to V_2$ such that $T(1,2) = (3,0)$ and $T(2,1) = (1,2)$

5. $T : V_3 \to V_3$ such that $T(0,1,2) = (3,1,2)$ and $T(1,1,1) = (2,2,2)$.

6. $T : V_2(R) \to V_3(R)$ such that $T(1,2) = (3,-1,5)$ and $T(0,1) = (2,1,-1)$

7. $T : R^2 \to R^3$ such that $T(2,-5) = (-1,2,3)$ and $T(3,4) = (0,1,5)$

8. Find a linear transformation. $T : R^2 \to R^2$, such that $T(1,0) = (1,1)$ and $T(0,1) = (-1,2)$

Prove that T maps the square with vertices (0,0), (1,0), (1,1), (0,1) into a parallelogram. ***(S. V. U. 2001/0)***

SUM OF LINEAR TRANSFORMATONS

3.9. Definition. *Let* T_1 *and* T_2 *be two linear transformations from U (F) into V (F). Then their sum* $T_1 + T_2$ *is defined by* $(T_1 + T_2)(\alpha) = T_1(\alpha) + T_2(\alpha)\ \forall\ \alpha \in U$

3.10. Theorem. ***Let U(F) and V(F) be two vector spaces. Let*** T_1 ***and*** T_2 ***be two linear transformations from U into V. Then the mapping*** $T_1 + T_2$ ***defined by*** $(T_1 + T_2)(\alpha) = T_1(\alpha) + T_2(\alpha)\ \forall\ \alpha \in U$ ***is a linear tranformation.***

Proof. Given $T_1 : U \to V$ and $T_2 : U \to V$, $(T_1 + T_2)(\alpha) = T_1(\alpha) + T_2(\alpha)\ \forall\ \alpha \in U$

$T_1(\alpha) \in V$ and $T_2(\alpha) \in V \Rightarrow T_1(\alpha) + T_2(\alpha) \in V$. Hence $(T_1 + T_2) : U \to V$

Let $a, b \in F$ and $\alpha, \beta \in U$. Then $(T_1 + T_2)(a\alpha + b\beta) = T_1(a\alpha + b\beta) + T_2(a\alpha + b\beta)$ (by def.)

$$= a\,T_1(\alpha) + b\,T_1(\beta) + a\,T_2(\alpha) + b\,T_2(\beta) = a[T_1(\alpha) + T_2(\alpha)] + b[T_1(\beta) + T_2(\beta)]$$

$$= a(T_1 + T_2)(\alpha) + b(T_1 + T_2)(\beta)$$

$\therefore$ $T_1 + T_2$ is a L. T. from U into V.

3.11. SCALAR MULTIPLICATION OF A L. T.

Theorem. ***Let*** $T : U(F) \to V(F)$ ***be a linear transformation and*** $a \in F$. ***Then the function*** aT ***defined by*** $(aT)(\alpha) = a\,T(\alpha)\ \forall\ \alpha \in U$ ***is a linear transformation.***

Proof. Given $T : U(F) \to V(F)$ and $(aT)(\alpha) = a\,T(\alpha)$, $a \in F$, $\alpha \in U$

Now $T(\alpha) \in V \Rightarrow a\,T(\alpha) \in V$. $\therefore$ (aT) is a mapping from U into V

For $c, d \in F$ and $\alpha, \beta \in U$

$(aT)[c\alpha + d\beta] = a\,T(c\alpha + d\beta)$ (by def)

$$= a[cT(\alpha) + d\,T(\beta)] = acT(\alpha) + ad\,T(\beta) = c(aT)(\alpha) + d(aT)(\beta)$$

Hence aT is a L.T. from U into V.

SOLVED PROBLEMS

Ex. 1. *Let* $T : V_3(R) \to V_2(R)$ *and* $H : V_3(R) \to V_2(R)$ *be the two linear transformations defined by* $T(x,y,z) = (x-y, y+z)$ *and* $H(x,y,z) = (2x, y-z)$

Find (*i*) H + T (*ii*) *a*H

Sol. (*i*) $(H+T)(x,y,z) = H(x,y,z) + T(x,y,z) = (2x, y-z) + (x-y, y+z) = (3x-y, 2y)$

(*ii*) $(aH)(x,y,z) = aH(x,y,z) = a(2x, y-z) = (2ax, ay-az)$

Ex. 2. *Let* $G: V_3 \to V_3$ *and* $H: V_3 \to V_3$ *be two linear operators defined by*

$G(e_1) = e_1 + e_2, G(e_2) = e_3, G(e_3) = e_2 - e_3$ *and* $H(e_1) = e_3, H(e_2) = 2e_2 - e_3, H(e_3) = 0$

where $\{e_1, e_2, e_3\}$ *is the standard basis of* $V_3(R)$.

Find (*i*) G + H (*ii*) 2 G **(*A.N. U. M 14*)**

Sol. Let S = {(1,0,0), (0,1,0), (0,0,1)} be the standard basis of $V_3(R)$ so that

$e_1 = (1,0,0)$, $e_2 = (0,1,0)$, $e_3 = (0,0,1)$

$G(e_1) = e_1 + e_2 \Rightarrow G(1,0,0) = (1,1,0)$, $G(e_1) = e_3 \Rightarrow G(0,1,0) = (0,0,1)$

$G(e_3) = e_2 - e_3 \Rightarrow G(0,0,0) = (0,1,-1)$

Again $H(e_1) = e_3 \Rightarrow H(1,0,0) = (0,0,1)$, $H(e_2) = 2e_2 - e_3 \Rightarrow H(0,1,0) = (0,2,-1)$

$H(e_3) = \bar{o} \Rightarrow H(0,0,1) = (0,0,0)$

(*i*) $(G + H)(e_1) = G(e_1) + H(e_1) = e_1 + e_2 + e_3 \Rightarrow (G + H)(1,0,0) = (1,1,1)$

$(G + H)(e_2) = G(e_2) + H(e_2) = 2e_2 \Rightarrow (G + H)(0,1,0) = (0,2,0)$

$(G + H)(e_3) = G(e_3) + H(e_3) = e_2 - e_3 \Rightarrow (G + H)(0,0,1) = (0,1,-1)$

(*ii*) $2G(e_1) = 2G(e_1) = 2e_1 + e_2$, $2G(e_2) = 2G(e_2) = 2e_3$, $2G(e_3) = 2G(e_3) = 2e_2 - 2e_3$ etc.

PRODUCT OF LINEAR TRANSFORMATIONS

3.12. Theorem. ***Let U(F), V (F) and W(F) are three vector spaces and*** $T: U \to W$ ***and*** $H: U \to V$ ***are two linear transformations. Then the composite function TH (called the product of linear transformations) defined by*** $(TH)(\alpha) = T[H(\alpha)] \forall \alpha \in U$ ***is a linear transformation from U into W.***

Proof. Given $H: U(F) \to W(F)$ and $T: V(F) \to W(F)$

For $a \in U \Rightarrow H(\alpha) \in V$

Again $H(\alpha) \in V \Rightarrow T[H(\alpha)] \in W \Rightarrow (TH)(\alpha) \in W$

$\therefore$ TH is a mapping from U into W

Now Let $a, b \in F$, $\alpha, \beta \in U$. Then $(TH)[a\alpha + b\beta] = T[H(a\alpha + b\beta)]$ (by def.)

$= T[aH(\alpha) + bH(\beta)]$ (H is L.T.)

$= a(TH)(\alpha) + b(TH)(\beta)$

$\therefore$ TH is a LT. from U to W.

Note. The range of H is the domain of T.

3.13. Theorem. ***Let H, H' be two linear transformations from U(F) to V(F). Let T, T' be the linear transformations from V(F) to W(F) and*** $a \in F$.

Then (*i*) $T(H+H') = TH + TH'$ (*ii*) $(T + T')H = TH + T'H$

(*iii*) $a(TH) = (aT)H = T(aH)$

Proof. (*i*) Let $\alpha \in U$. Then $T(H+H')(\alpha) = T[H(\alpha)+H'(\alpha)]$

$= TH(\alpha)+TH'(\alpha) = (TH+TH')(\alpha)$

(*ii*) Similar to (*i*)

(*iii*) Let at $\alpha \in U$. $a(TH)(\alpha) = aT[H(\alpha)] = [(aT)H](\alpha)$

Again $[T(aH)](\alpha) = T[aH(\alpha)] = [a(TH)](\alpha)$. $\therefore\ a(TH)] = (aT)H = T(aH)$

3.14. ALGEBRA OF LINEAR OPERATORS

Theorem. ***Let A, B, C be linear operators on a vector space V(F). Also let O be the zero operator and I the identity operator on V. Then***

(*i*) $AO = OA = O$ **(*ii*) $AI = IA = A$**

(*iii*) $A(B+C) = AB + AC$ **(*iv*) $(A+B)C = AC + BC$**

(*v*) $A(BC) = (AB)C$

Proof. Let $\alpha \in V$

(*i*) $AO(\alpha) = A[O(\alpha)] = A(\bar{O})$ (by def. of 0)

$= \bar{O}$ (A is L.T.) $= O(\alpha)$ $(\forall\ \alpha \in U)$

Similarly $OA(\alpha) = O[A(\alpha) = \bar{O} = O(\alpha) \Rightarrow OA = O$. Thus $AO = OA = O$.

(*ii*) Similar to (*i*)

(*iii*) $[A(B+C)](\alpha) = A[(B+C)(\alpha)] = A[B(\alpha)+C(\alpha)] = AB(\alpha)+AC(\alpha) = (AB+AC)(\alpha)$

$\therefore\ A(B+C) = AB+AC$

(*iv*) Similar to (*iii*)

(*v*) $[A(BC)](\alpha) = A[(BC)(\alpha)] = A[B\{C(\alpha)\}] = (AB)[C(\alpha)] = [(AB)C](\alpha)$

$\therefore\ A(BC) = (AB)C$

SOLVED PROBLEMS

Ex. 1. *Let* $T: R^3 \to R^2$ *and* $H: R^3 \to R^2$ *be defined by* $T(x, y, z) = (3x, y+z)$ *and* $H(x, y, z) = (2x-z, y)$. *Compute* (*i*) $T+H$ (*ii*) $4T-5H$ (*iii*) TH (*iv*) HT

Sol. Since T and H map V, the linear transformations $T+H$ and $4T-5H$ are defined.

(*i*) $(T+H)(x, y, z) = T(x, y, z)+H(x, y, z) = (3x, y+z)+(2x-z, y) = (5x-z, 2y+z)$

(*ii*) $(4T-5H)(x, y, z) = 4T(x, y, z)-5H(x, y, z)$

$= 4(3x, y+z)-5(2x-z, y) = (2x+5z, -y+4z)$

(*iii*) and (iv) both TH and HT are not defined because the range of T is not equal to the domain of H and vice versa.

Ex. 2. *Let* $T_1: R^3 \to R^2$ *and* $T_2: R^2 \to R^2$ *are two linear transformations defined by* $T_1(x, y, z) = (3x, 4y-z)$, $T_2(x, y) = (-x, y)$. *Compute* T_1T_2 *and* T_2T_1.

Sol. (*i*) Since the range of T_2 *i.e.* R^2 is not equal to the domain of T_1 *i.e.*, R^3, T_1T_2 is not defined.

(*ii*) But the range of T_1 *i.e.* R^3 is equal to the domain of T_2, T_2T_1 is defined.

$\therefore \quad (T_2T_1)(x,y,z) = T_2[T_1(x,y,z)] = T_2(3x, 4y-z) = (-3x, 4y-z)$

Ex. 3. *Let P(R) be the vector space of all polynomials in x and D, T be two linear operators on P defined by* $D[f(x)] = \frac{df}{dx}$ *and* $T[f(x)] = x f(x) \; \forall \; f(x) \in V$

Show (*i*) $TD \neq DT$ (*ii*) $(TD)^2 = T^2D^2 + TD$

Sol. (*i*) $(DT) f(x) = D[T f(x)] = D[x f(x)] = f(x) + x f'(x)$

$$(TD) f(x) = T[D f(x)] = T\left[\frac{df}{dx}\right] = x f'(x)$$

Clearly $DT \neq TD$. Also $(DT) f(x) - (TD) f(x) = f(x)$

$\Rightarrow (DT - TD) f(x) = I f(x)$ (I is identity) $\Rightarrow DT - TD = I$

(*ii*) $$(TD)^2 f(x) = (TD)[(TD) f(x)] = (TD)\left[x\frac{df}{dx}\right] = T\left[D\left(x\frac{df}{dx}\right)\right]$$

$$= T\left[\frac{df}{dx} + x\frac{d^2f}{dx^2}\right] = x\frac{df}{dx} + x^2\frac{d^2f}{dx^2}$$

Now $$(T^2D^2) f(x) = T^2D\{Df(x)\} = T^2\left[D\left(\frac{df}{dx}\right)\right] = T^2\left(\frac{d^2f}{dx^2}\right)$$

$$= T\left[T\left(\frac{d^2f}{dx^2}\right)\right] = T\left[x\frac{d^2f}{dx^2}\right] = x^2\frac{d^2f}{dx^2}$$

$$\therefore \quad (T^2D^2 + TD) f(x) = (T^2D^2 + TD) f(x) + (TD) f(x) = x^2\frac{d^2f}{dx^2} + x\frac{df}{dx}$$

$\therefore \quad (TD)^2 f(x) = (T^2D^2 + TD) f(x) \quad \forall \; f(x) \in P$. Hence $(TD)^2 = T^2D^2 + TD$.

TRANSFORMATIONS AS VECTORS

3.15. Theorem. ***Let L (U, V) be the set of all linear transformations from a vector space U (F) into a vector space V (F). Then L (U, V) be a vector space relative to the operations of vector addition and scalar multiplication defined as***

(i) $(T+H)(\alpha) = T(\alpha) + H(\alpha)$ ***(ii)*** $(aT)(\alpha) = aT(\alpha)$ ***for all*** $\alpha \in U$, $a \in F$ ***and*** $T, H \in L(U, V)$. ***The set L (U, V) is also denoted by Hom (U, V)*** ***(S. V. U. 2001/0)***

Proof. We have already proved that for all $T, H \in L(U, V)$ and $a \in F$,

$(T + H)$ and (aT) or (aH) are linear transformations and hence $\in L(U, V)$.

Now we verify the remaining properties of vector space.

(*i*) Associativity.

$[T+(H+G)](\alpha) = T(\alpha)+(H+G)(\alpha) = T(\alpha)+[H(\alpha)+G(\alpha)]$

$= [T(\alpha)+H(\alpha)]+G(\alpha)$ (addition in V associative)

$= (T+H)(\alpha)+G(\alpha) = [(T+H)+G](\alpha)$ $\therefore\ T+(H+G) = (T+H)+G$

(*ii*) Additive identity in L (U, V)

Let O be the zero transformation from U into V *i.e.*, $O(\alpha) = \hat{\hat{O}}$ $\forall\ \alpha \in U, \hat{\hat{O}} \in V$

$\therefore\ O \in L(U, V)$

Now $(O+T)(\alpha) = O(\alpha)+T(\alpha) = \hat{\hat{O}}+T(\alpha) = T(\alpha)$ ($\hat{\hat{O}}$ is additive indentity in V)

$\therefore\ O+T = T\ \ \forall\ T \in L(U, V)$

$\Rightarrow$ O is the additive identity in L (U, V)

(*iii*) Additive inverse.

For $T \in L(U, V)$ let us define $(-T)$ as $(-T)(\alpha) = -T(\alpha)\ \ \forall\ \alpha \in U$

Then $(-T) \in L(U, V)$.

Now $(-T+T)(\alpha) = (-T)(\alpha)+T(\alpha) = -T(\alpha)+T(\alpha) = \hat{\hat{O}}$ $(\hat{\hat{O}} \in V)$

$\Rightarrow -T+T = O$ for all $T \in L(U, V)$.

(*iv*) Commutativity

$\Rightarrow (T+H)(\alpha) = T(\alpha)+H(\alpha) = H(\alpha)+T(\alpha)$ (addition in V is commutative)

$= (H+T)(\alpha) \Rightarrow (T+H) = (H+T)$

$\therefore$ L (U, V) is an abelian group w.r.t. to addition

(*v*) Let $a \in F$ and $T, H \in L(U, V), \alpha \in U$

Then $[a(T+H)](\alpha) = a[(T+H)(\alpha)$ (by def.)

$= a[T(\alpha)+H(\alpha)]$ (by def.)

$= a\,T(\alpha)+a\,H(\alpha) = (a\,T)(\alpha)+(a\,H)(\alpha)$ (by def.)

$= (a\,T+a\,H)(\alpha) \Rightarrow a(T+H) = a\,T+a\,H$

(*vi*) Let $a, b \in F$ and $T \in L(U, V)$. Then $[(a+b)\,T](\alpha) = (a+b)\,T(\alpha)$ (by def.)

$= a\,T(\alpha)+b\,T(\alpha)$ ($\because$ V is a vector space)

$= (a\,T)(\alpha)+(b\,T)(\alpha)$ (by def.) $= (a\,T+b\,T)(\alpha) \Rightarrow (a+b)\,T = a\,T+b\,T$

(*vii*) $[(a\,b)\,T]\alpha = (a\,b)\,T(\alpha) = a[b\,T(\alpha)]$ ($\because$ V is vector space)

$= a[(b\,T)(\alpha)]$ (by def.) $= [a(b\,T)(\alpha)] \Rightarrow (a\,b)\,T = a(b\,T)$

(viii) Let $1 \in F$ and $T \in L(U, V)$.

Then $(1.T)(\alpha) = 1.T(\alpha) = T(\alpha)$ (by def.) $\Rightarrow 1.T = T$.

Hence L (U, V) is a vector space over the field F.

3.16. Theorem. ***Let L (U, V) be the vector space of all linear transformations from U(F) to V (F) such that dim U = n and dim V = m. Then dim L (U, V) = mn.***

(A.U. M2014)

Proof. Let $B_1 = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ and $B_2 = \{\beta_1, \beta_2, ..., \beta_n\}$

be the ordered basis of U and V respectively. By the previous theorems there exists uniquely a linear transformation T_{ij} from U to V such that

$T_{ij}(\alpha_1) = \beta_1, T_{ij}(\alpha_2) = \hat{\hat{O}}, ..., T_{ij}(\alpha_1) = \hat{\hat{O}}$ where $\beta_1, \hat{\hat{O}} \in V$.

i.e. $T_{ij}(\alpha_i) = \beta_j$, $i = 1, 2 ... n$ $j = 1, 2 ... m$ and $T_{ij}(\alpha_k) = \hat{\hat{O}}, k \neq i$.

Thus there are *mn* such T_{ij}'s $\in L(U, V)$. We shall show that $S = \{T_{ij}\}$ of *mn* elements is a basis for L (U, V)

(*i*) *To prove S is L.I.*

For a_{ij}'s $\in F$ let us suppose that $\sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} T_{ij} = 0$ $(O \in L(U, V))$

For $\alpha_k \in U$, $k = 1, 2 ... n$ we get $\left[\sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} T_{ij}\right](\alpha_k) = O(\alpha_k)$

$\Rightarrow \sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} T_{ij}(\alpha_k) = \hat{\hat{O}} \Rightarrow \sum_{j=1}^{m} a_{kj} T_{kj}(\alpha_k) = \hat{\hat{O}}$ $(\hat{\hat{O}} \in V)$

$\Rightarrow a_{k1} \alpha_1 + a_{k2} \alpha_2 + ... + a_{km} \alpha_m = \hat{\hat{O}} \Rightarrow a_{k1} = 0, a_{k2} = 0 ... a_{km} = 0$ $(\because B_1$ is L.I.)

Hence $S = \{T_{ij}\}$ is an L.I. set.

(*ii*) *To show that* L (S) = L (U,V).

Let $T \in L(U, V)$. The vector $T(\alpha_1) \in V$ can be expressed as

$T(\alpha_1) = b_{11}\beta_1 + b_{21}\beta_2 + ... + b_{m1}\beta_m$

In general for $i = 1, 2 ... m$

$$T(\alpha_i) = b_{i1}\beta_1 + b_{2i}\beta_2 + ... + b_{mi}\beta_m \quad ... (1)$$

Consider the linear transformation $H = \sum_{i=1}^{n} \sum_{j=1}^{m} b_{ij} T_{ij}$

Clearly H is a linear combination of $S = \{T_{ij}\}$; therefore $H \in L(U, V)$.

Let $\alpha_k \in U$ for $k = 1, 2 \ldots n$. Since $T_{ij}(\alpha_k) = \hat{\hat{O}}$ for $k \neq i$, We have $T_{kj}(\alpha_k) = \beta_j$

Consider $H(\alpha_k) = \sum_{i=1}^{n} \sum_{j=1}^{m} b_{ij} T_{ij}(\alpha_k) = \sum_{j=1}^{m} b_{kj} T_{kj}(\alpha_k) = \sum_{j=1}^{m} b_{kj} \beta_j$

i.e. $H(\alpha_k) = b_{k1}\beta_1 + b_{k2}\beta_2 + \ldots + b_{km}\beta_m = T(\alpha_k)$ [by (1)]

Hence $H(\alpha_k) = T(\alpha_k)$ for each k. $\therefore$ $H = T$

Thus T is a linear combination of elements of S. *i.e.,* $L(S) = L(U, V)$.

$\therefore$ S is a basis set of $L(U, V)$ $\therefore$ dim $L(U, V) = mn$.

EXERCISE 3 (*b*)

1. Let $T: R^3 \to R^2$ and $H: R^2 \to R^3$ be two linear transformation defined by

$T(x, y, z) = (x - 3y - 2z, y - 4z)$ and $H(x, y) = (2x, 4x - y, 2x + 3y)$

Find HT and TH. Is product commutative ?

2. Define on R^2 linear operators H and T as follows $H(x, y) = (0, x)$ and $T(x, y) = (x, 0)$ and show that $TH = 0$, $HT \neq TH$ and $T^2 = T$.

3. Give an example of a linear operator T on R^3 such that $T \neq O$, $T^2 \neq O$ but $T^2 = O$.

[**Hint:** $T: R^3 \to R^3$ where $T(x, y, z) = (0, x, y)$]

4. Let P be the polynomial space in one indeterminate x with real coefficients. Let $D: P \to P$ and $S: P \to P$ be two linear operators defined by

$$D f(x) = \frac{df}{dx} \text{ and } S f(x) = \int_0^x f(x)\, dx \quad \forall\, f(x) \in P$$

Show that $DS = I$ and $SD \neq I$ where I is the identity transformation.

5. Let $T: V_3(R) \to V_3(R)$ be defined by $T(a, b, c) = (3a, a - b, 2a + b + c)$

Prove that $(T^2 - I)(T - 3I) = O$.

3.17. RANGE AND NULL SPACE OF A LINEAR TRANSFORMATION *(O.U. M 2015)*

RANGE. Definition. *Let U (F) and V (F) be two vector spaces and let* $T: U \to V$ *be a linear transformation. The range of T is defined to be the set* $\text{Range}\ (T) = R(T) = \{T(\alpha) : \alpha \in U\}$.

Obviously the range of T is a subset of V. *i.e.* $R(T) \subseteq V$.

3.18. Theorem. *Let U(F) and V(F) be two vector spaces. Let* $T: U(F) \to V(F)$ *be a linear transformation. Then the range set R (T) is a subspace of V (F).*

(S. V. U. M 2015, K.U. 11,S. V. U. O 2000, S. K. U. S01, O. U. S01)

Proof. For $\bar{O} \in U \Rightarrow T(\bar{O}) = O \in R(T)$ $\therefore$ R (T) is non-empty set and $R(T) \subseteq V$

Let $\alpha_1, \alpha_2 \in U$ and $\beta_1, \beta_2 \in R(T)$ be such that $T(\alpha_1) = \beta_1$ and $T(\alpha_2) = \beta_2$

For $a, b \in F$, $a\alpha_1 + b\alpha_2 \in U$ $(\because U$ is V.S.) $\Rightarrow T(a\alpha_1 + b\alpha_2) \in R(T)$

But $T(a\alpha_1 + b\alpha_2) = aT(\alpha_1) + bT(\alpha_2) = a\beta_1 + b\beta_2 \in R(T)$ $(\because$ T is L.T.)

Thus $a, b \in F$ and $\beta_1, \beta_2 \in R(T) \Rightarrow a\beta_1 + b\beta_2 \in R(T)$

$\therefore$ R (T) is a subspace of V(F). R (T) is called the range space.

3.19. NULL SPACE OR KERNEL

(O.U. M 2015)

Definition. *Let U (F) and V (F) be two vector spaces and* $T: U \to V$ *be a linear transformation. The null space denoted by N (T) is the set of all vectors* $\alpha \in U$ *such that* $T(\alpha) = \bar{O}$ *(zero vector of V).*

The null space N (T) is also called the kernel of T *i.e.,* $N(T) = \{\alpha \in U : T(\alpha) = \bar{O} \in V\}$.

Obviously the null space $N(T) \subseteq U$. *(S.V. U. 2011)*

3.20. Theorem. *Let U (F) and V (F) be two vector spaces and* $T: U \to V$ *is a linear transformation. Then null space N (T) is a subspace of U (F).*

(K. U. M14, S. V. U. M13, O2000, 11, S. K. U. O2001, M14, O. U. M13, 2001/O)

Proof. Let $N(T) = \{\alpha \in U : T(\alpha) = \bar{O} \in V\}$

$\because$ $T(\bar{o}) = \hat{\bar{O}} \Rightarrow \hat{\bar{O}} \in N(T)$ $(\bar{O} \in U, \hat{\bar{O}} \in V)$

$\therefore$ N(T) is a non-empty subset of U.

Now $\alpha, \beta \in N(T) \Rightarrow T(\alpha) = \hat{\bar{O}}, T(\beta) = \hat{\bar{O}}$

For $a, b \in F$. $T(a\alpha + b\beta) = aT(\alpha) + bT(\beta) = a.\hat{\bar{O}} + b.\hat{\bar{O}} = \hat{\bar{O}}$ (T is L.T.)

$\therefore$ $T(a\alpha + b\beta) = \hat{\bar{O}}$

By definition $a\alpha + b\beta \in N(T)$

Thus $a, b \in F$ and $\alpha, \beta \in N(T) \Rightarrow a\alpha + b\beta \in N(T)$

$\therefore$ Null space N (T) is a subspace of U (F).

3.21. Theorem. *Let* $T: U(F) \to V(F)$ *be a linear transformation. If U is finite dimensional then the range space (T) is a finite dimensional subspace of V (F).*

(S.V. U. 2012)

Proof. Given U is finite dimensional

$\therefore$ Let $S = \{\alpha_1, \alpha_2 \ldots \alpha_n\}$ be the basis set of U (F).

Let $\beta \in R(T)$, the range space of T.

Then there exists $\alpha \in U$ such that $T(\alpha) = \beta$.

$\therefore \quad \alpha = a_1\alpha_1 + a_2\alpha_2 \ldots a_n\alpha_n$ for $a's \in F$

$\Rightarrow T(\alpha) = T(a_1\alpha_1 + a_2\alpha_2 \ldots a_n\alpha_n) \Rightarrow \beta = a_1T(\alpha_1) + a_2T(\alpha_2) \ldots + a_nT(\alpha_n)$

But $S' = \{T(\alpha_1), T(\alpha_2) \ldots T(\alpha_n)\} \in R(T)$

Now $\beta \in R(T)$ and $\beta = l.c$ of elements of $S' \Rightarrow \beta \in L(S')$.

Thus R(T) is spanned by a finite set S'.

$\therefore$ R(T) is finite dimensionalsubspace of V(F).

DIMENSION OF RANGE AND KERNEL

3.22. Definition. Let $T : U(F) \to V(F)$ be *a linear transformation where U is finite dimensional vector space.*

Rank : *Then the rank of T denoted by* $\rho(T)$ *is the dimension of range space R(T).*

$$\rho(T) = \dim R(T)$$

Nullity : *The nullity of T denoted by* $\nu(T)$ *is the dimension of null space N (T).*

$$\nu(T) = \dim N(T)$$

RANK - NULLITY THEOREM :

3.23. Theorem. ***Let U(F) and V(F) be two vector spaces and*** $T : U \to V$ ***be a linear transformation. Let U be finite dimensional then*** $\rho(T) + \nu(T) = \dim U$

i.e. rank (T) + nullity (T) = dim U.

(O. U. A 14, A.N. U. M 14, S. V. U. M 15, M 14, M13, A2000, S. K. U. O01, M13, M14, M15)

Proof. The null space N (T) is a subspace of finite dimensional space U (F).

$\Rightarrow$ N (T) is finite dimensional

Let $S = \{\alpha_1, \alpha_2 \ldots \alpha_k\}$ be a basis set of N (T). $\quad \therefore \quad \dim N(T) = \nu(T) = k.$

$\therefore T(\alpha_1) = \hat{\hat{O}}, T(\alpha_2) = \hat{\hat{O}} \ldots T(\alpha_k) = \hat{\hat{O}} \quad \ldots (1) \quad (\hat{\hat{O}} \in V)$

As S is L.I. it can be extended to form a basis of U (F)

Let $S_1 = \{\alpha_1, \alpha_2 \ldots \alpha_k, \theta_1, \theta_2 \ldots \theta_m\}$ be the extended basis of U (F). $\therefore \quad \dim U = k + m$

Now we show that the set of images of additional vectors

$S_2 = \{T(\theta_1), T(\theta_2) \ldots T(\theta_m)\}$ is a basis of R (T). Clearly $S_2 \subseteq R(T)$

(i) To prove S_2 is L. I.

Let $a_1, a_2 \ldots a_m \in F$ be such that, $a_1 T(\theta_1) + a_2 T(\theta_2) + \ldots + a_m T(\theta_m) = \hat{\hat{O}}$

$\Rightarrow \quad T[a_1\theta_1 + a_2\theta_2 + \ldots + a_m\theta_m] = \hat{\hat{O}}$

$\Rightarrow \quad a_1\theta_1 + a_2\theta_2 \ldots + a_m\theta_m \in N(T)$, null space of T.

But each vector in N(T) is a *l.c.* of elements of basis S

$\therefore$ For some $b_1, b_2 \ldots b_m \in F$, let $a_1\theta_1 + a_2\theta_2 \ldots + a_m\theta_m = b_1\alpha_1 + b_2\alpha_2 + \ldots b_k\alpha_k$

$\Rightarrow \quad a_1\theta_1 + \ldots + a_m\theta_m - b_1\alpha_1 - \ldots - b_k\alpha_k = \hat{\hat{O}}$

$\Rightarrow \quad a_1 = 0, a_2 = 0 \ldots a_m = 0, b_1 = 0 \ldots b_k = 0 \qquad (\because S_1 \text{ is L.I.}) \quad \Rightarrow S_2$ is L.I. set

(*ii*) *To prove* $L(S_2) = R(T)$

Let $\beta \in$ range space R (T), then there exists $\alpha \in U$ such that $T(\alpha) = \beta$

Now $\alpha \in U \Rightarrow$ there exist $c's, d's \in F$ such that

$$\alpha = c_1\alpha_1 + c_2\alpha_2 \ldots + c_k\alpha_k + d_1\theta_1 + d_2\theta_2 \ldots + d_m\theta_m$$

$$\Rightarrow T(\alpha) = T[c_1\alpha_1 + \ldots + c_k\alpha_k + d_1\theta_1 + \ldots + d_m\theta_m]$$

$$= c_1 T(\alpha_1) + \ldots + c_k T(\alpha_k) + d_1 T(\theta_1) + \ldots + d_m T(\theta_m)$$

$\Rightarrow \beta = d_1 T(\theta_1) + d_2 T(\theta_2) \ldots + d_m T(\theta_m) \qquad$ (by (1)) $\qquad \Rightarrow \beta \in L(S_2)$

$\therefore S_2$ is a basis of R(T) and dim R(T) = m.

Thus dim R(T) + dim N(T) = $m + k$ = dim U *i.e.*, $\rho(T) + \nu(T) = \dim U$.

SOLVED PROBLEMS

Ex. 1. *If* $T: V_4(R) \to V_3(R)$ *is a linear transformation defined by*

$T(a,b,c,d) = (a-b+c+d, a+2c-d, a+b+3c-3d)$ *for* $a,b,c,d \in R$,

then verify $\rho(T) + \nu(T) = \dim V_4(R)$ ***(O. U. M06)***

Sol. Let $S = \{(1,0,0,0), (0,1,0,0), (0,0,1,0), (0,0,0,1)\}$ be the basis set of $V_4(R)$.

$\therefore$ The transformation T on B will be $T(1,0,0,0) = (1,1,1)$, $T(0,1,0,0) = (-1,0,1)$

$T(0,0,1,0) = (1,2,3)$, $T(0,0,0,1) = (1,-1,-3)$

Let $S_1 = \{(1,1,1), (-1,0,1), (1,2,3), (1,-1,-3)\}$ $\qquad \therefore S_1 \subseteq R(T)$

Now we verify whether S_1 is L.I. or not,. If not, we find the least L.I. set by forming

the matrix, $S_1 = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 2 & 3 \\ 1 & -1 & -3 \end{bmatrix}$ applying $R_2 + R_1, R_3 - R_1, R_4 - R_1$

$S_1 \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & -2 & -4 \end{bmatrix}$ Again apply $R_4 + 2R_3, R_3 - R_2$, $S_1 \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\therefore$ The non - zero rows of vectors $\{(1,1,1),(0,1,2)\}$

Constitute the L.I. set forming the basis of R (T) $\Rightarrow$ dim R (T) = 2

Basis for null space of T. $\alpha \in N(T) \Rightarrow T(\alpha) = \hat{\hat{O}}$

$\therefore$ $T(a,b,c,d) = \hat{\hat{O}}$ where $\hat{\hat{O}} = (0,0,0) \in V_3(R)$

$\Rightarrow (a-b+c+d, a+2c-d, a+b+3c-3d) = (0,0,0)$

$\Rightarrow a-b+c+d=0$, $a+2c-d=0$

$a+b+3c-3d=0$, we have to solve these for *a, b, c, d.*

Coefficient matrix $= \begin{bmatrix} 1 & -1 & 1 & 1 \\ 1 & 0 & 2 & -1 \\ 1 & 1 & 3 & -3 \end{bmatrix}$, by $R_2 - R_1, R_3 - R_1 = \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 2 & 2 & -4 \end{bmatrix}$

by $R_3 - 2R_2$, the echelon form is $\begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\therefore$ The equivalent system of equations are

$a-b+c+d=0$, $b+c-2d=0$ $\Rightarrow b = 2d-c$, $a = d-2c$

The number of free variables is 2 namely *c, d* and the values of *a, b* depend on these and hence nullity (T) = dim N(T) = 2.

Choosing $c=1, d=0$, we get $a=-2, b=-1$, $\therefore$ $(a,b,c,d) = (-2,-1,1,0)$

Choosing $c=0, d=1$, we get $a=1, b=2$. $\therefore$ $(a,b,c,d) = (1,2,0,1)$

$\therefore$ $\{(-2,-1,1,0),(1,2,0,1)\}$ constitute a basis of N(T)

$\therefore$ dim R(T) + dim N(T) = 2 + 2 = 4 = dim $V_4(R)$.

Ex. 2. *Find a linear transformation* $T: R^3 \to R^3$ *whose range is spanned by* $(1,2,0,-4)$, $(2,0,-1,-3)$.

SoL Given R (T) is spanned by $\{(1,2,0,-4),(2,0,-1,-3)\}$

Let us include a vector (0, 0, 0, 0) in this set which will not effect the spanning property so that

$S = \{(1,2,0,-4),(2,0,-1,-3),(0,0,0,0)\}$

Let $B = \{\alpha_1, \alpha_2, \alpha_3\}$ be the basis of R^3. We know there exists a transformation T such that

$T(\alpha_1) = (1,2,0,-4)$; $T(\alpha_2) = (2,0,-1,-3)$ and $T(\alpha_3) = (0,0,0,0)$

Now if $\alpha \in R^3 \Rightarrow \alpha = (a,b,c) = a\,\alpha_1 + b\,\alpha_2 + c\,\alpha_3$

$\therefore$ $T(a,b,c) = T(a\,\alpha_1 + b\,\alpha_2 + c\,\alpha_3) = a\,T(\alpha_1) + b\,T(\alpha_2) + c\,T(\alpha_3)$

$= a(1,2,0,-4) + b(2,0,-1,-3) + c(0,0,0,0)$

$\therefore$ $T(a,b,c) = (a+2b, 2a, -b, -4a-3b)$ is the required transformation.

Ex. 3: *Find* $F: R^3 \to R^4$ *is a linear transformation whose range is spanned by* $(1,-1,2,3)$ *and* $(2,3,-1,0)$.

Sol. Consider the standard basis for R^3 e_1, e_2, e_3 where $e_1 = (1,0,0)$, $e_2 = (0,1,0)$, $e_3 = (0,0,1)$. Then $F(e_1) = (1,-1,2,3)$, $F(e_2) = (2,3,-1,0)$ and $F(e_3) = (0,0,0,0)$.

We know that, $(x,y,z) = xe_1 + ye_2 + ze_3$

$\therefore$ $F(x,y,z) = F(xe_1 + ye_2 + ze_3) = xF(e_1) + yF(e_2) + zF(e_3)$ ($\because$ F is a linear transformation)

$= (x,-x,2x,3x) + (2y,3y,-y,0) + (0,0,0,0) \quad = (x+2y, -x+3y, 2x-y, 3x)$

Ex. 4. *Let V be a vector space of* 2×2 *matrices over reals. Let P be a fixed matrix of V ;* $P = \begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix}$ *and* $T: V \to V$ *be a linear operator defined by T(A) = PA,* $A \in V$. *Find the nullity T.*

Sol. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in V$

The null space N (T) is the set of all 2×2 matrices whose T - image is $\hat{\hat{O}}$

$$\Rightarrow T(A) = PA = \hat{\hat{O}} \Rightarrow T\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a-c & b-d \\ -2a+2c & -2b+2d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} a-c & b-d \\ a-c & b-d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$\Rightarrow a-c=0, b-d=0 \Rightarrow a=c, b=d$,

The free variables are c and d. Hence dim N(T) = 2.

Ex. 5. *Find the null space, range, rank and nullity of the transformation* $T: R^2 \to R^3$ *defined by* $T(x,y) = (x+y, x-y, y)$. ***(S. V. U. M12, O2000, K. U. M 2013)***

Sol. (*i*) Let $\alpha = (x,y) \in R^2$. Then $\alpha \in N(T) \Rightarrow T(a) = \hat{\hat{O}}$ $(\hat{\hat{O}} \in R^3)$

i.e., $T(x,y) = (0,0,0) \Rightarrow (x+y, x-y, y) = (0,0,0)$

$\Rightarrow \quad x+y=0, x-y=0, y=0 \Rightarrow \quad x=0, y=0 \qquad \therefore \quad \alpha=(0,0)=(\hat{\bar{O}} \in R^2)$

Thus the null space of T consists of only zero vector of R^2

$\therefore$ nullity T = dim N(T) = 0

(*ii*) Range space of $T=\{\beta \in R^2 : T(\alpha)=\beta \text{ for } \alpha \in R^2\}$

$\therefore$ The range space consists of all vectors of the type $(x+y, x-y, y)$ for all $(x, y) \in R^2$

(*iii*) dim R(T) + dim N(T) = dim $R^2 \Rightarrow$ dimR(T) + 0 = 2 $\Rightarrow$ rank of T = 2.

Ex. 6. *Let V (F) be a vector space and T be a linear operator on V. Prove that the following statements are true*

(*i*)*The intersection of the range of T and null space of T is the zero subspace of T.*

i.e., $R(T) \cap N(T)=\{\bar{O}\}$ (*ii*) *If* $T[T(\alpha)]=\bar{O}$, *then* $T(\alpha)=\bar{O}$

Sol. (*i*) $\Rightarrow$ (*ii*)

Let $R(T) \cap N(T)=\{\bar{O}\}$. Let $T(\alpha)=\beta \quad \therefore \quad \beta \in R(T)$ ... (1)

Now $T[T(\alpha)]=\bar{O} \Rightarrow T(\beta)=\bar{O} \Rightarrow \beta \in N(T)$ ... (2)

From(1) and (2) $\beta \in R(T) \cap N(T)$. But $R(T) \cap N(T)=\{\bar{O}\} \Rightarrow \beta=\bar{O} \quad \Rightarrow \quad T(\alpha)=\bar{O}$

Thus $T[T(\alpha)]=\bar{O} \Rightarrow T(\alpha)=\bar{O}$

(*ii*) $\Rightarrow$ (*i*) Given $T[T(\alpha)]=\bar{O} \Rightarrow T(\alpha)=\bar{O}$

Let $\beta \in R(T) \cap N(T)$. $\therefore \quad \beta \in R(T)$ and $\beta \in N(T)$

Now $\beta \in R(T) \Rightarrow T(\alpha)=\beta$ for some $\alpha \in V$ and $\beta \in N(T) \Rightarrow T(\beta)=\bar{O} \Rightarrow T[T(\alpha)]=\bar{O}$

$\Rightarrow T(\alpha)=\bar{O} \Rightarrow \beta=\bar{O}$. Thus $R(T) \cap N(T)=\bar{O}$

Ex. 7. *Verify the Rank-Nullity theorem for the linear map* $T: V_4 \to V_3$ *defined by* $T(e_1)=f_1+f_2+f_3$, $T(e_2)=f_1-f_2+f_3$, $T(e_3)=f_1$, $T(e_4)=f_1+f_3$ *when* $\{e_1, e_2, e_3\}$ *and* $\{f_1, f_2, f_3\}$ *are standard basis* V_4 *and* V_3 *respectively* ***(S.K.U. 11, S. V. U. 2001/O)***

Sol.:Let $e_1=(1,0,0,0), e_2=(0,1,0,0)$ $e_3=(0,0,1,0), e_4=(0,0,0,1)$ and

$f_1=(1,0,0), f_2=(0,1,0), f_3=(0,0,1)$

$\{e_1, e_2, e_3, e_4\}, \{f_1, f_2, f_3\}$ are the standard basis of V_4 and V_3 respectively.

we have $T(e_1)=f_1+f_2+f_3 \Rightarrow T(1,0,0,0)=(1,0,0)+(0,1,0)+(0,0,1)=(1,1,1)$

$T(e_2)=f_1-f_2+f_3 \Rightarrow T(0,1,0,0)=(1,0,0)-(0,1,0)+(0,0,1)=(1,-1,1)$

$T(e_3)=f_1 \Rightarrow T(0,0,1,0)=(1,0,0)$; $T(e_4)=f_1+f_3 \Rightarrow T(0,0,0,1)=(1,0,0)+(0,0,1)$

Let $\alpha \in V_4$. The α can be written as $\alpha = ae_1 + be_2 + ce_3 + de_4$

Then $T(\alpha) = a\,T(e_1) + b\,T(e_2) + c\,T(e_3) + d\,T(e_4)$

$= a(1,1,1) + b(1,-1,1) + c(1,0,0) + d(1,0,1) = (a+b+c+d, a-b, a+b+d)$

Consider $B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{matrix} R_4 = R_4 - R_1 \\ R_3 = R_3 - R_1 \\ R_2 = R_2 - R_1 \end{matrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & 0 \\ 0 & -1 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

$R_4 = R_4 - 2R_2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix} R_1 + R_3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} R_3 - R_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

Then $\{(1,0,0)\ (0,1,0)\ (0,0,1)\}$ is a basis set for R(T).

Thus dim R(T) = 3. Suppose $T(\alpha) = \bar{0} \Rightarrow (a+b+c+d, a-b, a+b+d) = (0,0,0)$

Which gives $a+b+c+d = 0, a-b = 0, a+b+d = 0$.

From this we have $c = 0, b = a$ and $d = -2a$. Thus $[a,b,c,d] = [1,1,0,-2]$

$\therefore$ Rank of the null space N(T) = 1. $\therefore$ Rank of T = dim R(T) = 3

Nullity = dim N(T) = 1. Dim $V_4 = 4$

Thus Rank + Nullity = Dimension, is verified

Ex. 8. *Verify Rank - nullity theorem for the linear transformation* $T: R^3 \to R^3$ *defined by* $T(x,y,z) = (x-y, 2y+z, x+y+z)$ ***(A. N.U. 2011)***

Sol. Given $T: R^3 \to R^3$ defined by $T(x,y,z) = (x-y, 2y+z, x+y+z)$ is a linear transformation.

We know that dimension of $R^3 = 3$ ... (1)

Let $\alpha = (x,y,z) \in R^3$. If $\alpha \in N(T)$ then $T(\alpha) = \bar{0}$

$\Rightarrow T(x,y,z) = \bar{0} \quad \Rightarrow (x-y, 2y+x, x+y+z) = (0,0,0)$

Comparing the components, $x - y = 0; 2y + z = 0; x + y + z = 0$

taking $y = k$ we get $x = k$ and $z = -2k$

$\therefore (x,y,z) = (k,k,-2k) = k(1,1,-2)$

Thus every element in N(T) is generated by the vector $(1,1,-2)$.

Thus dim (N(T)) = 1 ... (2)

Again $T(x,y,z) = (x-y, 2y+z, x+y+z)$

from this $T(1,0,0)=(1,0,1); T(0,1,0)=(-1,2,1)$ and $T(0,0,1)=(0,1,1)$

Let $S=\{(1,0,0),(-1,2,1),(0,1,1)\}$ and let $A=\begin{bmatrix}1 & 0 & 1\\ -1 & 2 & 1\\ 0 & 1 & 1\end{bmatrix}$

$$R_2+R_1 \text{ gives } \sim \begin{bmatrix}1 & 0 & 1\\ 0 & 2 & 2\\ 0 & 1 & 1\end{bmatrix}$$

$$\frac{R_2}{2} \text{ gives } \sim \begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\\ 0 & 1 & 1\end{bmatrix}$$

$$R_3-R_2 \text{ gives } \sim \begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\\ 0 & 0 & 0\end{bmatrix}$$

Thus the set $\{(1,0,0),(0,1,1)\}$ constitute the basis of R(T) *i.e.* range of T.

Thus, $\dim(R(T))=2$ .... (3)

Substituting in rank - nullity theorem, rank + nullity = dimension

$\Rightarrow 1+2=3$. This verifies the theorem.

Ex. 9. *$T:R^3\to R^2$ be the linear transformation defined as $T(x_1,x_2,x_3)=(x_1-x_2,x_1+x_3)$. Find the Rank (T), and Nullity (T).* ***(O. U. 2011)***

Sol. Let $T(x_1,x_2,x_3)=(0,0)=(x_1-x_2,x_1+x_3)$

$x_1-x_2=0$ and $x_1+x_3=0$; $x_1=x_2$ and $x_1=-x_3$. $\therefore x_1=-x_3$

Let $x_1=k, x_2=k$ and $x_3=-k$

$\therefore (x_1,x_2,x_3)=(k,k,-k)=k(1,1,-1)$ $\therefore (1,1,-1)$ constitutes a basis for N(T)

$\dim\{N(T)\}=1=$ Nullity. We know that, $\dim(R^3)=3$

from rank - nullity theorem, we have rank + nullity = dimension $\Rightarrow$ rank (T) $=3-1=2$

Ex. 10. *Find the Kernal of the linear transformation $T:R^2\to R^2$ defined as $T(1,0)=(1,1)$ are $T(0,1)=(-1,2)$* ***(O. U. 2011)***

Sol. We know that $\{(1,0),(0,1)\}$ is a standard basis for R^2

Let $(x,y)\in R^2$. Let $\alpha=(x,y)=x(1,0)+y(0,1)$

Now $T(x,y) = xT(1,0) + yT(0,1) = x(1,1) + y(-1,2)$

Thus, $T(x,y) = (x-y, x+2y)$

To find the null space

If $(x,y) \in N(T), T(x,y) = \bar{0} \Rightarrow (x-y, x+2y) = (0,0)$

Comparing components, $x+y=0, x+2y=0$

Solving $x=0, y=0$. $\therefore$ $(0,0)$ is the only element in null space.

Thus $N(T) = \{(0,0)\}$

Ex. 11. *If $T: R^3 \to R^3$ be a linear transformation defined by $T(x,y,z) = (x-y, y-z, z-x)$ then show that T is a linear transformation and find its rank.* ***(O. U. M 14)***

Sol. Given $T: R^3 \to R^3$ is defined as $T(x,y,z) = (x-y, y-z, z-x)$

Let $\alpha = (x_1, y_1, z_1)$ and $\beta = (x_2, y_2, z_2)$

$$T(\alpha+\beta) = T(x_1+x_2, y_1+y_2, z_1+z_2)$$

$$= (x_1+x_2-\overline{y_1+y_2}, y_1+y_2-\overline{z_1+z_2}, z_1+z_2-\overline{x_1+x_2})$$

$$= [(x_1-y_1)+(x_2-y_2), (y_1-z_1)+(y_2-z_2), (z_1-x_1, z_2-x_2)]$$

$$= [(x_1-y_1, y_1-z_1, x_1-x_1)] + [(x_2-y_2, y_2-z_2, z_2-x_2)]$$

$$= T(\alpha) + T(\beta)$$

$$T(c\alpha) = T(cx_1, cy_1, cz_1) = (cx_1-cy_1, cy_1-cz_1, cz_1-cx_1)$$

$$= c[(x_1-y_1), (y_1-z_1), (z_1-x_1)]$$

$$= cT(\alpha)$$

$\therefore$ $T: R^3 \to R^3$ is a linear transformation.

We have $T(\alpha) = (x-y, y-z, z-x)$

$$= x(1,0,-1) + y(-1,1,0) + z(0,-1,1)$$

$\therefore$ $R(T) = L\{(1,0,-1), (-1,1,0), (0,-1,1)\}$

Let $(1,0,-1) = a(-1,1,0) + b(0,-1,1) = (-a, a-b, b)$

Comparing $a=-1, b=-1$

$\therefore$ $(1,0,-1) = 1(-1,1,0) - 1(-1,1,0)$

$\therefore$ $R(T)$ is generated by $\{(-1,1,0), (0,-1,1)\}$

Rank of $T = 2$.

EXERCISE 3 (c)

1. Let $T : V_4 \to V_3$ be a linear transformation defined by $T(\alpha_1) = (1,1,1)$; $T(\alpha_2) = (1,-1,1)$; $T(\alpha_3) = (1,0,0)$, $T(\alpha_4) = (1,0,1)$. Then verify that $\rho(T) + \nu(T) = \dim V_4$

2. Describe explicitly the linear transformation $T : R^3 \to R^3$ whose range space is spanned by $\{(1,0,-1),(1,2,2)\}$.

3. Let $T : R^3 \to R^2$ be a linear transformation defined by $T(x,y,z) = (x+y, y+z)$

Find a basis, dimension of each of the range and null space of T.

4. Let $T : R^3 \to R^3$ be the linear mapping defined by

$T(x,y,z) = (x+2y-z, y+z, x+y-2z)$

Find the rank, nullity and find a basis for each of the range and null space of T.

(K.U. M 2014, M 2013, 2011)

5. Let $T : R^3 \to R^3$ be defined by $T(x,y,z) = (x-y+2z, 2x+y-z, -x-2y)$

Find the null space of T.

6. $T : V_3(R) \to V_2(R)$ is defined by $T(a,b,c) = (a,b) \forall (a,b,c) \in R^3$

Prove that T is a linear transformation. Find the kernel of T.

7. Let V (F) be an n dimensional vector space and let T be a L.T. from V into V such that range and null space of T are identical. Prove that n is even.

4

Vector Space Isomorphism

(Optional)

4.1. Definition. *Let U(F) and V(F) be two vector spaces. The one-one onto transformation* $T: U \to V$ *is called the isomorphism and is denoted by* $U(F) \cong V(F)$.

Now we prove some more properties of vector space isomorphism.

4.2. Theorem. ***Two finite dimensional vector spaces U and V over the same field F are isomorphic if and only if they have the same dimension.***

i.e. $U(F) \cong V(F) \Leftrightarrow \dim U = \dim V$ ***(S. V. U. M 15, A. U. M 2014, S. K. U. M 07)***

Proof. Let U(F) and V(F) be finite dimensional and $U(F) \cong V(F)$. Then there exists an one-one onto transformation $T: U \to V$

To prove that $\dim U = \dim V$.

Let $S = \{\alpha_1, \alpha_2 \ldots \alpha_n\}$ be a basis of U. $\therefore \dim U = n$

Let $S' = \{T(\alpha_1), T(\alpha_2) \ldots T(\alpha_n)\}$ be the set of T-images of S $\Rightarrow S' \subseteq V$

(*i*) To show that S' is L.I.

Consider the equation $a_1 T(\alpha_1) + a_2 T(\alpha_2) \ldots + a_n T(\alpha_n) = \hat{O}$ $a's \in F$ $\hat{O} \in V$

$\Rightarrow T[a_1\alpha_1 + a_2\alpha_2 \ldots + a_n\alpha_n] = T(\bar{O})$ (T is L.T. ; $\bar{O} \in U$)

$\Rightarrow a_1\alpha_1 + a_2\alpha_2 \ldots + a_n\alpha_n = \bar{O}$ ($\because$ T is one-one)

$\Rightarrow a_1 = 0, a_2 = 0, \ldots a_n = 0$ ($\because$ S is L.I.)

$\Rightarrow$ S' is L.I. set.

(*ii*) To show that L (S') = V

Let $\beta \in V$. Since T is onto there exists; $\alpha \in U$ such that $T(\alpha) = \beta$

But $\alpha \in U \Rightarrow \alpha = l.c$ of elements of basis

$= b_1\alpha_1 + b_2\alpha_2 \ldots + b_n\alpha_n$ ($b's \in F$)

$\Rightarrow T(\alpha) = T(b_1\alpha_1 + b_2\alpha_2 \ldots + b_n\alpha_n)$

$\Rightarrow \beta = b_1 T(\alpha_1) + b_2 T(\alpha_2) \ldots + b_n T(\alpha_n)$ (T is L.T.)

$\Rightarrow$ $= l.c.$ of elements of S' $\Rightarrow \beta \in L(S')$

$\therefore$ L(S') = V. As S' is also L..I. ; S' forms a basis of V. $\therefore \dim V = n = \dim U$

Converse. Let dim U = dim V. To prove $U \cong V$.

Let $S = \{\alpha_1, \alpha_2 \ldots \alpha_n\}$ and $S' = \{\beta_1, \beta_2 \ldots \beta_n\}$ be the bei of U and V respectively so that dim U = n = dim V.

$\therefore \alpha \in U \Rightarrow \alpha = c_1\alpha_1 + c_2\alpha_2 \ldots + c_n\alpha_n$ for some $c's \in F$

Now define $T' : U \to V$ such that $T'(\alpha) = c_1\beta_1 + c_2\beta_2 \ldots + c_n\beta_n = \sum c_i\beta_i$

(*a*) *To show T' is one - one*

Let $\theta \in U$ such that , $\theta = d_1\alpha_1 + \ldots + d_n\alpha_n \quad d's \in F$

$\Rightarrow T'(\theta) = d_1\beta_1 + \ldots + d_n\beta_n$ (by def.)

Now $T'(\alpha) = T'(\theta) \Rightarrow c_1\beta_1 + c_2\beta_2 \ldots + c_n\beta_n = d_1\beta_1 + d_2\beta_2 \ldots + d_n\beta_n$

$\Rightarrow (c_1 - d_1)\beta_1 + \ldots + (c_n - d_n)\beta_n = \hat{O}$ $(\hat{O} \in V)$

$\Rightarrow c_1 - d_1 = 0, \ldots c_n - d_n = 0$ (S' is L.I.)

$\Rightarrow c_1 = d_1, \ldots c_n = d_n$

$\therefore c_1\alpha_1 + c_2\alpha_2 \ldots + c_n\alpha_n = d_1\alpha_1 + d_2\alpha_2 \ldots + d_n\alpha_n$

$\Rightarrow \alpha = \theta$ $\therefore$ T' is one-one.

(*b*) *To show T' is onto*

For $\delta \in V$, we can express $\delta = c_1\beta_1 + c_2\beta_2 \ldots + c_n\beta_n$ for $c's \in F$

If $\gamma = e_1\alpha_1 + e_2\alpha_2 \ldots + e_n\alpha_n$ by definition $T'(\gamma) = \delta$

$\therefore$ For $\delta \in V$ there exists $\gamma \in U$ such that $T'(\gamma) = \delta$ $\therefore$ T' is onto.

(*c*) *To show that T' is L. T.*

Let $a, b \in F$ and $\alpha, \theta \in U$, then $T'(a\alpha + b\theta) = T'[a \Sigma c_i\alpha_i + b \Sigma d_i\alpha_i]$

$= T'[\Sigma (a c_i + b \Sigma d_i)\alpha_i] = \Sigma (a c_i + b d_i)\beta_i$ (by def.)

$= a \Sigma c_i\beta_i + b \Sigma d_i\beta_i = a T'(\alpha) + b T'(\theta)$

$\therefore$ T' is a linear transformation.

Thus $T' : U \to V$ is one-one onto linear transformation.

Hence T' is an isomorphism $\therefore U \cong V$

Corollary. ***The image set of a basis set under an isomorphism is a basis set.***

The proof of this statement is the first part of the above proof.

4.3. FUNDAMENTAL THEOREM OF HOMOMORPHISM

Theorem. *Let U(F) and V(F) be two vector spaces and* $T : U \to V$ *is a onto linear transformation. N is the null space of T. Then* $U/N \cong V$

Proof. N is the null space of T $\Rightarrow$ $N \subseteq U$

$\therefore$ The quotient space, $U/N = \{ N+\alpha : \alpha \in U \}$ is the set of all cosets of N in U.

We know that $(N+\alpha_1)+(N+\alpha_2) = N+(\alpha_1+\alpha_2)$ $\forall\ \alpha_i \in U$, $a(N+\alpha) = N+a\alpha$ $\forall\ a \in F$

Let a function $f : U/N \to V$ be defined such that $f(N+\alpha) = T(\alpha)$ $\forall\ \alpha \in U$

Now to prove f is an homomorphism.

Let $\alpha_1, \alpha_2 \in U$ and $a, b \in F$, then $f[a(N+\alpha_1)+b(N+\alpha_2)] = f[(N+a\alpha_1)+(N+b\alpha_2)]$

$= f[(N+(a\alpha_1+b\alpha_2)] = T(a\alpha_1+b\alpha_2)$

$= aT(\alpha_1)+bT(\alpha_2) = af(N+\alpha_1)+b(N+\alpha_2)$

$\therefore$ f is a homomorphism.

(*i*) *To prove f is one-one*

$f(N+\alpha_1) = f(N+\alpha_2) \Rightarrow T(\alpha_1) = T(\alpha_2) \Rightarrow T(\alpha_1)-T(\alpha_2) = \hat{0}$

$\Rightarrow T(\alpha_1-\alpha_2) = \hat{0} \Rightarrow \alpha_1-\alpha_2 \in N \Rightarrow N+\alpha_1 = N+\alpha_2$

(*ii*) *To prove f is onto*

Since $T : U \to V$ is onto for every $\beta \in V$, there exists some $\alpha \in U$ such that $T(\alpha) = \beta$

$\therefore$ For this $\alpha \in U$, $(N+\alpha) \in U/N$. Hence $f(N+\alpha) = T(\alpha) = \beta$.

$\therefore$ For all $\beta \in V$, there exists $N+\alpha \in U/N$, so that $f(N+\alpha) = \beta$

$\Rightarrow$ f is onto, thus f is one-one homomorphism

$\therefore$ f is an isomorphism from U/N to V *i.e.* $U/N \cong V$.

Note. In the above theorem, if T is not given as onto then the statement will be as :

Let U(F) and V(F) be two vector spaces and $T : U \to V$ be a linear transformation.

Then $\frac{U}{N} \cong T(U)$ where N is the null space of T.

4.4. Theorem. ***Every n-dimensional vector space V(F) is isomorphic to*** F^n. ***(S. K. U. M 07)***

Proof. Let $S = \{\alpha_1, \alpha_2 \ldots \alpha_n\}$ be the basis of the n-dimensional vector space V(F)

For $\alpha \in V$ there exist $a_1, a_2 \ldots a_n \in F$ such that $\alpha = a_1\alpha_1 + a_2\alpha_2 \ldots + a_n\alpha_n$

Let a mapping $T : U \to F^n$ be defined by $T(\alpha) = (\alpha_1, \alpha_2 \ldots \alpha_n)$

i.e., then the T-image of α is the *n*-tuple of the coordinates of α

(*i*) *To show that T is one-one*

For $\alpha, \beta \in V$, $a's, b's \in F$

Let $\alpha = a_1\alpha_1 + a_2\alpha_2 \ldots + a_n\alpha_n = \Sigma\, a_i\alpha_i$; $\beta = b_1\alpha_1 + b_2\alpha_2 \ldots + b_n\alpha_n = \Sigma\, b_i\alpha_i$

Now $T(\alpha) = T(\beta) \Rightarrow (a_1, a_2 \ldots a_n) = (b_1, b_2 \ldots b_n)$

$\Rightarrow a_1 = b_1, a_2 = b_2, \ldots a_n = b_n \Rightarrow \Sigma a_i\alpha_i = \Sigma b_i\alpha_i \Rightarrow \alpha = \beta$. $\therefore$ T is one-one

(*ii*) To show that T is onto

For given any $(c_1, c_2 \ldots c_n) \in F^n$ there exists $\gamma \in V$ such that $\gamma = (c_1\alpha_1 + c_2\alpha_2 \ldots + c_n\alpha_n)$

$\therefore T(\gamma) = (c_1, c_2 \ldots c_n)$ (by def.) $\therefore$ T is onto

(*iii*) *To show that T is linear*

Let $a, b \in F$ and $\alpha, \beta \in V$

$$T(a\alpha + b\beta) = [a \Sigma a_i\alpha_i + b \Sigma b_i\alpha_i] = T[\Sigma (aa_i + bb_i)\alpha_i]$$

$$= (aa_1 + bb_1, aa_2 + bb_2, \ldots aa_n + bb_n)$$

$$= a(a_1, a_2 \ldots a_n) + b(b_1, b_2 \ldots b_n) = aT(\alpha) + bT(\beta)$$

$\therefore$ T is a linear transformation.

Thus T is an isomorphism from U to F^n *i.e.,* $U \cong F^n$

SOLVED PROBLEMS

Ex. 1. *Let* $T_k : V_3(R) \to V_3(R)$ *defined by* $T(x, y, z) = (x, y, kz), k \neq 0, k \in R$ *show that* T_k *is an isomorphism. What about* T_o *if* $T_o : V_3(R) \to V_3(R)$ *defined by* $T(x, y, z) = (x, y, 0)$.

Sol. Let $\alpha, \beta \in V_3(R)$ where $\alpha = (x_1, y_1, z_1)$, $\beta = (x_2, y_2, z_2)$

$\therefore T(\alpha) = (x_1, y_1, z_1), T(\beta) = (x_2, y_2, z_2)$

(*i*) For $a, b \in R$, $T_k(a\alpha + b\beta) = T_k[a(x_1, y_1, z_1) + b(x_2, y_2, z_2)]$

$= T_k(ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2) = (ax_1 + bx_2, ay_1 + by_2, k(az_1 + bz_2))$

$= (ax_1, ay_1, akz_1) + (bx_2, by_2, bkz_2) = a(x_1, y_1, kz_1) + b(x_2, y_2, kz_2)$

$= aT_k(\alpha) + bT_k(\beta)$. $\therefore$ T_k is a linear transformation

(*ii*) *To prove* T_k *is one one*

Now $T_k(x_1, y_1, z_1) = T_k(x_2, y_2, z_2) \Rightarrow (x_1, y_1, kz_1)(x_2, y_2, kz_2)$

$\Rightarrow kz_1 = kz_2 \Rightarrow z_1 = z_2$ $\because k \neq 0$

$\Rightarrow (x_1, y_1, z_1) = (x_2, y_2, z_2)$ $\therefore$ T is one one.

(*iii*) *To prove T is onto*

Since $k \neq 0$, for every vector $(x_1, y_1, z_1) \in R^3$ there exists a vector $\left(x_1, y_1, \frac{z_1}{k}\right)$

in R^3 such that $T_k\left(x_1, y_1, \frac{z_1}{k}\right) = (x_1, y_1, z_1)$ $\therefore$ T is onto.

Thus T is one-one onto linear transformation from R^3 onto itself and hence t is an isomorphism.

Clearly T is L.T. $T_0(x_1, y_1, z_1) = T_0(x_2, y_2, z_2) \Rightarrow (x_1, y_1, 0) = (x_2, y_2, 0)$

does not $\Rightarrow (x_1, y_1, z_1) = (x_2, y_2, z_2)$ $\therefore$ T_0 is not one one

Also T_0 is not onto and hence T_0 is not an isomorphism

Ex. 2. *If A and B are subspaces of a vector space V over a field F, then prove that* $\frac{A+B}{B} \cong \frac{A}{A \cap B}$.

Sol. We know that $A+B$ is a subspace of V containing B

$\therefore$ $\frac{A+B}{B}$ is a vector space over F.

Also $A \cap B$ is a subspace of A. $\therefore$ $\frac{A}{A \cap B}$ is a vector space over F.

An element of $\frac{A+B}{B}$ is of the form $\alpha+\beta+B$ where $\alpha \in A, \beta \in B$.

But $\beta + B = B$. $\therefore$ An element of $\frac{A+B}{B}$ is of the form $\alpha + \beta$

Define a map $T: A \to \frac{A+B}{B}$ by $T(\alpha) = \alpha + B \; \forall \; \alpha \in A$

Clearly T is well defined and onto.

Let $a, b \in F$ and $\alpha_1, \alpha_2 \in A$. $\therefore$ $a\alpha_1 + b\alpha_2 \in A$

Now $T(a\alpha_1 + b\alpha_2) = (a\alpha_1 + b\alpha_2) + B = (a\alpha_1 + B) + (b\alpha_2 + B)$

$$= a(\alpha_1 + B) + b(\alpha_2 + B) = aT(\alpha_1) + bT(\alpha_2)$$

$\therefore$ T is a linear transformation.

Hence (By Theorem 4.3) $\frac{A}{\text{Ker } T} \cong \frac{A+B}{B}$

Now $\text{Ker } T = \left\{\alpha \in A / T(\alpha) = 0 \text{ of } \frac{A+B}{B}\right\} = \{\alpha \in A / \alpha + B = B\} = A \cap B$.

Thus $\frac{A}{A \cap B} \cong \frac{A+B}{B}$ *i.e.,* $\frac{A+B}{B} \cong \frac{A}{A \cap B}$

SINGULAR AND NON-SINGULAR TRANSFORMATIONS

4. 5. SINGULAR TRANSFORMATION

Definition. *A linear transformation* $T: U(F) \to V(F)$ *is said to be singular if the null space of T consists of atleast one non-zero vector.*

i.e. If there exists a vector $\alpha \in U$ *such that* $T(\alpha) = \bar{O}$ *for* $\alpha \neq \bar{O}$ *then T is singular.*

4. 6. NON - SINGULAR TRANSFORMATION

Definition. *A linear transformation $T: U(F) \to V(F)$ is said to be non-singular if the null space consists of one zero vector alone.*

i.e., $\alpha \in U$ and $T(\alpha) = \hat{O} \Rightarrow \alpha = \hat{O} \Rightarrow N(T) = \{\bar{O}\}$

4. 7. Theorem. ***Let U(F) and V(F) be two vector spaces and $T: U \to V$ be a linear transformation. Then T is non-singular if f, the set of images of a linearly independent set, is linearly independent.***

Proof. (*i*) Let T be non-singular and let $S = \{\alpha_1, \alpha_2, \ldots \alpha_n\}$

be a L.I. subset of U. Then its T - images set be $S' = \{T(\alpha_1), T(\alpha_2), \ldots T(\alpha_n)\}$

Now to prove S' is L.I. For some $a_1, a_2 \ldots a_n \in F$

Let $a_1 T(\alpha_1) + a_2 T(\alpha_2) + \ldots + a_n T(\alpha_n) = \hat{O}$ $(\hat{O} \in V)$

$\Rightarrow T[a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n] = \hat{O}$ $(\because$ T is L.T.)

$\Rightarrow a_1\alpha_1 + a_2\alpha_2 + \ldots + a_n\alpha_n = \hat{O}$ $(\because$ T is non- singular)

$\Rightarrow a_1 = 0, a_2 = 0, \ldots, a_n = 0$ $(\because$ S is L.I.)

Thus S' is linearly independent.

(*ii*) Let the T - image of any L.I. set be L.I., then to prove T is non-singular.

Let $\alpha \in U$ and $\alpha \neq \bar{O}$. Then the set $B = \{\alpha\}$ is L.I. set and its image set $B' = \{T(\alpha)\}$ given to be L.I.

$\Rightarrow T(\alpha) \neq \hat{O}$ $(\because \{\hat{O}\}$ vectors is L.D.)

Thus $\alpha \neq \bar{O} \Rightarrow T(\alpha) \neq \hat{O}$. $\therefore$ T is non-singular.

4.8. Theorem. ***Let U(F) and V(F) be two finite dimensional vector spaces. The linear transformation $T: U \to V$ is an isomorphism iff T is non singular.***

Proof. (*i*) Let $T: U(F) \to V(F)$ be an isomorphism so that T is one- one onto.

To prove T is non-singular

Let $\alpha \in U$, then $T(\alpha) = \hat{O} \Rightarrow T(\alpha) = T(\bar{O})$ $(\because T(\bar{O}) = \hat{O}$ in any L.T.)

$\Rightarrow \alpha = \bar{O}$ (T is one-one)

$\therefore$ T is non-singular

(*it*) Let T be non-singular .

i.e. $\alpha \in U, T(\alpha) = \hat{O} \Rightarrow \alpha = \hat{O}$; $N(T) = \{\bar{O}\}$, $\therefore$ dim N(T) = 0

For $\alpha_1, \alpha_2 \in U, T(\alpha_1) = T(\alpha_2) \Rightarrow T(\alpha_1) - T(\alpha_2) = \hat{O}$ $[\hat{O} \in V]$

$\Rightarrow T(\alpha_1 - \alpha_2) = \hat{O}$ ($\because$ T is L.T.)

$\Rightarrow \alpha_1 - \alpha_2 = \bar{O} \Rightarrow \alpha_1 = \alpha_2$ ($\because$ T is non-singular)

$\therefore$ T is one-one

(*iii*) dim U = dim R(T) + dim N(T) = dim R(T) ($\because$ dim N(T) = 0)

Also $T : U \to V$ is one-one by (*ii*) $\Rightarrow$ V = R(T). $\Rightarrow$ T is an onto mapping

Again dim U = dim V. Hence T is an isomorphism.

SOLVED PROBLEMS

Ex. 1. *A linear mapping* $T : R^3 \to R^3$ *is defined by*

$T(x,y,z) = (x\cos\theta - y\sin\theta, x\sin\theta + y\cos\theta, z)$. *Show that T is non-singular.*

Sol. Let $T(x,y,z) = \hat{O}$

$\Rightarrow (x\cos\theta - y\sin\theta, x\sin\theta + y\cos\theta, z) = (0,0,0) \Rightarrow x\cos\theta - y\sin\theta = 0$

$x\sin\theta + y\cos\theta = 0, \quad z = 0$

Squaring and adding $x^2 + y^2 = 0 \Rightarrow x = 0, y = 0$

Thus $T(x,y,z) = \hat{O} \Rightarrow (x,y,z) = (0,0,0)$. $\therefore$ T is non singular.

Ex. 2. *Show that a linear transformation* $T : U \to V$ *over the field F is non-singular if and only if T is one-one.* ***(O. U. M 15, S. K. U. M 07, S. U. M 07)***

Sol. (*i*) Let T be non-singular *i.e.*, $\alpha \in U, T(\alpha) = \bar{O} \Rightarrow \alpha = \bar{O}$

Now for $\alpha_1, \alpha_2 \in U, T(\alpha_1) = T(\alpha_2)$

$\Rightarrow T(\alpha_1) - T(\alpha_2) = \hat{O}$ ($\hat{O} \in V$)

$\Rightarrow T(\alpha_1 - \alpha_2) = \hat{O}$ ($\because$ T is L.T.)

$\Rightarrow \alpha_1 - \alpha_2 = 0$ ($\because$ T is non-singular)

$\Rightarrow \alpha_1 = \alpha_2$

$\therefore$ T is one-one.

(*ii*) Let T be one-one

$\therefore$ zero element $\hat{O}$ of V is the T - image of only one element $\in U$.

$\Rightarrow$ null space of U consists of only one element.

$\because$ null space $N(T) \subseteq U$, it must consist of $\bar{O}$.

$\Rightarrow$ null space N(T) consists of only one $\bar{O}$ element.

$\Rightarrow N(T) = \{\bar{O}\} \Rightarrow$ T is non-singular.

Ex. 3. *Let* $T: U \to V$ *be a linear transformation of* $U(F)$ *into* $V(F)$ *where* $U(F)$ *is finite dimensional. Prove that U and the range space of T have the same dimnesion iff T is non-singular.*

Sol. (*i*) Let dim U = dim (Range T) = dim R(T)

$\because$ dim U = dim R(T) + dim N(T) $\Rightarrow$ dim N(T) = 0

$\Rightarrow$ The null space of T is the zero space $\{\bar{O}\}$

$\Rightarrow$ Hence T is non-singular

(*ii*) Let T be non-singular. Then $N(T) = \{\bar{O}$ and nullity T = 0.

As dim U = dim R(T) + dim N(T) = dim R(T) + 0

$\Rightarrow$ dim U = dim R(T).

Ex. 4. *If U and V are finite dimensional vector spaces of the same dimension, then a linear mapping* $T: U \to V$ *is one-one iff it is onto.*

Sol. T is one-one $\Leftrightarrow$ $N(T) = \{\bar{O}\}$ $\Leftrightarrow$ dim N(T) = 0

$\Leftrightarrow$ dim R(T) + dim N(T) = dim U = dim V

$\Leftrightarrow$ R (T) = V $\Leftrightarrow$ T is onto.

Note. In view of the above examples, we can state the following theorem.

Theorem. Let U and V are vector spaces of equal (finite) dimension, and let $T: U \to V$ is a linear transformation. Then the following are equivalent.

(*a*) T is one - to one (*b*) T is onto (*c*) rank (T) = dim (U)

We note that linearity of T is essential in the above theorems.

4. 9. INVERSE FUNCTION

Definition. *Let* $T: U \to V$ *be a one-one onto mapping. Then the mapping* $T^{-1}: U \to V$ *defined by* $T^{-1}(\beta) = \alpha \Leftrightarrow T(\alpha) = \beta$, $\alpha \in U, \beta \in V$ *is called the inverse mapping of T.*

Note: If $T: U \to V$ is one-one onto mapping then the mapping $T^{-1}: V \to U$ is also one-one onto.

4. 10. Theorem. ***Let U (F) and V (F) be two vector spaces and*** $T: U \to V$ ***be a one-one onto linear transformation. Then*** T^{-1} ***is a linear transformation and that T is said to be invertihle.***

Proof. Let $\beta_1, \beta_2 \in V$ and $a, b, \in F$

Since T is one-one onto function there exist unique vectors $\alpha_1, \alpha_2 \in U$ such that

$T(\alpha_1) = \beta_1$ and $T(\alpha_2) = \beta_2$. Hence by the definition T^{-1},

we have $\alpha_1 = T^{-1}(\beta_1)$ and $\alpha_2 = T^{-1}(\beta_2)$. Also $\alpha_1, \alpha_2 \in U$ and $a, b, \in F \Rightarrow a\alpha_1 + b\alpha_2 \in U$

$\therefore$ $T(a\alpha_1 + b\alpha_2) = aT(\alpha_1) + bT(\alpha_2) = a\beta_1 + b\beta_2$ ($\because$ T is L.T.)

$\therefore$ By the definition of inverse $T^{-1}(a\beta_1 + b\beta_2) = a\alpha_1 + b\alpha_2 = aT^{-1}(\beta_1) + bT^{-1}(\beta_2)$

$\because$ T^{-1} is a linear transformation from V into U.

4. 11. Theorem. ***A linear transformation T on a finite dimensional vector space is invertible if and only if T is non-singular.***

Proof. Let U (F) and V (F) be two vector spaces.

Let $T : U \to V$ be a linear transformation.

(*i*) Let T be non-singular *i.e.* For $\alpha \in U, T(\alpha) = \hat{0} \Rightarrow \alpha = \bar{0}$

Now to prove T is invertible, it is enough to show that T is one - one onto. For this refer proof of theorem 4.8.

(*ii*) Let T be invertible so that T is one - one onto. Now to prove T is non-singular.

For $\alpha \in U, T(\alpha) = \hat{0} = T(\bar{0})$ ($\because$ T is L. T.)

$\Rightarrow \alpha = \bar{0}$ T is one-one. $\therefore$ T is non-singular.

4.12. Theorem. ***Let U(F) and V(F) be two finite dimensional vector spaces such that dim U = dim V. If*** $T : U \to V$ ***is a linear transformation then the following are equivalent : -***

(1) ***T is invertible*** **(2)** ***T is non-singular*** **(3)** ***The range of T is V***

(4) ***If*** $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ ***is any basis of U, then*** $\{T(\alpha_1), T(\alpha_2), \ldots, T(\alpha_n)\}$ ***is a basis of V.***

(5) ***There is some basis*** $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ ***of U such that*** $\{T(\alpha_1), T(\alpha_2), \ldots, T(\alpha_n)\}$ ***basis of V.***

Proof. Here we shall prove a series of implications

viz., (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (5) $\Rightarrow$ (1)

Now (1) $\Rightarrow$ (2)

If T is invertible then T is one-one and therefore $T(\alpha) = \hat{0} = T(\bar{O}) \Rightarrow \alpha = 0$.

Hence T is nonsingular.

(*ii*) $\Rightarrow$ (*iii*)

Let T be non -singular. Let $S = \{\alpha_1, \alpha_2 \ldots \alpha_n\}$ be a basis of U.

Then S is L.I. set. Since T is non-singular the set $S' = \{T(\alpha_1), T(\alpha_2) \ldots T(\alpha_n)\}$ is a linearly independent set in V. But dim V = n, hence the set S' is a basis of V. Then a vector $\beta \in V$ can be expressed as $\beta = a_1 T(\alpha_1) + a_2 T(\alpha_2) \ldots + a_n T(\alpha_n)$ for some a_i'$s \in F$

$= T[a_1\alpha_1 + a_2\alpha_2 \ldots + a_n\alpha_n] = T(\alpha)$, $\alpha \in V$ ($\because$ T is L.T.)

$\Rightarrow \beta \in$ range of T.

Thus every vector in V is in the range of T. $\therefore$ R (T) = V

(3) $\Rightarrow$ (4)

Let the range of T be V *i.e.* T is onto. If $S=\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ is a basis of U, the T-images of these vectors $S' = \{T(\alpha_1), T(\alpha_2) \ldots T(\alpha_n)\}$ span the range of T *i.e.* V.

$\therefore$ L(S') = V.

Since S' is a L.I. set of n vectors and dim V = n, the set S' is a basis of V.

(4) $\Rightarrow$ (5). This is obvious in the above proof. (5) $\Rightarrow$ (1)

Let $S = \{\alpha_1, \alpha_2 \ldots \alpha_n\}$ be a basis of U such that $S' = \{T(\alpha_1), T(\alpha_2) \ldots T(\alpha_n)\}$ is a basis of V.

Since L (S') = R (T) it is clear that R (T) = V. *i.e.* T is onto.

Let $\alpha \in$ null space of T *i.e.* N(T) then $\alpha \in U$

$\therefore \alpha = b_1\alpha_1 + b_2\alpha_2 \ldots + b_n\alpha_n$ for some b_i's $\in F$. Hence $T(\alpha) = \hat{\bar{O}}$.

$\Rightarrow T[b_1\alpha_1 + b_2\alpha_2 \ldots + b_n\alpha_n] = \hat{\bar{O}}$

$\Rightarrow b_1T(\alpha_1) + b_2T(\alpha_2) \ldots + b_nT(\alpha_n) = \hat{\bar{O}}$ ($\because$ T is L.T.)

$\Rightarrow b_1 = 0, b_2 = 0 \ldots b_n = 0 \Rightarrow \alpha = \bar{O}$ ($\because$ S' is L.I.)

$\therefore$ This shows that T is non-singular and one-one. Hence T is invertible.

SOLVED PROBLEMS

Ex. 1. *If* $T: R^3 \to R^3$ *is invertible operator defined by*

$T(x, y, z) = (2x, 4x - y, 2x + 3y - z)$. *Find* T^{-1}. ***(A.N.U. 11, S. V. U. M 15, M 14, M 07)***

Sol. Since T is invertible. $T(x, y, z) = (a, b, c) \Rightarrow T^{-1}(a, b, c) = (x, y, z)$

$\therefore (2x, 4x - y, 2x + 3y - z) = (a, b, c) \Rightarrow 2x = a, 4x - y = b, 2x + 3y - z = c$

Solving $x = \frac{a}{2}, y = 2a - b, z = 7a - 3b - c$. Hence $T^{-1}(a, b, c) = \left(\frac{a}{2}, 2a - b, 7a - 3b - c\right)$

Ex. 2. *The set* $\{e_1, e_2, e_3\}$ *is the standard basis of* $V_3(R)$. $T: V_3(R) \to V_3(R)$ *is a linear operator defined by* $T(e_1) = e_1 + e_2$, $T(e_2) = e_2 + e_3$, $T(e_3) = e_1 + e_2 + e_3$. *Show that T is non-singular and find its inverse.* ***(S. K. U. S01)***

Sol. Let $e_1 = (1, 0, 0), e_2 = (0, 1, 0), e_3 = (0, 0, 1)$

Now $T: R^3 \to R^3$ is defined by $T(e_1) = e_1 + e_2 \Rightarrow T(1, 0, 0) = (1, 1, 0)$;

$T(e_2) = e_2 + e_3 \Rightarrow T(0, 1, 0) = (0, 1, 1)$; $T(e_3) = e_1 + e_2 + e_3 \Rightarrow T(0, 0, 1) = (1, 1, 1)$

Let $\alpha = (x, y, z) \in V_3(R)$

$\therefore \alpha = x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1)$

$\therefore$ $T(\alpha) = x\,T(1,0,0) + y\,T(0,1,0) + z\,T(0,0,1) = x(1,1,0) + y(0,1,1) + z(1,1,1)$

$\therefore$ The transformation is given by $T(x,y,z) = (x+z, x+y+z, y+z)$

Now If $T(x,y,z) = \hat{\bar{O}}$, then $(x+z, x+y+z, y+z) = (0,0,0)$

$\Rightarrow x+z=0, x+y+z=0, y+z=0 \Rightarrow x=y=z=0$.

$\therefore$ $T(\alpha) = \hat{\bar{O}} \Rightarrow \alpha = \bar{O}$. Hence T is non-singular and therfore T^{-1} exists.

Let $T(x,y,z) = (a,b,c)$ $\Rightarrow (x+z, x+y+z, y+z) = (a,b,c)$

$\Rightarrow x+z=a$ Solving $x=b-c$

$x+y+z=b$, $y+z=c$, $y=b-a$, $z=a-b+c$

$\therefore$ $T^{-1}(a,b,c) = (x,y,z) = (b-c, b-a, a-b+c)$

Ex. 3. *Show that the linear operator T defined on R^3 by $T(x,y,z) = (x+z, x-z, y)$ is invertible and hence find T^{-1}* **(O. U. 2011)**

Sol. Let $(x,y,z) \in N(T) \Rightarrow T(x,y,z) = \bar{0} \Rightarrow (x+z, x-z, y) = (0,0,0)$

$\Rightarrow x+z=0, x-z=0, y=0$. Solving $x=0, y=0, z=0 \Rightarrow N(T) = \{\bar{0}\}$

Hence T is non-singular and so it is invertible.

(2) Let $T(x,y,z) = (a,b,c) \Rightarrow T^{-1}(a,b,c) = (x,y,z)$

$\therefore$ $(x+z, x-z, y) = (a,b,c) \Rightarrow x+z=a; x-z=b; y=c$

from this we get, $2x = a+b \Rightarrow x = \dfrac{a+b}{2}$. $z = a - \left(\dfrac{a+b}{2}\right) = \dfrac{a-b}{2}$

$$T^{-1}(a,b,c) = \left(\frac{a+b}{2}, c, \frac{a-b}{2}\right)$$

4. 13. Theorem. ***The necessary and sufficient condition for a linear operator T on a vector space V (F) to be invertible is that there exists a linear transformation H on V such that TH = HT = I.*** **(K. U. 2014)**

Proof. Given $T : V(F) \to V(F)$ is a L.T.

(*i*) Let T be invertible. Hence T^{-1} exists and is one-one onto.

Let $\alpha \in V$ and $T(\alpha) = \beta$ so that $\alpha = T^{-1}(\beta)$

$(TT^{-1})(\beta) = T[T^{-1}(\beta)] = T(\alpha) = \beta = I(\beta) \Rightarrow TT^{-1} = I$

Again $(T^{-1}T)(\alpha) = T^{-1}[T(\alpha)] = T^{-1}(\beta) = \alpha = I(\alpha) \Rightarrow T^{-1}T = I$

$\therefore$ $T^{-1}T = T^{-1}T = I$

Taking $H = T^{-1}$ we get $TH = HT = I$ such that $H : V \to V$ is one - one onto L.T.

(*ii*) Let there exist a linear operator $H : V \to V$ such that $TH = HT = I$.

To prove T is invertible.

For $\alpha_1, \alpha_2 \in V$ $\quad T(\alpha_1) = T(\alpha_2)$

$\Rightarrow H[T(\alpha_1)] = H[T(\alpha_2)] \Rightarrow (HT)(\alpha_1) = (HT)(\alpha_2)$

$\Rightarrow I(\alpha_1) = I(\alpha_2) \Rightarrow \alpha_1 = \alpha_2$ $\quad (\because\ HT = I = TH)$

$\therefore$ T is one-one

For $\beta \in V$ there exists $\alpha \in V$ such that $H(\beta) = \alpha$ $\quad (\because\ H : V \to V)$

$\Rightarrow TH(\beta) = T(\alpha) \Rightarrow I(\beta) = T(\alpha) \quad \Rightarrow \beta = T(\alpha)$

$\therefore$ For any $\alpha \in V$ there exists $\beta \in V$ such that $T(\alpha) = \beta$, therefore T is onto.

Thus T is one-one onto and hence T is invertible.

Problems for Practicals

I. VECTOR SPACES

1. Let V denote the set of ordered pairs of real numbers. If $(a_1, a_2), (b_1, b_2) \in V$ and $c \in R$ define $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2\ b_2)$ and $c\,.\,(a_1, a_2) = (c\,a_1, a_2)$.
Is V a vector space over R with the said operations.

2. Let $V = \{(a_1, a_2) : a_1, a_2 \in F\}$ where F is a field with the definition.
$(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2),\ \forall\ a's, b's \in R$ and
$c\,.\,(a_1, a_2) = (a_1, 0)$ for $c \in R$. Is V(F) a vector space.

3. V is a set of ordered pairs of elements of the field R, with the definition
$(a_1, a_2) + (b_1, b_2) = (a_1 + 2b_1, a_2 + 3b_2),\ \forall\ a's, b's \in R$ and
$c\ (a_1, a_2) = (c\,a_1, c\,a_2)\ \forall\ c \in R$. Is V(R) a vector space.

4. Let $V = \{(a,b) : a,b \in R\}$ For all $a,b,c \in R$ if we define
$(a_1, b_1) + (a_2, b_2) = (a_1 + a_2, b_1 - b_2)$ and $(a_1, a_2) = (c\,a_1, c\,a_2)$
Is V(R) a vector space.

ANSWERS

1. No **2.** No **3.** No **4.** Yes

II. SUBSPACES

1. Determine whether the following sets are subspaces of R^3 under the operations of addition and scalar multiplication defined as R^3.

(*a*) $W = \{(a,b,c) \in R^3 : a = 3b \text{ and } c = -b\}$ (*b*) $W = \{(a_1, a_2, a_3) \in R^3 : a_1 = 2 + a_3\}$

(*c*) $W = \{(a,b,c) \in R^3 : 2a - 7b + c = 0\}$ (*d*) $W = \{(a,b,c) \in R^3 : a + 2b - 3c = 1\}$

(*e*) $W = \{(a,b,c) \in R : a - 4b - c = 0\}$

2. Prove that a subset W of a vector space V is a subspace of V $\Leftrightarrow$ $\bar{O} \in W$ and $ax + y \in W$ for all $a \in F$ and $x, y \in W$.

3. Prove that $W_1 = \{(a_1, a_2, \ldots a_n) \in F^n : a_1 + a_2 \ldots + a_n = 1\}$ is not a subspace of F^n.

4. If a vector space is the set of real valued. Continuous functions over R, then show that the set W of solutions of differential equation $2\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 2y = 0$ is a subspace of V.

5. A non-empty subset W of a vector space V(F) is a subspace of V $\Leftrightarrow$
(*i*) $\alpha \in W, \beta \in W \Rightarrow \alpha - \beta \in W$ (*ii*) $a \in F, \alpha \in W \Rightarrow a\,\alpha \in W$.

6. Let W_1 denote the set of all polynomials $f(x)$ in P(F) such that in the representation $f(x) = a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x + a_0$. We have $a_i = 0$ whenever i is even. Similarly W_2 denote the set of all polynomials $g(x)$ in P(F) such that in the representation $g(x) = b_n x^n + b_{n-1}x^{n-1} + ... + b_1 x + b_0$. we have $b_i = 0$ whenever i is odd. Prove that $P(F) = W_1 + W_2$.

ANSWERS

1(*a*) Yes (*c*) Yes (*d*) No

III. LINEAR COMBINATION

1. Find the scalars a, b, c such that $(x, y, z) \in R^3$ is a linear combination of the vectors (1,1,0), (1,0,1) and (0,1,1). Where $(x, y, z) = a(1,1,0) + b(1,0,1) + c(0,1,1)$

2. Let α, β, γ be the polynomials given by $\alpha = x^2 + 3x - 2$, $\beta = 2x^2 + 5x - 3$, $\gamma = -x^2 - 4x + 4$ α, β, γ generate $P_2(R)$. Express $ax^2 + bx + c \in R$ as a linear combination of α, β, γ.

3. The following are the vectors $\in R^3$. Determine whether α can be expressed as a linear combination of the other two β and γ.

(*a*) $\alpha = (-2,0,3)$, $\beta = (1,3,0)$, $\gamma = (2,4,-1)$ (*b*) $\alpha = (1,2,-3)$, $\beta = (2,-1,-1)$, $\gamma = (-3,2,1)$

(*c*) $\alpha = (3,4,1)$, $\beta = (1,-2,1)$, $\gamma = (-2,-1,1)$ (*d*) $\alpha = (-2,2,2)$, $\beta = (1,2,-1)$, $\gamma = (-3,-3,3)$

(*e*) $\alpha = (5,1,-5)$, $\beta = (1,-2,-3)$, $\gamma = (-2,3,-4)$

4. $\alpha, \beta, \gamma \in P_3(R)$. Can α be expressed as a linear combinarion of β and γ, given

(*a*) $\alpha = x^3 - 3x + 5$, $\beta = x^3 + 2x^2 - x + 1$, $\gamma = x^3 + 3x^2 - 1$

(*b*) $\alpha = x^3 - 8x^2 + 4x$, $\beta = x^3 - 2x + 3$, $\gamma = x^3 - 2x^2 + 3x - 1$

(*c*) $\alpha = -2x^3 - 11x^2 + 3x + 2$, $\beta = 2x^3 + x^2 + 3x - 2$, $\gamma = x^3 - 2x^2 + 3x - 1$

(*d*) $\alpha = 6x^3 - 3x^2 + x + 2$, $\beta = 2x^3 - 3x + 1$, $\gamma = x^3 - x^2 + 2x + 3$

5. Determine whether vector α is a span of the set S given

(*a*) S = $\{(1,0,2), (-1,1,1)\}$, $\alpha = (2,-1,1)$ (*b*) S = $\{(1,0,1,-1), (0,1,1,1)\}$, $\alpha = (-1,1,1,2)$

(*c*) S = $\{(1,0,1,-1), (0,1,1,1)\}$, $\alpha = (2,-1,1,-3)$

(*d*) S = $\{x^3 + x^2 + x + 1, x^2 + x + 1, x + 1\}$, $\alpha = -x^3 + 2x^2 + 3x + 3$

6. Determine whether vector α, β are spans of the set S given

$$S = \left\{\begin{bmatrix}1 & 0\\ -1 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\\ 0 & 1\end{bmatrix}, \begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix}\right\} \quad \alpha = \begin{pmatrix}1 & 2\\ -3 & 4\end{pmatrix}, \beta = \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$$

7. Show that the vector α, β, γ generate R^3 given $\alpha = (1,1,0)$, $\beta = (1,0,1)$, $\gamma = (0,1,1)$

8. Show that the set $\{M_1, M_2, M_3\}$ generate the V(F) of all 2×2 matrices given

$$M_1 = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}, \quad M_2 = \begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}, \quad M_3 = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$$

ANSWERS

1. $a = \frac{1}{2}(x+y-z)$, $b = \frac{1}{2}(x-y+z)$, $c = \frac{1}{2}(-x+y+z)$

2. $(-8a+5b+3c)\alpha + (4a-2b-c)\beta + (-a+b+c)\gamma$

3. (*a*) Yes (*b*) Yes (*c*) No (*d*) Yes (*e*) No.

4. (*a*) Yes (*b*) No (*c*) Yes **5.** (*a*) Yes (*b*) No (*c*) Yes (*d*) Yes

6. (*a*) Yes

IV. LINEAR DEPENDENCY & LINEAR INDEPENDENCE

1. Shows that the set S = $\{(-1,0,1,0), (1,-3,2,-4), (1,3,-4,-2), (2,2,-4,0)\}$ in R^4 is linearly dependent. ***(K. U. M 2013)***

2. Let $\alpha = \begin{pmatrix}1 & -3 & 2\\ -4 & 0 & 5\end{pmatrix}$, $\beta = \begin{pmatrix}-3 & 7 & 4\\ 6 & -2 & -7\end{pmatrix}$, $\gamma = \begin{pmatrix}-2 & 3 & 11\\ -1 & -3 & 2\end{pmatrix}$

Show that the set (α, β, γ) is linearly dependent.

3. Let $\alpha = (1,0,0,-1)$, $\beta = (0,1,0,-1)$, $\gamma = (0,0,1,-1)$ and $\delta = (0,0,0,1)$ are vectors $\in R^4$. Show the set S = $(\alpha, \beta, \gamma, \delta)$ is linearly in dependent. ***(O.U. 2011)***

4. Determine whether the set S = $\{\alpha, \beta, \gamma\}$ in R^3 is linearly dependent or linearly independent. given

(*a*) $\alpha = \begin{bmatrix}1 & -3\\ -2 & 4\end{bmatrix}$, $\beta = \begin{bmatrix}-2 & 6\\ 4 & -8\end{bmatrix}$, $\gamma = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

(*b*) $\alpha = x^3 - x, \beta = 2x^2 + 4, \gamma = -2x^3 + 3x^2 + 2x + 6$

(*c*) $\alpha = (-1,2,-1), \beta = (1,-1,2), \gamma = (2,0,1)$ (*d*) $\alpha = (1,-1,2), \beta = (1,-2,1), \gamma = (1,1,4)$

5. Determine whether the following set S = $\{\alpha, \beta, \gamma, \delta\}$ in $M_{2\times 2}(R)$ is linearly dependent or linearly in dependent

(*a*) $\alpha = \begin{bmatrix}1 & 0\\ -2 & 1\end{bmatrix}$, $\beta = \begin{bmatrix}0 & -1\\ 1 & 1\end{bmatrix}$, $\gamma = \begin{bmatrix}-1 & 2\\ 1 & 0\end{bmatrix}$, $\delta = \begin{bmatrix}2 & 1\\ -4 & 4\end{bmatrix}$

(*b*) $\alpha = \begin{bmatrix} 2 & 1 \\ 2 & -2 \end{bmatrix}, \beta = \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}, \gamma = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, \delta = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$

6. Show that the set $\{\alpha, \beta, \gamma, \delta\}$ in $M_{3\times 2}(R)$ is L.D. given

$$\alpha = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \beta = \begin{bmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}, \gamma = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}, \delta = \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}$$

7. Prove that a set S of vectors is linearly independent if and if each of finite subset of s is linearly independent.

ANSWERS

4. (*a*) L.D. (*d*) L.D. (*c*) L.I. (*b*) L.D. **5.** (*a*) L.D.

V. BASIS AND DIMENSION

1. Show that whether the set $\{\alpha, \beta, \gamma\}$ is a basis of $V_3(R)$ or not where

(*a*) $\alpha = (2,-3,5), \beta = (1,0,-2), \gamma = (0,2,-1)$

(*b*) $\alpha = (1,-3,-2), \beta = (-3,1,3), \gamma = (-2,-10,-2)$

(*c*) $\alpha = (1,1,2), \beta = (2,0,1), \gamma = (-1,2,1)$ (*d*) $\alpha = (-1,3,0), \beta = (-1,0,6), \gamma = (1,-4,2)$

2. Do the set of polynomials $\{x^3 - 2x^2 + 1, 4x^3 - x + 3, 3x - 2\}$ generate $P_3(R)$.

3. The set of solutions to the system of linear equations $x - 2y + z = 0, 2x - 3y + z = 0$ is a subspace of R^3. Find a basis for this subspace.

4. Determine which of the following sets are bases for $P_2(R)$.

(*a*) $\{1 + 2x - x^2, 4 - 2x + x^2, -9x^2 + 18x - 1\}$ (*b*) $\{1 - 2x - 2x^2, -2 + 3x - x^2, 1 - x + 6x^2\}$

(*c*) $\{-1 - x + 2x^2, 2 + x - x^2, 1 - 2x + 4x^2\}$ (*d*) $\{1 + 2x + x^2, 3 + x^2, x + x^2\}$

5. The vectors $u_1 = (2,-3,1), u_2 = (1,4,-2), u_3 = (-8,12,-4), u_4 = (1,37,-17), u_5 = (-3,-5,8)$ generate R^3. Find a subset of the set $\{u_1, u_2, u_3, u_4, u_5\}$ that is a basis of R^3.

6. The vectors $u_1 = (2,-3,4,-5,2), u_2 = (-6,9,12,15,-6), u_3 = (3,-2,7,-9,1)$, $u_4 = (2,-8,2,-2,6), u_5 = (-1,1,2,1,-3), u_6 = (0,-3,-18,9,12), u_7 = (1,0,-2,3,-2)$, $u_8 = (2,-1,1,-9,7)$ generate W which is a subspace of R^5. In each vector the sum of the coordinates is zero. Find a subset of the set $\{u_1, u_2, u_3, u_4, u_5, u_6, u_7, u_8\}$ that is a basis of W.

7. The vectors $\alpha = (1,1,1,1), \alpha = (0,1,1,1), \gamma = (0,0,1,1)$ and $\delta = (0,0,0,1)$ form a basis for R^4. Find the unique representation of an arbitrary vectors (a_1, a_2, a_3, a_4) in R^4 as a linear combination of $\alpha, \beta, \gamma, \delta$.

8. Let α, β be distinct vectors of a vector space V. Show that if $\{\alpha, \beta\}$ is a basis of V and a, b are non zero scalars, then both $\{\alpha+\beta, a\alpha\}$ and $\{a\alpha+b\beta\}$ are also the bases for V.

ANSWERS

1. (*a*) Yes (*b*) No (*c*) Yes (*d*) No **2.** No. **3.** (1,1,1)

4. (*a*) No (*b*) Yes (*c*) No **5.** $\{u_1, u_2, u_5\}$

7. $(\alpha, \beta, \gamma, \delta) = a_1\alpha + (a_2 - a_1)\beta + (a_3 - a_2)\gamma + (a_4 - a_3)\delta$

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LINEAR ALGEBRA

UNIT - IV

Matrix

Matrices, Elementary Properties of Matrices, Inverse Matrices, Rank of Matrix, Linear Equations, Characteristic Roots, Characteristic Values & Vectors of square Matrix, Cayley – Hamilton Theorem.

5. **Fundamentals of Matrices**
6. **Characteristic Values and Characteristic Vectors**

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5

Fundamentals of Matrices

5.1. INTRODUCTION

The student is familiar with matrices and their properties. In the coming chapters we define rank and using this we solve the systems of linear equations. Before doing that we list some of the properties of matrix (without proofs).

5.2. MATRIX

Definition. *A system of mn entries from a field F arranged in the form of an ordered set of m rows, each row consisting of an ordered set of n entries between* [ ] *or* ( ). $\| \ \|$ *is called a matrix of order or type* $m \times n$.

Each of mn entries constituting the $m \times n$ *matrix is called an element of the matrix.*

Thus we write a matrix,

$$A = \begin{Bmatrix} a_{11} & a_{12} & \ldots\ldots\ldots & a_{1n} \\ a_{21} & a_{22} & \ldots\ldots\ldots & a_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots & \ldots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{Bmatrix}_{m \times n} = [a_{ij}]_{m \times n}, 1 \le i \le m, 1 \le j \le n$$

In relation to a matrix, we call the entries as scalars.

Note. 1. A set of *mn* entries gives rise to many matrices.

2. *If the elements of the matrix A are real (complex) numbers, then the matrix A is said to be defined over the field of real (complex) numbers,* $R\,(C)$.

5.3. SOME TYPES OF MATRICES

Definitions 1. *If* $A = [a_{ij}]_{m \times n}$ *and* $m = n$, *then A is called a* **square matrix.**

A square matrix A of order $n \times n$ *is sometimes called as a n-rowed matrix A or simply a square matrix of order n.*

If A is a square matrix of order *n*, it consists of n^2 elements a_{ij} $(i, j = 1,2,3,\ldots\ldots,n)$. Here the elements $a_{i1}, a_{i2}, \ldots\ldots\ldots, a_{in}$ constitute the ith row and the elements $a_{1j}, a_{2j}, \ldots\ldots\ldots, a_{nj}$ constitute the jth column. Thus a_{ij} is the element of the ith row (first suffix) and the jth column (second suffix) of the square matrix A.

2. *A matrix which is not a square matrix is called a* **rectangular matrix.**

3. *A matrix of order* $1 \times m$ *is called a* **row matrix.**

4. *A matrix of order* $n \times 1$ *is called a* **column matrix.**

5. *If* $A = [a_{ij}]_{n \times n}$ *such that* $a_{ij} = 1$ *for* $i = j$ *and* $a_{ij} = 0$ *for* $i \neq j$, *then* A *is called a* **unit matrix**. *It is denoted by* 1.

6. *If* $A = [a_{ij}]_{m \times n}$ *such that* $a_{ij} = 0 \; \forall \; i$ *and* j, *then* A *is called a* **zero matrix** *or a* **null matrix**. *It is denoted by* O.

7. *Each of the elements* a_{ij} *for* $i = j$ *is called a diagonal element.*

8. *The rows of* $A_{m \times n}$ *are regarded as vectors in* F^n.

9. *The columns of* $A_{m \times n}$ *are regarded as vectors of* F^m

5.4. EQUAL MATRICES

Definition. *Two matrices* $A = [a_{ij}]$ *and* $B = [b_{ij}]$ *are said to be equal if and only if* (*i*) A *and* B *are of the same type (or order) and* (*ii*) $a_{ij} = b_{ij}$ *for every* i *and* j.

5.5. DIAGONAL ELEMENTS OF A SQUARE MATRIX. PRINCIPAL DIAGONAL

Definition. *If a matrix* $A = [a_{ij}]_{n \times n}$, *the elements* a_{ij} *of* A *for which* $i = j$ (*i.e.* $a_{11}, a_{22}, \ldots\ldots, a_{nn}$) *are called the diagonal elements of* A. *The line along which the diagonal elements lie is called the Principal diagonal of* A.

5.6. THE TRANSPOSE OF A MATRIX.

Definition. *The matrix obtained from any given matrix* A, *by interchanging its rows and columns is called the transpose of* A. *It is denoted by* A' *or* A^T.

If $A = [a_{ij}]_{m \times n}$ *then the transpose of* A *is* $A' = [b_{ij}]_{n \times m}$ *where* $b_{ij} = a_{ji}$.

Also $(A')' = A$.

Note. A, B *are matrices. Then* $A = B \Leftrightarrow A' = B'$.

5.7. THE CONJUGATE OF A MATRIX.

Definition. *The matrix obtained from any given matrix* A, *on replacing its elements by the corresponding conjugate complex numbers is called the conjugate of* A *and is denoted by* $\bar{A}$.

Thus if $A = [a_{ij}]_{m \times n}$, then $\bar{A} = [b_{ij}]_{n \times m}$ where $b_{ij} = \overline{a_{ij}}$, the conjugate complex number of a_{ij}.

Note. 1. If the matrix A is defined over R, then $\bar{A} = A$. **2.** $\overline{(\bar{A})} = A$.

e.g. $A = \begin{bmatrix} 2 & 3i & 2-5i \\ -i & 0 & 4i+3 \end{bmatrix}_{2 \times 3} \Rightarrow \bar{A} = \begin{bmatrix} 2 & -3i & 2+5i \\ i & 0 & -4i+3 \end{bmatrix}_{2 \times 3}$

5. 8. THE CONJUGATE TRANSPOSE OF A MATRIX.

Definition. *The conjugate of the transpose of the matrix A is called the conjugate transpose of A and is denoted by* A^{θ}.

Thus $A^{\theta} = (\bar{A}')$ *where* A' *is the transpose of A.*

Now $A = [a_{ij}]_{m \times n} \Rightarrow A^{\theta} = [b_{ij}]_{n \times m}$ where $b_{ij} = \overline{a_{ji}}$

e.g. $A = \begin{bmatrix} 5 & 3-i & -2i \\ 0 & 1+i & 4-i \end{bmatrix}_{2 \times 3} \Rightarrow A^{\theta} = \begin{bmatrix} 5 & 0 \\ 3+i & 1-i \\ 2i & 4+i \end{bmatrix}_{3 \times 2}$

Note. (*i*) $A^{\theta} = (\bar{A}') = (\bar{A})'$ (*ii*) $(A^{\theta})^{\theta} = A$

(*iii*) $(KA)^{\theta} = \bar{K}A^{\theta}$ where K is a complex number

(*iv*) $(A \pm B)^{\theta} = A^{\theta} \pm B^{\theta}$ (*v*) $(AB)^{\theta} = B^{\theta}A^{\theta}$

5.9. DIAGONAL MATRIX

Definition. *A square matrix A is said to be a diagonal matrix if each of its non-diagonal elements is zero.*

Thus if $A = [a_{ij}]_{n \times n}$ is a diagonal matrix, then $a_{ij} = 0$ for $i \neq j$.

If $d_1, d_2, ..., d_n$ are the diagonal elements of a n - rowed diagonal matrix, then we write the diagonal matrix A as : $A = \text{diag}\,(d_1, d_2, ..., d_n)$

A diagonal matrix whose diagonal elements are all equal is called a scalar matrix.

5.10. TRIANGULAR MATRICES

Definition. *A matrix* $A = [a_{ij}]_{m \times n}$ *is said to be an upper triangular matrix if* $a_{ij} = 0$ *for every* $i > j$ *and a lower triangular matrix if* $a_{ij} = 0$ *for every* $i < j$.

e.g. $\begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 4 & 2 & 1 \\ 0 & 0 & -6 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is an upper triangular matrix.

and $\begin{bmatrix} 7 & 0 & 0 & 0 & 0 \\ 5 & 3 & 0 & 0 & 0 \\ -4 & 6 & 0 & 0 & 0 \\ 2 & 1 & -8 & 5 & 0 \\ 2 & 0 & 4 & 1 & 6 \end{bmatrix}$ is a lower triangular matrix.

5.11. SYMMETRIC MATRIX

Definition. *A square matrix* $A=[a_{ij}]$ *is said to be symmetric if* $a_{ij}=a_{ji}$ *for every* i *and* j.

Thus : A is a symmetric matrix $\Leftrightarrow A = A'$

SKEW SYMMETRIC MATRIX

Definition. *A square matrix* $A=[a_{ij}]$ *is said to be skew symmetric if* $a_{ij}=-a_{ji}$ *for every* i *and* j.

Thus A is a skew symmetric matrix $\Leftrightarrow A = -A'$

Note. Every diagonal element of a skew symmetric matrix is necessarily zero since $a_{ii}=-a_{ii} \Rightarrow a_{ii}=0$.

e.g. $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$ is a symmetric matrix

and $\begin{bmatrix} 0 & a & -b \\ -a & 0 & c \\ b & -c & 0 \end{bmatrix}$ is a skew symmetric matrix.

Note. $O_{n\times n}$ and I_n are symmetric matrices.

5.12. COMPLEX MATRICES

HERMITIAN MATRIX.

A square matrix A such that $A^T = \bar{A}$ is called a Hermitian matrix. This can also be written as $(\overline{A^T}) = A$.

e.g. If $A=\begin{bmatrix} 4 & 1+3i \\ 1-3i & 7 \end{bmatrix}$ then $\bar{A}=\begin{bmatrix} 4 & 1-3i \\ 1+3i & 7 \end{bmatrix}$

and $A^T=\begin{bmatrix} 4 & 1-3i \\ 1+3i & 7 \end{bmatrix}$ then $A^T = \bar{A}$ and A is Hermitian.

Note. 1. The elements of the leading diagonal of a Hermitian matrix must be real.

2. A Hermitian matrix over the field of real numbers is nothing but a real symmetric matrix.

3. A necessary and sufficient condition for a matrix A to be Hermitian is that $A^\theta = A$.

SKEW - HERMITIAN MATRIX

A square matrix A such that $A^T = -\bar{A}$ is called a skew - hermitian matrix. This can also be written as $(\overline{A^T}) = -A$.

Let $A = \begin{bmatrix} -3i & 2+i \\ -2+i & -i \end{bmatrix}$ then $\bar{A} = \begin{bmatrix} 3i & 2-i \\ -2-i & i \end{bmatrix}$

and $A^T = \begin{bmatrix} -3i & -2+i \\ 2+i & -i \end{bmatrix}$ and $-\bar{A} = \begin{bmatrix} -3i & -2+i \\ 2+i & -i \end{bmatrix}$ which shows $A^T = -\bar{A}$

Note. 1. The elements of the leading diagonal must be all zero or all are purely imaginary.

2. A Skew - Hermitian matrix over the field of real numbers is nothing but a real skew symmetric matrix.

3. A necessary and sufficient condition for a matrix A to be Skew - Hermitian is that $A^\theta = -A$.

Unitary Matrix : A square matrix A such that $(\bar{A})^T = \bar{A}^T$

That is $(\bar{A})^T A = A(\bar{A})^T = I$ or $A^\theta A = I$ is called a unitary matrix.

e.g. $\begin{bmatrix} \frac{1}{2}i & \frac{1}{2}\sqrt{3} \\ \frac{1}{2}\sqrt{3} & \frac{1}{2}i \end{bmatrix}$ is a unitary matrix.

Note. 1. Every real symmetric matrix is Hermitian. Similarly a real Skew - Symmetric matrix is Skew - Hermitian and a real orthogonal matrix is unitary.

2. Thus Hermitian, Skew - Hermitian and unitary matrices generalize symmetric, Skew - Symmetric and orthogonal matrices respectively.

3. If A is a Hermitian matrix then iA is a Skew - Hermitian matrix.

4. Every square matrix is uniquely expressible as the sum of a Hermitian matrix and a Skew - Hermitian matrix.

5. Every Hermitian matrix can be written as $A + iB$ where A is real and symmetric and B is real and Skew - Symmetric.

6. If A is a Skew - Hermitian matrix then iA is a Hermitian matrix.

5.13. SUBMATRICES OF A MATRIX

Definition. *Let A be a matrix. If some rows or some columns or both are deleted from A, then the resulting matrix is called a submatrix of A.*

Thus we can have many submatrices of a matrix A.

5.14. $\delta_{ij} = 1, i = j; \delta_{ij} = 0, i \neq j$ IS CALLED KRONECKER DELTA.

e.g. If $I_n = [a_{ij}]_{n \times n}$ is a unit matrix, then $a_{ij} = \delta_{ij}$.

5.15. THE SET OF MATRICES OVER A FIELD

If all the elements of a matrix belong to a field F, we say that the matrix is over the field F. From now on, we consider the set of all matrices over field F.

5.16. MULTIPLICATION OF A MATRIX BY A SCALAR

Let A be a matrix. The matrix obtained by multiplying every element of A by K, a scalar, is called the product of A by K and is denoted by KA or AK.

Thus : If $A=[a_{ij}]_{m\times n}$, then $KA=[Ka_{ij}]_{m\times n}$ and $[Ka_{ij}]_{m\times n}=K[a_{ij}]_{m\times n}=KA$

Properties.

(*i*) $OA=O$ (null matrix), $(-1)A=-A$, called the negative of A.

(*ii*) $K_1(K_2A)=(K_1K_2)A=K_2(K_1A)$ where K_1, K_2 are scalars.

(*iii*) A is symmetric $\Rightarrow$ KA is symmetric.

(*iv*) A is skew symmetric $\Rightarrow$ KA is skew symmetric.

(*v*) $(KA)'=KA'$ (*vi*) $KA=0\Leftrightarrow A=0$ if $K\neq 0$.

(*vii*) $K_1A=K_2A$ and A is not a null matrix. $\Rightarrow K_1=K_2$

5.17. SUM OF MATRICES OR MATRIX ADDITION

Let $A=[a_{ij}]_{m\times n}, B=[b_{ij}]_{m\times n}$ *be two matrices.*

The matrix $C=[c_{ij}]_{m\times n}$ *where* $c_{ij}=a_{ij}+b_{ij}$, *is called the sum of the matrices A and B. The sum of A and B is denoted by* $A+B$.

Thus $[a_{ij}]_{m\times n}+[b_{ij}]_{m\times n}=[a_{ij}+b_{ij}]_{m\times n}$ and $[a_{ij}+b_{ij}]_{m\times n}=[a_{ij}]_{m\times n}+[b_{ij}]_{m\times n}$

Properties. Let M be the set of all $m\times n$ matrices and ' + ' be the matrix addition (composition) in M. Then

(*i*) **Closure.** $A,B\in M\Rightarrow A+B\in M$.

(*ii*) **Commutativity.** $A,B\in M\Rightarrow A+B=B+A$

(*iii*) **Associativity.** $A,B,C\in M\Rightarrow (A+B)+C=A+(B+C)$

Thus $(A+B)+C=A+(B+C)=A+B+C$

(*iv*) **Identity.** O (null matrix), $A\in M\Rightarrow O+A=A+O=A$

Thus the null matrix is the additive identity in M.

(*v*) **Inverse.** $A\in M\Rightarrow A+(-A)=(-A)+A=O$,

$-A$ is called the additive inverse of A in M.

(*vi*) $A,B\in M$ and $K\in F\Rightarrow K(A+B)=KA+KB$

(*vii*) $(A+B)'=A'+B'$ for $A,B\in M$.

Note : If $A, B, C\in M$

1. $A+B=0\Rightarrow B=-A$ or $A=-B$ 2. $A+B=A+C\Rightarrow B=C$

3. $A+B=C\Rightarrow A=C-B$ or $B=C-A$.

5.18. THE DIFFERENCE (SUBTRACTION) OF TWO MATRICES.

If A, B are two matrices of the same type (order), then $A+(-B)$ *is taken as* $A-B$.

Thus : If $A=[a_{ij}]_{m\times n}, B=[b_{ij}]_{m\times n}$ then $A-B=[a_{ij}]-[b_{ij}]=[a_{ij}-b_{ij}]$ and

$[a_{ij}-b_{ij}]=[a_{ij}]-[b_{ij}]$

Properties. $(A-B)'=A'-B'$, $K(A-B)=KA-KB$ where $K\in F$.

Note. If A and B are matrices of the same type (order), then we say that A and B are *conformable* for addition or subtraction.

5.19. Theorem. ***Every square matrix can be expressed as the sum of a symmetric and a skew - symmetric matrix in one and only one way (uniquely).***

5. 20. MATRIX MULTIPLICATION.

Let $A=[a_{ij}]_{m\times n}, B=[b_{ij}]_{n\times p}$. The matrix $C=[c_{ij}]_{m\times p}$ where $c_{ij}=\sum_{k=1}^{n} a_{ik}\, b_{kj}$ is called the product of the matrices A and B in that order.

In the product AB, the matrix A is called the prefactor and B the post factor.

If the number of columns of A is equal to the number of rows in B, then the matrices are said to be conformable for multiplication in that order.

Note. 1. If A, B are matrices, then

(*i*) AB may be possible and BA may not be possible.

(*ii*) BA may be possible and AB may not be possible.

(*iii*) AB, BA may both be possible.

(*iv*) Even if AB and BA are possible, they may not be equal.

(*v*) AB and BA are possible when A, B are each of order $n\times n$. In this case also they may not be equal.

(*vi*) Multiplication of matrices is not commutative. If A and B are matrices such that $AB=BA$ then we say that A and B commute.

2. If A, B are conformable for multiplication and k, l are scalars,

then $(kA)B=k(AB)=A(kB)$ and $(kA)(lB)=(kl)(AB)$

5.21. Theorem. ***Matrix multiplication is associative i.e. if A, B, C are matrices, then*** **(AB) C = A (BC).**

Cor. Generalised Associative Law. The Associative law for the product of any number of matrices can be proved to be true using mathematical induction. Hence the product of any number of matrices is independent of any particular manner of bracketing, while the order of the matrix factors is not changed.

5.22. POSITIVE INTEGRAL POWERS OF SQUARE MATRICES

Let A be a square matrix. Then A^2 is defined as AA.

Now, by the associative law,

$A^2A = (AA)A = A(AA) = AA^2$ so that we write $A^2A = AA^2 = A^3$

Similarly we have $AA^{m-1} = A^{m-1}A = A^m$ where m is a positive integer.

Further we have $A^mA^n = A^{m+n}$ and $(A^m)^n = A^{mn}$ where m, n are positive integers.

Note : $I^n = I, 0^n = 0$

5.23. Theorem. ***Multiplication of matrices is distributive w.r.t. addition of matrices i.e.*** $A(B+C) = AB+AC$ ***and*** $(B+C)A = BA+CA$.

5. 24. Theorem. ***If A is a matrix of order*** $m \times n$, ***then*** $AI_n = I_mA = A$.

5.25. ***A, B are matrices such that AB = O. Then AB = O does not necessarily mean that atleast one of the matrices A and B must be O.***

e.g. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}_{2\times 2}$ and $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}_{2\times 2}$

Now $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2\times 2} = O$ even though $A \neq O$ and $B \neq O$

5.26. TRACE OF A SQUARE MATRIX

Definition. *Let* $A = [a_{ij}]_{n\times n}$. *The trace of the square matrix A is defined as* $\sum_{i=1}^{n} a_{ii}$ *and is denoted by* '*tr* A'. *Thus* $tr\ A = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + ... + a_{nn}$.

Properties. *If A and B are square matrices of order n and* λ *is any scalar, then*

(*i*) $tr(\lambda A) = \lambda\ tr\ A$ (*ii*) $tr(A+B) = tr\ A + tr\ B$ (*iii*) $tr(AB) = tr(BA)$

5.27. Theorem. ***If A is*** $m \times n$ ***matrix and B is a*** $n \times p$ ***matrix, then*** $(AB)' = B'A'$.

Cor. $(ABC....Z)' = Z'Y'....C'B'A'$

5.28. 1. ***If A is a square matrix such that*** $A^2 = A$, ***then A is called 'idempotent'.***

2. ***If A is a square matrix such that*** $A^m = O$, ***where m is a positive integer, then A is called 'nilpotent'. If m is least positive integer such that*** $A^m = O$, ***then A is called 'nilpotent' of index, m.***

3. ***If A is a square matrix such that*** $A^2 = I$, ***then A is called involutory.***

4. ***If*** $AB = A$ ***and*** $BA = B$ ***then A and B are idempotent.***

5.29. ADJOINT OF A SQUARE MATRIX

Let $A = [a_{ij}]_{n\times n}$ be a matrix. The matrix $[b_{ij}]_{n\times n}$ where b_{ij} is the cofactor of a_{ji} in A, is called the adjoint of A and is denoted by the symbol adj A. Thus adjoint of A is the transpose of the matrix formed by the cofactor of A.

i.e. $adj\ A = \begin{bmatrix} A_{11} & A_{21} \ldots\ldots & A_{n1} \\ A_{12} & A_{22} \ldots\ldots & A_{n2} \\ \ldots\ldots & \ldots\ldots & \ldots\ldots \\ A_{1n} & A_{2n} \ldots\ldots & A_{nn} \end{bmatrix}$ where A_{ij} is the cofactor of a_{ij}.

e. g. If $A = \begin{bmatrix} 2 & -1 & 4 \\ 4 & 2 & -3 \\ 1 & 1 & 2 \end{bmatrix}$, then $adj\ A = \begin{bmatrix} 7 & 6 & -5 \\ -11 & 0 & 22 \\ 2 & -3 & 8 \end{bmatrix}$

Note. $adj\ I_n = I_n, adj\ O = O$. $adj\ (KA) = K^{n-1} adj\ A$, if A is of order n and K is a scalar.

5.30. Theorem. ***If*** $A = [a_{ij}]_{n \times n}$ ***then*** $A\ (adj\ A) = (adj\ A)\ A = |A| I_n$

5.31. INVERSE OF A MATRIX

Definition. *Let A be any matrix, then a matrix B, if exists such that* $AB = BA = I$ *is called inverse of A.*

For AB, BA to be both defined and equal, it is necessary that A and B are both square matrices of same order. Thus a non-square matrix cannot have inverse.

Definition. *A matrix is said to be invertible, if it possesses inverse.*

5.32. Theorem. ***Every invertible matrix possesses a unique inverse.***

Note. 1. The inverse of A is denoted by A^{-1}.

Thus $AA^{-1} = A^{-1}A = I$. Also $A^{-1}A = AA^{-1} = I \Rightarrow (A^{-1})^{-1} = A$

$\Rightarrow$ The inverse of a matrix is invertible and the inverse of the inverse of the matrix is itself.

2. Since $I_n\ I_n = I_n$, we have $I^{-1} = I$ *i.e.* the inverse of a unit matrix is itself.

3. If A is an invertible matrix and if $A = B$, then $A^{-1} = B^{-1}$

5.33. Theorem. ***The necessary and sufficient condition for a square matrix to possess inverse is that*** $|A| \neq 0$.

Note. If $|A| \neq 0$, then $A^{-1} = \frac{1}{|A|}(adj\ A)$

SINGULAR AND NON-SINGULAR MATRICES

Definition. *A square matrix is said to be singular if* $|A| = 0$. *If* $|A| \neq 0$ *then A is said to be non-singular.*

Thus only non-singular matrices possess inverses.

Note : If A, B are non-singular, then AB, the product is also non-singular, Thus the product of non-singular matrices is also non-singular.

5.34. Theorem. *If A is non-singular and $AB = AC$, then $B = C$*

5.35. Result. *If A and B are square matrices of order n such that $AB = I$, then $BA = I$.*

5.36. Result. *If A is singular, then* $A(adj\ A) = (adj\ A)A = 0$.

5.37. Result. *If the product of non-zero square matrices is a zero matrix, both of them must be singular matrices.*

5.38. Result. *If* $|A| = 0$, *then* $|adj\ A| = 0$

5.39. Result. *If A is a symmetric matrix, then adj A is also symmetric.*

5.40. Result. *If A be a square matrix, then* $adj\ A' = (adj\ A)'$.

5.41. Result. *If A is a non-singular matrix, then* $\det(A^{-1}) = \dfrac{1}{\det A}$

5.42. Result. *If A is a square matrix of order n,* $|adj\ A| = |A|^{n-1}$.

Note. If A is non-singular, then $adj\ A$ is also non-singular.

5.43. Theorem. *If A and B are two non-singular matrices of the same type, then* $(adj\ AB) = (adj\ B)\ (adj\ A)$.

5.44. Theorem. *If A,B are invertible matrices of the same order, then*

$(i)\ (AB)^{-1} = B^{-1}A^{-1}$ $(ii)\ (A^1)^{-1} = (A^{-1})^1$

5.45. Result. *If A is a square matrix of order n and B is a non-singular matrix of order n, then there exists one and only one matrix X such that $A = BX$ and there exists one and only one matrix Y such that $A = YB$.*

5.46. RIGHT AND LEFT DIVISORS

Definition. *If A, B are square matrices of the same order such that $AB = 0$ and $A \neq 0$ then A is called a left zero divisor. If $AB = 0$ and $B \neq 0$ then B is called a right zero divisor.*

5.47. ORTHOGONAL MATRIX

Definition. *A square matrix A is said to be orthogonal if* $A'A = I$ **(O. U. M 08)**

A is an orthogonal matrix $\Rightarrow A'A = I$

$\Rightarrow |A^1A| = |I| \Rightarrow |A^1||A| = 1 \Rightarrow |A^2| = 1$

$\Rightarrow |A| = \pm 1 \Rightarrow |A| \neq 0 \Rightarrow A$ is invertible. Also $AA^1 = I$, if $A^1A = I$.

$\therefore\ A^1A = AA^1 = I \Rightarrow A^1 = A^{-1}$

Thus : A is an orthogonal matrix $\Leftrightarrow A'A = I = AA'$

5.48. Theorem. ***If A, B are orthogonal matrices, each of order n then AB and BA are orthogonal matrices.***

5.49. Result. ***The inverse of an orthogonal matrix is orthogonal.***

5.49. ELEMENTARY TRANSFORMATIONS OR OPERATIONS ON A MATRIX

(*i*) R_{ij} . *Interchange of* i^{th} *and* j^{th} *rows termed as Type I*

(*ii*) $R_i(k)$. *Multiplying every element of* i^{th} *row by k, termed as Type II*

(*iii*) $R_{ij}(k)$. *Multiplying every element of* j^{th} *row by k and adding to the corresponding element of* i^{th} *row, termed as Type III*

Similarly we understand C_{ij}, $C_i(k)$, $C_{ij}(k)$ for columns.

Elementary transformations on a matrix do not change the order of the matrix.

5.50. ELEMENTARY MATRIX.

Definition. *A matrix obtained from a unit matrix* I_n *by subjecting the unit matrix to anyone of the elementary transformations is called an elementary matrix. The elementary matrix is said to be of type 1, 2 or 3 according as whether the elementary operations performed on unit matrix* I_n *is a type 1, 2 or 3 operation, respectively.*

5.51. SYMBOLS

1. E_{ij}. Elementary matrix obtained by interchanging i^{th} and j^{th} row or i^{th} and j^{th} columns in I.
2. $E_i(k)$. Elementary matrix obtained by multiplying every element of i^{th} row or i^{th} column with k in I.
3. $E_{ij}(k)$. Elementary matrix obtained by multiplying every element of j^{th} row with k and then adding them to the corresponding elements of i^{th} row in I.
4. $E'_{ij}(k)$. Elementary matrix obtained by multiplying every element of j^{th} column with k and then adding them to the corresponding elements of i^{th} column in I.

5.52. Theorem. ***Every elementary row (column) transformation of a matrix can be obtained by pre-multiplication (post multiplication) with the corresponding elementary matrix.***

5.53. Definition. *Let A belong to* $M_{m\times n}(F)$, *we define the rank of A, denoted as rank (A) or* $\rho(A)$ *to be the rank of the linear transformation* $L_A : F^n \to F^m$.

5.54. Theorem. ***Let*** $T: V \to W$ ***be a linear transformation between finite dimensional vector spaces, and let*** β ***and*** γ ***be ordered bases for V and W respectively. Then rank (T) = rank*** $([T]_\beta^\gamma)$

5.55. Theorem. *Let A be an $m \times n$ matrix. If P and Q are invertible $m \times m$ and $n \times n$ matrices, respectively, then*

1. *rank* (AQ) = *rank (A)*

2. *rank* (PA) = *rank (A), and therefore*

3. *rank* (PAQ) = *rank (A)*

Theorem. *The rank of a matrix equals the maximum number of its linearly independent columns (or) The rank of a matrix is the dimnesion of subspace generated by its columns.*

5.56. REDUCTION TO NORMAL FORM

Theorem. *Every non-zero matrix can be reduced to the form* $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ *by a finite number of elementary transformations, where* I_r *is the unit matrix of order r and r is the rank of the matrix.*

5.57. Theorem . *If A be a $m \times n$ matrix of rank r, there exist non-singular matrices P and Q such that* $PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$

5.58. Theorem. *The rank of a matrix does not change by pre-multiplication or post-multiplication with a non-singular matrix.*

5.59. Theorem. *Let* $T: V \to W$ *and* $U: W \to Z$ *be linear transformations of finite - dimensional vector spaces V, W and Z, and let A and B be matrices such that the product AB is defined. Then*

(i) rank (UT) $\leq$ rank (U) (*O. U. M 08*) *(ii) rank (AB) $\leq$ rank (A)*

(iii) rank (AB) $\leq$ rank (B) *(iv) rank (UT) $\leq$ rank (T)*

5.60. ECHELON FORM OF A MATRIX.

If all the elements in a row of a matrix are zeros, then it is called a zero row and if there is atleast one non-zero eleement in a row, then it is called a non-zero row.

Row Echelon form or simply Echelon form of a matrix.

A matrix is said to be in Echelon form if it has the following properties.

(*i*) Zero rows, if any, must follow non-zero rows.

(*ii*) The first non-zero element in each non-zero row must be 1.

(*iii*) The number of zeros before the first non-zero element in a row is less than the number of such zeros in the next row.

Some authors ignore the property (*ii*) to consider a matrix to be in Echelon form.

The rank of a matrix in Echelon form is equal to the number of non-zero rows of the matrix.

5.61. INVERSE OF A MATRIX

We have remarked that an $n \times n$ matrix is invertible if and only if its rank is n. As we know how to compute the rank of any matrix we can always test a matrix to determine whether it is invertible or not.

5.62. Def. Augmented Matrix :

Let A and B are $m \times n$, $m \times p$ matrices respectively. By the augmented matrix [A B] *or* [A | B] *or* (A | B) *we mean $m \times (n+p)$ matrix whose first n columns are the columns of A, whose last p columns are the columns of B.*

We now define the rank of a matrix using the minors of that matrix.

1. MINOR OF A MATRIX

Let A be a matrix of the type $m \times n$ defined over the field F. Let t be a positive integer such that $t \leq$ minimum of m and n. If we delete from A, $(m-t)$ rows, leaving certain t rows and $(n-t)$ columns leaving a certain t columns and if $t > 1$, then the remaining elements of A form a t - rowed square matrix whose determinant is a *minor* of the matrix A of order t. Thus there corresponds a system of minors of A to each possible value of t. Each single element of A may be regarded as a minor of order 1 $(t = 1)$

2. RANK OF A MATRIX

If A is a null matrix, then the rank of A is defined as zero.

Definiton. *If A is not a null matrix; then the rank of the matrix A is the positive integer, if it possesses the following two properties* : -

(*i*) Each minor of order $n+1$ of the matrix A vanishes (*i.e.* is zero)

(*ii*) There exists at least one minor of order n of the matrix A which is not zero.

Note.

1. Rank of A is denoted by $\rho(A)$ and is unique.

2. Every Matrix will have a rank.

3. If A is a matrix of order $m \times n$, then $\rho(A) \leq m$ or n (smaller of the two)

4. If $\rho(A) = n$, then every minor of order $n+1, n+2$ etc., is zero.

5. A is a matrix of order $n \times n$. A is non-singular $\Leftrightarrow \rho(A) = n$

6. Rank of $I_n = n$

7. A is a matrix of order $m \times n$. If every k^{th} order minor $(k < m, k < n)$ is zero, then $\rho(A) < k$.

8. A is matrix of order $m \times n$. If there is a minor of order k $(k < m, k < n)$ which is not zero, then $\rho(A) > k$

3. RANK OF A PRODUCT OF MATRICES.

Theorem. ***The rank of a product of two matrices cannot exceed the rank of either matrix.***

OR: ***If A, B are matrices conformable for multiplication , then*** $\rho(AB) \leq \rho(A)$ ***and*** $\rho(AB) \leq \rho(B)$.

4. VECTOR

Definition. *An ordered n- tuple of numbers is called an n-vector. The n numbers of an n-vector are called its components. The components may be written in a horizontal line or in a vertical line. Thus a vector may be taken as a row or a column matrix.*

If the components of a n-vector belong to a field F, then the vector is said to be over the field F.

If $\xi = (a_1, a_2, \ldots a_n)$ and $K \in F$, then

$K\xi = (ka_1, ka_2, \ldots ka_n)$ -- called scalar multiplication of a vector.

If $\xi_1 = (a_1, a_2, \ldots a_n)$ and $\xi_2 = (b_1, b_2, \ldots b_n)$, then $\xi_1 + \xi_2 = (a_1 + b_1, a_2 + b_2 + \ldots a_n + b_n)$ called vector addition.

If V_n is a set of x-vectors over a field F and $\xi_1, \xi_2 \ldots V \in V_n$, $K \in F \Rightarrow \xi_1 + \xi_2 \in V_n$ and $K\xi \in V_n$.

The set of all n-vectors of a field of F is a vector space over the field F for the operations of vector addition and scalar multiplication. It is denoted by $V_n(F)$ or simply V_n.

A vector X, expressed in the form $X = k_1x_1 + k_2x_2 \ldots + k_rx_r$ where $k_1, k_2, \ldots k_r \in F$ and $x_1, x_2, \ldots, x_r \in V_n$ is called a linear combination of vectors $x_1, x_2, \ldots, x_r$.

5. SUBSPACE OF AN *n* - VECTOR SPACE V_n.

Definition. *A non-empty subset, S of* $V_n(F)$ *is called a subspace of* $V_n(F)$ *if it follows vector addition and scalar multiplication.*

Every subspace of $V_n(F)$ contains the zero vector.

6. SUBSPACE SPANNED BY A SET OF VECTORS

Definition. *A subspace which arises as a set of all linear combinations of any given set of vectors, is said to be spanned by the given set of vectors.*

7. BASIS OF A SUBSPACE

Definition. *A set of vectors is said to be a basis of a subspace if*

(i) the subspace is spanned by the set and (ii) the set is L.I.

8. Theorem. ***A basis of a subspace S, can always be selected from a set of vectors which span S.***

Note. It may also be shown that $V_n(F)$ possesses a basis and that the number of vectors in every basis of S, is the same.

9. DIMENSION OF A SUBSPACE

Definition. *The number of vectors in any basis of a subspace is called the dimension of the subspace.*

10. ROW SPACE AND COLUMN SPACE OF A MATRIX

Let A be a matrix of order $m \times n$ over a field F. Every row of A consists of n elements and every column of A consists of m elements. So each row may be taken as an n-vector and each column may be taken as a m-vector. Clearly every row of A is an element of $V_n(F)$ and every column of A is an element of $V_m(F)$.

The space spanned by the m rows which is a subspace of $V_n(F)$ is called the *row space* of the matrix A.

The space spanned by the n columns which is a subspace of $V_m(F)$ is called the *column space* of the $m \times n$ matrix A.

The dimension of the row space of A is called the *row rank* of A and the dimension of the column space of A is called *column rank* of A.

11. EQUALITY OF ROW RANK, COLUMN RANK AND RANK.

Theorem. ***The row rank of a matrix is equal to its rank.***

12. Theorem. ***The column rank of a matrix is equal to its rank.***

RANK OF A SUM.

13. Theorem. ***Rank of the sum of two matrices cannot exceed the sum of their ranks. (OR) If A,B are two matrices, conformable for addition, then***

rank of $(A+B) \leq$ ***rank of*** A + ***rank of*** B.

14. Theorem. ***If A, B are two n-rowed square matrices,***

then $\rho(AB) \geq \rho(A) + \rho(B) - n$. ***(S. V. U. 93 S)***

15. Result. If A be any non-singular matrix and B a matrix such that AB exists, then show that AB and B have the same rank.

Sol. Since A is non-singular, A^{-1} exists.

$\therefore C = AB \Rightarrow A^{-1}C = A^{-1}(AB) = (A^{-1}A)B$

$\Rightarrow IB = A^{-1}C \Rightarrow B = A^{-1}C$.

$\therefore \rho(C) = \rho(AB) \leq \rho(B)$ ... (1) and $\rho(B) = \rho(A^{-1}C) \leq \rho(C)$ ... (2)

From (1) and (2), $\rho(B) = \rho(C) = \rho(AB)$.

16. Result. If A,B are square matrices of the same type n and I is a unit matrix of order n, then show that $\rho(AB - I) \leq \rho(A - I) + \rho(B - I)$.

Sol. Now $(AB - I) = AB - A + A - I = AB - AI + A - I = A(B - I) + (A - I)$

$\therefore \rho(AB - I) = \rho[A(B - I) + (A - I)] \Rightarrow \rho(AB - I) \leq \rho[A(B - I)] + \rho(A - I)$

But $\rho[A(B - I)] \leq$ min. of $\{\rho(A), \rho(B - I)\} = \rho(B - I)$.

$\therefore \rho(AB - I) \leq \rho(B - I) + \rho(A - I) \Rightarrow \rho(AB - I) \leq \rho(A - I) + \rho(B - I)$.

17. SYSTEM OF NON-HOMOGENEOUS LINEAR EQUATIONS

Consider a system of m non-homogeneous linear equations in n unknowns.

$a_{11}x_1 + a_{12}x_2 + \ldots\ldots + a_{1n}x_n = b_1,$

$a_{21}x_1 + a_{22}x_2 + \ldots\ldots + a_{2n}x_n = b_2,$

$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$

$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$

$a_{m1}x_1 + a_{m2}x_2 + \ldots\ldots + a_{mn}x_n = b_m$

The system can be expressed as the matrix equation.

$A_{m\times n}\, X_{n\times 1} = B_{m\times 1}$ *i.e.* $AX = B$

18. Theorem. ***If A is a non-singular matrix of order n, then the linear system AX = B in n unknowns has a unique solution.***

19. PROPERTIES OF DETERMINANTS

Theorem. ***For any*** $\mathbf{A,B} \in \mathbf{M}_{n\times n}(\mathbf{F})$, ***then*** $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$.

Theorem. ***A matrix*** $\mathbf{A} \in \mathbf{M}_{n\times n}(\mathbf{F})$ ***is invertible if and only if*** $\det(\mathbf{A}) = \mathbf{0}$.

Further if A is invertible then $\det(\mathbf{A}^{-1}) = \dfrac{1}{\det \mathbf{A}}$.

Theorem. ***For any*** $\mathbf{A} \in \mathbf{M}_{n\times n}(\mathbf{F})$, $\det(\mathbf{A}^{\mathbf{T}}) = \det(\mathbf{A})$

LINEAR EQUATIONS

5.63. SYSTEM OF NON-HOMOGENEOUS LINEAR EQUATIONS

Consider the system of m equations in n unknowns given by

$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1$
$a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n = b_2$
$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$
$a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n = b_m.$

(*i.e.*) $AX = B$

The augmented matrix of the above system is $[A|B] = \left(\begin{array}{cccc|c} a_{11} & a_{12} & a_{13}\ldots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23}\ldots & a_{2n} & b_2 \\ \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots \\ \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots \\ a_{m1} & a_{m2} & a_{m3}\ldots & a_{mn} & b_m \end{array}\right)$

Find the ranks of A, $[A|B]$.

Working Rule : The system $AX = B$ is consistent *i.e.* it has a solution (unique or infinite) if and only if

rank of A = rank of $[A/B]$

(*i*) If rank of A = rank of $[A/B] = r <$ number of unknowns, the system is consistent, but there exists infinite number of solutions. Giving arbitrary values to $n - r$ of the unknowns, we may express the other r unknowns in terms of these.

(*ii*) If rank of A = rank of $[A/B] = r = n$ the system has unique solution.

(*iii*) If rank of $A \neq$ rank of $[A/B]$ the system is inconsistent. It has no solution.

To obtain solutions, set $(n-r)$ variables any arbitrary value and solve for the remaining unknowns.

Thus irrespective of $m < n$ or $m = n$, or $m > n$, we have decided about the existence and uniqueness of solutions of a system of linear simultaneous equations $AX = B$.

This procedure is illustrated through the following examples.

SOLVED PROBLEMS

Ex. 1. *Show that the equations* $x + y + z = 4$, $2x + 5y - 2z = 3$, $x + 7y - 7z = 5$ *are not consistent.*

Sol. We write the given equations in the form $AX = B$, where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & -2 \\ 1 & 7 & -7 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$$

Consider $[A|B] = \begin{bmatrix} 1 & 1 & 1 & 4 \\ 2 & 5 & -2 & 3 \\ 1 & 7 & -7 & 5 \end{bmatrix}$

Applying $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$, we get

$$[A/B] \sim \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 3 & -4 & -5 \\ 0 & 6 & -8 & 1 \end{bmatrix}$$

Applying $R_3 \to R_3 - 2R_2$, we get $[A/B] \sim \begin{bmatrix} 1 & 1 & 1 & 4 \\ 0 & 3 & -4 & -5 \\ 0 & 0 & 0 & 11 \end{bmatrix}$

We can see that rank $[A|B] = 3$, since the number of non-zero rows is 3.

Applying the same row operations on A, we get from above

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & -4 \\ 0 & 0 & 0 \end{bmatrix}$$

Here the number of non-zero rows is 2 so the rank of $A = 2$.

Here we have rank $(A) \neq$ rank $[A/B]$.

$\therefore$ The given system is not consistent.

Ex. 2. *Discuss for what values of* λ, μ *the simultaneous equations* $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ *have (i) no solution (ii) a unique solution (iii) an infinite number of solutions.*

Sol. The matrix form of given system of equations is

$$AX = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix} = B$$

We have the augmented matrix is $[A/B] = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{bmatrix}$

Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$, we get $[A/B] \sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda - 1 & \mu - 6 \end{bmatrix}$

Applying $R_3 \to R_3 - R_2$, we get $[A/B] \sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda - 3 & \mu - 10 \end{bmatrix}$

Case I. Let $\lambda \neq 3$ then rank of $A = 3$ and rank of $[A/B] = 3$, so that they have same rank. Then the system of equations is consistent. Here the number of unknowns is 3 which is same as the rank of A. The system of equations will have a unique solution. This is true for any value of μ.

Thus if $\lambda \neq 3$ and μ has any value, the given system of equations will have a unique solution.

Case II. Suppose $\lambda = 3$ and $\mu \neq 10$, then we can see that rank of $A = 2$ and rank of $[A/B] = 3$. Since the ranks of A and $[A/B]$ are not equal, we say that the system of equations has no solution (inconsistent).

Case III. Let $\lambda = 3$ and $\mu = 10$. Then we have rank of A = rank of $[A/B] = 2$.

$\therefore$ The given system of equations will be consistent.

But here the number of unknowns = 3 > rank of A.

Hence the system has infinitely many solutions.

Ex. 3. *Find whether the following equations are consistent, if so solve them.*
$x + y + 2z = 4;\ 2x - y + 3z = 9;\ 3x - y - z = 2$

Sol. The given equations can be written in the matrix form as $\begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 3 \\ 3 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 2 \end{bmatrix}$

i.e. AX = B

The augmented matrix $[A|B] = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 2 & -1 & 3 & 9 \\ 3 & -1 & -1 & 2 \end{bmatrix}$

$[A \mid B] \sim \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & -3 & -1 & 1 \\ 0 & -4 & -7 & -10 \end{bmatrix}$ (Applying $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$)

$$[A \mid B] \sim \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & -3 & -1 & 1 \\ 0 & 0 & -17 & -34 \end{bmatrix} \text{(Applying } R_3 \to 3R_3 - 4R_2\text{)}$$

Since Rank of A = 3 and Rank of $[A \mid B] = 3$ $\therefore$ Rank of A = Rank of $[A \mid B]$

The given system is consistent. So it has a solution.

Since Rank of A = Rank of $[A/B]$ = number of unknowns,

$\therefore$ The given system has a unique solution.

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -3 & -1 \\ 0 & 0 & -17 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ -34 \end{bmatrix}$$

$\Rightarrow$ $x + y + 2z = 4$ .... (1) $-3y - z = 1$ .... (2) $-17z = -34$ or $z = 2$ .... (3)

Substituting $z = 2$ in (2) $\Rightarrow -3y - 2 = 1 \Rightarrow -3y = 3 \Rightarrow y = -1$

Substituting $y = -1, z = 2$ in (1), we get

$x - 1 + 4 = 4 \Rightarrow x = 1$

$\therefore$ $x = 1, y = -1, z = 2$ is the solution.

Ex. 4. *Find the value of* λ *for which the system of equations* $3x - y + 4z = 3$, $x + 2y - 3z = -2$, $6x + 5y + \lambda z = -3$ *will have infinite number of solutions and solve them with that* λ *value.*

Sol. The given system of equations can be written in the matrix form as AX = B

$$(i.e.) \quad \begin{bmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ -3 \end{bmatrix}$$

The Augmented matrix $[A, B] = \begin{bmatrix} 3 & -1 & 4 & 3 \\ 1 & 2 & -3 & -2 \\ 6 & 5 & \lambda & -3 \end{bmatrix}$

Applying $R_2 \to 3R_2 - R_1, R_3 \to R_3 - 2R_1$, $[A, B] \sim \begin{bmatrix} 3 & -1 & 4 & 3 \\ 0 & 7 & -13 & -9 \\ 0 & 7 & \lambda - 8 & -9 \end{bmatrix}$

Applying $R_3 \to R_3 - R_2$, $[A, B] \sim \begin{bmatrix} 3 & -1 & 4 & 3 \\ 0 & 7 & -13 & -9 \\ 0 & 0 & \lambda + 5 & 0 \end{bmatrix}$

If $\lambda = -5$, Rank of A = 2 and Rank of $[A, B] = 2$

Number of unknowns = 3

$\therefore$ Rank of A = Rank of $[A, B]$ $\neq$ number of unknowns

Hence when $\lambda = -5$, the given system is consistent and it has an infinite number of solutions.

If $\lambda = -5$ the given system becomes $\begin{bmatrix} 3 & -1 & 4 \\ 0 & 7 & -13 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -9 \\ 0 \end{bmatrix}$

$3x - y + 4z = 3$ ..... (1) and $7y - 13z = -9$ ..... (2)

From (2), Let $z = k$ · Then

$7y - 13k = -9 \Rightarrow 7y = 13k - 9 \Rightarrow y = (13k - 9)/7$

Substituting the value of y in (1), we get

$$3x - \frac{1}{7}(13k - 9) + 4k = 3 \Rightarrow 3x = \frac{13}{7}k - 4k + 3 - \frac{9}{7}$$

$$3x = -\frac{15}{7}k + \frac{12}{7} \Rightarrow x = \frac{1}{7}(-5k + 4)$$

$\therefore\ x = \frac{1}{7}(-5k + 4), y = \frac{1}{7}(13k - 9), z = k$ is the solution.

Ex. 5. Find the values of p and q so that the equations $2x + 3y + 5z = 9$, $7x + 3y + 2z = 8$, $2x + 3y + pz = q$ have

(*i*) No solution (*ii*) Unique solution

(*iii*) An infinite number of solutions.

Sol. Given equations are $2x + 3y + 5z = 9, 7x + 3y + 2z = 8,\ 2x + 3y + pz = q$

The equations can be written in the matrix form $AX = B$

$$i.e., \begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & 2 \\ 2 & 3 & p \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 8 \\ q \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 3 & 5 \\ 0 & -15 & -31 \\ 0 & 0 & p-5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ -47 \\ q-9 \end{bmatrix} \quad (\text{Applying } 2R_2 - 7R_1, R_3 - R_1)$$

Applying same operations,

$$[A,\ B] \sim \begin{bmatrix} 2 & 3 & 5 & 9 \\ 0 & -15 & -31 & -47 \\ 0 & 0 & p-5 & q-9 \end{bmatrix}$$

We have $\det A = \begin{vmatrix} 2 & 3 & 5 \\ 7 & 3 & 2 \\ 2 & 3 & p \end{vmatrix}$

$$= 2(3p-6)-3(7p-4)+5(21-6)$$
$$= 6p-12-21p+12+75 \quad = -15p+75$$

$\det A = 0 \Rightarrow p = 5$

Case I : When $p = 5, q \neq 9$

The rank (A) = 2 and rank [A, B] = 3

The system will be inconsistent. The system will not have any solution.

Case II : When $p \neq 5$, $\det A \neq 0$

The system will have unique solution.

Case III : $p = 5, q = 9$

rank A = 2 and rank [A, B] = 2

Number of variables = 3

The system will be consistent and will have infinite number of solutions.

5.64. SYSTEM OF HOMOGENEOUS LINEAR EQUATIONS

Consider a system of m homogeneous linear equations in n unknowns, namely

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots\ldots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \ldots\ldots + a_{2n}x_n &= 0 \\ a_{31}x_1 + a_{32}x_2 + \ldots\ldots + a_{3n}x_n &= 0 \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots & \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots & \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots & \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= 0 \end{aligned}\right\} \quad \ldots (1)$$

We write

$$A = \begin{bmatrix} a_{11} & a_{12}\ldots a_{1n} \\ a_{21} & a_{22}\ldots a_{2n} \\ \vdots & \\ a_{m1} & a_{m2}\ldots a_{mn} \end{bmatrix}_{m\times n}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}, \quad O = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{m\times 1} \quad \ldots (2)$$

Then (1) can be written as $AX = O$ which is the matrix equation.

Here A is called the *coefficient matrix.* It is clear that

$x_1 = 0 = x_2 = x_3 = \ldots = x_n$ is a solution of (1) *i.e.,* $X = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$ is a solution of (*2*).

This is called **trivial solution** of $AX = O$.

This $AX = O$ is always **consistent** (*i.e.*) it has a solution.

The ***trivial solution*** is called the **zero solution**.

A zero solution is not linearly independent (*i.e.),* it is linearly dependent.

Theorem : ***The number of linearly independent solutions of the linear system $AX = O$ is n–r, r being the rank of the matrix $A_{m\times n}$ and n being the number of variables.***

Note :

1. If A is a non-singular matrix. (*i.e.*, det $A \neq 0$) then the linear system $AX = O$ has only the zero solution.
2. The system $AX = O$ possesses a non-zero solution if and only if A is a non-singular matrix.

Working Rule for Finding the Solutions of the Equation $AX = O$:

Let rank of $A = r$ and rank of $[A/B] = r_1$.

Since all b's are zero, $r = r_1$, then

I.(*i*) If $r = n$(number of variables) $\Rightarrow$ the system of equations have only trivial solution (*i.e.*, zero solution).

(*ii*) If $r < n \Rightarrow$ the system of equations have an infinite number of non-trivial solutions, we shall have $n - r$ linearly independent solutions.

To obtain infinite solutions, set $(n - r)$ variables any arbitrary value and solve for the remaining unknowns.

If the number of equations is less than the number of unknowns, it has a non-trivial solution. The number of solutions of the equation $AX = O$ will be infinite.

II. If the number of equations is less than number of variables, the solution is always other than a trivial solution.

III. If the number of equations = number of variables, the necessary and sufficient condition for solutions other than a trivial solution is that the determinant of the coefficient matrix is zero.

SOLVED PROBLEMS

Ex. 1. *Solve completely the system of equations*

$x + y - 2z + 3w = 0$; $x - 2y + z - w = 0$; $4x + y - 5z + 8w = 0$; $5x - 7y + 2z - w = 0$

Sol. The given system of equations in matrix form is

$$AX = \begin{bmatrix} 1 & 1 & -2 & 3 \\ 1 & -2 & 1 & -1 \\ 4 & 1 & -5 & 8 \\ 5 & -7 & 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = O$$

Applying $R_2 - R_1$, $R_3 - 4R_1$ and $R_4 - 5R_1$, we get $A \sim \begin{bmatrix} 1 & 1 & -2 & 3 \\ 0 & -3 & 3 & -4 \\ 0 & -3 & 3 & -4 \\ 0 & -12 & 12 & -16 \end{bmatrix}$

Applying $R_4 \to \frac{R_4}{4}$, we get $A \sim \begin{bmatrix} 1 & 1 & -2 & 3 \\ 0 & -3 & 3 & -4 \\ 0 & -3 & 3 & -4 \\ 0 & -3 & 3 & -4 \end{bmatrix}$

Applying $R_3 - R_2$ and $R_4 - R_2$, we get $A \sim \begin{bmatrix} 1 & 1 & -2 & 3 \\ 0 & -3 & 3 & -4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

This is in the Echelon form. We have

rank A = the number of non-zero rows in this Echelon form = 2

Since rank $A(=2)$ is less than the number of unknowns (= 4), therefore, the given system has infinite number of non-trivial solutions.

$\therefore$ Number of independent solutions = 4 – 2 = 2.

Now, we shall assign arbitrary values to 2 variables and the remaining 2 variables shall be found in terms of these. The given system of equations is equivalent to

$$\begin{bmatrix} 1 & 1 & -2 & 3 \\ 0 & -3 & 3 & -4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives the equations $x + y - 2z + 3w = 0, -3y + 3z - 4w = 0$

Taking $z = \lambda$ and $w = \mu$, we see that $x = \lambda - \frac{5}{3}\mu, y = \lambda - \frac{4}{3}\mu, z = \lambda, w = \mu$ constitutes the general solution of the given system.

Ex. 2. *Show that the only real number λ for which the system*

$x + 2y + 3z = \lambda x;\ 3x + y + 2z = \lambda y\ ;\ 2x + 3y + z = \lambda z$

has non-zero solution is 6 and solve them, when $\lambda = 6$.

Sol. Given system can be expressed as $AX = O$ where

$$A = \begin{bmatrix} 1-\lambda & 2 & 3 \\ 3 & 1-\lambda & 2 \\ 2 & 3 & 1-\lambda \end{bmatrix},\ X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } O = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Here number of variables = $n = 3$.

The given system of equations possess a non-zero (non-trivial) solution, if

Rank of A < number of unknowns *i.e.*, Rank of $A < 3$

For this we must have det. $A = 0$

$$\therefore \begin{vmatrix} 1-\lambda & 2 & 3 \\ 3 & 1-\lambda & 2 \\ 2 & 3 & 1-\lambda \end{vmatrix} = 0$$

Applying $R_1 \to R_1 + R_2 + R_3$, we have $\begin{vmatrix} 6-\lambda & 6-\lambda & 6-\lambda \\ 3 & 1-\lambda & 2 \\ 2 & 3 & 1-\lambda \end{vmatrix} = 0$

$$\Rightarrow (6-\lambda) \begin{vmatrix} 1 & 1 & 1 \\ 3 & 1-\lambda & 2 \\ 2 & 3 & 1-\lambda \end{vmatrix} = 0$$

Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$, we get $(6-\lambda)\begin{vmatrix} 1 & 0 & 0 \\ 3 & -2-\lambda & -1 \\ 2 & 1 & -1-\lambda \end{vmatrix} = 0$

$\Rightarrow (6-\lambda)\left[(-2-\lambda)(-1-\lambda)+1\right]=0$

$\Rightarrow (6-\lambda)(\lambda^2+3\lambda+3)=0 \Rightarrow \lambda = 6$ is the only real value and other values are complex.

When $\lambda = 6$, the given system becomes

$$\begin{bmatrix} -5 & 2 & 3 \\ 3 & -5 & 2 \\ 2 & 3 & -5 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_2 \to 5R_2 + 3R_1, R_3 \to 5R_3 + 2R_1 \Rightarrow \begin{bmatrix} -5 & 2 & 3 \\ 0 & -19 & 19 \\ 0 & 19 & -19 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3 \to R_3 + R_2 \Rightarrow \begin{bmatrix} -5 & 2 & 3 \\ 0 & -19 & 19 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\Rightarrow -5x+2y+3z=0$ and $-19y+19z=0 \Rightarrow y=z$

Since Rank of A < number of unknowns (Rank of A = 2, number of unknowns = 3), therefore, the given system has infinite number of non-trivial solutions.

Let $z=k \Rightarrow y=k$ and $-5x+2k+3k=0 \Rightarrow x=k$

$\therefore\ x=k, y=k, z=k$ is the solution.

Ex. 3. *Solve the system* $2x-y+3z=0, 3x+2y+z=0$ *and* $x-4y+5z=0$.

Sol. The Given system can be written as AX = O

where $A = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & -4 & 5 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $O = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Consider $A \sim \begin{bmatrix} 1 & -4 & 5 \\ 3 & 2 & 1 \\ 2 & -1 & 3 \end{bmatrix}$ (Applying $R_3 \leftrightarrow R_1$)

$\sim \begin{bmatrix} 1 & -4 & 5 \\ 0 & 14 & -14 \\ 0 & 7 & -7 \end{bmatrix}$ (Applying $R_2 - 3R_1$ and $R_3 - 2R_1$)

$\sim \begin{bmatrix} 1 & -4 & 5 \\ 0 & 14 & -14 \\ 0 & 0 & 0 \end{bmatrix}$ (Applying $2R_3 - R_2$)

$$\sim \begin{bmatrix} 1 & -4 & 5 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \text{ (Applying } \frac{R_2}{14}\text{)}$$

Since the number of non-zero rows is 2, we have rank (A) = 2

Here number of variables = 3

The system will have $n-r=3-2=1$ non-zero solution.

From the matrix, we have $y-z=0 \Rightarrow y=z$

Let $y=z=k$. Then

$x-4y+5z=0 \Rightarrow x=4y-5z=4k-5k=-k$

$\therefore$ The solution is given by $X=\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} -k \\ k \\ k \end{bmatrix}=k\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

Ex. 4. Find all the solutions of the system of equations :

$x+2y-z=0, 2x+y+z=0, x-4y+5z=0$

Sol. Let $A=\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 1 \\ 1 & -4 & 5 \end{bmatrix}$. Then

$\det A=1(5+4)-2(10-1)-1(-8-1)=9-18+9=0$

$\therefore$ The rank of A is <3.

Applying $R_2 \to R_2-2R_1$ and $R_3 \to R_3-R_1$, we get

$$A \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 3 \\ 0 & -6 & 6 \end{bmatrix}$$

Applying $R_3 \to R_3-2R_2$, we get

$$A \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus rank (A) = 2. $\therefore$ Number of non-zero solutions = $n-r=3-2=1$.

From the above matrix, $-3y+3z=0 \Rightarrow y=z$

Let $z=k$. Then $y=k$.

$x+2y-z=0 \Rightarrow x=z-2y=k-2k=-k$

$\therefore$ The solutions are given by $X=k\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$.

6

Characteristic values and Characteristic vectors, Cayley-Hamilton Theorem

6.1. INTRODUCTION

In this chapter we introduce important notions of Characteristic values and Characteristic vectors of a linear operator on a finite dimensional vector space and a square matrix defined over a field. Using these concepts we will discuss the diagonalization and diagonalizability of linear operators and matrices.

CHARACTERISTIC VECTOR, CHARACTERISTIC VALUE OF A LINEAR OPERATOR

6.2. Definition. *Let T be a linear operator on a finite dimensional vector space V(F). A non zero vector* $\alpha \in V$ *is called a characteristic vector of T if there exists a scalar c such that* $T(\alpha) = c\alpha$. *Each such non-zero vector is called a characteristic vector of T corresponding to a characteristic value c. For* $T(\alpha)$ *we may write sometimes* $T\alpha$. *The scalar c is called characteristic value of T corresponding to the characteristic vector* α.

Also if c is a characteristic value of T, then any non-zero vector α in V such that $T(\alpha) = c\alpha$, is called a *characteristic vector of T* belonging the characteristic value c.

Characteristic values are sometimes also called proper values, **eigen values** or **spectral values**. Similarly characteristic vectors are called *proper vectors*, *eigen vectors* or *spectral vectors*. The set of all characteristic values of T is called the *spectrum of T*.

6.3. CHARACTERISTIC VECTOR, CHARACTERISTIC VALUE OF A MATRIX

Definition. *Any non-zero vector X is said to be a characteristic vector of a square matrix A if there exists a scalar* λ *such that* $AX = \lambda X$.

Here A can be a $n \times n$ matrix and X can be a $n \times 1$ matrix.

Then λ is said to be a characteristic value of the matrix A corresponding to the characteristic vector X. Also X is said to be characteristic vector corresponding to the characteristic value λ of the matrix A.

Characteristic vectors are sometimes called *Proper* or *Latent* or *Eigen* vectors.

Note. 1. (*a*) If X is a characteristic vector of a matrix A, X cannot correspond to more than one characteristic value of A. ***(K. U. M2014)***

For : Let the characteristic vector X of A correspond to two distinct characteristic values λ_1 and λ_2. Then $AX = \lambda_1 X$ and $AX = \lambda_2 X$.

$$\therefore \lambda_1 X = \lambda_2 X \Rightarrow (\lambda_1 - \lambda_2)X = \bar{0} \quad \Rightarrow \lambda_1 - \lambda_2 = 0 \quad (\because X \neq \bar{0}) \quad \Rightarrow \lambda_1 = \lambda_2 .$$

(*b*) *If* α *is a characteristic vector of T, then* α *cannot correspond to more than one characteristic value of T.*

For : Let α be a characteristic vector of T corresponding to two distinct characteristic values c_1 and c_2 of T. Then

$$T(\alpha) = c_1\alpha \text{ and } T(\alpha) = c_2\alpha .$$

$$\therefore\ c_1\alpha = c_2\alpha \Rightarrow (c_1 - c_2)\alpha = \mathbf{0} \Rightarrow c_1 - c_2 = 0\ (\because \alpha \neq \mathbf{0}) \Rightarrow c_1 = c_2 .$$

2. (*a*) If X is a characteristic vector of a matrix A corresponding to the characteristic value λ, then kX is also a characteristic vector of A corresponding to the same characteristic value λ. Here k is a non - zero scalar.

For : X is a characteristic vector of the matrix A corresponding to the characteristic value λ. Then $X \neq \bar{0}$ and $AX = \lambda X$.

(*b*) *If* α *is a characteristic vector of T corresponding to the characteristic value c, then* $k\alpha$ *is also a characteristic vector of T corresponding to the characteristic value c. Here k is any non-zero scalar.*

For : α is a characteristic vector of T corresponding to the characteristic value $c \Rightarrow \alpha \neq 0$ and $T(\alpha) = c\,\alpha$. Now k is a non-zero scalar $\Rightarrow k\alpha \neq 0$

Also $T(k\alpha) = kT(\alpha) = k(c\,\alpha) = (kc)\alpha = (c\,k)\alpha = c(k\,\alpha)$

$\therefore\ k\alpha$ is the characteristic value of T corresponding to characteristic value c.

If k is any non-zero scalar, then $kX \neq \bar{O}$. Also $A(kX) = k(AX) = k(\lambda k) = \lambda(kX)$

Since kX is a non-zero vector, kX is a characteristic vector of A, corresponding to the characteristic value λ of A.

Thus, from Note. 1(*a*) and Note.2(*a*), we have :

A characteristic vector of a matrix A corresponds to a unique characteristic value and a characteristic value of a matrix A does not correspond to unique characteristic vector.

3. If A, B are two square matrices such that $AB = \lambda B$, then every non-zero column of B is a characteristic vector of A corresponding to the characteristic value λ.

6.4. CHARACTERISTIC POLYNOMIAL, CHARACTERISTIC EQUATION OF A SQUARE MATRIX

Definition. *Let* $A = [a_{ij}]_{n\times n}$ *and* λ *any indeterminate scalar. The matrix* $A - \lambda I$ *is called the characteristic matrix of A, where I is the unit matrix of order n.*

$$\text{Also } |A - \lambda I| = \begin{vmatrix} a_{11}-\lambda & a_{12} & ..... & a_{1n} \\ a_{21} & a_{22}-\lambda & .... & .... \\ .... & .... & .... & .... \\ a_{n1} & a_{n2} & ... & a_{nn}-\lambda \end{vmatrix},$$

a polynomial in λ of degree n, is called the *characteristic polynomial of A.*

e.g. The characteristic polynomial of $A = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 2 & 6 \\ 3 & 1 & 4 \end{bmatrix}$ is $\det(A - xI) = 0$

i.e., $\begin{vmatrix} 1-x & 0 & 5 \\ 0 & 2-x & 6 \\ 3 & 1 & 4-x \end{vmatrix} = -x^3 + 7x^2 + 7x - 28$ ***(A. U. 2012)***

It is denoted by $f(\lambda)$. The equation $|A - \lambda I| = 0$ is called the *characteristic equation of A.*

The coefficients of λ in this equation will be $(-1)^n$. It can be $f(\lambda) = 0$ where λ is a root of the characterstic equation of A.

6.5. Theorem 1. ***A scalar λ is a characteristic root of a square matrix A if and only if $|A - \lambda I| = 0$.***

If λ is a characteristic root and X, a corresponding characteristic vector of a matrix A, then we have $AX = \lambda X = \lambda IX \Rightarrow (A - \lambda I)X = 0$

Since $X \neq \bar{0}, (A - \lambda I)$ is singular *i.e.*, $|A - \lambda I| = 0$

Thus every characteristic root λ of a matrix A is a root of its characteristic equation $|A - \lambda I| = 0$.

Conversely, if λ be any root of $|A - \lambda I| = 0$, then the matrix equation $(A - \lambda I)X = 0$ necessarily possesses a non-zero solution X so that there exists a vector $X \neq \bar{0}$ such that

$AX = \lambda IX = \lambda X$.

Thus every root of the characteristic equation of a matrix is a characteristic root of the matrix.

Note. If A be *n*-rowed, then the characteristic equation $|A - \lambda I| = 0$ is of *n*th degree polynomial in λ equated to 0 so that every *n* - rowed square matrix possesses *n* characteristic roots and these *n* roots may not all be distinct.

6.6. Theorem. 2. ***(i) The characteristic vectors corresponding to distinct characteristic roots of a matrix are linearly independent (L. I.)***

(O. U. 2011, N. U. 94, M, S. V. U. 2001/S)

Proof. Let A be a square matrix. Let $X_1, X_2, \ldots, X_m$ be characteristic vectors of A corresponding to respective distinct characteristic roots $\lambda_1, \lambda_2, \ldots, \lambda_m$.

Then $AX_i = \lambda_i X_i$ for $i = 1, 2, \ldots, m$ .... (1)

Now we prove that the set of vectors $\{X_1, X_2, \ldots, X_m\}$ is L. I.

Since $x_1 \neq \bar{0}$, the set $\{x_1\}$ is L. I.

If $\{X_1, X_2, ..., X_m\}$ is linearly dependent (L. D.), then we can choose $r\,(r < m)$ such that $\{X_1, X_2, ..., X_r\}$ is L.I. and $\{X_1, X_2, ..., X_r, X_{r+1}\}$ is L. D. Hence we can choose scalars $k_1, k_2, ..., k_{r+1}$ not all zeros such that

$$k_1X_1 + ... + k_rX_r + k_{r+1}X_{r+1} = \bar{0} \qquad .... (2)$$

$$\Rightarrow A(k_1X_1 + ... + k_rX_r + k_{r+1}X_{r+1}) = A\bar{0}$$

$$\Rightarrow k_1(AX_1) + ... + k_r(AX_r) + k_{r+1}(AX_{r+1}) = \bar{0}$$

$$\Rightarrow k_1(\lambda_1X_1) + ... + k_r(\lambda_rX_r) + k_{r+1}(\lambda_{r+1}X_{r+1}) = \bar{0} \qquad .... (3). \qquad \text{using (1)}$$

$$\therefore (3) - \lambda_{r+1}(2) : k_1(\lambda_1 - \lambda_{r+1})X_1 + ... + k_r(\lambda_r - \lambda_{r+1})X_r = \bar{0} \qquad .... (4)$$

Since $\{X_1, ..., X_r\}$ is L.I. and $\lambda_1, ..., \lambda_{r+1}$ are distinct, we have $k_1 = 0, ..., k_r = 0$

Putting $k_1 = 0, ..., k_r = 0$ in (2), we obtain $k_{r+1}X_{r+1} = \bar{0}$

But $X_{r+1} \neq \bar{0}$. So, $k_{r+1} = 0$.

Thus (2) implies that $k_1 = 0, ..., k_r = 0, k_{r+1} = 0$

But this contradicts our assumption that the scalars $k_1, ..., k_r, k_{r+1}$ are all not zeros.

Hence our assumption that $\{X_1, X_2, ..., X_m\}$ is L. D. is wrong.

$\therefore \{X_1, X_2, ..., X_m\}$, which correspond to distinct characteristic roots of A is L. I.

Theorem.(*ii*). ***Distinct characteristic vectors of T corresponding to distinct characteristic values of T are L.I.*** **(K. U. M 2011, 2012, O.U. 2012)**

Proof. Let T be a linear operator defined on V(F). Let $\alpha_1, \alpha_2, ..., \alpha_m$ be the characteristic vectors of T corresponding to the respective distinct characteristic values $c_1, c_2, ..., c_m$ of T. Then $T(\alpha_i) = c_i\alpha_i$ for $i = 1,2, ..., m$ *i.e.* $1 \leq i \leq m$ ... (1)

Now we prove that the set of vectors $\{\alpha_1, \alpha_2, ..., \alpha_m\}$ is L.I.

Since $\alpha_1 \neq 0$, $\{\alpha_1\}$ is L.I.

If $\{\alpha_1, \alpha_2, ..., \alpha_m\}$ is L.D., then we can choose $r(r < m)$ such that $\{\alpha_1, \alpha_2, ..., \alpha_r\}$ is L.I. and $\{\alpha_1, \alpha_2, ..., \alpha_r, \alpha_{r+1}\}$ is L.D. Hence we can choose scalars

$a_1, a_2, ..., a_{r+1}$ not all zeros such that $a_1\alpha_1 + a_2\alpha_2 + ... + a_r\alpha_r + a_{r+1}\alpha_{r+1} = \bar{0}$ ... (2)

$$\Rightarrow T(a_1\alpha_1 + a_2\alpha_2 + ... + a_r\alpha_r + a_{r+1}\alpha_{r+1}) = T(\bar{0})$$

$$\Rightarrow a_1T(\alpha_1) + a_2T(\alpha_2) + ... + a_rT(\alpha_r) + a_{r+1}T(\alpha_{r+1}) = T(\bar{0})$$

$$\Rightarrow a_1(c_1\alpha_1) + a_2(c_2\alpha_2) + ... + a_r(c_r\alpha_r) + a_{r+1}(c_{r+1}\alpha_{r+1}) = \bar{0} \qquad ... (3)$$

$$\therefore (3) - c_{r+1}(2) : \Rightarrow a_1(c_1 - c_{r+1})\alpha_1 + ... + a_r(c_r - c_{r+1})\alpha_r = \bar{0}$$

Since $\{\alpha_1, \alpha_2, ..., \alpha_r\}$ is L.I. and $c_1, ..., c_{r+1}$ are distinct . we have $a_1 = 0, ..., a_r = 0$

Putting $a_1 = 0, \ldots a_r = 0$ in (2), we obtain $a_{r+1}\alpha_{r+1} = \bar{0}$

But $\alpha_{r+1} \neq \bar{0}$. So, $a_{r+1} = 0$

Thus (2) implies that $a_1 = 0, \ldots a_r = 0, a_{r+1} = 0$

But this contradicts our assumption that the scalars $a_1, a_2, \ldots, a_r, a_{r+1}$ are all not zeros.

Hence our assumption that $\{\alpha_1, \alpha_2, \ldots \alpha_m\}$ is L.D. is wrong.

$\therefore \{\alpha_1, \alpha_2, \ldots \alpha_m\}$, which corresponds to distinct characteristic values of T is L.I.

Note. If dim V = n and T is a linear operator on V, then T cannot have more than n distinct values.

For: Assume that T has more than n distinct values. Then the corresponding set of distinct vectors of T will be L.I. implying that V has a L.I. subset with more than n vectors. It is absurd since dim V = n. Hence T cannot have more than n distinct characteristic values.

6. 7. CHARACTERISTIC SPACE OF A SQUARE MATRIX

Let λ be a characteristic root of an n - rowed square matrix A.

Consider the matrix equation $(A - \lambda I)X = 0$ .... (1)

Every non-zero solution of (1) is a characteristic vector of the matrix A corresponding to the characteristic root λ.

Let the rank of $A - \lambda I$ be r.

Then (1) possesses $(n - r)$ L. I. solutions. Every non-zero linear combination of these solutions is also a solution of (1) and hence is a characteristic vector corresponding to λ. The set of all these linear combinations, including the zero vector is a subspace of $V_n(F)$ called the characteristic space of A, corresponding to the characteristic root λ.

Thus the characteristic space of a matrix A, corresponding to a characteristic root λ is just the column null space of the matrix $A - \lambda I$.

CHARACTERISTIC SPACE OF A LINEAR OPERATOR

Let c be a characteristic value of a linear operator T defined on a finite dimensional vector space V(F).

Consider the set $W_c = \{\alpha \in V : T(\alpha) = c\alpha\}$

If $\alpha, \beta \in W_c$, then $T(\alpha) = c\alpha$ and $T(\beta) = c\beta$.

If $a, b \in F$, then $a\alpha + b\beta \in V$ and

$$T(a\alpha + b\beta) = T(a\alpha) + T(b\beta) = aT(\alpha) + bT(\beta)$$

$$= a(c\alpha) + b(c\beta) = c(a\alpha + b\beta)$$

$\Rightarrow a\alpha + b\beta \in W_c \Rightarrow W_c$ is a subspace of V.

If zero vector of V is included in W_c, then W_c of V is called the *characteristic space* of the characteristic value c of the linear operator T.

6. 8. CHARACTERISTIC POLYNOMIAL OF A LINEAR OPERATOR

Definition. *Let T be a linear operator on an n - dimensional vector space V with ordered basis* β. *We define the characteristic polynomial* $f(\lambda)$ *of T to be the characteristic polynomial of* $A = [T]_\beta$ *i.e.* $f(\lambda) = \det(T - \lambda I) = \det(A - \lambda I)$.

The equation $\det(T - \lambda I) = 0$ is called **characteristic equation of T.**

Note. If T is a linear operator on a finite dimensional vector space V and β is an ordered basis for V, then λ is an eigen value of T if and only if λ is an eigen value of $[T]_\beta$. We denote the characteristic polynomial of an operator T by $\det(T - \lambda I)$ *i.e.* $\det(A - \lambda I)$.

At a later stage we know that the above definition is independent of the choice of ordered basis β. Here the polynomial of T is unique. If F is the field of real numbers, then T may or may not have a characteristic value.

Def. Algebraic multiplicity . If A is $n \times n$ matrix and λ_1 is a charactristic root of order t of the charactristic equation of A, then t is called the algebriac multiplicy of λ_1.

Def. Geomatric multiplicity. If s is the number of linearly independent characteristic vectors corresponding to the charactristic vector λ_1, then s is called the geometric multiplicity of λ_1.

SOLVED PROBLEMS

Ex. 1. *Show that the characteristic roots of any diagonal matrix are same as its elements in the diagonal.*

Sol. Let $A = \text{diag}(d_1, d_2, \ldots\ldots, d_n) = \begin{bmatrix} d_1 & 0 & \ldots\ldots & 0 \\ 0 & d_2 & \ldots\ldots & 0 \\ \ldots & \ldots & \ldots\ldots & \ldots \\ \ldots & \ldots & \ldots\ldots & \ldots \\ 0 & 0 & \ldots\ldots & d_n \end{bmatrix}$

Its characteristic equation is $|A - \lambda I| = 0$ *i.e.* $\det \begin{bmatrix} d_1 - \lambda & 0 & \ldots\ldots & 0 \\ 0 & d_2 - \lambda & \ldots\ldots & 0 \\ \ldots & \ldots & \ldots\ldots & \ldots \\ \ldots & \ldots & \ldots\ldots & \ldots \\ 0 & 0 & \ldots\ldots & d_n - \lambda \end{bmatrix} = 0$

i.e. $(d_1 - \lambda)(d_2 - \lambda)\ldots\ldots\ldots\ldots\ldots\ldots(d_n - \lambda) = 0$

Hence $d_1, d_2, \ldots\ldots, d_n$ *i.e.* the elements in the diagonal of A are its characteristic roots.

Ex. 2. *Let V(F) be a vector space of dimension n. Find the characteristic polynomial of (i) the identity operator on V (ii) the zero operator on V.*

Sol. Let β be any ordered basis of V(F).

(i) Now $[I]_\beta = I_{n\times n}$.

The characteristic polynomial of I (identity operator)

$$= \det(I-\lambda I) = \begin{vmatrix} 1-\lambda & 0 & : & : & 0 \\ 0 & 1-\lambda & : & : & 0 \\ 0 & 0 & : & : & 0 \\ : & : & : & : & : \\ 0 & 0 & : & : & 1-\lambda \end{vmatrix} = (1-\lambda)^n .$$

(ii) If 0 is the zero operator on V, then $[0]_\beta = 0_{n\times n}$, the null matrix of order n.

The characteristic polynomial of 0 operator

$$= \det(0-\lambda I) = \begin{vmatrix} -\lambda & 0 & : & : & 0 \\ 0 & -\lambda & : & : & 0 \\ 0 & 0 & : & : & 0 \\ : & : & : & : & : \\ 0 & 0 & : & : & -\lambda \end{vmatrix} = (-\lambda)^n = (-1)^n \cdot \lambda^n .$$

Ex. 3. *Prove that the characteristic roots of a triangular matrix are just the diagonal elements of the matrix.*

Sol. Let $A = \begin{bmatrix} a_{11} & a_{12} & ..... & a_{1n} \\ 0 & a_{22} & ..... & a_{2n} \\ ... & ... & ..... & ... \\ ... & ... & ..... & ... \\ 0 & 0 & ..... & a_{nn} \end{bmatrix}$ be a triangular matrix.

Its characteristic equation is $|A-\lambda I| = 0$ *i.e.* $\begin{bmatrix} a_{11}-\lambda & a_{12} & ..... & a_{1n} \\ 0 & a_{22}-\lambda & ..... & a_{2n} \\ ... & ... & ..... & ... \\ ... & ... & ..... & ... \\ 0 & 0 & ..... & a_{nn}-\lambda \end{bmatrix} = 0$

i.e. $(a_{11}-\lambda)(a_{22}-\lambda).....(a_{nn}-\lambda) = 0$

$\Rightarrow a_{11}, a_{22}, ..., a_{nn}$ *i.e.* the diagonal elements of A are the characteristic roots of A.

Ex. 4. *Prove that the square matrices A and A′ have the same characteristic values.*

Sol. If λ is any scalar, then $(A-\lambda I)' = A' - \lambda I' = A' - \lambda I$

$\Rightarrow |(A-\lambda I)'| = |A' - \lambda I|$

$\Rightarrow |A - \lambda I| = |A' - \lambda I|$ ($\because$ det of a matrix = det of its transpose)

$\Rightarrow |A - \lambda I| = 0$ if and only if $\Rightarrow |A' - \lambda I| = 0$.

i.e. λ is a characteristic value of A $\Leftrightarrow \lambda$ is a characteristic value of A'.

Ex. 5. *Show that 0 is a characteristic root of a matrix if and only if the matrix is singular.*

Sol. 0 is a characteristic value of A

$\Leftrightarrow \lambda = 0$ satisfies the equation $|A - \lambda I| = 0$

$\Leftrightarrow |A - 0I| = 0 \Leftrightarrow |A| = 0 \Rightarrow$ A is singular.

Note. 1. λ is a characteristic root of a non-singular matrix. $\lambda \neq 0$.

2. At least one characteristic root of every singular matrix is zero.

Ex. 6. *T is a linear operator on a finite dimensional vector space V(F). Show that T is not invertible iff 0 is a characteristic value of T.*

Sol. Let T be not invertible. Now T is not invertible means T is singular.

Therefore, there exists a non - zero vector α in V such that $T\alpha = 0 = 0\alpha$.

$\therefore$ 0 is a characteristic value of T.

Let 0 be a characteristic value of T.

Then there exists a non-zero vector α in V such that $T\alpha = 0\alpha$.

$\Rightarrow T\alpha = 0 \Rightarrow$ T is singular $\Rightarrow$ T is not invertible.

Ex. 7. *If λ is a characteristic root of the matrix A, and k is a scalar, show that $k + \lambda$ is a characteristic root of the matrix $A + kI$*

Sol. Let λ be a characteristic root of the matrix A and X be a corresponding characteristic vector.

Then $AX = \lambda X$ and $(A + kI)X = AX + k(IX) = \lambda X + kX = (\lambda + k)X$

Since $X \neq 0, \lambda + k$ is a characteristic root of the matrix $A + kI$ and X is a corresponding characteristic vector.

Ex. 8. *If $\lambda_1, \lambda_2, \ldots\ldots, \lambda_n$ are the characteristic values of a n - rowed square matrix A and k is a scalar , show $k\lambda_1, k\lambda_2, \ldots\ldots, k\lambda_n$ are the characteristic values of kA.*

(S. V. U. M 06)

Sol. Let $k \neq 0$. Now $|kA - \lambda kI| = |k(A - \lambda I)| = k^n |A - \lambda I|$

$\Rightarrow |kA - (k\lambda) I| = 0$ if and only if $|A - \lambda I| = 0$

i.e. $k\lambda$ is a characteristic value of kA if and only if λ is a characteristic value of A. Thus $k\lambda_1, k\lambda_2, ..., k\lambda_n$ are the characteristic values of kA if $\lambda_1, \lambda_2, ..., \lambda_n$ are the characteristic values of A.

Ex. 9. *The product of the Eigen values of a square matrix A is its determinant.*

Sol. A is a square matrix of order n. Let $\lambda_1, \lambda_2, ..., \lambda_n$ be eigen values of A.

Now $|A - \lambda I| = (-1)^n (\lambda - \lambda_1).....(\lambda - \lambda_n)$. Put $\lambda = 0$.

$\therefore \quad |A| = (-1)^n (-1)^n \lambda_1 . \lambda_2 .....\lambda_n = \lambda_1 . \lambda_2 .....\lambda_n$

$\Rightarrow$ Product of the eigen values = det of the matrix.

Note. A is a singular matrix $\Leftrightarrow$ A has a zero eigen value.

Also Matrix A is non-singular $\Leftrightarrow$ Matrix A has non-zero eigen values.

Ex. 10. *If* $\lambda_1, \lambda_2, ..., \lambda_n$ *are the characteristic roots of a n -rowed square matrix A and k is a scalar, show that the characteristic roots of* $A - kI$ *are*

$\lambda_1 - k, \lambda_2 - k, ..., \lambda_n - k$.

Sol. $\lambda_1, \lambda_2, ..., \lambda_n$ are the characteristic roots of the n - rowed square matrix A.

$\therefore \quad \lambda_1, \lambda_2, ..., \lambda_n$ are the n roots of $|A - \lambda I| = 0$, which is a nth degree equation in λ.

$\therefore (\lambda_1 - \lambda)(\lambda_2 - \lambda).....(\lambda_n - \lambda) = 0 \qquad .... (1)$

The characteristic equation of $A - kI$ is $|(A - kI) - \lambda I| = 0$ *i.e.* $|A - (k + \lambda)I| = 0$

$\therefore$ From (1), $(\lambda_1 - k + \lambda)(\lambda_2 - k + \lambda).....(\lambda_n - k + \lambda) = 0$

$\Rightarrow (\overline{\lambda_1 - k} - \lambda)(\overline{\lambda_2 - k} - \lambda).....(\overline{\lambda_n - k} - \lambda) = 0$

$\Rightarrow \lambda_1 - k, \lambda_2 - k, ..., \lambda_n - k$ are the characteristic roots of the matrix $A - k$ I.

Ex. 11. *If* λ *is an eigen value of the matrix A, show that* $k + \lambda$ *is an eigen value of* $A + kI$

Sol. Let λ be an eigen value of A and X the corresponding eigen vector.

Then, by definition $AX = \lambda X$

Now $(A + kI)X = AX + kIX = \lambda X + kX = (\lambda + k)X$

From this we can see that the scalar $\lambda + k$ is an eigen value of the matrix $A + k$ I.

Ex. 12. *If the characteristic roots of a n - rowed square matrix A are* $\lambda_1, \lambda_2, ..., \lambda_n$ *then prove that the characteristic roots of* A^2 *are* $\lambda_1^2, \lambda_2^2, ..., \lambda_n^2$. **(*S. V. U. M 12, M06*)**

Sol. Let λ be a characteristic root of A and X be a corresponding characteristic vector of A.

Then $AX = \lambda X \Rightarrow A(AX) = A(\lambda X)$

$\Rightarrow A^2X = \lambda(AX) \Rightarrow A^2X = \lambda(\lambda X) \quad (\because AX = \lambda X)$

$\Rightarrow A^2X = \lambda^2 X$

$\therefore$ λ^2 is a characteristic root of the matrix A^2 corresponding to the characteristic vector X of A^2.

$\therefore$ If $\lambda_1, \lambda_2, ..., \lambda_n$ are the characteristic roots of A, then $\lambda_1^2, \lambda_2^2, ..., \lambda_n^2$ are the characteristic roots of A^2.

Ex. 13. *If the matrix A is non-singular, show that the eigen values of A^{-1} are the reciprocals of the eigen values of A.*

Sol. A is non-singular $\Rightarrow A^{-1}$ exists.

Let λ be an eigen value of A and X be a corresponding eigen vector of A.

Then $AX = \lambda X \Rightarrow A^{-1}(AX) = A^{-1}(\lambda X) \Rightarrow X = \lambda(A^{-1}X)$

$\Rightarrow A^{-1}X = \frac{1}{\lambda}X \quad (\lambda \neq 0,$ since A is non-singular)

$\Rightarrow \frac{1}{\lambda}$ is an eigen value of A^{-1} and X is a corresponding eigen vector.

Converse : Let k be an eigen value of A^{-1}

A is non-singular $\Rightarrow A^{-1}$ is non- singular and $(A^{-1})^{-1} = A$.

$\therefore$ $\frac{1}{k}$ is an eigen value of A.

Thus each eigen value of A^{-1} is the reciprocal of some eigen value of A.

Hence the eigen values of A^{-1} are nothing but the reciprocals of the eigen values of A.

Ex. 14. *c is a characteristic value of invertible linear operator T. Show that c^{-1} is a characteristic value of T^{-1}.*

Sol. c is a characteristic value of an invertible linear operator

$\Rightarrow c$ is a non - zero scalar and c^{-1} exists.

Now there exists a non-zero vector α in V such that $T\alpha = c\alpha$.

$\therefore$ $T^{-1}(T\alpha) = T^{-1}(c\alpha) \Rightarrow (T^{-1}T)\alpha = cT^{-1}(\alpha)$

$\Rightarrow I(\alpha) = cT^{-1}(\alpha) \Rightarrow \alpha = cT^{-1}(\alpha)$

$\Rightarrow c^{-1}\alpha = T^{-1}(\alpha) \Rightarrow T^{-1}(\alpha) = c^{-1}(\alpha), \alpha \neq 0$

$\Rightarrow c^{-1}$ is a characteristic value of T^{-1}.

Ex. 15. *Show that the two matrices* A, $C^{-1}AC$ *have the same characteristic roots.*

Sol. Let $B = C^{-1}AC$.

Then $B - \lambda I = C^{-1}AC - \lambda I = C^{-1}AC - C^{-1}\lambda IC = C^{-1}(A - \lambda I)C$

$\therefore |B - \lambda I| = |C^{-1}(A - \lambda I)C| = |C^{-1}||A - \lambda I||C| = |A - \lambda I|$ $\left(\because |C^{-1}| = \frac{1}{|C|}\right)$

$\therefore$ B and A *i.e.* $C^{-1}AC$ and A have the same characteristic polynomials.

i.e. $C^{-1}AC$ and A have the same characteristic equations.

i.e. A and $C^{-1}AC$ have the same characteristic roots.

Note : A and B are matrices $M_{n\times n}(F)$. Then we say that B is *similar* to A if there exists an invertible matrix C such that $B = C^{-1}AC$. Observe that *similar matrices have the same chatacteristic polynomial.*

Ex. 16. *Find the eigen roots and the corresponding eigen vectors of the matrix* $A = \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$. ***(O. U. M 13, S. V. U. M 07, M 08, S97, S. K. U. 2001/O)***

Sol. The characteristic equation of A is

$$|A - \lambda I| = 0 \text{ i. e. } \begin{vmatrix} 1-\lambda & 4 \\ 3 & 2-\lambda \end{vmatrix} = 0 \text{ i.e. } (\lambda + 2)(\lambda - 5) = 0$$

Hence the eigen roots of A are $-2, 5$.

Case (*i*) Let $\lambda = -2$. Eigen vectors X corresponding to the eigen root -2 are given by

$$(A - (-2)I)X = 0 \text{ i.e. } \begin{bmatrix} 1+2 & 4 \\ 3 & 2+2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} i.e. \begin{bmatrix} 3 & 4 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{By } R_{21}(-1)\begin{bmatrix} 3 & 4 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow 3x_1 + 4x_2 = 0.$$

Let $x_2 = k \ (\neq 0)$ $\therefore x_1 = -4k/3$

$\therefore$ Eigen vectors corresponding to eigen root -2 are given by $k\begin{bmatrix} -4/3 \\ 1 \end{bmatrix}$ where k is a non-zero parameter. Clearly the sub-space generated by $\begin{bmatrix} -4/3 \\ 1 \end{bmatrix}$ is a one-dimensional characteristic space of R^2.

Case (*ii*) Let $\lambda = 5$. Eigen vectors X corresponding to the eigen root 5 are given by

$$(A - 5I)X = 0 \text{ i.e. } \begin{bmatrix} -4 & 4 \\ 3 & -3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{By } R_{21}\left(\frac{3}{4}\right)\begin{bmatrix}-4 & 4\\ 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}\Rightarrow -4x_1+4x_2=0$$

Let $x_2=k\ (\neq 0)$. $\quad\therefore\ x_1=k$.

$\therefore$ Eigen vectors corresponding to eigen root 5 are given by $k\begin{bmatrix}1\\ 1\end{bmatrix}$ where k is a non-zero parameter. Clearly the subspace generated by $\begin{bmatrix}1\\ 1\end{bmatrix}$ is a one-dimensional characteristic space of R^2.

Ex. 17. *Find the characteristic roots and the corresponding characteristic vectors of the matrix* $A=\begin{bmatrix}8 & -6 & 2\\ -6 & 7 & -4\\ 2 & -4 & 3\end{bmatrix}$ ***(S.K.U. 2011, A. U. 2008)***

Sol. The characteristic equation of the matrix A is $|A-\lambda I|=0$

i.e. $$\begin{vmatrix}8-\lambda & -6 & 2\\ -6 & 7-\lambda & -4\\ 2 & -4 & 3-\lambda\end{vmatrix}=0$$

$$\Rightarrow (8-\lambda)[21-10\lambda+\lambda^2-16]+6[-18+6\lambda+8]+2[24-14+2\lambda]=0$$

$$\Rightarrow -\lambda^3+10\lambda^2-5\lambda+8\lambda^2-80\lambda+40-60+36\lambda+20+4\lambda=0$$

$$\Rightarrow -\lambda^3+18\lambda^2-45\lambda=0\Rightarrow \lambda(\lambda-3)(\lambda-15)=0 \qquad \Rightarrow \lambda=0,3,15$$

$\therefore$ The characteristic roots of A are 0, 3, 15.

Case (*i*) $\lambda=0$

Characteristic vectors corresponding to the characteristic root 0 are given by

$$(A-0I)X=0 \text{ } i.e. \begin{bmatrix}8 & -6 & 2\\ -6 & 7 & -4\\ 2 & -4 & 3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=0$$

$$\text{By } R_{13}\begin{bmatrix}2 & -4 & 3\\ -6 & 7 & -4\\ 8 & -6 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=0, \text{ By } R_{21}(3), R_{31}(-4)\begin{bmatrix}2 & -4 & 3\\ 0 & -5 & 5\\ 0 & 10 & -10\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=0$$

$$\text{By } R_{32}(2)\begin{bmatrix}2 & -4 & 3\\ 0 & -5 & 5\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=0$$

$\Rightarrow 2x_1 - 4x_2 + 3x_3 = 0, -5x_2 + 5x_3 = 0$

Let $x_3 = k$. $\therefore x_2 = k$ and $2x_1 = k$ *i.e.* $x_1 = k/2$

$\therefore$ Characteristic vectors corresponding to the characteristic root 2 are given by

$k\begin{bmatrix} 1/2 \\ 1 \\ 1 \end{bmatrix}$ where k is a non-zero parameter.

Similarly, by considering characteristic equations $(A - 3I)X = 0, (A - 15I)X = 0$, we get characteristic vectors $k_1\begin{bmatrix} -1 \\ -1/2 \\ 1 \end{bmatrix}, k_2\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$ for non-zero parameters k_1, k_2 respectively corresponding to characteristic roots 3,15.

Note. The subspaces of V_3 or R^3 spanned by these three vectors $\begin{bmatrix} 1/2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1/2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$ separately are the three characteristic spaces.

Ex. 18. *Find the characteristic roots and the corresponding characteristic vectors of the matrix* $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

Sol. The characteristic equation of the matrix A is $|A - \lambda I| = 0$

i.e. $\det\begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix} = 0$

$\Rightarrow -\lambda^3 + 12\lambda^2 - 36\lambda + 32 = 0 \Rightarrow (\lambda - 2)^2(\lambda - 8) = 0 \qquad \Rightarrow \lambda = 2, 2, 8$

$\therefore$ The characteristic roots of A are 2, 2, 8 and only two roots are distinct.

(*i*) $\lambda = 8$

Characteristic vectors corresponding to the characteristic root 8 are given by

$(A - 8I)X = 0$ *i. e.* $\begin{bmatrix} 6-8 & -2 & 2 \\ -2 & 3-8 & -1 \\ 2 & -1 & 3-8 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$ etc.

We observe that characteristic vectors corresponding to characteristic root 8 are given by $k\begin{bmatrix}2\\-1\\1\end{bmatrix}$ where k is a non-zero scalar.

(*ii*) $\lambda = 2$

Characteristic vectors corresponding to the characteristic root 2 are given by

$$(A-2\lambda)X = 0 \quad i.e. \quad \begin{bmatrix}6-2 & -2 & 2\\ -2 & 3-2 & -1\\ 2 & -1 & 3-2\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = 0$$

$$i.e. \quad \begin{bmatrix}4 & -2 & 2\\ -2 & 1 & -1\\ 2 & -1 & 1\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = 0 \Rightarrow \begin{bmatrix}2 & -1 & 1\\ -2 & 1 & -1\\ 4 & -2 & 2\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = 0 \quad (\text{by } R_{13})$$

$$\Rightarrow \begin{bmatrix}2 & -1 & 1\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = 0 \Rightarrow 2x_1 - x_2 + x_3 = 0$$

Let $x_2 = k_1, x_3 = k_2$. $\quad \therefore\ x_1 = \frac{1}{2}k_1 - \frac{1}{2}k_2$

We observe that characteristic vectors corresponding to characteristic root 2 are given by $k_1\begin{bmatrix}1/2\\1\\0\end{bmatrix} + k_2\begin{bmatrix}-1/2\\0\\1\end{bmatrix}$ where k_1, k_2 are parameters, not both being 0 simultaneously.

Note. The sub-space spanned by $\begin{bmatrix}1/2\\1\\0\end{bmatrix}, \begin{bmatrix}-1/2\\0\\1\end{bmatrix}$ is the characteristic space of the characteristic root 2.

Similarly sub-space spanned by $\begin{bmatrix}2\\-1\\1\end{bmatrix}$ is the characteristic space of the characteristic root 8.

Ex. 19. *Find the characteristic roots and the corresponding characteristic vectors of the matrix* $A = \begin{bmatrix}2 & 1 & 0\\ 0 & 2 & 1\\ 0 & 0 & 2\end{bmatrix}$

Sol. The characteristic equation of the matrix is given by $|A - \lambda I| = 0$

$$i.e. \begin{vmatrix} 2-\lambda & 1 & 0 \\ 0 & 2-\lambda & 1 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 0 \quad i.e. (2-\lambda)^3 = 0$$

$\therefore$ The characteristic roots of A are 2, 2, 2 (not all distinct).

Characteristic vectors corresponding to the characteristic root 2 are given by

$$(A-2I)X = 0 \ \ i.e. \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \Rightarrow x_2 = 0, x_3 = 0. \quad \text{Let } x_1 = k.$$

$\therefore$ Characteristic vectors corresponding to the characteristic root 2 are given by $k \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ where k is a non-zero parameter.

Ex. 20. *Find the Eigen values and the corresponding Eigen vectors of the Matrix*

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{bmatrix}$$

(S. V. U. M 15, M07, S. K. D. M 07)

Sol. If λ is Eigen value of A and X is the corresponding Eigen vector, then $(A-\lambda I)X = O$

$$\begin{bmatrix} 3-\lambda & 1 & 1 \\ 2 & 4-\lambda & 2 \\ 1 & 1 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \ldots. (1)$$

Charectristic equation of A is $|A-\lambda I| = 0$

$$\Rightarrow \begin{bmatrix} 3-\lambda & 1 & 1 \\ 2 & 4-\lambda & 2 \\ 1 & 1 & 3-\lambda \end{bmatrix} = 0$$

$$\Rightarrow (3-\lambda)[(4-\lambda)(3-\lambda)-2]-1(6-2\lambda-2)+1(2-4+\lambda) = 0$$

$$\Rightarrow (\lambda-2)(\lambda-2)(\lambda-6) = 0$$

Eigen values of A are $\lambda = 2, 2, 6$

Eigen vector of A corresponding to $\lambda = 2$

Put $\lambda = 2$ in (1)

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Performing $R_2 \to R_2 - 2R_1$

$R_3 \to R_3 - R_1$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ and hence we have, } x_1 + x_2 + x_3 = 0$$

let $x_2 = k_1, x_3 = k_2 \Rightarrow x_1 = -k_1 - k_2$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -k_1 - k_2 \\ k_1 \\ k_2 \end{bmatrix} = k_1 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - k_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Eigen vectors corresponding to $\lambda = 2$ are $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

Put $\lambda = 6$ in (1)

$$\begin{bmatrix} -3 & 1 & 1 \\ 2 & -2 & 2 \\ 1 & 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 1 & 1 \\ 2 & -2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \; R_3 \to R_1 + R_2 + R_3$$

$$\begin{bmatrix} -3 & 1 & 1 \\ 0 & -4 & 8 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \; R_2 \to 3R_2 + 2R_1$$

$\Rightarrow -3x_1 + x_2 + x_3 = 0$

$-x_2 + 2x_3 = 0$

Let $x_3 = k, x_2 = 2k \Rightarrow -3x_1 = -3k \Rightarrow x_1 = k$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} k \\ 2k \\ k \end{bmatrix} \Rightarrow k \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Eigen vector corresponding to $\lambda = 6$ is $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

Ex. 21. *Find the eigen values and corresponding eigen vectors of* $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

(K. U. M 14)

Sol. Given matrix is $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

The characteristic equation is $|A-\lambda I|=0$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 2-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{vmatrix} = 0$$

$\Rightarrow (1-\lambda)(2-\lambda)(3-\lambda)=0$

$\Rightarrow \lambda = 1, \lambda = 2, \lambda = 3$ are the eigen values.

Eigen vector corresponding to eigen value $\lambda = 1$

$(A-\lambda I)X = 0$

$\Rightarrow (A-I)X = 0$

$$\Rightarrow \begin{bmatrix} 1-1 & 0 & 0 \\ 0 & 2-1 & 0 \\ 0 & 0 & 3-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From the matrix, $x_2 = 0$ and $x_3 = 0$

We take $x_1 = k$

Then the eigen vector is $X_1 = \begin{bmatrix} k \\ 0 \\ 0 \end{bmatrix} = k \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

Then $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is the characteristic vector corresponding to $\lambda = 1$

Characteristic vector corresponding to $\lambda = 2$

$$\Rightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From this we get $x_1 = 0$ and $x_3 = 0$

The eigen vector can be takes as $X_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

Eigen vector corresponding to $\lambda = 3$

$(A - 3I)X = 0$

$$\Rightarrow \begin{bmatrix} -2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\Rightarrow x_1 = 0$ and $x_2 = 0$

We get $X_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ is the eigen vector corresponding to the value $\lambda = 3$

Ex. 22. *Let* $T : R^3 \to R^3$ *be a linear operator defined by*

$T(x_1, x_2, x_3) = (4x_1 + 2x_2 + 5x_3,\ 5x_1 + x_2 + 7x_3, 9x_3)$. *Then find*

(*i*) *The characteristic equation of T.*

(*ii*) *The characteristic roots of T and characteristic vectors corresponding to them*

(*iii*) *Algebraic and geometric multiplicity of each characteristic root.* (***A. U. M 12***)

Sol. $T : R^3 \to R^3$ is a linear operator defined by

$T(x_1, x_2, x_3) = (4x_1 + 2x_2 + 5x_3, 5x_1 + x_2 + 7x_3, 9x_3)$

Consider the standard basis, $B = \{(1,0,0),(0,1,0),(0,0,1)\}$

$T(1,0,0) = (4+0+0, 5, 0) = (4,5,0) = 4(1,0,0) + 5(0,1,0) + 0(0,0,1)$

$T(0,1,0) = (0+2+0, 0+1+0, 0) = (2,1,0) = 2(1,0,0) + 1(1,0,0) + 0(0,0,1)$

$T(0,0,1) = (0+0+5, 0+0+7, 9) = (5,7,9) = 5(1,0,0) + 7(0,1,0) + 9(0,0,1)$

$$[T]_B = \begin{bmatrix} 4 & 2 & 5 \\ 5 & 1 & 7 \\ 0 & 0 & 9 \end{bmatrix} = A \text{ (say)}$$

The characteristic equation, characteristic values and the corresponding characteristic vector of thee linear operation are relating to the matrix A.

Characteristic equation is $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 4-\lambda & 2 & 5 \\ 5 & 1-\lambda & 7 \\ 0 & 0 & 9-\lambda \end{vmatrix} = 0$$

$\Rightarrow (4-\lambda)[(1-\lambda)(9-\lambda)] - 2[5(9-\lambda)] + 5[0] = 0$

$\Rightarrow (9-\lambda)[(4-\lambda)(1-\lambda)-10]=0$

$\Rightarrow (9-\lambda)(\lambda^2-5\lambda-6)=0$

$\Rightarrow (9-\lambda)(\lambda-6)(\lambda+1)=0$

$\therefore \lambda=-1,6,9$ are the chatacteristic values

Chatacteristic vector corresponding to the chatacteristic value $\lambda=-1$

$(A-\lambda I)X=O$

$\Rightarrow (A+I)X=O$

$$\begin{bmatrix}5 & 2 & 5\\ 5 & 2 & 7\\ 0 & 0 & 10\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

From the matrix, $x_3=0$ and $5x_1+2x_2+5x_3=0 \Rightarrow 5x_1+2x_2=0$

$\Rightarrow x_1=\frac{-2}{5}x_2$. Let $x_2=k \Rightarrow x_1=\frac{-2}{5}k$

Required chatacteristic vector is $\begin{bmatrix}-\frac{2}{5}k\\ k\\ 0\end{bmatrix}=\frac{-1}{5}k\begin{bmatrix}2\\ -5\\ 0\end{bmatrix}$

$\therefore \begin{bmatrix}2\\ -5\\ 0\end{bmatrix}$ chatacteristic vector with respect to $\lambda=-1$

Chatacteristic vector relating to the eigen value $\lambda=6$

$$\begin{bmatrix}-2 & 2 & 5\\ 5 & -5 & 7\\ 0 & 0 & 3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

From the matrix $x_3=0$ and $-2x_1+2x_2=0 \Rightarrow x_1=x_2$

Let $x_1=x_2=k$.

Then the chatacteristic vector is $\begin{bmatrix}k\\ k\\ 0\end{bmatrix}=k\begin{bmatrix}1\\ 1\\ 0\end{bmatrix}$

$\therefore \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ is the chatacteristic vector relating to $\lambda = 6$

Chatacteristic vector relating to $\lambda = 9$

$$\begin{bmatrix} -5 & 2 & 5 \\ 5 & -8 & 7 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_2 + R_1$ gives $\begin{bmatrix} -5 & 2 & 5 \\ 0 & -6 & 12 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$-6x_2 + 12x_3 = 0 \Rightarrow 6x_2 = 12x_3 \Rightarrow x_2 = 2x_3$

Let $x_3 = k \Rightarrow x_2 = 2k$

$-5x_1 + 2x_2 + 5x_3 = 0$

$\Rightarrow -5x_1 + 4k + 5k = 0$

$\Rightarrow -5x_1 + 9k = 0 \Rightarrow x_1 = \frac{9}{5}k$

Characteristic vector is $\begin{bmatrix} \frac{9}{5}k \\ 2k \\ k \end{bmatrix} = \frac{k}{5}\begin{bmatrix} 9 \\ 10 \\ 5 \end{bmatrix}$

Ex. 23. *Find the characteristic polynomial of the matrix* $A = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 2 & 6 \\ 3 & 1 & 4 \end{bmatrix}$ ***(A. U. M 12)***

Sol. Given matrix is $A = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 2 & 6 \\ 3 & 1 & 4 \end{bmatrix}$

The characteristic polynomial is, $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} (1-\lambda) & 0 & 5 \\ 0 & 2-\lambda & 6 \\ 3 & 1 & (4-\lambda) \end{vmatrix} = 0$$

$\Rightarrow (1-\lambda)[(2-\lambda)(4-\lambda) - 6] + 5(3\lambda - 6) = 0$

$\Rightarrow (1-\lambda)(\lambda^2 - 6\lambda + 2) + (15\lambda - 30) = 0$

$\Rightarrow \lambda^2 - 6\lambda + 2 - \lambda^3 + 6\lambda^2 - 2\lambda + 15\lambda - 30 = 0$

$\Rightarrow -\lambda^3 + 7\lambda^2 + 7\lambda - 28 = 0$

Ex. 24. *Let T be the linear operator on $P_2(R)$ defined as $T(f(x)) = f(x) + (x+1)f'(x)$. Find the characteristic values of T.*

Sol. Given binary operator T is defined on $P_2(R)$ by $T(f(x)) = f(x) + (x+1)f'(x)$

Let $\beta = \{1, x, x^2\}$ is the standard basis for $P_2(R)$

$f(x) = 1$, $T(1) = 1 + (x+1)(0) = 1.1 + 0.x + 0.x^2$;

$f(x) = x$, $T(x) = x + (x+1)1 = 2x + 1 = 1.1 + 2.x + 0.x^2$;

$f(x) = x^2$, $T(x^2) = x^2 + (x+1)(2x) = 3x^2 + 2x = 0.1 + 2.x + 3.x^2$

Then $A = [T]_\beta = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$. The characteristic polynomial of T is $|T - \lambda I|$

i.e. $|A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 & 0 \\ 0 & 2-\lambda & 2 \\ 0 & 0 & 3-\lambda \end{vmatrix} = -(\lambda - 1)(\lambda - 2)(\lambda - 3)$

Hence the eigen values of A and hence of T are 1, 2, 3.

EXERCISE 6 (*a*)

1. Show that *a, b, c* are the eigen values of the matrix $\begin{bmatrix} a & h & g \\ 0 & b & 0 \\ 0 & c & c \end{bmatrix}$

2. Find the eigen values of the matrix $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{bmatrix}$ ***(S. V. U. S97, O. U. M15, S. K. U. M15)***

3. Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

4. Find the characteristic roots of the matrix

(*i*) $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{bmatrix}$ (*ii*) $A = \begin{bmatrix} 5 & 6 & 8 \\ 0 & 7 & 2 \\ 0 & 0 & 4 \end{bmatrix}$ ***(O. U. M 2012)***

Hence find the corresponding characteristic vectors of the matrix A.

5. Find the characteristic roots and the corresponding characteristic vectors of the matrix

(*a*) $\begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$ (*b*) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ ***(K. U. M13)*** (*c*) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ ***(K. U. M11)***

Hence find the characteristic subspaces of V_3.

6. Find eigen values and eigen vectors of the following matrices

(*i*) $A=\begin{bmatrix}1 & 2\\ 3 & 2\end{bmatrix}$ defined over R. (*ii*) $A=\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$ defined over R. ***(O. U. M14)***

(*iii*) $A=\begin{bmatrix}i & 1\\ 2 & -i\end{bmatrix}$ defined over C. (*iv*) $A=\begin{bmatrix}1 & 1\\ 0 & i\end{bmatrix}$ defined over C. ***(K. U. M12)***

7. For each linear operator T on V, find the eigen values of T.

(*i*) $V=R^2, T(a,b)=(-2a+3b,-10a+9b)$

(*ii*) $V=R^3, T(a,b,c)=(7a-4b+10c,4a-3b+8c,-2a+b-2c)$

(*iii*) $V=P_2(R), T(f(x))=x.f'(x)+f(2).x+f(3)$

ANSWERS

2. $2,-1+\sqrt{3},-1-\sqrt{3}$ **3.** 6,1 **4.** (*i*) 1,−4,7 (*ii*) 5,7,4 **5.** (*a*) 2,2,3 (*b*) 0,0,3 (*c*) 1,1,1

6. (*i*) $4,-1.\left\{\begin{pmatrix}2\\3\end{pmatrix},\begin{pmatrix}1\\-1\end{pmatrix}\right\}$ (*ii*) $0.\left\{\begin{pmatrix}1\\0\end{pmatrix}\right\}$ (*iii*) $1,-1.\left\{\begin{pmatrix}1\\1-i\end{pmatrix},\begin{pmatrix}1\\-1-i\end{pmatrix}\right\}$

(*iv*) $1,\begin{bmatrix}k_1\\0\end{bmatrix};i,\begin{bmatrix}k_2\\(i-1)k_2\end{bmatrix}$ where k_1,k_2 are non-zero complex numbers.

7. (*i*) $\lambda=3,4$ (*ii*) $-1,1,2$ (*iii*) 1,2,3

6. 9. CAYLEY - HAMILTON THEOREM

In this section we will discuss about a very important theorem namely Cayley - Hamilton theorem.

1. MATRIC POLYNOMIAL

Definition. *An expression of the form* $F(x)=A_0+A_1x+A_2x^2+....A_mx^m, A_m\neq 0$, *where* $A_0,A_1,A_2,.....,A_m$ *are matrices each of order* $n\times n$ *over a field F, is called a matric polynomial of degree m.*

The symbol x is called indeterminate and it will be assumed that it is commutative with every matrix coefficient.

The matrices themselves are matric polynomials of zero degree.

2. EQUALITY OF MATRIC POLYNOMIALS

Definition. *Two matric polynomials are equal if and only if the coefficients of like powers of x are the same.*

3. ADDITION AND MULTIPLICATION OF POLYNOMIALS

Let $G(x)=A_0+A_1x+A_2x^2+.....+A_mx^m$ and $H(x)=B_0+B_1x+B_2x^2+.....+B_kx^k$

We define : If $m > k$, then

$$G(x)+H(x)=(A_0+B_0)+(A_1+B_1)x+.....+(A_k+B_k)x^k+A_{k+1}x^{k+1}+....+A_mx^m$$

Similarly we have $G(x)+H(x)$ where $m=k$ and $m<k$.

Also $G(x)\,H(x)=A_0B_0+(A_0B_1+A_1B_0)x+(A_0B_2+A_1B_1+A_2B_0)\,x^2+....+A_kB_mx^{k+m}$

Note that the degree of the product of two matric polynomials is less than or equal to the sum of their degrees.

4. Theorem 3. ***Every square matrix, whose elements are polynomials in x, can be expressed as a matric polynomial in x of degree m, where m is the highest power of x occuring in any element of the matrix.***

We illustrate the theorem by an example.

Consider the matrix $A=\begin{bmatrix} 2x & 3x^2+4 & x-x^2+x^3 \\ x^3-5 & 0 & 4+7x^2 \\ 6 & 8x^2 & -3x^2+2 \end{bmatrix}$

We write $A=\begin{bmatrix} 0+2x+0x^2+0x^3 & 4+0x+3x^2+0x^3 & 0+1x-1x^2+1x^3 \\ -5+0x+0x^2+x^3 & 0+0x+0x^2+0x^3 & 4+0x+7x^2+0x^3 \\ 6+0x+0x^2+0x^3 & 0+0x+8x^2+0x^3 & 2+0x-3x^2+0x^3 \end{bmatrix}$

$$=\begin{bmatrix} 0 & 4 & 0 \\ -5 & 0 & 4 \\ 6 & 0 & 2 \end{bmatrix}+\begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}x+\begin{bmatrix} 0 & 3 & -1 \\ 0 & 0 & 7 \\ 0 & 8 & -3 \end{bmatrix}x^2+\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}x^3 = A_0+A_1x+A_2x^2+A_3x^3$$

5. CAYLEY - HAMILTON THEOREM (MATRICES).

Theorem 4 (*i*). ***Every square matrix satisfies its characteristic equation.***

(S. V. U. M15, A. U. M14, S. K. U. M 15, M 14, K. U. M 2013, N. U. S93)

Proof. Let A be a n - rowed square matrix *i.e.*, Let $A=[a_{ij}]_{n\times n}$.

The characteristic equation of A is given by

$$|A-\lambda I|=f(\lambda)=\begin{vmatrix} a_{11}-\lambda & a_{12} & ..... & a_{1n} \\ a_{21} & a_{22}-\lambda & .... & a_{2n} \\ . & . & . & . \\ . & . & . & . \\ . & . & . & . \\ a_{n1} & a_{n2} & .... & a_{nn}-\lambda \end{vmatrix}$$

$$=(-1)^n[\lambda^n+a_1\lambda^{n-1}+a_2\lambda^{n-2}+....+a_n] \text{ where } a_i's \in F.$$

Since all the elements of $(A-\lambda I)$ are at most of Ist degree in λ, all the elements of *adj* $(A-\lambda I)$ are polynomials in λ of degree $(n-1)$ or less and hence *adj* $(A-\lambda I)$ can be expressed as a matrix polynomial in λ

Let $adj\,(A-\lambda I) = B_0\lambda^{n-1} + B_1\lambda^{n-2} + ... + B_{n-2}\lambda^1 + B_{n-1}$ where $B_0, B_1, ..., B_{n-1}$ are n-rowed square matrices.

Now $(A-\lambda I)\,adj\,(A-\lambda I) = |A-\lambda I|\,I$ $\qquad (\because A\,(adj\,A) = |A|\,I)$

$$\Rightarrow (A-\lambda I)(B_0\lambda^{n-1} + B_1\lambda^{n-2} + ... + B_{n-2}\lambda + B_{n-1}) = (-1)^n[\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + ... + a_n]\,I$$

Comparing coefficients of like powers of λ, we obtain

$$-B_0 = (-1)^n I,$$

$$AB_0 - B_1 = (-1)^n a_1\, I,$$

$$AB_1 - B_2 = (-1)^n a_2\, I,$$

.....

.....

$$AB_{n-1} = (-1)^n a_n\, I.$$

Premultiplying the above equations successively by $A^n, A^{n-1}, ..., I$ and adding, we obtain

$$0 = (-1)^n A^n + (-1)^n a_1 A^{n-1} + (-1)^n a_2 A^{n-2} + ... + (-1)^n a_n I$$

$$\Rightarrow (-1)^n[A^n + a_1 A^{n-1} + a_2 A^{n-2} + ... + a_n I] = 0$$

$$\Rightarrow A^n + a_1 A^{n-1} + a_2 A^{n-2} + ... + a_n I = O$$

$\Rightarrow$ A satisfies its characteristic equation. (O is zero matrix)

Cor. A satisfies its characteristic equation

$$\Rightarrow (-1)^n[A^n + a_1 A^{n-1} + a_2 A^{n-2} + ... + a_n I] = O \text{ (zero matrix)}$$

$$\Rightarrow A^n + a_1 A^{n-1} + a_2 A^{n-2} + ... + a_n I = O$$

$$\Rightarrow A^{-1}[A^n + a_1 A^{n-1} + a_2 A^{n-2} + ... + a_n I] = O \quad \text{if A is non-singular.}$$

$$\Rightarrow a_n A^{-1} = -A^{n-1} - a_1 A^{n-2} - a_{n-1} I$$

$$\Rightarrow A^{-1} = \left(-\frac{1}{a_n}\right)[A^{n-1} + A^{n-2} + ... + a_{n-1} I].$$

6. NOTATION.

We consider polynomials in a particular operator T and matrices which are polynomials in a particular matrix A. We use the following notation for convenience.

Definition. *Let* $f(x) = a_0 + a_1x + ... + a_nx^n$ *be a polynomial with coefficients from field* F *and* T *be a linear operator on a vector space* $V(F)$.

We define $f(T) = a_0I + a_1T + ... + a_nT^n$.

e.g. Let T be a linear operator on R^2 defined as $T(a,b) = (2a+b, a-b)$

Let $f(x) = x^2 + 2x - 3$. Then $T^2(a,b) = T(T(a,b)) = T(2a+b, a-b)$

$= (2(2a+b)+(a-b), 2a+b-(a-b)) = (5a+b, a+2b)$

Then $f(T)(a,b) = (T^2 + 2T - 3I)(a,b) = T^2(a,b) + 2T(a,b) - 3I(a,b)$

$= (5a+b, a+2b) + (4a+2b, 2a-2b) - 3(a,b) = (6a+3b, 3a-3b)$

PROPERTIES

Let $f(x)$ be a polynomial with coefficients from field F and T be a linear operator on a vector space V(F). Then

1. $f(T)$ is a linear operator on V.

2. If a finite ordered basis for V is β and $A = [T]_\beta$, then $[f(T)]_\beta = f(A)$.

3. T is a linear operator on a vector space V(F) and A is a square matrix with elements from F. $f_1(x)$ and $f_2(x)$ are polynmomials with coefficients from F. Then

(*a*) $f_1(T)f_2(T) = f_2(T)f_1(T)$. (*b*) $f_1(A)f_2(A) = f_2(A)f_1(A)$.

CAYLEY - HAMILTON THEOREM (LINEAR OPERATOR)

Theorem 4 (*ii*). ***T is a linear operator on a vector space V(F) of dimension n. If f(x) is the characteristic polynomial of T, then f(T)=0 (zero operator). i.e., T satisfies its characteristic equation.*** ***(O.U., A. U. II)***

Proof. T is a linear operator on V(F) of dim V = *n*.

Let B be an ordered basis of V(F) and let $A = [T]_B$.

Let $A = [a_{ij}]_{n\times n}$.

The characteristic polynomial of $T = f(\lambda) = \det(T - \lambda I)$

$$= \det(A - \lambda I) = \begin{vmatrix} a_{11}-\lambda & a_{12} & : & : & a_{1n} \\ a_{21} & a_{22}-\lambda & : & : & a_{2n} \\ : & : & : & : & : \\ : & : & : & : & : \\ a_{n1} & a_{n2} & : & : & a_{nn}-\lambda \end{vmatrix}$$

$= (-1)^n[\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + ... + a_n]$ where $a_i's \in F$.

Since all the elements of $A-\lambda I$ are atmost of first degree in λ, all the elements of adj $(A-\lambda I)$ are polynomials in λ of degree $n-1$ or less and hence adj $(A-\lambda I)$ can be expressed as a matrix polynomial in λ.

Let $\text{adj}(A-\lambda I) = B_0\lambda^{n-1} + B_1\lambda^{n-2} + ... + B_{n-2}\lambda^1 + B_{n-1}$ where $B_0, B_1, ..., B_{n-1}$ are n-rowed square matrices.

Now $(A-\lambda I)\text{adj}(A-\lambda I) = |A-\lambda I| I \quad (\because A(\text{adj}A) = |A| I)$

$$\Rightarrow (A-\lambda I)(B_0\lambda^{n-1} + B_1\lambda^{n-2} + ... + B_{n-2}\lambda^1 + B_{n-1})$$

$$= (-1)^n[\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + ... + a_n] I$$

Comparing coefficients of like powers of λ, we obtain

$$-B_0 = (-1)^n I$$

$$AB_0 - B_1 = (-1)^n a_1 I$$

$$AB_1 - B_2 = (-1)^n a_2 I$$

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$$AB_{n-1} = (-1)^n a_n I$$

Premultiplying the above equations successively by $A^n, A^{n-1}, ..., I$ and adding, we obtain

$$0 = (-1)^n A^n + (-1)^n a_1 A^{n-1} + (-1)^n a_2 A^{n-2} + ... + (-1)^n a_n I$$

$$\Rightarrow (-1)^n[A^n + a_1 A^{n-1} + a_2 A^{n-2} + ... + a_n I] = 0$$

$$\Rightarrow A^n + a_1 A^{n-1} + a_2 A^{n-2} + ... + a_n I = 0 \text{ (zero matrix)}$$

($\Rightarrow$ A satisfies its characteristic equation)

$\Rightarrow f(A) = 0$.

Now $f(T) = T^n + a_1 T^{n-1} + a_2 T^{n-2} + ... + a_n I$

$$\therefore [f(T)]_B = [T^n + a_1 T^{n-1} + a_2 T^{n-2} + ... + a_n I]_B$$

$$= [T^n]_B + a_1[T^{n-1}]_B + a_2[T^{n-2}]_B + ... + a_n[I]_B$$

$$= A^n + a_1 A^{n-1} + a_2 A^{n-2} + ... + a_n I = f(A)$$

$$= O = [O]_B$$

$\Rightarrow f(T) = O$ (zero operator) $\Rightarrow T^n + a_1 T^{n-1} + a_2 T^{n-2} + ... + a_n I = O$.

Cor. If T is non-singular, then T is invertible and $\det T \neq 0$ *i.e.*, $a_n \neq 0$.

We have $T^n + a_1T^{n-1} + a_2T^{n-2} + ... + a_nI = 0$

$\Rightarrow a_nI = -(a_{n-1}T + a_{n-2}T^2 + ... + a_1T^{n-1} + T^n)$

$$\Rightarrow I = -\left(\frac{a_{n-1}}{a_n}T + \frac{a_{n-2}}{a_n}T^2 + ... + \frac{a_1}{a_n}T^{n-1} + \frac{1}{a_n}T^n\right)$$

$$\Rightarrow T^{-1} = -\left(\frac{a_{n-1}}{a_n}I + \frac{a_{n-2}}{a_n}T + ... + \frac{a_1}{a_n}T^{n-2} + \frac{1}{a_n}T^{n-1}\right)$$

SOLVED PROBLEMS

Ex. 1. *Verify Cayley - Hamilton Theorem when T is a linear operator defined by* $T(a,b) = (a+2b, -2a+b)$.

Sol. Let $\beta = \{e_1, e_2\}$.

$T(e_1) = T(1,0) = (1+2(0), -2(1)+0) = (1,-2)$ and $T(e_2) = T(0,1) = (2,1)$

Thus $A = [T]_\beta = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$.

Let A be an $n \times n$ matrix and let $f(t)$ be the characteristic polynomial of A.

Then $f(A) = O$, the $n \times n$ zero matrix.

The characteristic polynomial of T is $f(T) = \det(A - \lambda I) = \det\begin{bmatrix} 1-\lambda & 2 \\ -2 & 1-\lambda \end{bmatrix} = \lambda^2 - 2\lambda + 5$

Now $f(T) = T^2 - 2T + 5I$

Given $T(a,b) = (a+2b, -2a+b)$

$\therefore T^2(a,b) = T(a+2b, -2a+b)$

$= (a+2b+2(-2a+b), -2(a+2b) - 2a + b) = (a+2b-4a+2b, -2a-4b-2a+b)$

$= (-3a+4b, -4a-3b)$

$2T(a,b) = (2a+4b, -4a+2b), \quad 5I = 5(a,b) = (5a,5b)$

$T^2 - 2T + 5I = (-3a+4b-2a-4b+5a, -4a-3b+4a-2b+5b) = (0,0) = T_0$

Thus T satisfies its characteristic equation.

Again, $f(A) = A^2 - 2A + 5I = \begin{bmatrix} -3 & 4 \\ -4 & -3 \end{bmatrix} + \begin{bmatrix} -2 & -4 \\ 4 & -2 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$

Ex. 2. *If* $A=\begin{bmatrix}2&1&2\\5&3&3\\-1&0&-2\end{bmatrix}$ *verify Cayley - Hamilton theorem. Hence find* A^{-1}.

(S. V. U. M 2014, O. U. M 2012, S.K.U. 2011)

Sol. The characteristic equation of A is $|A-\lambda I|=0$ *i.e.* $\begin{vmatrix}2-\lambda&1&2\\5&3-\lambda&3\\-1&0&-2-\lambda\end{vmatrix}=0$

i.e. $(2-\lambda)[-6-3\lambda+2\lambda+\lambda^2]-5[-2-\lambda]-1[3-6+2\lambda]=0$

i.e. $(2-\lambda)[-6-\lambda+\lambda^2]+10+5\lambda+3-2\lambda=0$

i.e. $-12-2\lambda+2\lambda^2+6\lambda+\lambda^2-\lambda^3+13+3\lambda=0$

i.e. $-\lambda^3+3\lambda^2+7\lambda+1=0$ *i.e.* $\lambda^3-3\lambda^2-7\lambda-1=0$

To prove that A satisfies $\lambda^3-3\lambda^2-7\lambda-1=0$, the characteristic equation of A.

Now $A^2=\begin{bmatrix}2&1&2\\5&3&3\\-1&0&-2\end{bmatrix}\begin{bmatrix}2&1&2\\5&3&3\\-1&0&-2\end{bmatrix}=\begin{bmatrix}7&5&3\\22&14&13\\0&-1&2\end{bmatrix}$

$A^3=A^2A=\begin{bmatrix}7&5&3\\22&14&13\\0&-1&2\end{bmatrix}\begin{bmatrix}2&1&2\\5&3&3\\-1&0&-2\end{bmatrix}=\begin{bmatrix}36&22&23\\101&64&60\\-7&-3&-7\end{bmatrix}$

Now A^3-3A^2-7A-I

$=\begin{bmatrix}36&22&23\\101&64&60\\-7&-3&-7\end{bmatrix}+\begin{bmatrix}-21&-15&-9\\-66&-42&-39\\0&3&-6\end{bmatrix}+\begin{bmatrix}-14&-7&-14\\-35&-21&-21\\7&0&14\end{bmatrix}+\begin{bmatrix}-1&0&0\\0&-1&0\\0&0&-1\end{bmatrix}=\begin{bmatrix}0&0&0\\0&0&0\\0&0&0\end{bmatrix}$

$\therefore$ A satisfies its characteristic equation.

$\therefore A^3-3A^2-7A-I=0\Rightarrow A^{-1}(A^3-3A^2-7A-I)=0$ (Here $|A|\neq 0$)

$\Rightarrow A^2-3A-7I-A^{-1}=0\Rightarrow A^{-1}=A^2-3A-7I$

$\therefore A^{-1}=\begin{bmatrix}7&5&3\\22&14&13\\0&-1&2\end{bmatrix}+\begin{bmatrix}-6&-3&-6\\-15&-9&-9\\3&0&6\end{bmatrix}+\begin{bmatrix}-7&0&0\\0&-7&0\\0&0&-7\end{bmatrix}=\begin{bmatrix}-6&2&-3\\7&-2&4\\3&-1&1\end{bmatrix}$

Note. Check $AA^{-1}=I$.

Ex.3. *Find the inverse of the matrix* $\begin{bmatrix}1&-1&0\\0&1&1\\2&1&2\end{bmatrix}$ *by using Cayley - Hamilton theorem.*

(S.K.U. M 2015)

Sol. Let $A=\begin{bmatrix}1 & -1 & 0\\ 0 & 1 & 1\\ 2 & 1 & 2\end{bmatrix}$ $\therefore$ $A^2=\begin{bmatrix}1 & -2 & -1\\ 2 & 2 & 3\\ 6 & 1 & 5\end{bmatrix}$

The characteristic equation of A is $\begin{vmatrix}1-\lambda & -1 & 0\\ 0 & 1-\lambda & 1\\ 2 & 1 & 2-\lambda\end{vmatrix}=0$

i.e. $\begin{vmatrix}1-\lambda & -1 & 0\\ 0 & 1-\lambda & 1\\ 2 & 1 & 2-\lambda\end{vmatrix}=0$ *i.e.* $(1-\lambda)[2-\lambda-2\lambda+\lambda^2-1]+2(-1)=0$

i.e. $(1-\lambda)(1-3\lambda+\lambda^2)-2=0$ *i.e.* $\lambda^3-4\lambda^2+4\lambda+1=0$

By Cayley - Hamilton theorem, A satisfies its characteristic equation.

$\therefore$ $A^3-4A^2+4A+I=0$

$\Rightarrow A^{-1}(A^3-4A^2+4A+I)=0$ $(\because |A|=-1\neq 0)$

$\Rightarrow A^2-4A+4I+A^{-1}=0 \Rightarrow A^{-1}=-4I+4A-A^2$

$$=\begin{bmatrix}-4 & 0 & 0\\ 0 & -4 & 0\\ 0 & 0 & -4\end{bmatrix}+\begin{bmatrix}4 & -4 & 0\\ 0 & 4 & 4\\ 8 & 4 & 8\end{bmatrix}+\begin{bmatrix}-1 & 2 & 1\\ -2 & -2 & -3\\ -6 & -1 & -5\end{bmatrix}=\begin{bmatrix}-1 & -2 & 1\\ -2 & -2 & 1\\ 2 & 3 & -1\end{bmatrix}$$

Note. Check : $AA^{-1}=I$

Ex. 4. *State Cayley - Hamilton theorem and use it to find the inverse of the matrix.*

$$A=\begin{bmatrix}1 & 2 & 3\\ 2 & -1 & 4\\ 3 & 1 & -1\end{bmatrix}$$

(S. K. U. M 2013, O. U. M13, S.V. U. 11)

Sol. Cayley-Hamilton theorem : Every square matrix satisfies its characteristic equation.

Given $A=\begin{bmatrix}1 & 2 & 3\\ 2 & -1 & 4\\ 3 & 1 & -1\end{bmatrix}$

The characteristic equation of A is $|A-\lambda I|=0$ *i.e.* $\begin{vmatrix}1-\lambda & 2 & 3\\ 2 & -1-\lambda & 4\\ 3 & 1 & -1-\lambda\end{vmatrix}=0$

$\Rightarrow (1-\lambda)[(1+\lambda^2)-4]-2[-2(1+\lambda)-12]+3[2+3(1+\lambda)]=0$

$\Rightarrow \lambda^3+\lambda^2-18\lambda-40=0$

By Cayley - Hamilton theorem "Every square matrix satisfies its characteristic equation".

Then we have $A^3 + A^2 - 18A - 40I = 0$ (Here $|A| \neq 0$)

Multiplying with A^{-1} on both sides $A^2 + A - 18I = 40A^{-1} \Rightarrow A^{-1} = \frac{1}{40}[A^2 + A - 18I]$

We have $A^2 = \begin{bmatrix} 14 & 3 & 8 \\ 12 & 9 & -2 \\ 2 & 4 & 14 \end{bmatrix}$

$$\therefore A^{-1} = \frac{1}{40}\left\{\begin{bmatrix} 14 & 3 & 8 \\ 12 & 9 & -2 \\ 2 & 4 & 14 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix} - \begin{bmatrix} 18 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 18 \end{bmatrix}\right\}$$

$$\Rightarrow A^{-1} = \frac{1}{40}\begin{bmatrix} -3 & 5 & 11 \\ 14 & -10 & 2 \\ 5 & 5 & -5 \end{bmatrix}$$

Ex. 5. *Using Cayley - Hamilton theorem find the inverse and A^4 of the matrix.*

$$A = \begin{bmatrix} 7 & 2 & -2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{bmatrix}$$

Sol. Let $A = \begin{bmatrix} 7 & 2 & -2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{bmatrix}$

The characteristic equation is given by $|A - \lambda I| = 0$ *i.e.* $\begin{vmatrix} 7-\lambda & 2 & -2 \\ -6 & -1-\lambda & 2 \\ 6 & 2 & -1-\lambda \end{vmatrix} = 0$

$R_1 - R_3, R_2 + R_3$ gives $\begin{vmatrix} 1-\lambda & 0 & \lambda-1 \\ 0 & 1-\lambda & 1-\lambda \\ 6 & 2 & -(1+\lambda) \end{vmatrix} = 0$

$$\Rightarrow (1-\lambda)^2 \begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 6 & 2 & -(1+\lambda) \end{vmatrix} = 0 \Rightarrow (1-\lambda)^2[(-1-\lambda-2) - 1(-6)] = 0$$

$$\Rightarrow (1-\lambda)^2[3-\lambda] = 0 \Rightarrow \lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

By Cayley - Hamilton theorem we must have $A^3 - 5A^2 + 7A - 3I = 0$ .... (*i*)

We have $A^2 = \begin{bmatrix} 25 & 8 & -8 \\ -24 & -7 & 8 \\ 24 & 8 & -7 \end{bmatrix}$, $A^3 = \begin{bmatrix} 79 & 26 & -26 \\ -78 & -25 & 26 \\ 78 & 26 & -25 \end{bmatrix}$

Multiply (i) with A^{-1}, $A^{-1}[A^3 - 5A^2 + 7A - 3I] = 0$

$$\Rightarrow A^2 - 5A + 7I - 3A^{-1} = 0 \Rightarrow 3A^{-1} = A^2 - 5A + 7I \Rightarrow A^{-1} = \frac{1}{3}[A^2 - 5A + 7I] \quad \ldots (ii)$$

$$\text{Now } A^2 - 5A + 7I = \begin{bmatrix} 25 & 8 & -8 \\ -24 & -7 & 8 \\ 24 & 8 & 7 \end{bmatrix} - \begin{bmatrix} 35 & 10 & -10 \\ -30 & -5 & 10 \\ 30 & 10 & -5 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{3}\begin{bmatrix} -3 & -2 & 2 \\ 6 & 5 & -2 \\ -6 & -2 & 5 \end{bmatrix} \text{ by (1)}$$

$A^3 - 5A^2 + 7A - 3I = 0$

By (i), Multiplying with A, $A^4 - 5A^3 + 7A^2 - 3A = 0 \Rightarrow A^4 = 5A^3 - 7A^2 + 3A$

$$= \begin{bmatrix} 395 & 130 & -130 \\ -390 & -125 & 130 \\ 390 & 130 & -125 \end{bmatrix} - \begin{bmatrix} 175 & 56 & -56 \\ 168 & -49 & 56 \\ 168 & 56 & -69 \end{bmatrix} + \begin{bmatrix} 21 & 6 & -6 \\ -18 & -3 & 6 \\ 18 & 6 & -3 \end{bmatrix} = \begin{bmatrix} 241 & 80 & -80 \\ -240 & -79 & 80 \\ 240 & 80 & -79 \end{bmatrix}$$

Ex. 6. *Find the characteristic polynomial of the matrix* $A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 5 \end{bmatrix}$

Verify Cayley - Hamilton theorem and hence find A^{-1}. **(S. V. U. M13)**

Sol. $A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 5 \end{bmatrix}$. The characteristic polynomial is $|A - \lambda I|$.

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 5-\lambda \end{vmatrix}$$

$$R_3 \to R_3 + R_2 = \begin{vmatrix} 3-\lambda & 1 & 1 \\ -1 & 5-\lambda & -1 \\ 0 & 4-\lambda & 4-\lambda \end{vmatrix} = (4-\lambda)\begin{vmatrix} 3-\lambda & 1 & 1 \\ -1 & 5-\lambda & -1 \\ 0 & 1 & 1 \end{vmatrix}$$

$$C_3 \to C_3 - C_2 = (4-\lambda)\begin{vmatrix} 3-\lambda & 1 & 0 \\ -1 & 5-\lambda & -6+\lambda \\ 0 & 1 & 0 \end{vmatrix}$$

$= (4-\lambda)[(3-\lambda)(6-\lambda) - 1(0)]$

$= (4-\lambda)[18 - 3\lambda - 6\lambda + \lambda^2] = (4-\lambda)(\lambda^2 - 9\lambda + 18) \qquad = -\lambda^3 + 13\lambda^2 - 54\lambda + 72$

Since A satisfies the characteristic equation we must have $-A^3+13A^2-54A+72I=0$

We have $A=\begin{bmatrix}3&1&1\\-1&5&-1\\1&-1&5\end{bmatrix}$ $\therefore A^2=\begin{bmatrix}9&7&7\\-9&25&-11\\9&-9&27\end{bmatrix}$ $\therefore A^3=\begin{bmatrix}27&37&37\\-63&127&-89\\63&-63&153\end{bmatrix}$

$-A^3+13A^2-54A+72I=$

$$=-\begin{bmatrix}27&37&37\\-63&127&-89\\63&-63&153\end{bmatrix}+13\begin{bmatrix}9&7&7\\-9&25&-11\\9&-9&27\end{bmatrix}-54\begin{bmatrix}3&1&1\\-1&5&-1\\1&-1&5\end{bmatrix}+72\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}=\begin{bmatrix}0&0&0\\0&0&0\\0&0&0\end{bmatrix}$$

$\therefore -A^3+13A^2-54A+72I=0$ · $\therefore$ Cayley - Hamilton theorem is verified .

Here $|A|\neq 0$.

Multiplying the above equation with A^{-1}, we have

$\therefore -A^2+13A-54I+72A^{-1}=0 \Rightarrow 72.A^{-1}=A^2-13A+54I$

$$\Rightarrow 72.A^{-1}=\begin{bmatrix}9&7&7\\-9&25&-11\\9&-9&27\end{bmatrix}-13\begin{bmatrix}3&1&1\\-1&5&-1\\1&-1&5\end{bmatrix}+54\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

$$\Rightarrow 72.A^{-1}=\begin{bmatrix}24&-6&-6\\4&14&2\\-4&4&16\end{bmatrix} \Rightarrow A^{-1}=\frac{1}{72}\begin{bmatrix}24&-6&-6\\4&14&2\\-4&4&16\end{bmatrix}$$

Ex. 7. $A=\begin{bmatrix}1&2&1\\3&-1&2\\2&5&0\end{bmatrix}$ *find charactristic equation and hence find* A^{-1}

(A. U. M 12)

Sol. charactristic equation is given by $|A-\lambda I|=0$

$$\Rightarrow\begin{vmatrix}1-\lambda&2&1\\3&-1-\lambda&2\\2&5&-\lambda\end{vmatrix}=0$$

$$\Rightarrow(1-\lambda)[(1+\lambda)\lambda-10]-2[-3\lambda-4]+1[15+2+2\lambda]=0$$

$$\Rightarrow(1-\lambda)[\lambda+\lambda^2-10]+6\lambda+8+17+2\lambda=0$$

$$\Rightarrow\lambda+\lambda^2-10-\lambda^2-\lambda^3+10\lambda+8\lambda+25=0$$

$$\Rightarrow-\lambda^3+19\lambda+15=0 \quad \ldots(1)$$

This is the charactristic equation of A

By Cayley-Hamilton theorem, we have $-A^3+19A+15I=O$ ... (2)

Multiplying with A^{-1}, we get $-A^2+19I+15A^{-1}=O$

We have, $A^2=\begin{bmatrix}1&2&1\\3&-1&2\\2&5&0\end{bmatrix}\begin{bmatrix}1&2&1\\3&-1&2\\2&5&0\end{bmatrix}=\begin{bmatrix}9&5&5\\4&17&1\\17&-1&11\end{bmatrix}$

$$\therefore A^{-1}=\frac{1}{15}\left[\begin{bmatrix}9&5&5\\4&17&1\\15&-1&12\end{bmatrix}-\begin{bmatrix}19&0&0\\0&19&0\\0&0&19\end{bmatrix}\right]=\frac{1}{15}\begin{bmatrix}-10&5&5\\4&-2&1\\15&-1&-7\end{bmatrix}$$

Ex. 8. *Find the inverse of the matrix* $\begin{bmatrix}1&-1&0\\0&1&1\\2&1&2\end{bmatrix}$ *using Cayley-Hamilton theorem.* ***(S.V. U. M 12)***

Sol. Given matrix is $A=\begin{bmatrix}1&-1&0\\0&1&1\\2&1&2\end{bmatrix}$

Charactristic equation is $|A-\lambda I|=0$

$$\Rightarrow\begin{vmatrix}1-\lambda&-1&0\\0&1-\lambda&1\\2&1&2-\lambda\end{vmatrix}=0$$

$$\Rightarrow(1-\lambda)[(1-\lambda)(2-\lambda)-1]+1[0-2]+0=0$$

$$\Rightarrow(1-\lambda)(\lambda^2-3\lambda+2-1)-2=0$$

$$\Rightarrow(1-\lambda)(\lambda^2-3\lambda+1)-2=0$$

$$\Rightarrow\lambda^2-3\lambda+1-\lambda^3+3\lambda^2-\lambda-2=0$$

$$\Rightarrow-\lambda^3+4\lambda^2-4\lambda-1=0$$

By Cayley - Hamilton theorem "Every square matrix satisfies its charactristic equation"

$\therefore\ -A^3+4A^2-4A-I=O$, multiplying with A^{-1}

$$-A^2+4A-4I-A^{-1}=0\ \Rightarrow A^{-1}\equiv A^2+4A-4I$$

We have, $A^2=\begin{bmatrix}1&-1&0\\0&1&1\\2&1&2\end{bmatrix}\begin{bmatrix}1&-1&0\\0&1&1\\2&1&2\end{bmatrix}=\begin{bmatrix}1&-2&-1\\2&2&3\\6&1&5\end{bmatrix}$

$$A^{-1} = -\begin{bmatrix} 1 & -2 & -1 \\ 2 & 2 & 3 \\ 6 & 1 & 5 \end{bmatrix} + \begin{bmatrix} 4 & -4 & 0 \\ 0 & 4 & 4 \\ 8 & 4 & 8 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} -1 & -2 & 1 \\ -2 & -2 & 1 \\ 2 & 3 & -1 \end{bmatrix}$$

Ex. 9. *If* $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$, *express* $A^6 - 4A^5 + 8A^4 - 12A^3 + 14A^2$ *as a polynomial in A.*

Sol. The characteristic equation of A is $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 \\ -1 & 3-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - 4\lambda + 5 = 0$$

$\therefore A^2 - 4A + 5I = 0 \Rightarrow A^2 = 4A - 5I$

$\therefore A^3 = 4A^2 - 5A, A^4 = 4A^3 - 5A^2, A^5 = 4A^4 - 5A^3, A^6 = 4A^5 - 5A^4$

$\Rightarrow A^6 - 4A^5 = -5A^4 \Rightarrow A^6 - 4A^5 + 8A^4 = 3A^4 = 12A^3 - 15A^2$

$\Rightarrow A^6 - 4A^5 + 8A^4 - 12A^3 + 14A^2 = -A^2 = -4A + 5I$ a linear polinomial in A.

Alternate Method :

The characteristic equation of A is $\lambda^2 - 4\lambda + 5 = 0$

$\therefore$ By Cayley - Hamilton theorem $A^2 - 4A + 5I = 0$

We can rewrite the given expression as $A^6 - 4A^5 + 5A^4 + 3A^4 - 12A^3 + 14A^2$

$= A^4(A^2 - 4A + 5I) + 3A^2(A^2 - 4A + 5I) - A^2$

$= 0 + 0 - A^2 = 5I - 4A$ using $A^2 - 4A + 5I = 0$.

Ex. 10. *If* $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ *write* $2A^5 - 3A^4 + A^2 - 4I$ *as a linear polynomial in A.*

Sol. We have $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$

Its characteristic equation is $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 3-\lambda & 1 \\ -1 & 2-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - 5\lambda + 7 = 0$

By Cayley - Hamilton theorem, A must satisfy it characteristic equation

Therefore we must have $A^2 - 5A + 7I = 0 \Rightarrow A^2 = 5A - 7I$ .... (1)

Multiplying both sides with A, we get $A^3 = 5A^2 - 7A$ .... (4)

$A^4 = 5A^3 - 7A^2$ .... (5), $A^5 = 5A^4 - 7A^3$ .... (6)

Consider $2A^5 - 3A^4 + A^2 - 4I = 2(5A^4 - 7A^3) - 3A^4 + A^2 - 4I$ using (6).

$= 7A^4 - 14A^3 + A^2 - 4I = 7(5A^3 - 7A^2) - 14A^3 + A^2 - 4I$ using (5)

$= 21A^3 - 48A^2 - 4I = 21(5A^2 - 7A) - 48A^2 - 4I$

$= 57A^2 - 147A - 4I = 57(5A - 7I) - 147A - 4I$

$= 138A - 403\,I$ which is a linear polynomial in A.

Alternate method : We have $A^2 - 5A + 7I = 0$

$2A^5 - 3A^4 + A^2 - 4I = 2A^5 - 10A^4 + 14A^3 + 7A^4 - 14A^3 + A^2 - 4I$

$= 2A^3(A^2 - 5A + 7I) + 7A^4 - 14A^3 + A^2 - 4I$

$= 0 + 7A^4 - 35A^3 + 49A^2 + 21A^3 - 48A^2 - 4I$

$= 7A^2(A^2 - 5A + 7I) + 21A^3 - 105A^2 + 147A + 57A^2 - 147A - 4I$

$= 0 + 21A\,(A^2 - 5A + 7I) + 57A^2 - 147A - 4I$

$= 0 + 57A^2 - 147A - 4I = 57A\,(A^2 - 5A + 7I) + 138A - 403I$

$= 57(0) + 138A - 403\,I = 138A - 403\,I$ which is a linear polynomial in A.

Ex.11. *Show that the matrix* $A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$ *satisfies Cayley - Hamilton theorem.*

Sol. We have $|A - \lambda| = \begin{vmatrix} 0-\lambda & c & -b \\ -c & 0-\lambda & a \\ b & -a & 0-\lambda \end{vmatrix}$

$= -\lambda\,(\lambda^2 + a^2) - c\,(c\lambda - ab) - b\,(ac + b\lambda) = -\lambda^3 - \lambda(a^2 + b^2 + c^2)$

$\therefore$ The characteristic equation of matrix A is given as $\lambda^3 + \lambda(a^2 + b^2 + c^2) = 0$

To verify Cayley - Hamilton theorem we have to prove that $A^3 + (a^2 + b^2 + c^2)\,A = 0$

We have $A^2 = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}\begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} = \begin{bmatrix} -c^2 - b^2 & ab & ac \\ ab & -c^2 - a^2 & bc \\ ac & bc & -b^2 - a^2 \end{bmatrix}$

$\therefore\ A^3 = A^2.A = \begin{bmatrix} -c^2 - b^2 & ab & ac \\ ab & -c^2 - a^2 & bc \\ ac & bc & -b^2 - a^2 \end{bmatrix}\begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$

$= \begin{bmatrix} 0 & -c^3 - b^2c - a^2c & bc^2 + b^3 + a^2b \\ c^3 + a^2c + b^2c & 0 & -ab^2 - ac^2 - a^3 \\ -bc^2 - b^3 - a^2b & ac^2 + ab^2 + c^3 & 0 \end{bmatrix}$

$= -(a^2 + b^2 + c^2)\begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} = -(a^2 + b^2 + c^2)\,A$

$\therefore\ A^3 + (a^2 + b^2 + c^2)\,A = 0.$ $\therefore$ A satisfies Cayley - Hamilton theorem.

EXERCISE 6 (*b*)

1. Show that the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ satisfies Cayley - Hamilton theorem. ***(O.U. 2011)***

2. Show that $A = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$ satisfies the characteristic equation. Hence find A^{-1}.

3. Show that $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ satisfies its characteristic equation. Hence find A^{-1}.

4. Find the inverse of the matrix $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}$ using Cayley - Hamilton theorem.

5. If $a_1 + a_2\lambda + a_3\lambda^2 = 0\ (a_1 \neq 0)$ is the characteristic equation of a matrix A, find A^{-1}.

ANSWERS

2. $\frac{1}{11}\begin{bmatrix} -2 & 5 & -1 \\ -1 & -3 & 5 \\ 7 & -1 & -2 \end{bmatrix}$ **3.** $A^3 - 6A^2 + 9A - 4I = O$; $A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$

4. $A^{-1} = \frac{1}{6}\begin{bmatrix} 4 & -4 & 2 \\ -2 & 2 & 2 \\ 1 & 2 & -1 \end{bmatrix}$ **5.** $-\frac{a_2}{a_1}I - \frac{a_3}{a_1}A$

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LINEAR ALGEBRA

UNIT - V

Inner Product Space

Inner product spaces, Euclidean and unitary spaces, Norm or length of a Vector, Schwartz inequality, Triangle in Inequality, Parallelogram law, Orthogonality, Orthonormal set, complete orthonormal set, Gram – Schmidt orthogonalisation process. Bessel's inequality and Parseval's Identity.

7. **Inner Product Spaces**
8. **Orthogonality**

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7

Inner Product Spaces

7.1. In the discussion of vector spaces, F is considered as an arbitrary field. In this section, we consider a vector space $V(F)$ with F as the field of real numbers or as the field of complex numbers. We define inner product in $V(F)$ and study the ideas of length and angle.

7.2. Before defining inner product in $V(F)$, we state below some important properties of complex numbers.

The set of all complex numbers is $C = \{x+iy \mid x, y \in R \text{ and } i = \sqrt{-1}\}$.

If Z is a complex number, then $Z = x+iy$ where $x, y \in R$ and $i^2 = -1$. x is called the Real part of Z and 'y' is called the imaginary part of Z. We write $x = \text{Re } Z$, $y = \text{Im } Z$ so that $Z = \text{Re } Z + i \text{ Im } Z$.

The modulus of $Z = x+iy$ denoted by $|Z|$ is the non-negative real number $\sqrt{x^2+y^2}$.

The conjugate of $Z = x+iy$, denoted by $\overline{Z}$ is the complex number $x-iy$.

1. $Z = x+iy = 0 \Leftrightarrow x = 0, y = 0$.

2. $Z_1 = x_1 + iy_1, Z_2 = x_2 + iy_2$ and $Z_1 = Z_2 \Leftrightarrow x_1 = x_2$ and $y_1 = y_2$.

3. If $Z_1 = x_1 + iy_1, Z_2 = x_2 + iy_2$, then $Z_1 + Z_2 = (x_1 + x_2) + i(y_1 + y_2)$.

4. If $Z_1 = x_1 + iy_1, Z_2 = x_2 + iy_2$, then $Z_1 Z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2)$

5. If $Z \in C$ then $\overline{\overline{Z}} = Z$ and $|Z| = |\overline{Z}|$

6. If $Z = x+iy$ and $Z = \overline{Z} \Leftrightarrow \text{Im } Z = 0$ or $y = 0$ or Z is real.

7. $|\overline{Z}| = 0 \Leftrightarrow Z = 0$.

8. $Z + \overline{Z} = 2 \text{Re } Z$ and $Z - \overline{Z} = 2i \text{ Im } Z$

9. If $Z = x+iy$ then $Z\overline{Z} = x^2 + y^2 = |Z|^2$.

10. If $Z = x+iy$ then $|Z| = \sqrt{x^2+y^2} \geq x$ *i.e.*, $|Z| \geq \text{Re } Z$.

11. If $Z_1, Z_2 \in C$ then $\overline{Z_1 \pm Z_2} = \overline{Z_1} \pm \overline{Z_2}$ and $\overline{Z_1 Z_2} = \overline{Z_1}\,\overline{Z_2}$.

12. If $Z_1, Z_2 \in C$ then $|Z_1 + Z_2| \leq |Z_1| + |Z_2|$

13. If $Z \in C$ then $\text{Im } Z = \text{Re}(-iZ)$.

7.3. INNER PRODUCT SPACE

Definition. 1. *Let $V(F)$ be a vector space where F is the field of real numbers or the field of complex numbers. The vector space $V(F)$ is said to be an inner product space if there is defined for any two vectors $\alpha, \beta \in V$ an element $\langle \alpha, \beta \rangle \in F$ such that*

(1) $\langle \alpha, \beta \rangle = \overline{\langle \beta, \alpha \rangle}$ (conjugate symmetry)

(2) $\langle \alpha, \alpha \rangle > 0$ (zero element in F) for $\alpha \neq \bar{0}$, the zero element in V (Positivity) and

(3) $\langle a\alpha + b\beta, \gamma \rangle = a\langle \alpha, \gamma \rangle + b\langle \beta, \gamma \rangle$ (Distributivity) for any $\alpha, \beta, \gamma \in V$ and $a, b \in F$.

(S. K. U. M 15, K.U. 08O. U. 04, 05, N. U. 03, 04, 08)

A function $f : V \times V \to F$ satisfying the above properties is called an **inner product.** If f is the **inner product** function then $f\langle \alpha, \beta \rangle = \langle \alpha, \beta \rangle$ or (α, β) $\forall$ $\alpha, \beta \in V$.

Note. 1. The vectors $\alpha, \beta \in V$ while the inner product is a scalar $\langle \alpha, \beta \rangle \in F$.

2. In the property (3) : the '+' sign' in $a\alpha + b\beta$ is addition in V while '+' sign' in $a\langle \alpha, \gamma \rangle + b\langle \beta, \gamma \rangle$ is addition in F.

3. If F is the field of complex numbers $\overline{\langle \beta, \alpha \rangle}$ denotes the complex conjugate of $\langle \beta, \alpha \rangle \in F$.

4. If F is the field of real numbers, then $\langle \beta, \alpha \rangle$ is a real number and $\overline{\langle \beta, \alpha \rangle} = \langle \beta, \alpha \rangle$.

5. By conjugate symmetry property; $\langle \alpha, \alpha \rangle = \overline{\langle \alpha, \alpha \rangle}$ for every $\alpha \in V$ $(Z = \bar{Z})$ and hence $\langle \alpha, \alpha \rangle$ **is always real.** Thus the order property (2) makes sense.

Definition. 2. *Let $V(F)$ be a vector space where F is the field of real numbers or the field of complex numbers. The vector space $V(F)$ is said to be inner product space if there is defined for any two vectors $\alpha, \beta \in V$ an element $\langle \alpha, \beta \rangle \in F$ such that*

(1) $\langle \alpha, \beta \rangle = \overline{\langle \beta, \alpha \rangle}$ (conjugate symmetry)

(2) $\langle \alpha, \alpha \rangle > 0$ (zero element in F) for $\alpha \neq \bar{0}$ the zero element in V (Positivity)

(3) $\langle \alpha + \beta, \gamma \rangle = \langle \alpha, \gamma \rangle + \langle \beta, \gamma \rangle$ (Distributivity) and

(4) $\langle a\alpha, \gamma \rangle = a\langle \alpha, \gamma \rangle$ for any $\alpha, \beta, \gamma \in V$ and $a \in F$.

Note. Definition (1) and Definition (2) are equivalent.

From the definition it is clear that a vector space V over F endowed with a specific inner product is an inner product space.

If $F = R$ the field of real numbers then $V(F)$ is called **Euclidean space or Real inner product space.**

If $F = C$, the field of complex numbers then $V(F)$ is called **Unitary space or Complex inner product space.**

An inner product space having only zero vector is called zero space or null space.

If $V(F)$ is an inner product space then $V(F)$ is a vector space. A sub space $W(F)$ of the vector space $V(F)$ is also inner product space with the same inner product as in $V(F)$.

ILLUSTRATIONS.

1. If $\alpha = (a_1, a_2, a_3), \beta = (b_1, b_2, b_3)$ are the elements of a vector space $\mathbf{R}^3$, then $\langle \alpha, \beta \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3$ defines an inner product on $\mathbf{R}^3$.

Let $\alpha = (a_1, a_2, a_3), \beta = (b_1, b_2, b_3)$ and $\gamma = (c_1, c_2, c_3) \in R^3$.

Then $a_1, a_2, a_3, b_1, b_2, b_3, c_1, c_2, c_3 \in R$.

(1) $\langle \alpha, \beta \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3 = b_1 a_1 + b_2 a_2 + b_3 a_3$ ($\because a$'s and b's are real numbers)

$= \langle \beta, \alpha \rangle = \overline{\langle \beta, \alpha \rangle}$.

(2) $\langle \alpha, \alpha \rangle = a_1 a_1 + a_2 a_2 + a_3 a_3 = a_1^2 + a_2^2 + a_3^2$

If $\alpha = (a_1, a_2, a_3) \neq (0,0,0)$ then at least one of a_1, a_2, a_3 is not zero.

So, $\langle \alpha, \alpha \rangle = a_1^2 + a_2^2 + a_3^2 > 0$.

(3) For $a, b \in R$ and $\alpha, \beta, \gamma \in R^3$ we have

$$a\alpha + b\beta = a(a_1, a_2, a_3) + b(b_1, b_2, b_3) = (aa_1 + bb_1, aa_2 + bb_2, aa_3 + bb_3)$$

$$\langle a\alpha + b\beta, \gamma \rangle = (aa_1 + bb_1)c_1 + (aa_2 + bb_2)c_2 + (aa_3 + bb_3)c_3$$

$$= (aa_1c_1 + aa_2c_2 + aa_3c_3) + (bb_1c_1 + bb_2c_2 + bb_3c_3)$$

$$= a(a_1c_1 + a_2c_2 + a_3c_3) + b(b_1c_1 + b_2c_2 + b_3c_3) = a\langle \alpha, \gamma \rangle + b\langle \beta, \gamma \rangle$$

$\therefore$ The product $\langle \alpha, \beta \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3$ is an inner product on the vector space R^3.

Hence R^3 is an inner product space with the above inner product and $R^3(R)$ is real inner product space.

Note. 1. The inner product of α and β namely, $\langle \alpha, \beta \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3$ is called the dot product of α and β and is denoted by $\alpha . \beta$. This is called the **standard inner product in R^3.**

Note. 2. If $\alpha = (a_1, a_2, ..., a_n), \beta = (b_1, b_2, ..., b_n)$ are any two vectors of the vector space $V_n(R)$ or R^n then $\langle \alpha, \beta \rangle = a_1 b_1 + a_2 b_2 + ... + a_n b_n$ is called the standard inner product.

2. If $\alpha = (a_1, a_2, ..., a_n)$ and $\beta = (b_1, b_2, ..., b_n)$ are the elements of the vector space $V_n(C)$ where C is the field of complex numbers, then

$$\langle \alpha, \beta \rangle = a_1 \overline{b_1} + a_2 \overline{b_2} + ... + a_n \overline{b_n} = \sum_{i=1}^{n} a_i \overline{b_i}$$ **defines an inner product on $V_n(C)$.**

Let $a, b \in C$ and $\alpha, \beta, \gamma \in V_n$ so that $\alpha = (a_1, a_2, ..., a_n), \beta = (b_1, b_2, ..., b_n)$, $\gamma = (c_1, c_2, ..., c_n)$ where a's, b's and c's are complex numbers.

(If Z is a complex number we know that $\overline{\overline{Z}} = Z$ and $Z\overline{Z} = |Z|^2, \overline{Z_1 + Z_2} = \overline{Z}_1 + \overline{Z}_2$)

(1) $\langle \overline{\beta, \alpha} \rangle = \overline{b_1\bar{a}_1 + b_2\bar{a}_2 + ... + b_n\bar{a}_n} = \overline{b_1\bar{a}_1} + \overline{b_2\bar{a}_2} + ... + \overline{b_n\bar{a}_n}$

$= \bar{b}_1\bar{\bar{a}}_1 + \bar{b}_2\bar{\bar{a}}_2 + ... + \bar{b}_n\bar{\bar{a}}_n = \bar{b}_1 a_1 + \bar{b}_2 a_2 + ... + \bar{b}_n a_n = a_1\bar{b}_1 + a_2\bar{b}_2 + ... + a_n\bar{b}_n = \langle \alpha, \beta \rangle$.

(2) $\langle \alpha, \alpha \rangle = a_1\bar{a}_1 + a_2\bar{a}_2 + ... + a_n\bar{a}_n = |a_1|^2 + |a_2|^2 + ... + |a_n|^2$

If $\alpha \neq 0$ then at least one of $a_1, a_2, ..., a_n$ is non-zero complex number.

So $\langle \alpha, \alpha \rangle = |a_1|^2 + |a_2|^2 + ... + |a_n|^2 > 0$.

(3) $a\alpha + b\beta = a(a_1, a_2, ..., a_n) + b(b_1, b_2, ..., b_n) = (aa_1 + bb_1, aa_2 + bb_2, ..., aa_n + bb_n)$

$\langle a\alpha + b\beta, \gamma \rangle = (aa_1 + bb_1)\bar{c}_1 + (aa_2 + bb_2)\bar{c}_2 + ... + (aa_n + bb_n)\bar{c}_n$

$= (aa_1\bar{c}_1 + aa_2\bar{c}_2 + ... + aa_n\bar{c}_n) + (bb_1\bar{c}_1 + bb_2\bar{c}_2 + ... + bb_n\bar{c}_n)$

$= a(a_1\bar{c}_1 + a_2\bar{c}_2 + ... + a_n\bar{c}_n) + b(b_1\bar{c}_1 + b_2\bar{c}_2 + ... + b_n\bar{c}_n) = a\langle \alpha, \gamma \rangle + b\langle \beta, \gamma \rangle$.

$\therefore$ the product $\langle \alpha, \beta \rangle = a_1\bar{b}_1 + a_2\bar{b}_2 + ... + a_n\bar{b}_n$ is an inner product on $V_n(C)$.

$\therefore$ $V_n(C)$ or $C^n(C)$ is the unitary space.

Note. 1. The product above defined is called **standard inner product** on $V_n(C)$.

2. We can always define an inner product on a finite dimensional vector space of real or complex numbers as follows :

Let $B = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ be a basis of $V_n(F)$.

$\alpha, \beta \in V_n \Rightarrow \alpha = a_1\alpha_1 + a_2\alpha_2 + ... + a_n\alpha_n$ and $\beta = b_1\alpha_1 + b_2\alpha_2 + ... + b_n\alpha_n$ where $a_1, a_2, ..., a_n, b_1, b_2, ..., b_n \in F$

Then $< \alpha, \beta >= a_1\bar{b}_1 + a_2\bar{b}_2 + ... + a_n\bar{b}_n = \sum_{i=1}^{n} a_i\bar{b}_i$ defines inner product on V_n.

3. If $\alpha = (1+i, 4)$ and $\beta = (2-3i, 4+5i)$ in $V_2(C)$ or C^2 then

$\langle \alpha, \beta \rangle = (1+i)\overline{(2-3i)} + 4\overline{(4+5i)} = (1+i)(2+3i) + 4(4-5i) = 15 - 15i$

$\langle \alpha, \alpha \rangle = (1+i)\overline{(1+i)} + 4(\bar{4}) = (1+i)(1-i) + 4(4) = 18$

$\langle \overline{\beta, \alpha} \rangle = \{\overline{(2-3i)(1-i) + (4+5i)4}\} = \{\overline{(-1-5i) + (16+20i)}\} = \overline{15+15i} = 15 - 15i = \langle \alpha, \beta \rangle$.

3. Let $V(C)$ be the vector space of all continuous complex valued functions on the closed interval $[0,1]$. For $f, g \in V$ if $\langle f, g \rangle = \int_0^1 f(t)\,\overline{g(t)}$, then V is an inner product space.

(O. U. 04, 05)

f is a continuous complex - valued function on [0,1] means that $f(t)$ is a complex number for $t \in [0,1]$.

Let $a,b \in C$ and $f,g,h \in V$ where $t \in [0,1]$.

(1) $\langle \overline{g,f} \rangle = \left\{ \overline{\int_0^1 g(t)\,\overline{f(t)}\,dt} \right\} = \int_0^1 \overline{g(t)\overline{f(t)}}\,dt$. $\left(\because \overline{\int_0^1 f(t)\,dt} = \int_0^1 \overline{f(t)}\,dt \right)$

$= \int_0^1 \overline{g(t)}\,\overline{\overline{f(t)}}\,dt = \int_0^1 \overline{g(t)} f(t)\,dt = \int_0^1 f(t)\overline{g(t)}\,dt = \langle f,g \rangle$. $(\because \overline{\overline{Z}} = Z)$

(2) $\langle f,f \rangle = \int_0^1 f(t)\overline{f(t)}\,dt = \int_0^1 |f(t)|^2\,dt$ $(\because \overline{Z}Z = |Z|^2)$

If $f(t) \neq 0$ then $|f(t)|^2 > 0$ *i.e.,* $\langle f,f \rangle > 0$.

(3) $\langle af+bg, h \rangle = \int_0^1 \{af(t)+bg(t)\}\,\overline{h(t)}\,\}\,dt = \int_0^1 \{af(t)\,\overline{h(t)} + bg(t)\,\overline{h(t)}\}\,dt$

$= \int_0^1 af(t)\,\overline{h(t)}\,dt + \int_0^1 bg(t)\,\overline{h(t)}\}\,dt = a\int_0^1 f(t)\,\overline{h(t)}\,dt + b\int_0^1 g(t)\,\overline{h(t)}\}\,dt$

$= a\langle f,h \rangle + b\langle g,h \rangle$ $\therefore$ $V(C)$ is an inner product space.

Note. We can define more than one inner product on a given vector space. Let $V(R)$ be the vector space of continuous real valued functions. We can define two inner products

(*i*) $\langle f,g \rangle = \int_0^1 f(t)\,g(t)\,dt$ and (*ii*) $\langle f,g \rangle = \int_{-1}^1 f(t)\,g(t)\,dt$ which result in two distinct inner product spaces.

Theorem. 1. ***Definition (1) and Definition (2) are equivalent.***

Proof. The properties of conjugate symmetry and positivity are same in both the definitions.

(*i*) Definition (1) $\Rightarrow$ Definition (2)

For any $\alpha,\beta,\gamma \in V$ and $a,b \in F$ we have $\langle a\alpha + b\beta, \gamma \rangle = a\langle \alpha,\gamma \rangle + b\langle \beta,\gamma \rangle$.

If we take $a = 1, b = 1 \in F$ then

$\langle 1\,\alpha + 1\,\beta, \gamma \rangle = 1\langle \alpha,\gamma \rangle + 1\langle \beta,\gamma \rangle \Rightarrow \langle \alpha+\beta,\gamma \rangle = \langle \alpha,\gamma \rangle + \langle \beta,\gamma \rangle$

If we take $b = 0 \in F$ the zero element, then $\langle a\alpha + 0\beta, \gamma \rangle = a\langle \alpha,\gamma \rangle + 0\,\langle \beta,\gamma \rangle$

$\Rightarrow \langle a\alpha + \bar{0}, \gamma \rangle = a\langle \alpha,\gamma \rangle + 0 \Rightarrow \langle a\alpha,\gamma \rangle = a\langle \alpha,\gamma \rangle$.

$(\because 0 \in F, \beta \in V \Rightarrow 0\beta = \bar{0} \in V$ and $a\alpha + \bar{0} = a\alpha \in V)$

(*ii*) Definiton (2) $\Rightarrow$ Definition (1)

$\langle a\alpha + b\beta, \gamma\rangle = \langle a\alpha, \gamma\rangle + \langle b\beta, \gamma\rangle$ using property (3) of def. (2)

$= a\langle \alpha, \gamma\rangle + b\langle \beta, \gamma\rangle$ using property (4) of def (2)

Theorem.2. ***In an inner product space*** $V(F)$ ***if*** $\alpha, \beta, \gamma, \bar{0} \in V$ ***and*** $a, 0 \in F$ ***then***

(*i*) $\langle \bar{0}, \beta\rangle = \langle \alpha, \bar{0}\rangle = 0$ **(*ii*)** $\langle \alpha, a\beta\rangle = \bar{a}\langle \alpha, \beta\rangle$ **(*iii*)** $\langle \alpha, \alpha\rangle = 0$ **iff** $\alpha = \bar{0}$

(*iv*) $\langle \alpha, \beta\rangle = 0 \ \forall\ \beta \in V \Rightarrow \alpha = \bar{0}$ ***and*** $\langle \alpha, \beta\rangle = 0 \ \forall\ \alpha \in V \Rightarrow \beta = \bar{0}$

(*v*) $\langle \alpha - \beta, \gamma\rangle = \langle \alpha, \gamma\rangle - \langle \beta, \gamma\rangle$ **and (*vi*)** $\langle \alpha, \beta - \gamma\rangle = \langle \alpha, \beta\rangle - \langle \alpha, \gamma\rangle$

Proof. (*i*) $\langle \bar{0}, \beta\rangle = \langle 0\alpha, \beta\rangle = 0\langle \alpha, \beta\rangle = 0$ (Using $\langle a\alpha, \beta\rangle = a\langle \alpha, \beta\rangle$)

$\langle \alpha, \bar{0}\rangle = \overline{\langle \bar{0}, \alpha\rangle}$ (Conjugate symmetry) $= \bar{0} = 0$ ($\because\ \bar{0} = 0 - 0i = 0$)

(*ii*) $\langle \alpha, a\beta\rangle = \overline{\langle a\beta, \alpha\rangle}$ (conjugate symmetry)

$= \overline{a\langle \beta, \alpha\rangle} = \bar{a}\,\overline{\langle \beta, \alpha\rangle} = \bar{a}\langle \alpha, \beta\rangle$ (using $\langle a\alpha, \beta\rangle = a\langle \alpha, \beta\rangle$ and $\overline{Z_1 Z_2} = \overline{Z_1}\,\overline{Z_2}$)

(*iii*) Let $\alpha = \bar{0}$. Then $\langle \alpha, \alpha\rangle = \langle \bar{0}, \bar{0}\rangle = \langle 0\bar{0}, \bar{0}\rangle = 0\langle \bar{0}, \bar{0}\rangle = 0$. (In a field F ; $0a = 0$)

Let $\langle \alpha, \alpha\rangle = 0$. By positivity $\langle \alpha, \alpha\rangle > 0 \ \forall\ \alpha \neq \bar{0}$ and hence $\langle \alpha, \alpha\rangle = 0 \Rightarrow \alpha = \bar{0}$.

(*iv*) $\langle \alpha, \beta\rangle = 0 \ \forall\ \beta \in V \Rightarrow \langle \alpha, \alpha\rangle = 0$ (putting $\beta = \alpha) \Rightarrow \alpha = \bar{0}$.

$\langle \alpha, \beta\rangle = 0 \ \forall\ \alpha \in V \Rightarrow \langle \beta, \beta\rangle = 0$ (putting $\alpha = \beta) \Rightarrow \beta = \bar{0}$.

(*v*) $\langle \alpha - \beta, \gamma\rangle = \langle \alpha + (-\beta), \gamma\rangle = \langle \alpha, \gamma\rangle + \langle -\beta, \gamma\rangle = \langle \alpha, \gamma\rangle + \langle (-1)\beta, \gamma\rangle = \langle \alpha, \gamma\rangle + (-1)\langle \beta, \gamma\rangle$

$= \langle \alpha, \gamma\rangle - \langle \beta, \gamma\rangle$ ($\because \langle a\alpha, \beta\rangle = a\langle \alpha, \beta\rangle$ and $(-1)\,a = -a$ in a field)

(*vi*) $\langle \alpha, \beta - \gamma\rangle = \overline{\langle \beta - \gamma, \alpha\rangle} = \overline{\langle \beta, \alpha\rangle - \langle \gamma, \alpha\rangle} = \overline{\langle \beta, \alpha\rangle} - \overline{\langle \gamma, \alpha\rangle} = \langle \alpha, \beta\rangle - \langle \alpha, \gamma\rangle$

Note. 1. $\langle \alpha, \beta\rangle = \langle \alpha, \gamma\rangle \ \forall\ \alpha \in V \Rightarrow \langle \alpha, \beta\rangle - \langle \alpha, \gamma\rangle = 0 \ \forall\ \alpha \in V$

$\Rightarrow \langle \alpha, \beta - \gamma\rangle = 0 \ \forall\ \alpha \in V \Rightarrow \beta - \gamma = \bar{O} \Rightarrow \beta = \gamma$.

This property is like cancellation law in inner product.

2. In the property $\langle a\alpha, \beta\rangle = a\langle \alpha, \beta\rangle$ by putting $a = -1$

we have $\langle (-1)\alpha, \beta\rangle = \langle -\alpha, \beta\rangle = (-1)\langle \alpha, \beta\rangle = -\langle \alpha, \beta\rangle$. Similarly, $\langle \alpha, -\beta\rangle = -\langle \alpha, \beta\rangle$.

Theorem. 3. ***In an inner product space*** $V(F)$ ***if*** $\bar{0}, \alpha, \beta, \gamma, \delta \in V$ ***and*** $0, a, b \in F$ ***then*** **(*i*)** $\langle a\alpha, b\beta\rangle = a\bar{b}\langle \alpha, \beta\rangle$ **(*ii*)** $\langle a\alpha - b\beta, \gamma\rangle = a\langle \alpha, \gamma\rangle - b\langle \beta, \gamma\rangle$

(*iii*) $\langle \alpha, a\beta \pm b\gamma\rangle = \bar{a}\langle \alpha, \beta\rangle \pm \bar{b}\langle \alpha, \gamma\rangle$ ***and***

(*iv*) $\langle a\alpha + b\beta, a\alpha + b\beta\rangle = a\bar{a}\langle \alpha, \alpha\rangle + a\bar{b}\langle \alpha, \beta\rangle + \bar{a}b\langle \beta, \alpha\rangle + b\bar{b}\langle \beta, \beta\rangle$

Proof. (*i*) $\langle a\alpha, b\beta\rangle = a\langle \alpha, b\beta\rangle = a\,\overline{\langle b\beta, \alpha\rangle}$

(Using Def. property and Conjugate symmetry)

$= a\{\overline{b\langle\beta,\alpha\rangle}\} = a\{\bar{b}\,\overline{\langle\beta,\alpha\rangle}\} = a\{\bar{b}\,\langle\alpha,\beta\rangle\} = \bar{a}b\langle\alpha,\beta\rangle$ ($\because\ \overline{Z_1Z_2} = \overline{Z_1}\,\overline{Z_2}$)

(*ii*) $\langle a\alpha - b\beta,\gamma\rangle = \langle a\alpha + (-b)\beta,\gamma\rangle = a\langle\alpha,\gamma\rangle + (-b)\langle\beta,\gamma\rangle = a\langle\alpha,\gamma\rangle - b\langle\beta,\gamma\rangle$.

(*iii*) $\langle\alpha, a\beta \pm b\gamma\rangle = \overline{\langle a\beta \pm b\gamma,\alpha\rangle} = \overline{a\langle\beta,\alpha\rangle \pm b\langle\gamma,\alpha\rangle} = \overline{a\langle\beta,\alpha\rangle} \pm \overline{b\langle\gamma,\alpha\rangle} = \bar{a}\,\overline{\langle\beta,\alpha\rangle} \pm \bar{b}\,\overline{\langle\gamma,\alpha\rangle}$

$= \bar{a}\,\langle\alpha,\beta\rangle \pm \bar{b}\,\langle\alpha,\gamma\rangle$ ($\because\ \overline{Z_1 \pm Z_2} = \overline{Z_1} \pm \overline{Z_2}$ and $\overline{Z_1Z_2} = \overline{Z_1}\,\overline{Z_2}$)

(*iv*) $\langle a\alpha + b\beta, a\alpha + b\beta\rangle = \langle a\alpha + b\beta, \delta\rangle$ where $\delta = a\alpha + b\beta$

$= a\langle\alpha,\delta\rangle + b\langle\beta,\delta\rangle = a\langle\alpha, a\alpha + b\beta\rangle + b\langle\alpha, a\alpha + b\beta\rangle$

$= a\{\bar{a}\langle\alpha,\alpha\rangle + \bar{b}\langle\alpha,\beta\rangle\} + b\{\bar{a}\langle\alpha,\alpha\rangle + \bar{b}\langle\alpha,\beta\rangle\}$

$= a\bar{a}\langle\alpha,\alpha\rangle + a\bar{b}\langle\alpha,\beta\rangle + b\bar{a}\langle\alpha,\alpha\rangle + b\bar{b}\langle\alpha,\beta\rangle$ (Distributive law in a field)

Note. 1. $\langle-\alpha,-\beta\rangle = (-1)\langle\alpha,-\beta\rangle = (-1)(\overline{-1})\langle\alpha,\beta\rangle = \langle\alpha,\beta\rangle$ ($\because\ \overline{-1} = -\overline{1+0i} = -1$)

2. $\langle\alpha,\beta \pm \gamma\rangle = \langle\alpha,\beta\rangle \pm \langle\alpha,\gamma\rangle$ and $\langle\alpha \pm \beta,\gamma\rangle = \langle\alpha,\gamma\rangle \pm \langle\beta,\gamma\rangle$

3. $\langle\alpha+\beta,\alpha+\beta\rangle = \langle\alpha,\alpha\rangle + \langle\alpha,\beta\rangle + \langle\beta,\alpha\rangle + \langle\beta,\beta\rangle$

Imp. 4. If $\alpha_1, \alpha_2, ..., \alpha_n, \beta \in V$ and $a_1, a_2, ..., a_n \in F$

then $\langle a_1\alpha_1 + a_2\alpha_2 + ... + a_n\alpha_n, \beta\rangle = a_1\langle\alpha_1,\beta\rangle + a_2\langle\alpha_2,\beta\rangle + ... + a_n\langle\alpha_n,\beta\rangle$

$$\Rightarrow \left\langle \sum_{i=1}^{n} a_i\,\alpha_i, \beta \right\rangle = \sum_{i=1}^{n} a_i\langle\alpha_i,\beta\rangle. \qquad \text{Also, } \left\langle \beta, \sum_{i=1}^{n} a_i\,\alpha_i \right\rangle = \sum_{i=1}^{n} \bar{a_i}\langle\beta,\alpha_i\rangle$$

SOME KEY POINTS IN MATRICES :

Let $A = [a_{ij}]_{n\times n} \in M_n(C)$, the vector space of $n \times n$ matrices over the field of complex numbers C.

1. If $tr\ A = a_{11} + a_{22} + ... + a_{nn} = \sum_{i=1}^{n} a_{ii}$, then (1) $tr\ (\lambda A) = \lambda\ tr\ A$ where $\lambda \in C$

(2) $tr\ (A+B) = tr\ A + tr\ B$ and (3) $tr\ (AB) = tr\ (BA)$ (4) $tr\ A = tr\ A^T$.

2. Conjugate transpose of A is denoted by A^* and is defined as $A^* = [a'_{ji}]_{n\times n}$ where $a'_{ji} = \overline{a_{ij}}$.

e.g.1. If $A = \begin{bmatrix} 3-i & -2i \\ 1+i & 4-i \end{bmatrix}_{2\times 2}$, then $A^* = \begin{bmatrix} 3+i & 1-i \\ 2i & 4+i \end{bmatrix}$. Also $(A^*)^* = A$.

If $A = [a_{ij}]_{n\times n} \in M_n(R)$ where R is the real number field then $A^* = A^T$.

e.g.2. Let $A = \begin{bmatrix} i & 1+2i \\ 2 & 3+4i \end{bmatrix} \in M_2(C)$ and α, β are column vectors in A.

Then $\alpha = \begin{bmatrix} i \\ 2 \end{bmatrix}, \beta = \begin{bmatrix} 1+2i \\ 3+4i \end{bmatrix}$ and $\beta^* = [1-2i \quad 3-4i]$

$\langle \alpha, \beta \rangle = \beta^* \alpha = [1-2i \quad 3-4i] \begin{bmatrix} i \\ 2 \end{bmatrix} = (1-2i)i + (3-4i)2 = 8-7i \in C$.

SOLVED PROBLEMS

Ex. 1. *If α, β are vectors in a complex inner product space prove that* $\langle \alpha, \beta \rangle = \text{Re}\langle \alpha, \beta \rangle + i \,\text{Re}\langle \alpha, i\beta \rangle$.

Sol. For $\alpha, \beta \in V$, $\langle \alpha, \beta \rangle \in C$, the field of complex numbers.

Let $\langle \alpha, \beta \rangle = x + iy$ so that $x = \text{Re}\langle \alpha, \beta \rangle$ and $y = \text{Im}\langle \alpha, \beta \rangle$.

$-i\langle \alpha, \beta \rangle = -i(x+iy) = y - ix$ so that $y = \text{Re}\{-i\langle \alpha, \beta \rangle\} = \text{Re}\{\bar{i}\langle \alpha, \beta \rangle\} = \text{Re}\langle \alpha, i\beta \rangle$

$\therefore \langle \alpha, \beta \rangle = \text{Re}\langle \alpha, \beta \rangle + i\,\text{Re}\langle \alpha, i\beta \rangle$.

Ex. 2. *In the innerproduct space $M_2(C)$, the vector space of 2×2 matrices over the complex field if $\langle A, B \rangle = tr(B^* A)$ $\forall$ $A, B \in M_2$ find $\langle A, B \rangle$*

given $A = \begin{bmatrix} 1 & 2+i \\ 3 & i \end{bmatrix}$ *and* $B = \begin{bmatrix} 1+i & 0 \\ i & -i \end{bmatrix}$.

Sol. $B^* = \begin{bmatrix} 1-i & -i \\ 0 & i \end{bmatrix}$ = the conjugate transpose of B.

$$B^* A = \begin{bmatrix} 1-i & -i \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 & 2+i \\ 3 & i \end{bmatrix} = \begin{bmatrix} 1-i-3i & 2+i-2i+1+1 \\ 0+3i & 0-1 \end{bmatrix} = \begin{bmatrix} 1-4i & 4-i \\ 3i & -1 \end{bmatrix}$$

so that $tr(B^* A) = 1-4i-1 = -4i$. $\quad \therefore \langle A, B \rangle = -4i \in C$.

Ex. 3. *In the inner product space C^2 for $x, y \in C^2$ and $A = \begin{bmatrix} 1 & i \\ -i & 2 \end{bmatrix}$ if the inner product $\langle x, y \rangle = x\, A\, y^*$ compute $\langle x, y \rangle$ for $x = [1-i, 2+3i]$ and $y = [2+i, 3-2i]$.*

Sol. $y = [2+i, 3-2i] \Rightarrow y^* = \begin{bmatrix} 2-i \\ 3+2i \end{bmatrix}$. $\qquad x\,A = [1-i, 2+3i] \begin{bmatrix} 1 & i \\ -i & 2 \end{bmatrix}$

$= [1-i-2i+3, i+1+4+6i] = [4-3i, 5+7i]$

$\therefore \langle x, y \rangle = x\, A\, y^* = [4-3i, 5+7i] \begin{bmatrix} 2-i \\ 3+2i \end{bmatrix} = (4-3i)(2-i) + (5+7i)(3+2i)$

$= (5-10i) + (1+31i) = 6+2i \in C$, the field.

Ex. 4. *For* $\alpha=(a,b),\beta=(c,d)\in R^2$, *the vector space, if* $\langle\alpha,\beta\rangle=ac-bd$ *then prove that it is not an inner product.*

Sol. Consider $\alpha=(3,4)\in R^2$ so that $\alpha\neq(0,0)$.

$\langle\alpha,\alpha\rangle=(3)(3)-(4)(4)=9-16=-7\not>0$. $\therefore$ the property of positivity is not true.

Ex. 5. *For* $f(x),g(x)\in P(R)$, *the vector space of polynomials over the field R defined on* $[0,1]$ *if* $\langle f(x),g(x)\rangle=\int_0^1 f'(t)\,g(t)\,dt$ *then prove that it is not an inner product.*

Sol. Take $f(x)=x,\ g(x)=x^2$ on $[0,1]$. $\langle f,g\rangle=\int_0^1 1.t^2\,dt=\frac{1}{3}$ and $\langle g,f\rangle=\int_0^1 2t.t\,dt=\frac{2}{3}$

$\therefore\ \langle f,g\rangle\neq\langle g,f\rangle$ and hence conjugate symmetry is not true.

Ex. 6. *In the vector space* $V_2(C)$, *the set of all ordered pairs over the field of complex numbers,* $\langle\alpha,\beta\rangle=2x_1\overline{y_1}+x_1\overline{y_2}+x_2\overline{y_1}+x_2\overline{y_2}\ \ \forall\,\alpha=(x_1,x_2),\beta=(y_1,y_2)\in V_2$. *Prove that* $V_2(C)$ *is inner product space.*

Sol. Conjugate symmetry.

$$\overline{\langle\beta,\alpha\rangle}=\overline{2y_1\overline{x_1}+y_1\overline{x_2}+y_2\overline{x_1}+y_2\overline{x_2}}=2\overline{y_1}\,\overline{\overline{x_1}}+\overline{y_1}\,\overline{\overline{x_2}}+\overline{y_2}\,\overline{\overline{x_1}}+\overline{y_2}\,\overline{\overline{x_2}}$$

$$=2\overline{y_1}x_1+\overline{y_1}x_2+\overline{y_2}x_1+\overline{y_2}x_2=2x_1\overline{y_1}+x_1\overline{y_2}+x_2\overline{y_1}+x_2\overline{y_2}=\langle\alpha,\beta\rangle.$$

Positivity . $\langle\alpha,\alpha\rangle=2x_1\overline{x_1}+x_1\overline{x_2}+x_2\overline{x_1}+x_2\overline{x_2}=x_1\overline{x_1}+(x_1+x_2)(\overline{x_1}+\overline{x_2})$

$$=x_1\overline{x_1}+(x_1+x_2)\overline{(x_1+x_2)}=|x_1|^2+|x_1+x_2|^2$$

For $\alpha=(x_1,x_2)\neq(0,0)$ either $x_1\neq0$ or $x_2\neq0$. $\therefore\ \langle\alpha,\alpha\rangle=|x_1|^2+|x_1+x_2|^2>0$.

Distributivity. Let $a,b\in C$ and $\gamma=(z_1,z_2)\in V_2$

$$a\alpha+b\beta=a(x_1,x_2)+b(y_1,y_2)=(ax_1+by_1,ax_2+by_2)$$

$$\langle a\alpha+b\beta,\gamma\rangle=2(ax_1+by_1)\overline{z}_1+(ax_1+by_1)\overline{z}_2+(ax_2+by_2)\overline{z}_1+(ax_2+by_2)\overline{z}_2$$

$$=a(2x_1\overline{z}_1+x_1\overline{z}_2+x_2\overline{z}_1+x_2\overline{z}_2)+b(2y_1\overline{z}_1+y_1\overline{z}_2+y_2\overline{z}_1+y_2\overline{z}_2)$$

$$=a\langle\alpha,\gamma\rangle+b\langle\beta,\gamma\rangle.\qquad\therefore\ V_2(C)\text{ is an inner product space.}$$

Ex. 7. *Let* $V(R)$ *be the vector space of real valued continuous functions on* $[0,1]$. *For* $f,g\in V$ *if we define* $\langle f,g\rangle=\int_0^1 f(t)\,g(t)\,dt$ *then prove that V is an inner product space.*

Sol. $f \in V$ is a continuous real valued function on $[0,1]$

$\Rightarrow f(t)$ is a real number $\forall\ t \in [0,1] \Rightarrow \overline{f(t)} = f(t)$.

Conjugate symmetry.

$$\overline{\langle g,f\rangle} = \overline{\int_0^1 g(t)f(t)\,dt} = \int_0^1 \overline{g(t)f(t)}\,dt = \int_0^1 g(t)f(t)\,dt = \int_0^1 f(t)g(t)\,dt = \langle f,g\rangle$$

Positivity. Let $f \neq O$, the zero function defined by $O(t) = 0 \in R$ for $t \in [0,1]$.

$$\langle f,f\rangle = \int_0^1 f(t)f(t)\,dt = \int_0^1 f(t)^2\,dt > 0 \quad (\because f \text{ is continuous on } [0,1])$$

Distributivity. Let $a,b \in R$ and $h \in V$. $\langle af+bg, h\rangle = \int_0^1 \{af(t)+bg(t)\}h(t)\,dt$

$$= \int_0^1 a\,f(t)h(t)\,dt + \int_0^1 b\,g(t)h(t)\,dt = a\int_0^1 f(t)h(t)\,dt + b\int_0^1 g(t)h(t)\,dt = a\langle f,h\rangle + b\langle g,h\rangle$$

$\therefore V(R)$ is an inner product space.

Ex. 8. *Let $V = M_2(R)$ be the vector space of 2×2 matrices over the field of real numbers. For $A,B \in V$ we define $\langle A,B\rangle = tr(B^T A)$. Prove that V is an inner product space.*

Sol. Let $A,B,C \in V$. Then A,B,C are 2×2 matrices with real numbers as elements.

Therefore for any matrix A, $\overline{A} = A$, $\overline{A^T} = A^T$.

Conjugate Symmetry. $\overline{\langle B,A\rangle} = \overline{tr(A^T B)} = tr(A^T B) = tr(BA^T)$

$$= tr(BA^T)^T = tr(B^T (A^T)^T) = tr(B^T A) = \langle A,B\rangle$$

Positivity. Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ where $a_{11}, a_{12}, a_{21}, a_{22} \in R$ and $A \neq O$.

Then atleast one of $a_{11}, a_{12}, a_{21}, a_{22}$ is not zero.

$\langle A,A\rangle = tr(A^T A) = a_{11}^2 + a_{12}^2 + a_{21}^2 + a_{22}^2 > 0$. $\therefore$ Positivity is true.

Distributivity. For $a,b \in R$; $\langle aA+bB, C\rangle = tr(C^T\{aA+bB\})$

$= tr(C^T(aA) + C^T(bB))$ (Distributity in matrices)

$= tr(a(C^T A) + b(C^T B)) = tr(a(C^T A) + tr(b(C^T B))$

$= a\,tr(C^T A) + b\,tr(C^T B) = a\langle A,C\rangle + b\langle B,C\rangle$

Hence $V = M_2(R)$ is an inner product space.

Ex. 9. *Let $P_n(x)$ be the vector space of all real valued polynomials of degree atmost n . Prove that $P_n(x)$ is an inner product space for the product $\langle f,g\rangle = a_0b_0 + a_1b_1 + \ldots + a_nb_n$ for $f(x) = a_0 + a_1x + \ldots + a_nx^n$, $g(x) = b_0 + b_1x + \ldots + b_mx^m \in P_n(x)$.*

Sol. Let $f(x) = a_0 + a_1x + \ldots + a_nx^n$, $g(x) = b_0 + b_1x + \ldots + b_mx^m$

and $h(x) = c_0 + c_1x + \ldots + c_px^p \in P_n(x)$ so that $m \le n$, $p \le n$.

Conjugate Symmetry. $\langle f,g\rangle = a_0b_0 + a_1b_1 + \ldots + a_nb_n = b_0a_0 + b_1a_1 + \ldots + b_na_n$

$= \langle g,f\rangle$ ($\because$ a_i, b_i 's are real numbers) $\Rightarrow \overline{\langle g,f\rangle} = \langle f,g\rangle$

Positivity. $\langle f,f\rangle = a_0a_0 + a_1a_1 + \ldots + a_na_n = a_0^2 + a_1^2 + \ldots + a_n^2 > 0$ for $f \neq O$.

Distributivity. For $\alpha, \beta \in R$;

$\langle \alpha f + \beta g, h\rangle = (\alpha a_0 + \beta b_0)\, c_0 + (\alpha a_1 + \beta b_1)\, c_1 + \ldots + (\alpha a_n + \beta b_n)\, c_n$

$= \alpha\,(a_0c_0 + a_1c_1 + \ldots + a_nc_n) + \beta\,(b_0c_0 + b_1c_1 + \ldots + b_nc_n)$

$= \alpha\langle f,h\rangle + \beta\langle g,h\rangle$ ($\because$ real numbers are commutative, associative and distributive)

$\therefore$ $P_n(x)$ with respect to the product is an inner product space.

EXERCISE 7 (*a*)

1. If α, β, γ are vectors in the inner product space $V(F)$ prove that

(*i*) $\langle \alpha, \beta \pm \gamma\rangle = \langle \alpha,\beta\rangle \pm \langle \alpha,\gamma\rangle$ (*ii*) $\langle \alpha - \beta, \alpha - \beta\rangle = \langle \alpha,\alpha\rangle - \langle \alpha,\beta\rangle - \langle \beta,\alpha\rangle + \langle \beta,\beta\rangle$

(*iii*) $\langle a\alpha - b\beta, a\alpha - b\beta\rangle = a\bar{a}\langle \alpha,\alpha\rangle - a\bar{b}\langle \alpha,\beta\rangle - b\bar{a}\langle \beta,\alpha\rangle + b\bar{b}\langle \beta,\beta\rangle$.

2. $\alpha = (1,2), \beta = (-1,1)$ are two vectors in the vector space R^2 with standard inner product. If γ is a vector such that $\langle \alpha,\gamma\rangle = -1$ and $\langle \beta,\gamma\rangle = 3$ find γ.

3. If $\alpha = (2, 1+i, i), \beta = (2-i, 2, 1+2i)$ are two vectors in C^3 with standard inner product find $\langle \alpha,\beta\rangle$.

4. If $M_2(R)$ is the vector space of 2×2 matrices over real field and $A, B \in M_2$ prove that $\langle A,B\rangle = tr\,(A+B)$ is not an inner product.

5. If $\alpha = (x_1, x_2, \ldots, x_n)$ and $\beta = (y_1, y_2, \ldots, y_n)$ are vectors in R^n vector space prove that $\langle \alpha,\beta\rangle = x_1y_1 + x_2y_2 + \ldots + x_ny_n$ defines inner product in R^n.

6. In R^2 show that $\langle \alpha,\beta\rangle = x_1 + x_2 + y_1 + y_2$ where $\alpha = (x_1, x_2), \beta = (y_1, y_2)$ does not define inner product.

7. In R^2 if we define $\langle \alpha, \beta \rangle = x_1 y_1 - x_2 y_1 + 4 x_2 y_2$ for $\alpha = (x_1, x_2), \beta = (y_1, y_2)$ prove that R^2 is an inner product space. ***(S. K. U 2011)***

8. In the vector space $R^3(R)$ if $\bar{x} = (x_1, x_2, x_3), \bar{y} = (y_1, y_2, y_3)$ are two vectors such that $\langle \bar{x}, \bar{y} \rangle = x_1 y_1 + x_2 y_2 - x_3 y_3$ then prove that it is not an inner product.

9. If f, g are two inner products defined on a vector space V(F) then show that *(i)* $(f+g)$ is also an inner product on V(F) *(ii)* (nf) where $n \in \mathrm{N}$ is also an inner product and *(iii)* $(f-g)$ need not be an inner product on V(F).

ANSWERS

2. $\left(-\frac{7}{3}, \frac{2}{3}\right)$ **3.** $8+5i$

7.4. NORM OR LENGTH OF A VECTOR

In what follows we generalise the concept of length in R^2 and R^3 to inner product space.

Definition. *Let V be an inner product space over the field F. The norm (length) of* $\alpha \in V$ *denoted by* $\| \alpha \|$ *is defined as the positive square root of* $\langle \alpha, \alpha \rangle$.

Norm or length of $\alpha \in V = \| \alpha \| = \sqrt{(\alpha, \alpha)} \Rightarrow \| \alpha \|^2 = \langle \alpha, \alpha \rangle$.

Note. 1. For $\alpha \in V, \langle \alpha, \alpha \rangle$ is a non- negative real number and hence the norm of α is always a non-negative real number.

2. $\alpha = \bar{0} \Leftrightarrow \| \alpha \| = 0$

e.g.1. In the inner product space $V_2(R) = R^2(R)$; if $\alpha = (a, b) \in V_2$

then $\| \alpha \| = \| (a, b) \| = \sqrt{a^2 + b^2} = \sqrt{\langle \alpha, \alpha \rangle}$.

2. In the inner product space $V_3(R) = R^3(R)$; if $\alpha = (a, b, c)$

then $\| \alpha \| = \| (a, b, c) \| = \sqrt{a^2 + b^2 + c^2} = \sqrt{\langle \alpha, \alpha \rangle}$.

3. In the inner product space $V_n(C) = C^n$ if $\alpha = (a_1, a_2, ..., a_n)$

then $\| \alpha \| = \| (a_1, a_2, ..., a_n) \| = \sqrt{| a_1 |^2 + | a_2 |^2 + | a_n |^2} = \sqrt{\sum_{i=1}^{n} | a_i |^2} = \sqrt{\langle \alpha, \alpha \rangle}$.

Theorem. 1. ***In an inner product space V(F) (1)*** $\| \alpha \| > 0$ ***if*** $\alpha \neq 0$ ***and (2)*** $\| a \alpha \| = | a | \, \| \alpha \|$ ***where*** $0, a \in F$ ***and*** $\bar{0}, \alpha \in V$. ***(S. K. D. 08, O.U. 03)***

Proof. (1) If $\alpha \neq \bar{0}$ then $\langle \alpha, \alpha \rangle > 0$.

$\therefore \|\alpha\| = \sqrt{\langle\alpha,\alpha\rangle} > 0$. For any $\alpha \in V$, $\|\alpha\| \geq 0$.

(2) By the definition of norm, $\|a\alpha\|^2 = \langle a\alpha, a\alpha\rangle$

$= a\langle\alpha, a\alpha\rangle = a\bar{a}\langle\alpha,\alpha\rangle$ (By (3) of the theorem)

$= |a|^2 \|\alpha\|^2 = (|a|\|\alpha\|)^2$ $(\because Z\bar{Z} = |Z|^2)$ $\therefore \|a\alpha\| = |a|\,\|\alpha\|$.

Note. If $\alpha \in V$ and $\alpha \neq \bar{0}$ by the above theorem $\|\alpha\| > 0$. Since $\|\alpha\| (> 0) \in F$ and F is a field, there exists $\frac{1}{\|\alpha\|} \in F$ such that $\|\alpha\| \frac{1}{\|\alpha\|} = 1$. Now, for $\frac{1}{\|\alpha\|} \in F$ and $\alpha \in V$ we have $\frac{1}{\|\alpha\|}\alpha \in V$, such that $\left\langle \frac{1}{\|\alpha\|}\alpha, \frac{1}{\|\alpha\|}\alpha \right\rangle = \frac{1}{\|\alpha\|}\overline{\left(\frac{1}{\|\alpha\|}\right)}\langle\alpha,\alpha\rangle = \left(\frac{1}{\|\alpha\|}\right)\left(\frac{1}{\|\alpha\|}\right)\|\alpha\|^2 = 1$.

Hence, for $\alpha \in V$ and $\alpha \neq \bar{0}$, $\frac{1}{\|\alpha\|}\alpha \in V$ is a vector of length 1.

Definition. *Let $V(F)$ be an inner product space. $\alpha \in V$ is called a unit vector if $\|\alpha\| = 1$. If $\alpha \in V$ then $\frac{1}{\|\alpha\|}\alpha \in V$ is unit vector.*

e.g.1. In the inner product space R^2, $i = (1,0)$, $j = (0,1)$ are unit vectors.

e.g.2. In the inner product space R^3 with standard inner product $i = (1,0,0)$, $j = (0,1,0)$ and $k = (0,0,1)$ are vectors of each length 1.

3. In the inner product space R^3, if $\alpha = (a_1, a_2, a_3)$ then $\|\alpha\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

and $\frac{1}{\|\alpha\|}\alpha = \left(\frac{a_1}{\sqrt{a_1^2 + a_2^2 + a_3^2}}, \frac{a_2}{\sqrt{a_1^2 + a_2^2 + a_3^2}}, \frac{a_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}}\right)$

Theorem. 2. ***In an inner product space $V(F)$, $|\langle\alpha,\beta\rangle| \leq \|\alpha\|\,\|\beta\|$ for all $\alpha,\beta \in V$.***

Cauchy - Schwarz's inequality.

(N. U. 11, 07, K. U. 11, 04, S. K. U. M 15, 08, A. U. 08, S. V. U. 07, A. U. 08, O. U. 03,05)

Proof. Case (1). Let $\alpha = \bar{0}$. Then $\langle\alpha,\beta\rangle = \langle\bar{0},\beta\rangle = 0$ and $\|\alpha\|^2 = \langle\alpha,\alpha\rangle = \langle\bar{0},\bar{0}\rangle = 0$.

$\therefore |\langle\alpha,\beta\rangle| = 0$ and $\|\alpha\|\,\|\beta\| = 0$ $(\|\beta\|) = 0$. $\therefore |\langle\alpha,\beta\rangle| = \|\alpha\|\,\|\beta\|$.

Case (2). Let $\alpha \neq \bar{0}$. Then $\|\alpha\| > 0$ so that $\frac{1}{\|\alpha\|} > 0$.

Take $\gamma \in V$ so that $\gamma = \beta - \frac{\langle \beta, \alpha \rangle}{\| \alpha \|^2} \alpha$. Now $(\gamma, \gamma) = \left\langle \beta - \frac{\langle \beta, \alpha \rangle}{\| \alpha \|^2} \alpha, \beta - \frac{\langle \beta, \alpha \rangle}{\| \alpha \|^2} \alpha \right\rangle$

$$= \langle \beta, \beta \rangle - \frac{\overline{\langle \beta, \alpha \rangle}}{\| \alpha \|^2} \langle \beta, \alpha \rangle - \frac{\langle \beta, \alpha \rangle}{\| \alpha \|^2} \langle \alpha, \beta \rangle + \frac{\langle \beta, \alpha \rangle}{\| \alpha \|^2} \frac{\overline{\langle \beta, \alpha \rangle}}{\| \alpha \|^2} \langle \alpha, \alpha \rangle$$

$$= \| \beta \|^2 - \frac{\overline{\langle \beta, \alpha \rangle} \langle \beta, \alpha \rangle}{\| \alpha \|^2} - \frac{\langle \beta, \alpha \rangle \langle \alpha, \beta \rangle}{\| \alpha \|^2} + \frac{\langle \beta, \alpha \rangle \overline{\langle \beta, \alpha \rangle}}{\| \alpha \|^2} \qquad \left(\because \frac{\langle \alpha, \alpha \rangle}{\| \alpha \|^2} = 1 \right)$$

$$= \| \beta \|^2 - \frac{\langle \beta, \alpha \rangle \langle \alpha, \beta \rangle}{\| \alpha \|^2} = \| \beta \|^2 - \frac{\overline{\langle \alpha, \beta \rangle} \langle \alpha, \beta \rangle}{\| \alpha \|^2} \qquad [\because \langle \beta, \alpha \rangle = \overline{\langle \alpha, \beta \rangle}]$$

But by the definition of the norm ; $(\gamma, \gamma) \geq 0$

$$\therefore \| \beta \|^2 - \frac{\langle \alpha, \beta \rangle \overline{\langle \alpha, \beta \rangle}}{\| \alpha \|^2} \geq 0 \Rightarrow \| \beta \|^2 \geq \frac{| \langle \alpha, \beta \rangle |^2}{\| \alpha \|^2} \qquad (\because Z \bar{Z} = | Z |^2)$$

$$\Rightarrow \| \beta \|^2 \| \alpha \|^2 \geq | \langle \alpha, \beta \rangle |^2 \quad \Rightarrow (\| \beta \| \ \| \alpha \|)^2 \geq | \langle \alpha, \beta \rangle |^2 \qquad (\because \| \alpha \|^2 > 0)$$

$\therefore \| \beta \| \ \| \alpha \| \geq | \langle \alpha, \beta \rangle |$ as $\| \beta \| \ \| \alpha \|$ and $| \langle \alpha, \beta \rangle |$ are non-negative.

Hence $| \langle \alpha, \beta \rangle | \leq \| \alpha \| \ \| \beta \|$.

Note. For $\gamma \in V$, $\langle \gamma, \gamma \rangle = 0 \Rightarrow \gamma = \bar{0} \Rightarrow \beta - \frac{\langle \beta, \alpha \rangle}{\| \alpha \|^2} \alpha = \bar{0} \Rightarrow \beta = \frac{\langle \beta, \alpha \rangle}{\| \alpha \|^2} \alpha$

$\Rightarrow \beta$ vector = scalar multiple of the vector $\alpha \Rightarrow \alpha, \beta$ are linearly dependent.

Hence α, β are linearly dependent vectors of $V \Leftrightarrow | \langle \alpha, \beta \rangle | = \| \alpha \| \ \| \beta \|$. ***(N. U. 04,08)***

SPECIAL CASES OF SCHWARZ INEQUALITY

1. Consider the inner product space $V_n(C)$, the space of all ordered n- tuples defined over the field of complex numbers with the standard inner product.

Let $\alpha = (a_1, a_2, ..., a_n), \beta = (b_1, b_2, ..., b_n) \in V_n$ where $a_1, a_2, ..., a_n, b_1, b_2, ..., b_n$ are complex numbers.

By the definition of standard inner product,

$$\langle \alpha, \beta \rangle = a_1 \bar{b_1} + a_2 \bar{b_2} + ... + a_n \bar{b_n}, \ \| \alpha \|^2 = \langle \alpha, \alpha \rangle = a_1 \bar{a_1} + a_2 \bar{a_2} + ... + a_n \bar{a_n}$$

$$= | a_1 |^2 + | a_2 |^2 + ... + | a_n |^2 \text{ and } \| \beta \|^2 = \langle \beta, \beta \rangle = | b_1 |^2 + | b_2 |^2 + ... + | b_n |^2$$

By Schwarz inequality, $| \langle \alpha, \beta \rangle |^2 \leq \| \alpha \|^2 \ \| \beta \|^2$ which implies that

$$| a_1 \bar{b_1} + a_2 \bar{b_2} + ... + a_n \bar{b_n} |^2 \leq (| a_1 |^2 + | a_2 |^2 + ... + | a_n |^2) (| b_1 |^2 + | b_2 |^2 + ... + | b_n |^2)$$

This is called **Cauchy's inequality.**

If the field is the set of all real numbers R, then $\overline{Z} = Z$ and $|\overline{Z}|^2 = Z^2$.

So, the above inequality takes the form

$$(a_1b_1 + a_2b_2 + ... + a_nb_n)^2 \le (a_1^2 + a_2^2 + ... + a_n^2)(b_1^2 + b_2^2 + ... + b_n^2)$$

2. Consider the inner product space $V(C)$, the space of all continuous, complex valued functions on [0,1] with the inner product defined by $\langle f, g\rangle = \int_0^1 f(t)\,\overline{g(t)}\,dt$ for all $f, g \in V$.

$$\|f\|^2 = \langle f, f\rangle = \int_0^1 f(t)\,\overline{f(t)}\,dt = \int_0^1 |f(t)|^2\,dt \text{ and } \|g\|^2 = \langle g, g\rangle = \int_0^1 |g(t)|^2\,dt$$

By Schwarz inequality, $|\langle f, g\rangle|^2 \le \|f\|^2\ \|g\|^2$

$$\Rightarrow \left|\int_0^1 f(t)\,\overline{g(t)}\,dt\right|^2 \le \left(\int_0^1 |f(t)|^2\,dt\right)\left(\int_0^1 |g(t)|^2\,dt\right). \text{ Also, } \|f\| = \sqrt{\int_0^1 f(t)^2\,dt} = \max|f(t)|$$

Theorem 3. (Triangle inequality) ***In an inner product space*** $V(F)$, $\|\alpha + \beta\| \le \|\alpha\| + \|\beta\|$ ***for all*** $\alpha, \beta \in V$ ***(N. U. 11, O. U. 11, S. K. 05, K. U. 04, S. K. U 05)***

Proof. By definition of norm, $\|\alpha+\beta\|^2 = \langle\alpha+\beta, \alpha+\beta\rangle = \langle\alpha,\alpha\rangle + \langle\alpha,\beta\rangle + \langle\beta,\alpha\rangle + \langle\beta,\beta\rangle$

$= \|\alpha\|^2 + \langle\alpha,\beta\rangle + \overline{\langle\alpha,\beta\rangle} + \|\beta\|^2 = \|\alpha\|^2 + 2\,\text{Re}\,(\alpha,\beta) + \|\beta\|^2$ $(\because Z + \overline{Z} = 2\,\text{Re}\,Z)$

$\le \|\alpha\|^2 + 2|\langle\alpha,\beta\rangle| + \|\beta\|^2$ $(\because \text{Re}\,Z \le |Z|)$

$\le \|\alpha\|^2 + 2\|\alpha\|\ \|\beta\| + \|\beta\|^2$ $(\because |\langle\alpha,\beta\rangle| \le \|\alpha\|\ \|\beta\|)$

$\le (\|\alpha\| + \|\beta\|)^2$ $\therefore \|\alpha+\beta\|^2 \le (\|\alpha\| + \|\beta\|)^2$

As both $\|\alpha+\beta\|$ and $\|\alpha\| + \|\beta\|$ are non-negative we have $\|\alpha+\beta\| \le \|\alpha\| + \|\beta\|$.

Corollary. If α, β are vectors of an inner product space then $|\|\alpha\| - \|\beta\|| \le \|\alpha - \beta\|$.

Proof. $\|\alpha\| = \|\alpha - \beta + \beta\| \le \|\alpha - \beta\| + \|\beta\|$ By the above theorem

$\therefore \|\alpha\| - \|\beta\| \le \|\alpha - \beta\|$ ... (1)

Again $\|\beta\| = \|\beta - \alpha + \alpha\| \le \|\beta - \alpha\| + \|\alpha\|$

$\therefore \|\beta\| - \|\alpha\| \le \|\beta - \alpha\| = \|\alpha - \beta\|$ ... (2)

From (1) and (2) : $|\|\alpha\| - \|\beta\|| < \|\alpha - \beta\|$.

GEOMETRICAL INTERPRETATION FOR TRIANGLE INEQUALITY

In the inner product space R^3 with the standard inner product, let α, β be two vectors. In the 3D space R^3 we can write $\alpha = \overline{AB}, \beta = \overline{BC}$ so that $\|\alpha\| = AB$ and $\|\beta\| = BC$ where A,B,C are three points.

By triangle law of vectors in 3 - D space

we have $\overline{AB}+\overline{BC}=\overline{AC}$ so that $\alpha+\beta=\overline{AC}$. Then $\|\alpha+\beta\|=AC$.

From triangle inequality proved above, $\|\alpha+\beta\|\leq\|\alpha\|+\|\beta\|$ which implies that $AC\leq AB+BC$ in 3D space.

Theorem 4. (Parallelogram law) : ***If*** α,β ***are two vectors in an inner product space*** $V(F)$ ***then*** $\|\alpha-\beta\|^2+\|\alpha+\beta\|^2=2(\|\alpha\|^2+\|\beta\|^2)$.

(N.U. 11, S. K. U. 11, A. U. 11, K. U. 11, O. U. 08, K. U. 07, A. U. 08, S. K. D. 05, 00, N. U. 00)

Proof. $\|\alpha-\beta\|^2=\langle\alpha-\beta,\alpha-\beta\rangle=\langle\alpha,\alpha\rangle-\langle\alpha,\beta\rangle-\langle\beta,\alpha\rangle+\langle\beta,\beta\rangle$

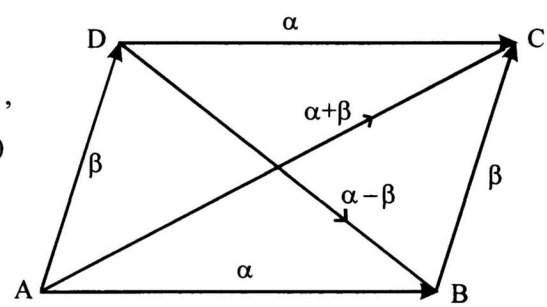
$$=\|\alpha\|^2-\langle\alpha,\beta\rangle-\langle\beta,\alpha\rangle+\|\beta\|^2$$

$\|\alpha+\beta\|^2=\langle\alpha+\beta,\alpha+\beta\rangle=\langle\alpha,\alpha\rangle+\langle\alpha,\beta\rangle+\langle\beta,\alpha\rangle+\langle\beta,\beta\rangle=\|\alpha\|^2+\langle\alpha,\beta\rangle+\langle\beta,\alpha\rangle+\|\beta\|^2$.

$\therefore\ \|\alpha-\beta\|^2+\|\alpha+\beta\|^2=2\|\alpha\|^2+2\|\beta\|^2=2(\|\alpha\|^2+\|\beta\|^2)$

GEOMETRICAL INTERPRETATION OF PARALLELOGRAM LAW

Let α,β be two vectors in the inner product space R^3 with standard inner product. In the 3D space R^3 we can take $\alpha=\overline{AB}$, $\beta=\overline{BC}$ so that $\|\alpha\|=AB,\|\beta\|=BC$ where A,B,C are points. Then the vectors $\alpha+\beta,\alpha-\beta$ represent the diagonals $\overline{AC},\overline{DB}$ of the parallelogram ABCD. So, $\|\alpha+\beta\|=AC$ and $\|\alpha-\beta\|=DB$



From the theorem of parallelogram law ,

$\|\alpha+\beta\|^2+\|\alpha-\beta\|^2=2(\|\alpha\|^2+\|\beta\|^2)$

$\Rightarrow\ AC^2+DB^2=2(AB^2+BC^2)$

$=AB^2+BC^2+CD^2+DA^2$

$\Rightarrow$ The sum of the squares on the diagonals of a parallelogram is equal to the sum of the squares on the four sides.

7.5. NORMED VECTOR SPACE AND DISTANCE

Definition. *Let* $V(F)$ *be an inner product space in which norm of a vector* $\alpha\in V$ *is defined as* $\|\alpha\|=\sqrt{\langle\alpha,\alpha\rangle}$. *The inner product space with this definition of norm is called a normed vector space if the following conditions are true :*

(*i*) $\|\alpha\|\geq 0$ and $\|\alpha\|=0\Leftrightarrow\alpha=\bar{0}$ (*ii*) $\|a\alpha\|=|a|\,\|\alpha\|$

and (*iii*) $\|\alpha+\beta\|\leq\|\alpha\|+\|\beta\|\ \forall\ \alpha,\beta\in V,\ a\in F$

As the above three conditions are true in every inner product space, every inner product space is a normed vector space.

Definition. *Let* α, β *be two vectors in an inner product space* $V(F)$. *The distance between the vectors* α, β *denoted by* $d\langle\alpha,\beta\rangle$ *is defined as* $\|\alpha-\beta\|$.

Note. 1. If $\alpha,\beta \in V$ then $d\langle\alpha,\beta\rangle = \|\alpha-\beta\| \Rightarrow d\left[\langle\alpha,\beta\rangle\right]^2 = \|\alpha-\beta\|^2 = \langle\alpha-\beta, \alpha-\beta\rangle$

2. $d\langle\alpha,\beta\rangle$ is a non-negative real number. **(A. U. 08)**

e.g. Let $\alpha = (a_1, a_2, a_3), \beta = (b_1, b_2, b_3)$ be two vectors in the inner product space R^3. Then $\alpha-\beta = (a_1-b_1, a_2-b_2, a_3-b_3)$ and $\|\alpha-\beta\|^2 = (a_1-b_1)^2 + (a_2-b_2)^2 + (a_3-b_3)^2$

$\therefore\ d\langle\alpha,\beta\rangle = \|\alpha-\beta\| = \sqrt{(a_1-b_1)^2 + (a_2-b_2)^2 + (a_3-b_3)^2}$.

Theorem. *If* $\alpha, \beta, \gamma \in V(F)$ *an inner product space then* **(1)** $d(\alpha,\beta) \geq 0$ *and* $d(\alpha,\beta) = 0 \Leftrightarrow \alpha = \beta$ **(2)** $d(\alpha,\beta) = d(\beta,\alpha)$ *and* **(3)** $d(\alpha,\beta) + d(\beta,\gamma) \geq d(\alpha,\gamma)$. **(A. U. 08)**

Proof. (1) By the definition, $d\langle\alpha,\beta\rangle = \|\alpha-\beta\| \geq 0$ since norm of a vector is a non-negative real number.

$d\langle\alpha,\beta\rangle = 0 \Leftrightarrow \|\alpha-\beta\| = 0 \Leftrightarrow \|\alpha-\beta\|^2 = 0 \Leftrightarrow \langle\alpha-\beta, \alpha-\beta\rangle = 0 \Leftrightarrow \alpha-\beta = 0$ *i.e.,* $\alpha = \beta$.

(2) $d\langle\alpha,\beta\rangle = \|\alpha-\beta\| = \|(-1)(\beta-\alpha)\| = |-1|\ \|\beta-\alpha\|$ $(\because \|\alpha\alpha\| = |\alpha|\ \|\alpha\|)$

$= (1)\|\beta-\alpha\| = d\langle\beta,\alpha\rangle$.

(3) $d\langle\alpha,\beta\rangle + d\langle\beta,\gamma\rangle = \|\alpha-\beta\| + \|\beta-\gamma\| \geq \|\alpha-\beta+\beta-\gamma\|$ By triangle inequality

$\geq \|\alpha-\gamma\| = d\langle\alpha,\gamma\rangle$.

Note. 1. In an inner product space $V(F)$ the distance function $d: V \to F$, defined as $d\langle\alpha,\beta\rangle = \|\alpha-\beta\|\ \forall\ \alpha,\beta \in V$ is satisfying the properties (1), (2), (3) of the metric space.

2. For $\alpha,\beta,\gamma \in V$; $d\langle\alpha+\gamma,\beta+\gamma\rangle = \|\alpha+\gamma-\beta-\gamma\| = \|\alpha-\gamma\| = d\langle\alpha,\gamma\rangle$.

SOLVED PROBLEMS

Ex. 1. *If* $\alpha = (2, 1, 1+i)$ *is a vector in* C^3 *with standard inner product find* $\|\alpha\|$ *and the unit vector of* α.

Sol. $\|\alpha\|^2 = \langle\alpha,\alpha\rangle = (2)(\bar{2}) + 1(\bar{1}) + (1+i)(\overline{1+i})$ $= (2)(2) + 1(1) + (1+i)(1-i) = 4+1+2 = 7$.

Unit vector of $\alpha = \dfrac{1}{\|\alpha\|}\alpha = \dfrac{1}{\sqrt{7}}(2, 1, 1+i)$.

Ex. 2. *If* $\alpha = (4, 1, 8), \beta = (1, 0, -3)$ *are two vectors in* R^3 *find the angle between* α *and* β.

Sol. $\|\alpha\| = \sqrt{4^2 + 1^2 + 8^2} = \sqrt{81} = 9$ and $\|\beta\| = \sqrt{1^2 + 0^2 + (-3)^2} = \sqrt{10}$.

$\langle \alpha, \beta \rangle = (4)(1) + (1)(0) + 8(-3) = -20$.

If $\theta =$ angle between α and β then $\cos\theta = \dfrac{|\langle \alpha, \beta \rangle|}{\|\alpha\| \|\beta\|} = \dfrac{|-20|}{9(\sqrt{10})} = \dfrac{20}{9(\sqrt{10})} = \dfrac{2}{9}\sqrt{10}$.

Ex. 3. *Let* $\alpha = (2, 1+i, i), \beta = (2-i, 2, 1+2i)$ *be two vectors in* $V_3(C)$ *or* C^3. *Compute* $\langle \alpha, \beta \rangle, \|\alpha\|, \|\beta\|$ *and* $\|\alpha + \beta\|$. *Also verity Cauchy - Schwarz inequality and Triangle inequality.*

Sol. $\langle \alpha, \beta \rangle = 2\overline{(2-i)} + (1+i)\bar{2} + i\overline{(1+2i)} = 2(2+i) + (1+i)2 + i(1-2i) = 8+5i$

$\langle \alpha, \alpha \rangle = 2(\bar{2}) + (1+i)\overline{(1+i)} + i(\bar{i}) = 2(2) + (1+i)(1-i) + i(-i) = 7$

$\langle \beta, \beta \rangle = (2-i)\overline{(2-i)} + 2(\bar{2}) + (1+2i)\overline{(1+2i)} = (2-i)(2+i) + 2(2) + (1+2i)(1-2i) = 14$

$\therefore \|\alpha\| = \sqrt{\langle \alpha, \alpha \rangle} = \sqrt{7}$ and $\|\beta\| = \sqrt{\langle \beta, \beta \rangle} = \sqrt{14}$.

$\alpha + \beta = (2+2-i, 1+i+2, i+1+2i) = (4-i, 3+i, 1+3i)$

$\langle \alpha+\beta, \alpha+\beta \rangle = (4-i)\overline{(4-i)} + (3+i)\overline{(3+i)} + (1+3i)\overline{(1+3i)}$

$= (4-i)(4+i) + (3+i)(3-i) + (1+3i)(1-3i) = (4^2+1^2) + (3^2+1^2) + (1^2+3^2) = 37$

$\therefore \|\alpha+\beta\| = \sqrt{\langle \alpha+\beta, \alpha+\beta \rangle} = \sqrt{37}$.

We have $|\langle \alpha+\beta \rangle| = |8+5i| = \sqrt{8^2+5^2} = \sqrt{89}$ and $\|\alpha\| \, \|\beta\| = \sqrt{7}\sqrt{14} = \sqrt{98}$

so that $|(\alpha+\beta)| < \|\alpha\| \, \|\beta\|$.

We have $\|\alpha+\beta\| = \sqrt{37}$ and $\|\alpha\| + \|\beta\| = \sqrt{7} + \sqrt{14}$ so that $\|\alpha+\beta\| < \|\alpha\| + \|\beta\|$.

Ex. 4. *In* $C([0,1])$, *if* $f(t) = t, g(t) = e^t$ *compute* $\langle f, g \rangle, \|f\|, \|g\|$ *and* $\|f+g\|$. *Also verify Cauchy - Schwarz and Triangle inequalities.*

Sol. Let $V = C([0,1])$ be the inner product space of real valued continuous functions on $[0,1]$ with the inner product of f and $g = \langle f, g \rangle = \int_0^1 f(t)g(t)\,dt$.

$$\langle f, g \rangle = \int_0^1 t\, e^t dt = e^t (t-1)\Big]_0^1 = 0 - (-1) = 1; \quad \langle f, f \rangle = \int_0^1 t \,.\, t\, dt = \int_0^1 t^2\, dt = \frac{1}{3}$$

$$\langle g, g \rangle = \int_0^1 e^t \,.\, e^t\, dt = \int_0^1 e^{2t}\, dt = \frac{1}{2} e^{2t} \Big]_0^1 = \frac{1}{2}(e^2 - 1)$$

$$\langle f+g, f+g \rangle = \int_0^1 (f(t)+g(t))^2\, dt = \int_0^1 (t+e^t)^2\, dt = \int_0^1 (t^2 + e^{2t} + 2t\, e^t)\, dt$$

$$= \left[\frac{t^3}{3} + \frac{e^{2t}}{2} + 2e^t(t-1)\right]_0^1 = \frac{1}{3} + \frac{1}{2}(e^2-1) + 2 = \frac{1}{6}(3e^2+11)$$

$$\| f \| = \sqrt{\langle f, f \rangle} = \sqrt{\frac{1}{3}}\ ;\ \| g \| = \sqrt{\langle g, g \rangle} = \sqrt{\frac{1}{2}(e^2-1)}\ ;$$

$$\| f+g \| = \sqrt{\langle f+g, f+g \rangle} = \sqrt{\frac{1}{16}(3e^2+11)}\ .$$

Ex. 5. *In the vector space $M_2(C)$, the set of all 2×2 matrices over the field of complex numbers if we define inner product as $\langle A, B \rangle = tr(B^* A)$ for $A, B \in M_2$; then compute $\| A \|$ and $\| B \|$ for $A = \begin{bmatrix} 1 & 2+i \\ 3 & i \end{bmatrix}$ and $B = \begin{bmatrix} 1+i & 0 \\ i & -i \end{bmatrix}$. Also find the angle between A and B.*

Sol. A^* = conjugate transpose of $A = \begin{bmatrix} 1 & 3 \\ 2-i & -i \end{bmatrix}$ and $B^* = \begin{bmatrix} 1-i & -i \\ 0 & i \end{bmatrix}$.

$$\langle A, A \rangle = tr(A^* A) = tr\left(\begin{bmatrix} 1 & 3 \\ 2-i & -i \end{bmatrix}\begin{bmatrix} 1 & 2+i \\ 3 & i \end{bmatrix}\right) = tr\begin{bmatrix} 10 & 2+4i \\ 2-4i & 6 \end{bmatrix} = 10+6 = 16\ .$$

$$\therefore \| A \| = \langle A, A \rangle = \sqrt{16} = 4\ .$$

$$\langle B, B \rangle = tr(B^* B) = tr\left(\begin{bmatrix} 1-i & -i \\ 0 & i \end{bmatrix}\begin{bmatrix} 1+i & 0 \\ i & -i \end{bmatrix}\right) = tr\begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix} = 3+1 = 4\ .$$

$$\therefore \| B \| = \sqrt{\langle B, B \rangle} = \sqrt{4} = 2\ .$$

If $\theta =$ angle between A and B then $\cos\theta = \dfrac{|\langle A, B \rangle|}{\| A \|\ \| B \|}$

$$\therefore \cos\theta = \frac{|-4i|}{(4)(2)} = \frac{4}{8} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$ (See Ex. 2 of the previous section)

Ex. 6. *By using Schwarz inequality prove that the cosine of an angle is of absoulte value at most* 1.

Sol. Consider the inner product space R^3 with standard inner product.

Let $\alpha = (a_1, a_2, a_3), \beta = (b_1, b_2, b_3) \in R^3$ and θ be the angle between the vectors α, β in the 3 D space. Then $\alpha \neq \bar{0}$ and $\beta \neq \bar{0}$.

From 3D geometry we know that

$$\cos\theta = \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}\ \sqrt{b_1^2 + b_2^2 + b_3^2}} = \frac{\langle \alpha, \beta \rangle}{\sqrt{\langle \alpha, \alpha \rangle}\ \sqrt{\langle \beta, \beta \rangle}} = \frac{\langle \alpha, \beta \rangle}{\| \alpha \| \| \beta \|}$$

By Schwarz inequality we have $|\langle \alpha, \beta \rangle| \leq \| \alpha \| \| \beta \| \Rightarrow \dfrac{\langle \alpha, \beta \rangle}{\| \alpha \| \| \beta \|} \leq 1$ $(\because \| \alpha \| . \| \beta \| > 0)$

But $|\cos\theta| = \dfrac{|\langle \alpha, \beta \rangle|}{\| \alpha \| \| \beta \|}$ and hence $|\cos\theta| \leq 1$.

Ex. 7. *If $V(R)$ is a vector space of polynomials with inner product defined by* $\langle f.g\rangle = \int_0^1 f(t)\,g(t)\,dt \;\forall\; f,g \in V$ *and* $t \in [0,1]$, *find* $\langle f.g\rangle$ *and* $\|g\|$ *for* $f(x) = x^2 + x - 4$, $g(x) = x - 1 \;\forall\; x \in [0,1]$. ***(S. V. U. 08)***

Sol. By definition of inner product ;

$$\langle f.g\rangle = \int_0^1 f(t)\,g(t)\,dt = \int_0^1 (t^2 + t - 4)(t-1)\,dt = \int_0^1 (t^3 - 5t + 4)\,dt = \frac{7}{4}$$

$$\|g\|^2 = \langle g,g\rangle = \int_0^1 g(t)\,g(t)\,dt = \int_0^1 (t-1)(t-1)\,dt = \int_0^1 (t-1)^2\,dt = \frac{1}{3} \qquad \therefore \|g\| = \frac{1}{\sqrt{3}}.$$

Ex. 8. *If α,β are two vectors in an inner product space, then α,β are linearly dependent if and only if* $|\langle\alpha,\beta\rangle| = \|\alpha\|\,\|\beta\|$. ***(N. U. 04, 07)***

Sol. Let α, β be linearly dependent.

Then either $\alpha = \bar{0}$ or $\beta = \bar{0}$ or $\alpha = a\beta$ where 'a' is a scalar.

When $\alpha = \bar{0}$: $\langle\alpha,\beta\rangle = \langle\bar{0},\beta\rangle = 0$ and $\|\alpha\| = 0$. $\therefore\; |\langle\alpha,\beta\rangle| = \|\alpha\|\,\|\beta\|$

When $\beta = \bar{0}$: $\langle\alpha,\beta\rangle = \langle\alpha,\bar{0}\rangle = \overline{\langle\bar{0},\alpha\rangle} = 0$ and $\|\beta\| = 0$ $\therefore\; |\langle\alpha,\beta\rangle| = \|\alpha\|\,\|\beta\|$

When $\alpha = a\beta$: $\langle\alpha,\beta\rangle = \langle a\beta,\beta\rangle = a\langle\beta,\beta\rangle = a\|\beta\|^2$ and $\|\alpha\| = \|a\beta\| = |a|\,\|\beta\|$

$\therefore\; |\langle\alpha,\beta\rangle| = |a|\,\|\beta\|^2 = (|a|\,\|\beta\|)(\|\beta\|) = \|\alpha\|\|\beta\|$.

Conversely, let $|\langle\alpha,\beta\rangle| = \|a\|\,\|\beta\|$.

When $\alpha = \bar{0}$ the vectors α, β are linearly dependent.

When $\alpha \neq \bar{0}$; we have $\|\alpha\| > 0$. Consider the vector $\gamma = \beta - \dfrac{\langle\beta,\alpha\rangle}{\|\alpha\|^2}\alpha$.

$$\langle\gamma,\gamma\rangle = \left\langle \beta - \frac{\langle\beta,\alpha\rangle}{\|\alpha\|^2}\alpha,\ \beta - \frac{\langle\beta,\alpha\rangle}{\|\alpha\|^2}\alpha \right\rangle = \langle\beta,\beta\rangle - \frac{\overline{\langle\beta,\alpha\rangle}}{\|\alpha\|^2}\langle\beta,\alpha\rangle - \frac{\langle\beta,\alpha\rangle}{\|\alpha\|^2}\langle\alpha,\beta\rangle + \frac{\langle\beta,\alpha\rangle}{\|\alpha\|^2}\frac{\overline{\langle\beta,\alpha\rangle}}{\|\alpha\|^2}\langle\alpha,\alpha\rangle$$

$$= \|\beta\|^2 - \frac{\overline{\langle\beta,\alpha\rangle}}{\|\alpha\|^2}\langle\beta,\alpha\rangle - \frac{\langle\beta,\alpha\rangle}{\|\alpha\|^2}\langle\alpha,\beta\rangle + \frac{\langle\beta,\alpha\rangle}{\|\alpha\|^2}\overline{\langle\beta,\alpha\rangle} \qquad (\because \langle\alpha,\alpha\rangle = \|\alpha\|^2)$$

$$= \|\beta\|^2 - \frac{|\langle\alpha,\beta\rangle|^2}{\|\alpha\|^2} = \|\beta\|^2 - \frac{\|\alpha\|^2\,\|\beta\|^2}{\|\alpha\|^2} = 0. \qquad \therefore \langle\gamma,\gamma\rangle = 0 \Rightarrow \gamma = \bar{0}$$

$$\Rightarrow \beta - \frac{\langle\beta,\alpha\rangle}{\|\alpha\|^2}\alpha = \bar{0} \Rightarrow \beta = \frac{\langle\beta,\alpha\rangle}{\|\alpha\|^2}\alpha \Rightarrow \beta = a\alpha \text{ where } a = \frac{\langle\beta,\alpha\rangle}{\|\alpha\|^2} \text{ is a scalar.}$$

$\therefore\; \alpha, \beta$ are linearly dependent.

Ex. 9. *Two vectors α, β in an unitary space $V(C)$ are such that* $\langle\alpha,\beta\rangle = 0$ *iff* $\|a\alpha + b\beta\|^2 = |a|^2\,\|\alpha\|^2 + |b|^2\,\|\beta\|^2 \;\forall\; a, b \in C$. ***(S. K. D. 11, 04)***

Sol. Let $\langle \alpha, \beta \rangle = 0$. Then $\overline{\langle \alpha, \beta \rangle} = 0$.

$$\| a\alpha + b\beta \|^2 = \langle a\alpha + b\beta, a\alpha + b\beta \rangle = a\langle \alpha, a\alpha + b\beta \rangle + b\langle \beta, a\alpha + b\beta \rangle$$

$$= a[\bar{a}\langle \alpha, \alpha \rangle + \bar{b}\langle \alpha, \beta \rangle] + b[\bar{a}\langle \beta, \alpha \rangle + \bar{b}\langle \beta, \beta \rangle] = a\bar{a}\langle \alpha, \alpha \rangle + a\bar{b}\langle \alpha, \beta \rangle + b\bar{a}\langle \beta, \alpha \rangle + b\bar{b}\langle \beta, \beta \rangle$$

$$= |a^2| \, \| \alpha \|^2 + a\bar{b}\langle \alpha, \beta \rangle + b\bar{a}\overline{\langle \alpha, \beta \rangle} + |b|^2 \| \beta \|^2 \quad \ldots. (1)$$

$$= |a|^2 \| \alpha \|^2 + 0 + 0 + |b|^2 \| \beta \|^2 = |a|^2 \| \alpha \|^2 + |b|^2 \| \beta \|^2$$

Conversely, Let $\| a\alpha + b\beta \|^2 = |a|^2 \| \alpha \|^2 + |b|^2 \| \beta \|^2 \ \forall \ a, b \in C$

Using (1) we have : $|a|^2 \| \alpha \|^2 + a\bar{b}\langle \alpha, \beta \rangle + b\bar{a}\overline{\langle \alpha, \beta \rangle} + |b|^2 \| \beta \|^2 = |a|^2 \| \alpha \|^2 + |b|^2 \| \beta \|^2$

$$\Rightarrow a\bar{b}\langle \alpha, \beta \rangle + b\bar{a}\overline{\langle \alpha, \beta \rangle} = 0 \ \forall \ a, b \in C \quad \ldots. (2)$$

Take $a = 1, b = 1$ so that $\bar{a} = 1, \bar{b} = 1$

Then (2) $\Rightarrow (1)(1)\langle \alpha, \beta \rangle + (1)(1)\overline{\langle \alpha, \beta \rangle} = 0 \Rightarrow \langle \alpha, \beta \rangle + \overline{\langle \alpha, \beta \rangle} = 0$

$$\Rightarrow 2\,\mathrm{Re}\langle \alpha, \beta \rangle = 0 \Rightarrow \mathrm{Re}\langle \alpha, \beta \rangle = 0 \qquad (\because \ z + \bar{z} = 2\,\mathrm{Re}\, z)$$

Take $a = i, b = 1$ so that $\bar{a} = -i, \bar{b} = 1$

Then (2) $\Rightarrow i\langle \alpha, \beta \rangle - i\overline{\langle \alpha, \beta \rangle} = 0 \Rightarrow i[\langle \alpha, \beta \rangle - \overline{\langle \alpha, \beta \rangle} = 0$

$$\Rightarrow \langle \alpha, \beta \rangle - \overline{\langle \alpha, \beta \rangle} = 0 \Rightarrow 2\,\mathrm{Im}\langle \alpha, \beta \rangle = 0 \qquad (\because z - \bar{z} = 2\,\mathrm{Im}\, z)$$

Thus we have $\mathrm{Re}\langle \alpha, \beta \rangle = 0$ and $\mathrm{Im}\langle \alpha, \beta \rangle = 0$. Hence $\langle \alpha, \beta \rangle = 0$.

Ex. 10. *If u, v are two vectors in a complex inner product space with standard inner product then prove that* $4\langle u, v \rangle = \| u + v \|^2 - \| u - v \|^2 + i \| u + iv \|^2 - i \| u - iv \|^2$.

Sol. $\| u + v \|^2 = \langle u + v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle$

$$= \| u \|^2 + \langle u, v \rangle + \langle v, u \rangle + \| v \|^2 \quad \ldots. (1)$$

$$\| u - v \|^2 = \langle u - v, u - v \rangle = \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle = \| u \|^2 - \langle u, v \rangle - \langle v, u \rangle + \| v \|^2 \ldots. (2)$$

$$\| u + iv \|^2 = \langle u + iv, u + iv \rangle = \langle u, u \rangle + \bar{i}\langle u, v \rangle + i\langle v, u \rangle + i\bar{i}\langle v, v \rangle$$

$$= \| u \|^2 - i\langle u, v \rangle + i\langle v, u \rangle + \| v \|^2 \qquad (\because \ i\bar{i} = i(-i) = 1)$$

$$i \| u + iv \|^2 = i \| u \|^2 + \langle u, v \rangle - \langle v, u \rangle + i \| v \|^2 \quad \ldots. (3)$$

$$\| u - iv \|^2 = \langle u - iv, u - iv \rangle = \langle u, u \rangle - \bar{i}\langle u, v \rangle - i\langle v, u \rangle + i\bar{i}\langle v, v \rangle = \| u \|^2 + i\langle u, v \rangle - i\langle v, u \rangle + \| v \|^2$$

$$i \| u - iv \|^2 = i \| u \|^2 - \langle u, v \rangle + \langle v, u \rangle + i \| v \|^2 \quad \ldots. (4)$$

From (1), (2), (3) and (4) : $\| u + v \|^2 - \| u - v \|^2 + i \| u + iv \|^2 - i \| u - iv \|^2$

$$= \{2\langle u, v \rangle + 2\langle v, u \rangle\} + \{2\langle u, v \rangle - 2\langle v, u \rangle\} = 4\langle u, v \rangle$$

EXERCISE 7 (*b*)

1. If *(i)* $\alpha = (1,-2,5) \in R^3(R)$ *(ii)* $\alpha = (3+i, 4i, -4) \in C^3(C)$ find $\| \alpha \|$ and unit vector of α. ***(S. V. U. 08)***

2. If $\alpha = (1-i, 2+3i), \beta = (2-5i, 3-i)$ are two vectors in $V(C) = C^2(C)$ with standard inner product find $\langle \alpha, \beta \rangle$ and $\| \alpha \|, \| \beta \|$.

3. If α, β are two vectors in Euclidean space $V(R)$ such that $\| \alpha \| = \| \beta \|$ prove that $\langle \alpha + \beta, \alpha - \beta \rangle = 0$. ***(S. V. U. 07)***

4. If α, β are two vectors in a unitary space find $\| \alpha + i\beta \|$ and $\| \alpha - i\beta \|$.

5. If α, β are two vectors in a unitary space prove that $\| \alpha + \beta \|^2 - \| \alpha - \beta \|^2 = 4 \operatorname{Re} \langle \alpha, \beta \rangle$.

6. If $M_2(R)$ is the inner product space and for $A = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}, B = \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \end{bmatrix} \in M_2$ the inner product is defined as $\langle A, B \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3 + a_4 b_4$ find the angle between $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$.

7. If $\alpha = (4,3,1,-2), \beta = (-2,1,2,3)$ be two vectors in the vector space $V_4(R)$ with standard inner product then find the angle between α and β.

8. In a real inner product space if α, β are two vectors such that $\| \alpha + \beta \| = \| \alpha \| + \| \beta \|$ then prove that α, β are linearly dependent.

ANSWERS

1. $\sqrt{30}, \sqrt{42}, \frac{1}{\sqrt{30}}(1,-2,5), \frac{1}{\sqrt{42}}(3+i, 4i, -4)$ **2.** $10+14i, \sqrt{27}, \sqrt{39}$

4. $\sqrt{\| \alpha \|^2 - i < \alpha, \beta > + i < \beta, \alpha > + \| \beta \|^2}$ $\sqrt{\| \alpha \|^2 + i < \alpha, \beta > - i < \beta, \alpha > + \| \beta \|^2}$

6. $\pi/2$ **7.** $\theta = \operatorname{Cos}^{-1} \sqrt{(3/20)}$

8
Orthogonality

8.1. We now introduce the concept of orthogonality in an inner product space which is analogous to perpendicularity in Geometry. In 3-D geometry we know that the three unit vectors $\bar{i}, \bar{j}$ and $\bar{k}$ along the three mutually perpendicular coordinate axes form a base for any vector. Just as bases are building blocks of vector spaces, the basis vectors forming an orthonormal set are the building blocks of inner product spaces. By Gram - Schmidt orthogonalisation process we now establish that every finite dimensional inner product space has an orthonormal basis.

8.2. ORTHOGONAL AND ORTHONORMAL VECTORS

Definition. *Let $V(F)$ be an inner product space and $\alpha, \beta \in V$. α is said to be orthogonal to β if $\langle \alpha, \beta \rangle = 0$.* ***(O. U. 00)***

α is orthogonal to $\beta \Leftrightarrow \langle \alpha, \beta \rangle = 0 \Leftrightarrow \langle \beta, \alpha \rangle = \overline{\langle \alpha, \beta \rangle} = 0$ ($\because$ the conjugate of 0 is 0)

$\Leftrightarrow \beta$ is orthogonal to α.

So, we say that $\alpha, \beta \in V$ are orthogonal, if and only if, $\langle \alpha, \beta \rangle = 0$.

Note. 1. If $\bar{0} \in V$ is the zero vector then $(\bar{0}, \alpha) = 0$ for all $\alpha \in V$.

Therefore the zero vector is orthogonal to every vector in V and $\bar{0}$ is the only vector in V with this property.

2. Since $(\bar{0}, \bar{0}) = 0$ for $\bar{0} \in V$, the zero vector in V is orthogonal to itself.

e.g. 1. Let $\alpha = (0, -2, 3), \beta = (7/2, 3, 2)$ be two vectors of the inner product space R^3 with standard inner product.

Then $\langle \alpha, \beta \rangle = (0)(7/2) + (-2)(3) + (3)(2) = 0$. $\therefore$ α, β are orthogonal vectors.

2. Consider the inner product space $V(C)$ of continuous complex valued functions on [0,1] with the inner product defined as : $\langle f, g \rangle = \int_0^1 f(t)\, \overline{g(t)}\, dt$ for $f(t), g(t) \in V$.

For $f(t) = 2\sin \pi t, g(t) = \cos \pi t$ we have

$$\langle f, g \rangle = \int_0^1 2\sin \pi t \cos \pi t \, dt = \int_0^1 \sin 2\pi t \, dt = -\frac{1}{2\pi}\left[\cos 2\pi t\right]_0^1 = 0.$$

$\therefore$ $2\sin \pi t, \cos \pi t$ are orthogonal vectors.

Theorem. 1. *If the vectors α, β of an inner product space $V(F)$ are orthogonal then (1) $a\alpha$ is orthogonal to β and (2) α is orthogonal $a\beta$ for any $a \in F$.*

Proof. α, β are orthogonal $\Rightarrow \langle \alpha, \beta \rangle = 0$.

$\langle a\alpha, \beta \rangle = a\langle \alpha, \beta \rangle = a\,0 = 0 \ \forall \ a \in F \Rightarrow a\alpha$ is orthogonal to β.

$\langle \alpha, a\beta \rangle = \bar{a}\langle \alpha, \beta \rangle = \bar{a}\,0 = 0 \ \forall \ a \in F \Rightarrow \alpha$ is orthogonal to $a\beta$.

Imp. Note. For $a \neq 0 \in F, \langle a\alpha, \beta \rangle = \langle \alpha, a\beta \rangle = 0$ whenever $\langle \alpha, \beta \rangle = 0$.

That is, the orthogonality of two vectors is not affected by multiplying vectors by non-zero scalar. The process of multiplying a nonzero vector by the reciprocal of its length is called **Normalising**.

Theorem. 2. *The vectors α, β of a real inner product space $V(F)$ are orthogonal if and only if, $\| \alpha + \beta \|^2 = \| \alpha \|^2 + \| \beta \|^2$.*

Proof. $\| \alpha + \beta \|^2 = \| \alpha \|^2 + \| \beta \|^2 \quad \Leftrightarrow \langle \alpha + \beta, \alpha + \beta \rangle = \| \alpha \|^2 + \| \beta \|^2$

$\Leftrightarrow \langle \alpha, \alpha \rangle + \langle \alpha, \beta \rangle + \langle \beta, \alpha \rangle + \langle \beta, \beta \rangle = \| \alpha \|^2 + \| \beta \|^2$

$\Leftrightarrow \| \alpha \|^2 + \langle \alpha, \beta \rangle + \overline{\langle \alpha, \beta \rangle} + \| \beta \|^2 = \| \alpha \|^2 + \| \beta \|^2$

$\Leftrightarrow \langle \alpha, \beta \rangle + \langle \alpha, \beta \rangle = 0 \qquad (\because \langle \alpha, \beta \rangle \text{ is real } \overline{\langle \alpha, \beta \rangle} = \langle \alpha, \beta \rangle)$

$\Leftrightarrow \langle \alpha, \beta \rangle = 0 \Leftrightarrow \alpha, \beta$ are orthogonal vectors.

GEOMETRICAL INTERPRETATION

Let α, β be two vectors in the inner product space R^3. In the 3-D space we can take $\alpha = \overline{AB}, \beta = \overline{BC}$ so that $\| \alpha \| = AB, \| \beta \| = BC$.

Then the vector $\alpha + \beta$ represent the third side $\overline{AC}$ of the triangle ABC so that $\| \alpha + \beta \| = AC$.

$\| \alpha + \beta \|^2 = \| \alpha \|^2 + \| \beta \|^2 \Leftrightarrow AC^2 = AB^2 + BC^2 \Leftrightarrow \angle ABC = 90^\circ$ (By Pythagoras theorem)

$\Leftrightarrow \overline{AB}, \overline{BC}$ are orthogonal $\Leftrightarrow \alpha, \beta$ are orthogonal.

$\therefore$ The above result is the Pythagoras theorem concerning a triangle in 3 - D space.

Note. The above theorem does not hold in complex inner product space.

If $\alpha = (0, i), \beta = (0,1)$ in $V_2(C)$ then

$$\| \alpha + \beta \|^2 = \langle \alpha + \beta, \alpha + \beta \rangle = \| \alpha \|^2 + \langle \alpha, \beta \rangle + \langle \beta, \alpha \rangle + \| \beta \|^2$$

$$= \| \alpha \|^2 + \| \beta \|^2 + \langle \alpha, \beta \rangle + \overline{\langle \alpha, \beta \rangle} = \| \alpha \|^2 + \| \beta \|^2 + 2\,\mathrm{Re}\,\langle \alpha, \beta \rangle \quad (\because z + \bar{z} = 2\,\mathrm{Re}\,z)$$

But $\langle \alpha, \beta \rangle = 0\ \bar{0} + i\bar{1} = 0 + i = i \neq 0$ and $\text{Re}(\alpha, \beta) = 0$ $(\because\ \bar{0} = 0, \bar{1} = 1)$

$\therefore\ \| \alpha + \beta \|^2 = \| \alpha \|^2 + \| \beta \|^2$ and $\langle \alpha, \beta \rangle \neq 0$.

Hence $\| \alpha + \beta \|^2 = \| \alpha \|^2 + \| \beta \|^2$ does not imply $\langle \alpha, \beta \rangle = 0$.

SOLVED PROBLEMS

Ex. 1. *Find a unit vector orthogonal to* $(4, 2, 3)$ *in* R^3.

(O. U. 11, N. U. 08, A. U. 08, S. K.D. 04, O. U. 01)

Sol. Let $\alpha = (4, 2, 3)$. Let $\beta = (x_1, x_2, x_3)$ be orthogonal to α.

Then $\langle \alpha, \beta \rangle = 0 \Rightarrow 4x_1 + 2x_2 + 3x_3 = 0$.

Any solution of this equation gives a vector orthogonal to α.

By inspection, a solution is $x_1 = 2, x_2 = -1, x_3 = -2$

We can take $\beta = (2, -1, -2)$. Then $\| \beta \| = \sqrt{4 + 1 + 4} = 3$.

$\therefore$ Unit vector orthogonal to $\alpha = \dfrac{1}{\| \beta \|} \beta = \left(\dfrac{2}{3}, -\dfrac{1}{3}, -\dfrac{2}{3} \right)$

Ex. 2. *In a real inner product space, if* u, v *are two vectors such that* $\| u \| = \| v \|$ *then prove that* $u - v, u + v$ *are orthogonal. Interpret the result geometrically.*

(S. V. U. 08, A. U. 03, O. U. 01)

Sol. $\langle u - v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle - \langle v, u \rangle - \langle v, v \rangle$

$= \| u \|^2 + \langle u, v \rangle - \overline{\langle u, v \rangle} - \| v \|^2 = \| u \|^2 - \| v \|^2$ $(\because\ \overline{\langle u, v \rangle} = \langle v, u \rangle = \langle u, v \rangle)$

$= 0$. $(\because\ \| u \| = \| v \|)$. $\therefore\ u - v, u + v$ are orthogonal vectors.

In the 3 - D space, let $u = \overline{AB}$ and $v = \overline{BC}$. Then $\| u \| = \| v \| \Rightarrow AB = BC$.

In the rhombus $ABCD$, $u + v = \overline{AC}$, $\overline{DB} = u - v$, where AC, BD are diagonals.

$u - v, u + v$ are orthogonal $\Rightarrow$ the diagonals are perpendicular.

Hence, in a rhombus, the diagonals are perpendicular.

Ex. 3. *If* α, β *be two orthogonal vectors in an inner product space* $V(F)$ *and* $\| \alpha \| = \| \beta \| = 1$ *then prove that* $\| \alpha - \beta \| = d \langle \alpha, \beta \rangle = \sqrt{2}$.

Sol. $\| \alpha - \beta \|^2 = \langle \alpha - \beta, \alpha - \beta \rangle = \langle \alpha, \alpha \rangle - \langle \alpha, \beta \rangle - \langle \beta, \alpha \rangle + \langle \beta, \beta \rangle$

$= \| \alpha \|^2 - 0 - 0 + \| \beta \|^2$ $(\because\ \alpha, \beta$ are orthogonal)

$= 1 + 1 = 2$ $(\because\ \| \alpha \| = \| \beta \| = 1)$ $\therefore\ d \langle \alpha, \beta \rangle = \| \alpha - \beta \| = \sqrt{2}$.

Ex. 4. *Let $V_3(R)$ be the inner product space with standard inner product. Find two linearly independent vectors each of which is orthogonal to* $(1,1,2)$. **(K. U. 2014)**

Sol. Let $\alpha = (1,1,2)$. Let each of $\beta = (x_1, x_2, x_3)$ and $\gamma = (y_1, y_2, y_3)$ be orthogonal to α.

Then $< \alpha, \beta > = 0 \Rightarrow x_1 + x_2 + 2x_3 = 0$ ... (1)

$< \alpha, \gamma > = 0 \Rightarrow y_1 + y_2 + 2y_3 = 0$ ... (2)

β, γ are linearly independent $\Rightarrow \beta \neq k\gamma$ where

$k \in R \Rightarrow (x_1, x_2, x_3) \neq k(\beta_1, \beta_2, \beta_3)$ ... (3)

We have to choose solutions of equations (1) and (2) by inspection satisfying condition (3).

$\therefore \beta = (-1,-1,1), \gamma = (4,-2,1)$

8.3. ORTHOGONAL AND ORTHONORMAL SETS OF INNER PRODUCT SPACE

Definition. *Let S be a non-empty subset of an inner product space $V(F)$.*

The set S is said to be an orthogonal set if every pair of distinct vectors are orthogonal.

Or, $S \subset V$ is an orthogonal set if $\langle \alpha_i, \alpha_j \rangle = 0 \ \forall \ \alpha_i, \alpha_j \in S$ and $i \neq j$. **(S.K.D. 05)**

Definition. *Let S be a non-empty subset of an inner product space $V(F)$.*

The set S is said to be an orthonormal set if (i) $\| \alpha_i \| = 1$ for each $\alpha_i \in S$ and (ii) $(\alpha_i, \alpha_j) = 0$ for $\alpha_i, \alpha_j \in S, i \neq j$.

$S \subset V$ is an orthonormal set if $(\alpha_i, \alpha_j) = 1$ for $i = j$ and $(\alpha_i, \alpha_j) = 0$ for $i \neq j$ where $\alpha_i, \alpha_j \in S$. That is $\langle \alpha_i, \alpha_j \rangle = \delta_{ij}$ (Kronecker delta). **(S. K. D. 05, N. U. 97)**

Note. 1. $S \subset V$ is an orthonormal set $\Leftrightarrow S$ contains mutually orthogonal unit vectors.

2. An orthonormal set is an orthogonal set with the property that each vector is of length 1.

3. An orthonormal set does not conatin zero vector.

e.g.1. If $\alpha \in V$ is a nonzero vector then $\frac{1}{\| \alpha \|} \alpha$ is unit vector. Therefore $\left\{ \frac{1}{\| \alpha \|} \alpha \right\}$ is an orthonormal set in V. Thus, every inner product space which is not a null space has an orthonormal subset.

e.g. 2. The standard basis of the inner product space R^3 or $V_3(R)$ is the set $\{e_1 = (1,0,0), e_2 = (0,1,0), e_3 = (0,0,1)\}$. **(S. K. D. 08)**

Then $\| e_1 \| = 1, \| e_2 \| = 1, \| e_3 \| = 1$ and $\langle e_1, e_2 \rangle = 0, \langle e_2, e_3 \rangle = 0, \langle e_1, e_3 \rangle = 0$.

Thus the standard basis form an orthonormal set.

Theorem 1. ***In an inner product space, any orthogonal set of non-zero vectors is linearly independent.*** ***(S. V. U. 07)***

Proof. Let S be a finite or infinite orthogonal set of non-zero vectors in an inner product spare $V(F)$.

Then $\alpha_i, \alpha_j \in S \Rightarrow \alpha_i \neq \bar{0}, \alpha_j \neq \bar{0}$ and $(\alpha_i, \alpha_j) = 0$ for $i \neq j$.

Let $T = \{\alpha_1, \alpha_2, ..., \alpha_m\}$ be a finite subset of S having m distinct vectors.

Let there exist $a_1, a_2, ..., a_m \in F$ such that $a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m = \bar{0}$.

For any $i \in \{1, 2, ..., m\}$, $(a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m, \alpha_i)$

$= a_1 (\alpha_1, \alpha_i) + a_2 (\alpha_2, \alpha_i) + ... + a_i (\alpha_i, \alpha_i) + ... + a_m(\alpha_m, \alpha_i)$

$= a_i (\alpha_i, \alpha_i)$ since $(\alpha_j, \alpha_i) = 0$ for $j \neq i$

$\therefore a_1, a_2, ..., a_m \in F$ and $\alpha_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m = \bar{0}$

$\Rightarrow \langle \bar{0}, \alpha_i \rangle = a_i \langle \alpha_i, \alpha_i \rangle$ for each $i \in \{1, 2, ..., m\}$

$\Rightarrow 0 = a_i \langle \alpha_i, \alpha_i \rangle$ for each $i \in \{1, 2, ..., m\}$

$\Rightarrow a_i = 0$ for each $i \in \{1, 2, ..., m\}$ $(\because \alpha_i \neq \bar{0}, \langle \alpha_i, \alpha_i \rangle \neq 0)$

Hence the set T is linearly independent.

Thus every non-zero finite subset T of S is linearly independent.

$\therefore$ The set S is linearly independent.

Note. If $\beta = a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m$ then for each $i \in \{1, 2, ..., m\}$

we have $\langle \beta, \alpha_i \rangle = a_i (\alpha_i, \alpha_i) = \alpha_i \| \alpha_i \|^2$ so that $a_i = \dfrac{\langle \beta, \alpha_i \rangle}{\| \alpha_i \|^2}$.

Therefore $\beta = a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m = \sum_{i=1}^{m} a_i\alpha_i = \sum_{i=1}^{m} \dfrac{\langle \beta, \alpha_i \rangle}{\| \alpha_i \|^2} \alpha_i$.

Corollary 1. ***In an inner product space any orthonormal set of vectors is linearly independent.*** ***(S. V. U. M 07, S. K. U. M 07, O. U. 99)***

Proof. Let S be a finite or infinite orthonormal set of vectors in an inner product space $V(F)$.

Then $\alpha_i, \alpha_j \in S \Rightarrow (\alpha_i, \alpha_j) = 1$ for $i = j$ and $(\alpha_i, \alpha_j) = 0$ for $i \neq j$.

Let $T = \{\alpha_1, \alpha_2, ..., \alpha_m\}$ be a finite subset of S having m distinct vectors.

$\therefore a_1, a_2, ..., a_m \in F$ and $a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m = \bar{0}$

$\Rightarrow (\bar{0}, \alpha_i) = a_i$ for each $i \in \{1, 2, ..., m\}$

$\Rightarrow a_i = 0 \ \forall \ i \in \{1, 2, ..., m\} \Rightarrow T$ is linearly independent

$\therefore S$ is linearly independent.

Corollary. 2. ***If S is an orthonormal set of vectors of an inner product space*** $V(F)$, ***then*** $\beta = a_1\alpha_1 + a_2\alpha_2 + ... + a_n\alpha_n \Rightarrow a_i = \langle \beta, \alpha_i \rangle$ ***for*** $i = 1, 2, ..., n$. **(N. U. 97)**

Proof. S is an orthonormal set $\Rightarrow$ for $\alpha_i, \alpha_j \in S$

$\| \alpha_i \| = 1, \| \alpha_j \| = 1$ and $(\alpha_i, \alpha_j) = 0$ when $i \neq j$.

Then, for each $i = 1, 2, ..., n$; $(\beta, \alpha_i) = (a_1\alpha_1 + a_2\alpha_2 + ... + a_n\alpha_n, a_i) = a_i(\alpha_i, \alpha_i) = a_i$

$\therefore \beta = a_1\alpha_1 + a_2\alpha_2 + ... + a_n\alpha_n = \sum_{i=1}^{n} a_i \alpha_i = \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \alpha_i$.

Theorem. 2. ***Let*** $V(F)$ ***be an inner product space and*** $S = \{\alpha_1, \alpha_2, ..., \alpha_m\}$ ***be an orthogonal subset of*** V ***consisting non - zero vectors. If*** $\beta \in$ ***span*** $S = L(S)$ ***then***

$$\beta = \sum_{i=1}^{m} \frac{\langle \beta, \alpha_i \rangle}{\| \alpha_i \|^2} \alpha_i \, .$$ **(K. U. 07, 08)**

Proof. $S = \{\alpha_1, \alpha_2, ..., \alpha_m\}$ is an orthogonal subset of $V \Rightarrow \langle \alpha_i, \alpha_j \rangle = 0$ for $i \neq j$.

Take $\beta = a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m = \sum_{i=1}^{m} a_i\alpha_i$ where $a_1, a_2, ..., a_m \in F$.

For $1 \leq j \leq m$ we have $\langle \beta, \alpha_j \rangle = \langle a_1\alpha_1 + a_2\alpha_2 + ... + a_j\alpha_j + ... + a_m\alpha_m \rangle$

$= a_1 \langle \alpha_i, \alpha_j \rangle + a_2 \langle \alpha_2, \alpha_j \rangle + ... + a_j \langle \alpha_j, \alpha_j \rangle + ... + a_m \langle \alpha_m, \alpha_j \rangle$

$= a_j \langle \alpha_j, \alpha_j \rangle = a_j \| \alpha_j \|^2$. $\quad \therefore a_j = \dfrac{\langle \beta, \alpha_j \rangle}{\| \alpha_j \|^2}$ for $j = 1, 2, ..., m$.

$$\therefore \beta = \frac{\langle \beta, \alpha_1 \rangle}{\| \alpha_1 \|^2} \alpha_1 + \frac{\langle \beta, \alpha_2 \rangle}{\| \alpha_2 \|^2} \alpha_2 + ... + \frac{\langle \beta, \alpha_m \rangle}{\| \alpha_m \|^2} \alpha_m = \sum_{i=1}^{m} \frac{\langle \beta, \alpha_i \rangle}{\| \alpha_i \|^2} \alpha_i \, .$$

Note. 1. We obtained the result of the theorem already in the note of Theorem (1).

2. If $\beta = a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m = \sum_{i=1}^{m} a_i \alpha_i = \bar{0}$

then $\forall j = 1, 2, ..., m$ we have $a_j = \dfrac{\langle \bar{0}, \alpha_j \rangle}{\| \alpha_j \|^2} = 0$.

Therefore $S = \{\alpha_1, \alpha_2, ..., \alpha_m\}$ is linearly independent which is the result of the Theorem (1) of Art. 2.3.

Corollary. ***Let*** $V(F)$ ***be an inner product space and*** $S = \{\alpha_1, \alpha_2, ..., \alpha_m\}$ ***be an orthonormal subset of*** V. ***If*** $\beta \in$ ***span*** $S = L(S)$ ***then*** $\beta = \sum_{i=1}^{m} \langle \beta, \alpha_i \rangle \alpha_i$. **(K. U. 08)**

Proof. $S = \{\alpha_1, \alpha_2, ..., \alpha_m\}$ is an orthonormal subset of $V \Rightarrow \langle \alpha_i, \alpha_j \rangle = 0$ for $i \neq j$ and $\langle \alpha_i, \alpha_j \rangle = 1$ for $i = j$.

In the proof of the above Theorem; we got $a_j = \langle \beta, \alpha_j \rangle$ for $j = 1, 2, ..., m$

$\therefore\ \beta = \sum_{i=1}^{m} \langle \beta, \alpha_i \rangle \alpha_i = \langle \beta, \alpha_1 \rangle \alpha_1 + \langle \beta, \alpha_2 \rangle \alpha_2 + ... + \langle \beta, \alpha_m \rangle \alpha_m$.

Note. 1. If $S = \{\alpha_1, \alpha_2, ..., \alpha_m\}$ is an orthonormal basis of the inner product space $V(F)$ and $\beta \in L(S)$ then $\beta = \sum_{i=1}^{m} \langle \beta, \alpha_i \rangle \alpha_i = \sum_{i=1}^{m} a_i \alpha_i$. So, the coefficients in the linear combination of $\beta = a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m$ can be computed by $a_i = \langle \beta, \alpha_i \rangle$ for $i = 1, 2, ..., m$.

Theorem. 3. ***If*** $\{\alpha_1, \alpha_2, ..., \alpha_n\}$ ***is an orthonormal set in an inner product space*** $V(F)$ ***and*** $\beta \in V$ ***then*** $\gamma = \beta - \langle \beta, \alpha_1 \rangle \alpha_1 - \langle \beta, \alpha_2 \rangle \alpha_2 .... - \langle \beta, \alpha_n \rangle \alpha_n$ ***is orthogonal to each of*** $\alpha_1, \alpha_2, ..., \alpha_n$. ***(O. U. 99)***

Proof. $\{\alpha_1, \alpha_2, ..., \alpha_n\}$ is an orthonormal set $\Rightarrow \langle \alpha_i, \alpha_i \rangle = 1$ and $\langle \alpha_i, \alpha_j \rangle = 0$ if $i \neq j$ for $i, j \in \{1, 2, ..., n\}$.

For any $i \in \{1, 2, ..., n\}$ we have $\langle \gamma, \alpha_i \rangle = \left\langle \beta - \sum_{j=1}^{n} (\beta, \alpha_j)\, \alpha_j, \alpha_i \right\rangle$

$= \langle \beta, \alpha_i \rangle - \left\langle \sum_{j=1}^{n} \langle \beta, \alpha_j \rangle \alpha_j, \alpha_i \right\rangle = \langle \beta, \alpha_i \rangle - \left\langle \sum_{j=1}^{n} \langle \beta, \alpha_j \rangle \langle \alpha_j, \alpha_i \rangle \right\rangle$

$= \langle \beta, \alpha_i \rangle - \langle \beta, \alpha_i \rangle \langle \alpha_i, \alpha_i \rangle \qquad (\because\ (\alpha_j, \alpha_i) = 0 \text{ for } j \neq i)$

$= \langle \beta, \alpha_i \rangle - \langle \beta, \alpha_i \rangle = 0 \qquad (\because\ (\alpha_i, \alpha_i) = 1)$

$\therefore\ \gamma$ is orthogonal to each of $\alpha_1, \alpha_2, ..., \alpha_n$.

Note. The vector γ is orthogonal to each vector of the sub-space spanned by S, that is, γ is orthogonal to each vector of $L(S)$.

If u is any vector in $L(S)$, then $u = a_1\alpha_1 + a_2\alpha_2 + ... + a_m\alpha_m$ where $a_1, a_2, ..., a_m \in F$, $\alpha_1, \alpha_2, ..., \alpha_m \in S$.

Then $\langle \gamma, u \rangle = \left\langle \gamma, \sum_{i=1}^{m} a_i \alpha_i \right\rangle = \sum_{i=1}^{m} \overline{a_i} \langle \gamma, \alpha_i \rangle = \sum_{i=1}^{m} \overline{a_i}\, 0 = 0$ so that γ is orthogonal to u.

SOLVED PROBLEMS

Ex. 1. *In the inner product space* $R^3(R)$ *w.r.t. the standard inner product, prove that the set* $X = \{(1,1,0), (1,-1,1), (-1,1,2)\}$ *is an orthogonal set* ***(A. U. 11)***

Sol. Let $\alpha = (1,1,0), \beta = (1,-1,1), \gamma = (-1,1,2)$

$< \alpha, \beta > = 1 - 1 + 0 = 0, < \beta, \gamma > = -1 - 1 + 2 = 0, < \alpha, \gamma > = -1 + 1 + 0 = 0$

$\therefore$ $X = \{\alpha, \beta, \gamma\}$ is an orthogonal set.

Ex. 2. *Prove that* $S = \left\{\left(\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3}\right), \left(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\right), \left(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right)\right\}$ *is an orthonormal set in* R^3 *with standard inner product.* **(*S. K. U. M 15, 05, O. U. M 15*)**

Sol. Let $\alpha = \left(\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3}\right)$, $\beta = \left(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\right)$, $\gamma = \left(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3}\right)$.

$\|\alpha\| = \sqrt{\frac{1}{9} + \frac{4}{9} + \frac{4}{9}} = 1$, $\|\beta\| = \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{4}{9}} = 1$, $\|\gamma\| = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9}} = 1$.

$\langle \alpha, \beta \rangle = \frac{2}{9} + \frac{2}{9} - \frac{4}{9} = 0$; $\langle \beta, \gamma \rangle = \frac{4}{9} - \frac{2}{9} - \frac{2}{9} = 0$ and $\langle \gamma, \alpha \rangle = \frac{2}{9} - \frac{4}{9} + \frac{2}{9} = 0$.

$\therefore$ S is an orthonormal set.

Ex. 3. *Normalise* $\alpha = \left(\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}\right)$ *in the inner product space* $R^3(R) = V_3(R)$.

Sol. $\|\alpha\|^2 = \langle \alpha, \alpha \rangle = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) + \left(\frac{1}{3}\right)\left(\frac{1}{3}\right) + \left(-\frac{1}{4}\right)\left(-\frac{1}{4}\right) = \frac{61}{144} \Rightarrow \|\alpha\| = \frac{\sqrt{61}}{12}$.

Normalising α means obtaining unit vector of α

$= \frac{1}{\|\alpha\|}\alpha = \frac{12}{\sqrt{61}}\left(\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}\right) = \left(\frac{6}{\sqrt{61}}, \frac{4}{\sqrt{61}}, \frac{-3}{\sqrt{61}}\right)$.

Ex. 4. *If* $S = \{(1, 1, 0), (1, -1, 1), (-1, 1, 2)\}$ *is an orthogonal set of the innerproduct space* $R^3(R)$ *find the orthonormal set.*

Sol. If $\alpha_1 = (1,1,0)$ then $\|\alpha_1\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$; $\alpha_2 = (1,-1,1)$ then

$\|\alpha_2\| = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3}$ and $\alpha_3 = (-1,1,2)$ then $\|\alpha_3\| = \sqrt{(-1)^2 + 1^2 + 2^2} = \sqrt{6}$.

$\therefore$ Orthonormal set of $S = \left\{\frac{\alpha_1}{\sqrt{2}}, \frac{\alpha_2}{\sqrt{3}}, \frac{\alpha_3}{\sqrt{6}}\right\} = \left\{\frac{1}{\sqrt{2}}(1, 1, 0), \frac{1}{\sqrt{3}}(1, -1, 1), \frac{1}{\sqrt{6}}(-1, 1, 2)\right\}$.

Ex. 5. *If* $S = \{u_1, u_2, ..., u_n\}$ *is an orthogonal set of an inner product space* $V(F)$ *then prove that* $\{a_1u_1, a_2u_2, ..., a_nu_n\}$ *is also an orthogonal set for any choice of non-zero scalars* $a_1, a_2, ..., a_n \in F$.

Sol. $S = \{u_1, u_2, ..., u_n\}$ is an orthogonal set $\Rightarrow \langle u_i, u_j \rangle = 0 \ \forall \ u_i, u_j \in S$ and $i \neq j$.

Consider $a_iu_i, a_ju_j \in S' = \{a_1u_1, a_2u_2, ..., a_nu_n\}$.

Then $\langle a_iu_i, a_ju_j \rangle = a_i \overline{a_j} \langle u_i, u_j \rangle = a_i \overline{a_j}(0) = 0$ for $i \neq j$.

$\therefore$ $\langle a_iu_i, a_ju_j \rangle = 0 \ \forall \ a_iu_i, a_ju_j \in S'$ and $i \neq j \Rightarrow S'$ is also an orthogonal set.

Ex. 6. *If* $S=\left\{\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}},0\right),\left(\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right),\left(-\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}}\right)\right\}$ *is an orthonormal subset of the inner product space* $R^3(R)$, *express the vector* $(2,1,3)$ *as a linear combination of the basis vectors of S.*

Sol. Let $S=\{\alpha_1,\alpha_2,\alpha_3\}$ and $\beta=(2,1,3)$.

Let $\beta=a_1\alpha_1+a_2\alpha_2+a_3\alpha_3$ where $a_1,a_2,a_3\in R$.

Using the note under corollary of the Theorem (2) :

$a_1=\langle\beta,\alpha_1\rangle=(2)(1/\sqrt{2})+(1)(1/\sqrt{2})+(3)(0)=3/\sqrt{2}$;

$a_2=\langle\beta,\alpha_2\rangle=(2)(1/\sqrt{3})+1(-1/\sqrt{3})+3(1/\sqrt{3})=4/\sqrt{3}$ and

$a_3=\langle\beta,\alpha_3\rangle=2(-1/\sqrt{6})+1(1/\sqrt{6})+3(2/\sqrt{6})=5/\sqrt{6}$.

$$\therefore\ \beta=(2,1,3)=\frac{3}{\sqrt{2}}\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}},0\right)+\frac{4}{\sqrt{3}}\left(\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)+\frac{5}{\sqrt{6}}\left(-\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}}\right).$$

Ex. 7. *Let* $V(C)$ *be the inner product space of continuous complex valued functions on* $[0,2\pi]$ *with inner product* $\langle f,g\rangle=\frac{1}{2\pi}\int_0^{2\pi}f(t)\,\overline{g(t)}\,dt$.

Prove that $S=\{f_n(t)=e^{\text{int}}\mid n\in Z\}$ *is an orthonormal subset of* V.

Sol. $\langle f_n,f_n\rangle=\frac{1}{2\pi}\int_0^{2\pi}e^{\text{int}}.\overline{e^{\text{int}}}\,dt=\frac{1}{2\pi}\int_0^{2\pi}e^{\text{int}}.e^{-\text{int}}\,dt=\frac{1}{2\pi}\int_0^{2\pi}1\,dt=\frac{1}{2\pi}.2\pi=1$.

$m\neq n\,;\langle f_m,f_n\rangle=\frac{1}{2\pi}\int_0^{2\pi}e^{imt}.\overline{e^{\text{int}}}\,dt=\frac{1}{2\pi}\int_0^{2\pi}e^{imt}.e^{-\text{int}}\,dt=\frac{1}{2\pi}\int_0^{2\pi}e^{i(m-n)t}\,dt$

$=\frac{1}{2\pi}\int_0^{2\pi}\{\cos(m-n)t+i\sin(m-n)t\}\,dt=\frac{1}{2\pi}\left[\frac{1}{(m-n)}\sin(m-n)t-\frac{i}{(m-n)}\cos(m-n)t\right]_0^{2\pi}$

$=\frac{1}{2\pi}(0)=0$. $\quad\therefore$ S is an orthonormal subset of V.

EXERCISE 8 (a)

1. Prove that the relation of orthogonality in an inner product space is a symmetric relation.
2. If $u+v$ and $u-v$ are two orthogonal vectors in a real inner product space prove that $\|u\|=\|v\|$.
3. If u, v are orthonormal vectors in $V(F)$ prove that $d\langle u,v\rangle=\sqrt{2}$.
4. Find a unit vector orthogonal to $(2,-1,6)$ in $R^3(R)$. ***(S. V. U. 08)***
5. Prove that $\{(3/5,0,4/5),(-4/5,0,3/5),(0,1,0)\}$ form an orthonormal subset of $R^3(R)$ space.

6. If $\{(1, 0, 1, 0), (0, 1, 0, 1), (-1, 0, 1, 0)\}$ is an orthogonal subset of $R^4(R)$ inner product space, obtain the orthonormal set by normalising.

7. Two vectors α, β in an unitary space $V(C)$ are orthogonal if and only if $\| a\alpha + b\beta \|^2 = |a|^2 \| \alpha \|^2 + |b|^2 \| \beta \|^2 \ \forall \ a, b \in C$. ***(S. K. D. 04)***

(Hint. See Ex. 9 in chapter 17)

8. Find two mutually orthogonal vectors each of which is orthogonal with $(4, 2, 3)$ in the inner product space $R^3(R)$.

9. If $S = \left\{\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right), (0, 1, 0)\right\}$ is an orthonormal subset of the space $R^3(R)$, express the vector $(2, 4, 6)$ as a linear combination of the basis vectors of S.

ANSWERS

4. $\frac{1}{\sqrt{21}}(4, 2, -1)$ **6.** $\frac{1}{\sqrt{2}}(1, 0, 1, 0), \frac{1}{\sqrt{2}}(0, 1, 0, 1), \frac{1}{\sqrt{2}}(-1, 0, 1, 0)$ **8.** $(1, 1, -2), (-7, 11, 2)$

9. $4\sqrt{2}\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) - 2\sqrt{2}\left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right) + 4(0, 1, 0)$.

8. 4. GRAM - SCHMIDT ORTHOGONALISATION PROCESS.

For a vector space we have seen earlier the special role of basis. We have learnt that, if S is a basis of vector space $V(F)$ then S is linearly independent and $L(S) = V$.

Definition. *A basis of an inner product space $V(F)$ which is also orthonormal is called* ***"orthonormal basis"*** *of the inner product space.*

e. g. 1. The basis $S = \{(1, 0), (0, 1)\}$ of the inner product space $R^2(R)$ is also orthonormal. So, S is an orthonormal basis of $R^2(R)$.

e. g. 2. The basis $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ of the inner product space $R^3(R)$ is also orthonormal. So, S is an orthonormal basis of $R^3(R)$

e.g. 3. For the inner product space $R^2(R)$, $S = \left\{\left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right), \left(\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right)\right\}$ is also an orthonormal basis.

Definition. *A finite dimensional vector space, in which an inner product is defined, is called a* ***finite dimensional inner product space.***

We now establish that every finite dimensional inner product space possesses an orthonormal basis. If S is a basis of the finite dimensional inner product space $V(F)$, we construct an orthonormal set S' from S such that $L(S) = L(S') = V$.

e.g. Let $V_2(F)$ be a 2 - dimensional inner product space and $S=\{\beta_1, \beta_2\}$ be a basis. Then S is linearly independent and $L(S)=V$.

Consider $S'=\{\alpha_1, \alpha_2\}$ where $\alpha_1=\beta_1$ and $\alpha_2=\beta_2-a\beta_1$ where $a \in F$ chosen such that α_2 is orthogonal to β_1.

Then $0=\langle\alpha_2, \beta_1\rangle=\langle\beta_2-a\beta_1, \beta_1\rangle=\langle\beta_2, \beta_1\rangle-a\langle\beta_1, \beta_1\rangle \Rightarrow a=\dfrac{\langle\beta_2, \beta_1\rangle}{\langle\beta_1, \beta_1\rangle}=\dfrac{\langle\beta_2, \beta_1\rangle}{\|\beta_1\|^2}$.

Thus $S'=\left\{\beta_1, \beta_2-\dfrac{\langle\beta_2, \beta_1\rangle}{\|\beta_1\|^2}\beta_1\right\}$ is an orthogonal set of non-zero vectors such that $L(S)=L(S')=V$.

Observe that each vector in $S'=\{\alpha_1, \alpha_2\}$ is a linear combination of β_1, β_2 of S.

Theorem. 1. ***Every finite dimensional inner product space has an orthonormal basis.*** *(N. U. 07,08, K. U. 05, 03, A. U. 04, 03, O. U. M 15, 04, 03)*

Proof. Let $V(F)$ be an inner product space of dimension n and $B=\{\beta_1, \beta_2, ..., \beta_n\}$ be a basis for V.

Now we shall obtain an orthonormal set $B'=\{\alpha_1, \alpha_2, ..., \alpha_n\}$ in $V(F)$ such that $L(B')=L(B)=V$ with the help of the vectors in B by means of a construction known as Gram - Schmidt orthogonalization process.

The main idea behind this construction is that each α_i is a linear combination of $\beta_1, \beta_2, ..., \beta_n$.

We now prove this construction by using induction on the dimension n of V.

Since B is a basis of V, B is linearly independent $\Rightarrow \beta_i \neq \bar{0}$ for $i=1, 2, ..., n$.

Let $\alpha_1=\dfrac{\beta_1}{\|\beta_1\|}$.

Then we have $(\alpha_1, \alpha_1)=1$ and α_1 is a linear combination of β_1

Thus $\{\alpha_1\}$ is an ortho-normal set which is a basis of V such that $L(\{\alpha_1\})=L(\{\beta_1\})=V$ when dim $V=n=1$.

Suppose that $\{\alpha_1, \alpha_2, ..., \alpha_m\}$ where $1 \leq m < n$ has been constructed as an ortho-normal basis of V when dim $V=m$.

Then $L(\{\alpha_1, \alpha_2, ..., \alpha_m\})=L(\{\beta_1, \beta_2, ..., \beta_m\})=V$ and each α_i $(i=1, 2, ..., m)$ is a linear combination of $\beta_1, \beta_2, ..., \beta_m$.

Consider the vector

$$\gamma_{m+1}=\beta_{m+1}-\langle\beta_{m+1}, \alpha_1\rangle\alpha_1-\langle\beta_{m+1}, \alpha_2\rangle\alpha_2-...-\langle\beta_{m+1}, \alpha_m\rangle\alpha_m \quad ... (1)$$

By theorem (3) of Art. 9.3; γ_{m+1} is orthogonal to each of $\alpha_1, \alpha_2, ..., \alpha_m$.

Suppose that $\gamma_{m+1} = \bar{0}$. Then $\beta_{m+1} = (\beta_{m+1}, \alpha_1)\alpha_1 + (\beta_{m+1}, \alpha_2)\alpha_2 + ... + (\beta_{m+1}, \alpha_m)\alpha_m$ so that β_{m+1} is a linear combination of $\alpha_1, \alpha_2, ..., \alpha_m$.

By our induction hypothesis, each α_i $(i = 1, 2, ..., m)$ is a linear combination of $\beta_1, \beta_2, ..., \beta_m$.

Therefore, β_{m+1} is a linear combination of $\beta_1, \beta_2, ..., \beta_m$.

This is a contradiction, as $\{\beta_1, \beta_2, ..., \beta_m, \beta_{m+1}\}$ is linearly independent.

$\therefore \gamma_{m+1} \neq \bar{0}$, Take $\alpha_{m+1} = \dfrac{\gamma_{m+1}}{\| \gamma_{m+1} \|}$.

Then we have $\| \alpha_{m+1} \| = 1$ and α_{m+1} is orthogonal to each of $\alpha_1, \alpha_2, ..., \alpha_m$.

Also, $\alpha_{m+1} \neq \alpha_i$ for $i = 1, 2, ..., m$.

Otherwise, from (1) we see that β_{m+1} will become a linear combination of $\beta_1, \beta_2, ..., \beta_m$.

Further from (1) we see that α_{m+1} is a linear combination of $\beta_1, \beta_2, ..., \beta_m, \beta_{m+1}$.

Thus $\{\alpha_1, \alpha_2, ..., \alpha_m, \alpha_{m+1}\}$ is an orthonormal basis of V such that

$L(\{\alpha_1, \alpha_2, ..., \alpha_{m+1}\}) = L(\{\beta_1, \beta_2, ..., \beta_{m+1}\}) = V$ when $\dim V = m+1$.

Therefore, by induction, $B' = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ is an orthonormal basis of V such that $L(B) = L(B') = V$ when $\dim V = n$.

The above theorem can also be stated as follows :

"Let $V(F)$ be an inner product space of dimension n and $B = \{\beta_1, \beta_2, ..., \beta_n\}$ be a linearly independent subset of V. If $B' = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ such that $\alpha_1 = \beta_1$ and

$$\alpha_k = \beta_k - \sum_{i=1}^{k-1} \frac{\langle \beta_k, \alpha_i \rangle}{\| \alpha_i \|^2} \alpha_i \text{ for } 2 \leq k \leq n$$ then B' is an orthogonal set of non-zero vectors such that $L(B') = L(B) = V$."

WORKING METHOD FOR FINDING ORTHOGONAL BASIS

Let $\{\beta_1, \beta_2, ..., \beta_n\}$ be the given L.I. basis of $V(F)$.

The orthogonal basis of $V(F)$, namely $\{\gamma_1, \gamma_2, ..., \gamma_n\}$ is given by the following relations:

$$\gamma_1 = \beta_1, \gamma_2 = \beta_2 - \frac{\langle \beta_2, \gamma_1 \rangle}{\| \gamma_1 \|^2} \gamma_1, \gamma_3 = \beta_3 - \frac{\langle \beta_3, \gamma_1 \rangle}{\| \gamma_1 \|^2} \gamma_1 - \frac{\langle \beta_3, \gamma_2 \rangle}{\| \gamma_2 \|^2} \gamma_2 .....$$

$$.... \gamma_n = \beta_n - \frac{\langle \beta_n, \gamma_1 \rangle}{\| \gamma_1 \|^2} \gamma_1 - .... - \frac{\langle \beta_n, \gamma_{n-1} \rangle}{\| \gamma_{n-1} \|^2} \gamma_{n-1}.$$

WORKING METHOD FOR FINDING ORTHONORMAL BASIS

Let $\{\beta_1, \beta_2, ..., \beta_n\}$ be a given basis of a finite dimensional inner product space $V(F)$.

The vectors $\alpha_1, \alpha_2, ..., \alpha_n$ of the orthonormal basis of $V(F)$ are given by

$$\alpha_1 = \frac{\beta_1}{\|\beta_1\|}, \quad \alpha_2 = \frac{\gamma_2}{\|\gamma_2\|} \text{ where } \gamma_2 = \beta_2 - \langle \beta_2, \alpha_1 \rangle \alpha_1, \quad \alpha_3 = \frac{\gamma_3}{\|\gamma_3\|}$$

where $\gamma_3 = \beta_3 - \langle \beta_3, \alpha_1 \rangle \alpha_1 - \langle \beta_3, \alpha_2 \rangle \alpha_2, ..., \alpha_n = \frac{\gamma_n}{\|\gamma_n\|}$

where $\beta_n = \beta_n - \langle \beta_n, \alpha_1 \rangle \alpha_1 - \langle \beta_n, \alpha_2 \rangle \alpha_2 - ... - \langle \beta_n, \alpha_{n-1} \rangle \alpha_{n-1}$.

SOLVED PROBLEMS

Ex. 1. *Given* $\{(2,1,3), (1,2,3), (1,1,1)\}$ *is a basis of* R^3*; construct an orthonormal basis.* ***(S. V. U. 2011, M14, N. U. 91, S. K. U. M 15, 00)***

Sol. Let $\beta_1 = (2,1,3), \beta_2 = (1,2,3), \beta_3 = (1,1,1)$.

By Gram - Schmidt orthogonalization process an ortho-normal basis $= \{\alpha_1, \alpha_2, \alpha_3\}$ is given by $\alpha_1 = \frac{\beta_1}{\|\beta_1\|}, \alpha_2 = \frac{\gamma_2}{\|\gamma_2\|}$ where $\gamma_2 = \beta_2 - \langle \beta_2, \alpha_1 \rangle \alpha_1$ and $\alpha_3 = \frac{\gamma_3}{\|\gamma_3\|}$.

where $\gamma_3 = \beta_3 - \langle \beta_2, \alpha_1 \rangle \alpha_1 - \langle \beta_3, \alpha_2 \rangle \alpha_2$.

$$\|\beta_1\|^2 = \langle \beta_1, \beta_1 \rangle = 2^2 + 1^2 + 3^2 = 14. \quad \therefore \alpha_1 = \frac{1}{\|\beta_1\|}\beta_1 = \frac{1}{\sqrt{14}}(2,1,3) = \left(\frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right)$$

$$\langle \beta_2, \alpha_1 \rangle = 1.\frac{2}{\sqrt{14}} + 2.\frac{1}{\sqrt{14}} + 3.\frac{3}{\sqrt{14}} = \frac{13}{\sqrt{14}}$$

$$\gamma_2 = (1,2,3) - \frac{13}{\sqrt{14}}\left(\frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right) = \left(\frac{-12}{14}, \frac{15}{14}, \frac{3}{14}\right)$$

$$\|\gamma_2\|^2 = \langle \gamma_2, \gamma_2 \rangle = \left(\frac{-12}{14}\right)^2 + \left(\frac{15}{14}\right)^2 + \left(\frac{3}{14}\right)^2 = \frac{378}{196}.$$

$$\therefore \alpha_2 = \frac{1}{\|\gamma_2\|}\gamma_2 = \frac{14}{\sqrt{378}}\left(\frac{-12}{14}, \frac{15}{14}, \frac{3}{14}\right) = \left(\frac{-12}{\sqrt{378}}, \frac{15}{\sqrt{378}}, \frac{3}{\sqrt{378}}\right).$$

$$\langle \beta_3, \alpha_1 \rangle = 1.\frac{2}{\sqrt{14}} + 1.\frac{1}{\sqrt{14}} + 1.\frac{3}{\sqrt{14}} = \frac{6}{\sqrt{14}}, \langle \beta_3, \alpha_2 \rangle = 1.\frac{-12}{\sqrt{378}} + 1.\frac{15}{\sqrt{378}} + 1.\frac{3}{\sqrt{378}} = \frac{6}{\sqrt{378}}.$$

$$\gamma_3 = (1,1,1) - \frac{6}{\sqrt{14}}\left(\frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right) - \frac{6}{\sqrt{378}}\left(\frac{-12}{\sqrt{378}}, \frac{15}{\sqrt{378}}, \frac{3}{\sqrt{378}}\right)$$

$$= (1,1,1) - \left(\frac{12}{14}, \frac{6}{14}, \frac{18}{14}\right) - \left(\frac{-72}{378}, \frac{90}{378}, \frac{18}{378}\right) = \left(\frac{126}{378}, \frac{126}{378}, \frac{-126}{378}\right) = \left(\frac{1}{3}, \frac{1}{3}, \frac{-1}{3}\right)$$

$$\|\gamma_3\|^2 = \langle \gamma_3, \gamma_3 \rangle = \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{-1}{3}\right)^2 = \frac{1}{3}.$$

$$\therefore \alpha_3 = \frac{1}{\| \gamma_3 \|} \gamma_3 = \sqrt{3}\left(\frac{1}{3}, \frac{1}{3}, -\frac{1}{3}\right) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)$$

Hence orthonormal basis $\left\{\left(\frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right), \left(\frac{-12}{\sqrt{378}}, \frac{15}{\sqrt{378}}, \frac{3}{\sqrt{378}}\right), \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right)\right\}$.

Ex. 2. *Find an orthonormal basis of a vector space $V(R)$ of all real polynomials of degree not greater than 2, in which the inner product is defined as* $\langle f(x), g(x) \rangle = \int_0^1 f(x)\, g(x)\, dx \;\; \forall \;\; f(x), g(x) \in V$ *and* $x \in [0,1]$, *from a given basis* $\{1, x, x^2\}$.

Sol. Let $\beta_1 = 1, \beta_2 = x, \beta_3 = x^2$. $\| \beta_1 \|^2 = \langle \beta_1, \beta_1 \rangle = \int_0^1 (1)\,(1)\, dx = 1$. $\therefore \alpha_1 = \frac{1}{\| \beta_1 \|} \beta_1 = 1$.

$$\langle \beta_2, \alpha_1 \rangle = \int_0^1 (x)\,(1)\, dx = \int_0^1 x \,.\, dx = \frac{1}{2} \cdot \qquad \gamma_2 = \beta_2 - \langle \beta_2, \alpha_1 \rangle \alpha_1 = x - \frac{1}{2}(1) = x - \frac{1}{2} \cdot$$

$$\| \gamma_2 \|^2 = \langle \gamma_2, \gamma_2 \rangle = \int_0^1 \left(x - \frac{1}{2}\right)\left(x - \frac{1}{2}\right) dx = \left[\frac{\left(x - \frac{1}{2}\right)^3}{3}\right]_0^1 = \frac{1}{12} \quad \therefore \alpha_2 = \frac{1}{\| \gamma_2 \|} \gamma_2 = \sqrt{12}\left(x - \frac{1}{2}\right)$$

$$\langle \beta_3, \alpha_1 \rangle = \int_0^1 x^2 \,.\, 1\, dx = \frac{1}{3} \text{ and } \langle \beta_3, \alpha_2 \rangle = \int_0^1 x^2 \,.\, \sqrt{12}\left(x - \frac{1}{2}\right) dx = \sqrt{12}\left[\frac{x^4}{4} - \frac{1}{2}\frac{x^3}{3}\right]_0^1 = \frac{1}{\sqrt{12}}$$

$$\gamma_3 = \beta_3 - (\beta_3, \alpha_1)\, \alpha_1 - (\beta_3, \alpha_2)\alpha_2 = x^2 - \frac{1}{3} \,.\, 1 - \frac{1}{\sqrt{12}}\sqrt{12}\left(x - \frac{1}{2}\right) = x^2 - x + \frac{1}{6}$$

$$\| \gamma_3 \|^2 = \langle \gamma, \gamma_3 \rangle = \int_0^1 \left(x^2 - x + \frac{1}{6}\right)\left(x^2 - x + \frac{1}{6}\right) dx$$

$$= \int_0^1 \left(x^4 - 2x^3 + \frac{4x^2}{3} - \frac{x}{3} + \frac{1}{36}\right) dx = \left[\frac{x^5}{5} - \frac{x^4}{2} + \frac{4x^3}{9} - \frac{x^2}{9} + \frac{x}{36}\right]_0^1 = \frac{1}{180}$$

$$\therefore \alpha_3 = \frac{1}{\| \gamma_3 \|} \gamma_3 = \sqrt{180}\left(x^2 - x + \frac{1}{6}\right).$$

Hence orthonormal basis $\{\alpha_1, \alpha_2, \alpha_3\} = \left\{1, \sqrt{12}\left(x - \frac{1}{2}\right), 3\sqrt{20}\left(x^2 - x + \frac{1}{6}\right)\right\}$

Theorem. 2. ***Let*** $B = \{\alpha_1, \alpha_2, \ldots\ldots, \alpha_k\}$ ***be an orthonormal set in n - dimensional inner product space V(F). Then B can be extended to an orthonormal basis*** $B' = \{\alpha_1, \alpha_2, \ldots\ldots, \alpha_k, \alpha_{k+1}, \ldots\ldots, \alpha_n\}$ ***for V.***

Proof. V(F) is an inner product space $\Rightarrow$ V(F) is a vector space.

By basis extension theorem of vector space, B can be extended to the set $B_1 = \{\alpha_1, \alpha_2, ..., \alpha_k, \beta_{k+1}, ..., \beta_n\}$. So as to form the basis of V(F).

By applying Gram - Schmidt orthogonalisation process to B_1 we can obtain orthonormal basis $B' = \{\alpha_1, \alpha_2, ..., \alpha_k, \alpha_{k+1}, ..., \alpha_n\}$ so that $L(B') = V$ of the inner product space.

Theorem. 3. ***If $V(F)$ is a non- zero finite dimensional inner product space of dimension n then*** $\gamma = \sum_{i=1}^{n} \langle \gamma, \alpha_i \rangle \alpha_i$ ***where*** $\alpha_1, \alpha_2, ..., \alpha_n, \gamma \in V$.

Proof. Let $B = \{\beta_1, \beta_2, ..., \beta_n\}$ be a linearly independent subset of V such that $L(B) = V$.

By Gram - Schmidt orthogonalisation process, we can obtain

$B' = \{\alpha_1, \alpha_2, ..., \alpha_n\}$, an orthonormal basis of V such that $L(B') = L(B) = V$

$\gamma \in V \Rightarrow \gamma \in L(B') \Rightarrow \gamma = \sum_{i=1}^{n} a_i \alpha_i$ for $a_i \in F$

For $1 \le j \le n$; $\langle \gamma, \alpha_j \rangle = \left\langle \sum_{i=1}^{n} a_i \alpha_i, \alpha_j \right\rangle = \sum_{i=1}^{n} a_i \langle \alpha_i, \alpha_j \rangle$

$= a_j \langle \alpha_j, \alpha_j \rangle = a_j \| \alpha_j \|^2$ $(\because \langle \alpha_i, \alpha_j \rangle = 0$ for $i \neq j)$

$\therefore a_j = \dfrac{\langle \gamma, \alpha_j \rangle}{\| \alpha_j \|^2} = \langle \gamma, \alpha_j \rangle$ $(\because \| \alpha_j \| = 1)$. Hence $\gamma = \sum_{i=1}^{n} \langle \gamma, \alpha_i \rangle \alpha_i$.

Corollary. ***Let $V(F)$ be a finite dimensional vector space with an orthonormal basis $B = \{\alpha_1, \alpha_2, ..., \alpha_n\}$. Let T be a linear operator on V and $A = [T]_B$. Then for any i and j, $a_{ij} = \langle T(\alpha_j), \alpha_i \rangle$.***

Proof. From the above theorem,

$T(\alpha_j) = \sum_{i=1}^{n} \langle T(\alpha_j), \alpha_i \rangle \alpha_i$ and hence $a_{ij} = \langle T(\alpha_j), \alpha_i \rangle$.

Definition. (Fourier Coefficients) *If B is an orthonormal basis of a finite dimensional inner product space $V(F)$ and $\beta \in V$ then the Fourier coefficients of β relative to basis B are the scalars $\langle \beta, \alpha \rangle$ where $\alpha \in B$.*

e.g. Let $B = \left\{ \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}} \right) \right\}$ be an orthonormal basis of $R^2(R)$.

Then the Fourier coefficients of $\beta = (3,4) \in R^2$ relative to B are as below :

Let $\alpha_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$ and $\alpha_2 = \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}} \right)$

The Fourier coefficients of β with respect to B are $\langle \beta, \alpha_1 \rangle$ and $\langle \beta, \alpha_2 \rangle$.

$$\langle \beta, \alpha_1 \rangle = 3\left(\frac{1}{\sqrt{2}}\right) + 4\left(\frac{1}{\sqrt{2}}\right) = \frac{7}{\sqrt{2}}; \quad \langle \beta, \alpha_2 \rangle = 3\left(\frac{1}{\sqrt{2}}\right) + 4\left(\frac{-1}{\sqrt{2}}\right) = \frac{-1}{\sqrt{2}}.$$

8.5. BESSEL'S INEQUALITY AND PARSEVAL'S IDENTITY

Theorem. 1. (Bessel's inequality) *If $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ is any orthonormal subset in an inner product space V(F) and β is any vector in V, then $\sum_{i=1}^{n} |\langle \beta, \alpha_i \rangle|^2 \leq \|\beta\|^2$.*

Further equality holds if and only if β is in the subspace generated by $\{\alpha_1, \alpha_2, ..., \alpha_n\}$ or if and only if $\{\alpha_1, \alpha_2, ..., \alpha_n\}$ is a basis for V.

(S.V. U, M14, 11, K. U. 05, 07, N. U. 00, 08, S. K. D. 01)

Proof. $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ is an orthonormal set

$\Rightarrow (\alpha_i, \alpha_j) = 1$ if $i \neq j$ and $(\alpha_i, \alpha_j) = 0$ if $i \neq j$.

Consider the vector $\gamma = \beta - \sum_{i=1}^{n} (\beta, \alpha_i)\, \alpha_i$.

$$\|\gamma\|^2 = \langle \gamma, \gamma \rangle = \left\langle \beta - \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \alpha_i, \beta - \sum_{j=1}^{n} \langle \beta, \alpha_j \rangle \alpha_j \right\rangle$$

$$= \langle \beta, \beta \rangle - \sum_{j=1}^{n} \overline{\langle \beta, \alpha_j \rangle} \langle \beta, \alpha_j \rangle - \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \langle \alpha_i, \beta \rangle + \left(\sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \right) \left(\sum_{j=1}^{n} \overline{\langle \beta, \alpha_j \rangle} \right) \langle \alpha_i, \alpha_j \rangle$$

$$= \langle \beta, \beta \rangle - \sum_{j=1}^{n} \overline{\langle \beta, \alpha_j \rangle} \langle \beta, \alpha_j \rangle - \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \overline{\langle \beta, \alpha_i \rangle} + \sum_{i=1}^{n} \sum_{j=1}^{n} \langle \beta, \alpha_i \rangle \overline{\langle \beta, \alpha_j \rangle} \langle \alpha_i, \alpha_j \rangle$$

$$= \langle \beta, \beta \rangle - 2 \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \overline{\langle \beta, \alpha_i \rangle} + \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \overline{\langle \beta, \alpha_i \rangle} \langle \alpha_i, \alpha_i \rangle$$

(On summing with respect to j and using $(\alpha_i, \alpha_j) = 0$ for $i \neq j$)

$$= \|\beta\|^2 - 2 \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \overline{\langle \beta, \alpha_i \rangle} + \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \overline{\langle \beta, \alpha_i \rangle} = \|\beta\|^2 - \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \overline{\langle \beta, \alpha_i \rangle}.$$

$$\therefore \|\gamma\|^2 = \|\beta\|^2 - \sum_{i=1}^{n} |\langle \beta, \alpha_i \rangle|^2 \qquad (\because z\bar{z} = |z|^2)$$

Since $\|\gamma\|^2 \geq 0$ we have $\|\beta\|^2 - \sum_{i=1}^{n} |\langle \beta, \alpha_i \rangle|^2 \geq 0 \Rightarrow \sum_{i=1}^{n} |\langle \beta, \alpha_i \rangle|^2 \leq \|\beta\|^2$

If the equality holds, then $\|\gamma\|^2 = 0 \Leftrightarrow \langle \gamma, \gamma \rangle = 0 \Leftrightarrow \gamma = \bar{0} \Leftrightarrow \beta = \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \alpha_i$

$\Leftrightarrow \beta$ is a linear combination of $\alpha_1, \alpha_2, ..., \alpha_n$

$\Leftrightarrow \beta$ is a vector in the subspace spanned by $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ *i.e.* $\beta \in L(S)$

$\Leftrightarrow \{\alpha_1, \alpha_2, ..., \alpha_n\}$ is a basis for V.

Corollary. *If* $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ *be an orthogonal set of non-zero vectors in an inner product space V (F) and* β *any vector in V, then* $\sum_{i=1}^{n} \frac{|\langle \beta, \alpha_i \rangle|^2}{\|\alpha_i\|^2} \leq \|\beta\|^2$

Proof. Write $u_i = \frac{\alpha_i}{\|\alpha_i\|}$ for $i = 1, 2, ..., n$.

Then $\langle u_i, u_i \rangle = \left\langle \frac{\alpha_i}{\|\alpha_i\|}, \frac{\alpha_i}{\|\alpha_i\|} \right\rangle = \frac{1}{\|\alpha_i\|^2} \langle \alpha_i, \alpha_i \rangle = 1$ for each $i = 1, 2, ..., n$.

Thus $\{u_1, u_2, ..., u_n\}$ is an orthonormal set in V.

Proceeding as in theorem (1) of Art. 18.5 we get

$$\sum_{i=1}^{n} |\langle \beta, u_i \rangle|^2 \leq \|\beta\|^2 \Rightarrow \sum_{i=1}^{n} \left| \left\langle \beta, \frac{\alpha_i}{\|\alpha_i\|} \right\rangle \right|^2 \leq \|\beta\|^2 \Rightarrow \sum_{i=1}^{n} \frac{|\langle \beta, \alpha_i \rangle|^2}{\|\alpha_i\|^2} \leq \|\beta\|^2$$

Theorem.2. (Parseval's Identity). *If* $B = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ *be an orthonormal basis of a finite dimensional inner product space V (F) then*

$\langle \alpha, \beta \rangle = \sum_{i=1}^{n} \langle \alpha, \alpha_i \rangle \langle \alpha_i, \beta \rangle$ ***for all*** $\alpha, \beta \in V$. **(*S. K. U. 11, A. U. 08, S. V. U. M 15, 08, N. U. 04, 07*)**

Proof. $B = \{\alpha_1, \alpha_2, ...\alpha_n\}$ is a basis of V and $\alpha, \beta \in V$

$\Rightarrow$ there exist $x_1, x_2, ...x_n$ and $y_1, y_2, ...y_n \in F$ so that

$$\alpha = x_1\alpha_1 + x_2\alpha_2 + ... + x_n\alpha_n = \sum_{i=1}^{n} x_i\alpha_i \quad \text{and} \quad \beta = y_1\alpha_1 + y_2\alpha_2 + ... + y_n\alpha_n = \sum_{j=1}^{n} y_j\alpha_j$$

$$\langle \alpha, \beta \rangle = \left(\sum_{i=1}^{n} x_i\alpha_i, \sum_{j=1}^{n} y_j\alpha_j \right) = \sum_{i=1}^{n} (x_i\alpha_i, y_i\alpha_i) \qquad (\because (\alpha_i, \alpha_j) = 0 \text{ for } i \neq j)$$

$$= \sum_{i=1}^{n} x_i\bar{y}_i(\alpha_i, \alpha_i) = \sum_{i=1}^{n} x_i\bar{y}_i \qquad (\because (\alpha_i, \alpha_j) = 1 \text{ for } i = j)$$

But $\langle \alpha, \alpha_i \rangle = \langle x_i\alpha_i, \alpha_i \rangle = x_i \langle \alpha_i, \alpha_i \rangle = x_i$ and $\langle \beta, \alpha_i \rangle = \langle y_i\alpha_i, \alpha_i \rangle = y_i \langle \alpha_i, \alpha_i \rangle = y_i$

$$\therefore \langle \alpha, \beta \rangle = \sum_{i=1}^{n} \langle \alpha, \alpha_i \rangle \overline{\langle \beta, \alpha_i \rangle} = \sum_{i=1}^{n} \langle \alpha, \alpha_i \rangle \langle \alpha_i, \beta \rangle.$$

SOLVED PROBLEMS

Ex. 3. *In* $R^4(R) = V_4(R)$ *if* $(1,0,1,1), (-1,0,-1,1), (0,-1,1,1)$ *are three linearly independent vectors, compute the orthonormal set of these vectors.* **(*S. K. U. 11*)**

Sol. Let $\beta_1 = (1,0,1,1), \beta_2 = (-1,0,-1,1), \beta_3 = (0,-1,1,1)$

If $\alpha_1, \alpha_2, \alpha_3$ are the orthonormal vectors then $\alpha_1 = \frac{\beta_1}{\|\beta_1\|}; \alpha_2 = \frac{\gamma_2}{\|\gamma_2\|}$

where $\gamma_2 = \beta_2 - \langle \beta_2, \alpha_1 \rangle \alpha_1$ and $\alpha_3 = \dfrac{\gamma_3}{\| \gamma_3 \|}$ where $\gamma_3 = \beta_3 - \langle \beta_3, \alpha_1 \rangle \alpha_1 - \langle \beta_3, \alpha_2 \rangle \alpha_2$

$$\| \beta_1 \|^2 = \langle \beta_1, \beta_1 \rangle = 1^2 + 0^2 + 1^2 + 1^2 = 3 \Rightarrow \| \beta_1 \| = \sqrt{3} . \quad \therefore \alpha_1 = \frac{1}{\sqrt{3}} \beta_1 = \left(\frac{1}{\sqrt{3}}, 0, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$$

$$\gamma_2 = \beta_2 - \langle \beta_2, \alpha_1 \rangle \alpha_1 = (-1, 0, -1, 1) - \left\{ -\frac{1}{\sqrt{3}} + 0 - \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}} \right\} \left(\frac{1}{\sqrt{3}}, 0, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$$

$$= (-1, 0, -1, 1) + \left(\frac{1}{3}, 0, \frac{1}{3}, \frac{1}{3} \right) = \left(-\frac{2}{3}, 0, -\frac{2}{3}, \frac{4}{3} \right). \quad \| \gamma_2 \| = \sqrt{\frac{4}{9} + 0 + \frac{4}{9} + \frac{16}{9}} = \sqrt{\frac{24}{9}} = \frac{2\sqrt{6}}{3}$$

$$\therefore \alpha_2 = \frac{1}{\| \gamma_2 \|} \gamma_2 = \frac{3}{2\sqrt{6}} \left(-\frac{2}{3}, 0, -\frac{2}{3}, \frac{4}{3} \right) = \left(-\frac{1}{\sqrt{6}}, 0, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right)$$

$$\gamma_3 = \beta_3 - \langle \beta_3, \alpha_1 \rangle \alpha_1 - \langle \beta_3, \alpha_2 \rangle \alpha_2$$

$$= \beta_3 - \left\{ 0 + 0 + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}} \right\} \alpha_1 - \left\{ 0 + 0 - \frac{1}{\sqrt{6}} + \frac{2}{\sqrt{6}} \right\} \alpha_2 = \beta_3 - \frac{2}{\sqrt{3}} \alpha_1 - \frac{1}{\sqrt{6}} \alpha_2$$

$$= (0, -1, 1, 1) - \left(\frac{2}{3}, 0, \frac{2}{3}, \frac{2}{3} \right) - \left(-\frac{1}{6}, 0, -\frac{1}{6}, \frac{2}{6} \right) = \left(-\frac{3}{6}, -1, \frac{3}{6}, 0 \right) = \left(-\frac{1}{2}, -1, \frac{1}{2}, 0 \right)$$

$$\therefore \alpha_3 = \frac{1}{\| \gamma_3 \|} \gamma_3 = \sqrt{\frac{2}{3}} \left(-\frac{1}{2}, -1, \frac{1}{2}, 0 \right) = \left(-\frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, 0 \right)$$

Hence $\{\alpha_1, \alpha_2, \alpha_3\} = \left\{ \left(\frac{1}{\sqrt{3}}, 0, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right), \left(-\frac{1}{\sqrt{6}}, 0, \frac{-1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right), \left(-\frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, 0 \right) \right\}$ is the orthonormal set of $\{\beta_1, \beta_2, \beta_3\}$

Ex. 4. *If $V = L(S)$ where $S = \{(1, i, 0), (1-i, 2, 4i)\}$ find the orthonormal basis B of V and compute the Fourier coefficients of $x = (3+i, 4i, -4)$ vector relative to B.*

Sol. Let $S = \{\beta_1, \beta_2\}$ so that $\beta_1 = (1, i, 0), \beta_2 = (1-i, 2, 4i)$. We shall first find orthogonal basis $\{\gamma_1, \gamma_2\}$ of V from the formulae : $\gamma_1 = \beta_1$ and $\gamma_2 = \beta_2 - \dfrac{\langle \beta_2, \gamma_1 \rangle}{\| \gamma_1 \|^2} \gamma_1$.

We have $\gamma_1 = (1, i, 0)$ and $\| \gamma_1 \|^2 = 1(1) + i(-i) + 0(0) = 2$

$$\langle \beta_2, \gamma_1 \rangle = \langle (1-i, 2, 4i), (1, i, 0) \rangle = (1-i)1 + 2(-i) + 4i(0) = 1 - i - 2i = 1 - 3i$$

$$\therefore \gamma_2 = (1-i, 2, 4i) - \left(\frac{1-3i}{2} \right) (1, i, 0) = \left(1 - i - \frac{(1-3i)}{2}, 2 - \frac{(1-3i)}{2} i, 4i - 0 \right) = \left(\frac{1+i}{2}, \frac{1-i}{2}, 4i \right)$$

Clearly $\langle \gamma_1, \gamma_2 \rangle = 0$ and hence $\{\gamma_1, \gamma_2\}$ is an orthogonal basis.

$$\| \gamma_2 \|^2 = \left(\frac{1+i}{2} \right) \left(\frac{1-i}{2} \right) + \left(\frac{1-i}{2} \right) \left(\frac{1+i}{2} \right) + (4i)(-4i) = \frac{2}{4} + \frac{2}{4} + 16 = 17$$

$\therefore$ $B =$ Orthonormal basis $= \{\alpha_1, \alpha_2\} = \left\{\frac{\gamma_1}{\|\gamma_1\|}, \frac{\gamma_2}{\|\gamma_2\|}\right\}$

$= \left\{\frac{1}{\sqrt{2}}(1,i,0), \frac{1}{\sqrt{17}}\left(\frac{1+i}{2}, \frac{1-i}{2}, 4i\right)\right\} = \left\{\left(\frac{1}{\sqrt{2}}, \frac{i}{\sqrt{2}}, 0\right), \left(\frac{1+i}{2\sqrt{17}}, \frac{1-i}{2\sqrt{17}}, \frac{4}{\sqrt{17}}i\right)\right\}$

Fourier coefficients of x relative to B are $\langle x, \alpha_1\rangle, \langle x, \alpha_2\rangle$.

$\langle x, \alpha_1\rangle = \left\langle (3+i, 4i, -4), \left(\frac{1}{\sqrt{2}}, \frac{i}{\sqrt{2}}, 0\right)\right\rangle$

$= (3+i)\left(\frac{1}{\sqrt{2}}\right) + 4i\left(\frac{-i}{\sqrt{2}}\right) + (-4)(0) = \frac{1}{\sqrt{2}}(3+i-4i^2) = \frac{7+i}{\sqrt{2}}$

$\langle x, \alpha_2\rangle = \left\langle (3+i, 4i, -4)\left(\frac{1+i}{2\sqrt{17}}, \frac{1-i}{2\sqrt{17}}, \frac{4i}{\sqrt{17}}\right)\right\rangle = (3+i)\left(\frac{1-i}{2\sqrt{17}}\right) + (4i)\left(\frac{1+i}{2\sqrt{17}}\right) + (-4)\left(\frac{-4i}{\sqrt{17}}\right)$

$= \frac{1}{2\sqrt{17}}[(4-2i) + (4i-4) + 32i] = \frac{34i}{2\sqrt{17}} = \sqrt{17}i$.

8. 6. ORTHOGONAL COMPLEMENT

Definition. *Let W be a non - empty subset of the inner product space $V(F)$. The* ***orthogonal complement*** *of W dentoed by $W^{\perp}$ is defined as the set*

$\{\alpha \in V \mid \langle \alpha, \beta\rangle = 0 \ \forall \ \beta \in W\}$.

Note. 1. $W^{\perp} \subset V$ and is read as "W perpendicular".

2. $W^{\perp}$ is the set of all vectors in V that are orthogonal to every vector in W.

3. $\alpha \in W^{\perp}$ and $\beta \in W \Rightarrow \langle \alpha, \beta\rangle = 0$.

4. For $\bar{0} \in V$ we have $\langle \bar{0}, \beta\rangle = 0 \ \forall \ \beta \in W$. So $\bar{0} \in V$ is an element of $W^{\perp}$ and hence $W^{\perp} \neq \phi$.

e.g. 1. If $W = \{\bar{0}\}$ then $W^{\perp} = \{\alpha \in V \mid \langle \alpha, \bar{0}\rangle = 0 \ \forall \ \bar{0} \in W\}$

We know that $\langle \alpha, \bar{0}\rangle = 0 \ \forall \ \alpha \in V$. Hence $W^{\perp} = \{0\}^{\perp} = V$ ***(O. U. 01)***

e.g. 2. $V^{\perp} = \{\alpha \in V \mid \langle \alpha, \beta\rangle = 0 \ \forall \ \beta \in V\}$.

We know that $\langle \bar{0}, \beta\rangle = 0 \ \forall \ \beta \in V$ and hence $V^{\perp} = \{\bar{0}\}$. ***(O. U. 01)***

e.g. 3. Let $R^3(R)$ be the inner product space and

$B = \{i = (1,0,0), j = (0,1,0), k = (0,0,1)\}$ be an orthonormal basis.

If we take $W = \{ai \mid a \in R\}$ then W is a sub space of $R^3(R)$.

Clearly $W = \{(a,0,0) \mid a \in R\}$ represents the set of all vectors along the X - axis.

By the definition, $W^{\perp} = \{\alpha \in R^3 \mid \langle \alpha, \beta\rangle = 0 \ \forall \ \beta \in W\}$

We know that $\langle j,i\rangle = 0, \langle k,i\rangle = 0$ and hence $\langle bj+ck,i\rangle = 0$ for $b,c \in R$.

$\therefore W^{\perp} = \{bj+ck \mid b,c \in R\} = \{(0,b,c) \mid b,c \in R\}$ which represents the YZ - plane.

Remark. $W \cap W^{\perp} = \{\bar{0}\}$

Theorem. 1. ***For any non-empty subset W of an inner product space V(F), the orthogonal complement $W^{\perp}$ is subspace of V.***

Proof. Let $\beta \in W$. Then $\beta \in W$ and $W \subset V \Rightarrow \beta \in V$.

For $\bar{0}, \beta \in V$ we have $\langle \bar{0}, \beta\rangle = 0$ and hence $\bar{0} \in W^{\perp}$.

$\therefore W^{\perp} \neq \phi$ and $W^{\perp} \subset V$. Let $\alpha, \beta \in W^{\perp}$ and $a,b \in F$.

From the definition, $\langle \alpha,\gamma\rangle = 0, \langle \beta,\gamma\rangle = 0 \ \forall \ \gamma \in W$.

For each $\gamma \in W; \langle a\alpha + b\beta,\gamma\rangle = a\langle \alpha,\gamma\rangle + b\langle \beta,\gamma\rangle = a0+b0 = 0 \Rightarrow a\alpha+b\beta \in W^{\perp}$

$\therefore W^{\perp}$ is a subspace of V.

Note. The non-empty subset W need not be a subspace of V.

Theorem. 2. ***If W is a subspace of the inner product space V(F) then $W^{\perp}$ is also subspace of V and $W \cap W^{\perp} = \{\bar{0}\}$.***

Proof. W is a subspace of $V \Rightarrow$ W is a non - empty subset of V

$\Rightarrow W^{\perp}$ is subspace of V (By theorem(1)). Let $\bar{0} \in V$ be the zero element.

Then $\bar{0} \in W$ and $\bar{0} \in W^{\perp}$ as $W, W^{\perp}$ are both subspaces of V. $\therefore \bar{0} \in W \cap W^{\perp}$.

Let $\alpha \in W \cap W^{\perp}$. $\alpha \in W^{\perp} \Rightarrow \langle \alpha,\beta\rangle = 0 \ \forall \ \beta \in W$

As $\alpha \in W$ we have $\langle \alpha,\alpha\rangle = 0 \Rightarrow \alpha = \bar{0}$

$\therefore$ Every $\alpha \in W \cap W^{\perp}$ is such that $\alpha = \bar{0}$. Hence $W \cap W^{\perp} = \{\bar{0}\}$.

SOLVED PROBLEMS

Ex. 5. *Let $V = P_3(R)$ be the inner product space of atmost 3rd degree polynomials continuous on $[-1,1]$ let the innerproduct be defined as $\langle f,g\rangle = \int_{-1}^{1} f(t)\, g(t)\, dt$ where $f,g \in V$. If $W = P_2(R)$ is the subspace of V with standard basis $B = \{1,x,x^2\}$, by Gram - Schmidt process obtain its orthonormal basis.*

Sol. Let $B = \{\beta_1,\beta_2,\beta_3\}$ so that $\beta_1 = 1, \beta_2 = x, \beta_3 = x^2$.

Let $B' = \{\alpha_1,\alpha_2,\alpha_3\}$ be the orthonormal basis.

$$\|\beta_1\|^2 = \langle \beta_1, \beta_1 \rangle = \int_{-1}^{1} (1)(1)\, dt = t]_{-1}^{1} = 2 \Rightarrow \alpha_1 = \frac{1}{\|\beta_1\|}\beta_1 = \frac{1}{\sqrt{2}}$$

$$\langle \beta_2, \alpha_1 \rangle = \int_{-1}^{1} t.\frac{1}{\sqrt{2}}\, dt = 0. \qquad \gamma_2 = \beta_2 - \langle \beta_2, \alpha_1 \rangle \alpha_1 = x - 0\alpha_1 = x$$

$$\|\gamma_2\|^2 = \langle \gamma_2, \gamma_2 \rangle = \int_{-1}^{1} t.t\, dt = 2\int_{0}^{1} t^2 = \frac{2}{3} \Rightarrow \alpha_2 = \frac{1}{\|\gamma_2\|}\gamma_2 = \sqrt{\frac{3}{2}}.x$$

$$\langle \beta_3, \alpha_1 \rangle = \int_{-1}^{1} t^2.\frac{1}{\sqrt{2}}\, dt = \frac{1}{\sqrt{2}}\left(\frac{2}{3}\right) = \frac{\sqrt{2}}{3}; \quad \langle \beta_3, \alpha_2 \rangle = \int_{-1}^{1} t^2.\sqrt{\frac{3}{2}}\, dt = \sqrt{\frac{3}{2}}\int_{-1}^{1} t^3 dt = 0$$

$$\gamma_3 = \beta_3 - \langle \beta_3, \alpha_1 \rangle \alpha_1 - \langle \beta_3, \alpha_2 \rangle \alpha_2 = x^2 - \frac{\sqrt{2}}{3}.\frac{1}{\sqrt{2}} - 0.\sqrt{\frac{3}{2}}x = x^2 - \frac{1}{3}$$

$$\|\gamma_3\|^2 = \int_{-1}^{1}\left(t^2 - \frac{1}{3}\right)^2 dt = \int_{-1}^{1}\left(t^4 - \frac{2}{3}t^2 + \frac{1}{9}\right) dt = 2\int_{0}^{1}\left(t^4 - \frac{2}{3}t^2 + \frac{1}{9}\right) dt = 2\left(\frac{1}{5} - \frac{2}{3}.\frac{1}{3} + \frac{1}{9}\right) = \frac{8}{45}$$

$$\therefore\ \alpha_3 = \frac{1}{\|\gamma_3\|}\gamma_3 = \left(x^2 - \frac{1}{3}\right)\sqrt{\frac{45}{8}} = \sqrt{\frac{5}{8}}(3x^2 - 1)$$

Hence orthonormal basis of the subspace $W = \{\alpha_1, \alpha_2, \alpha_3\} = \left\{\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}x, \sqrt{\frac{5}{8}}(3x^2 - 1)\right\}$

Ex. 6. *Represent the polynomial* $f(x) = 1 + 2x + 3x^2 \in P_2(R)$ *as linear combination of the vectors in the orthonormal basis* $\{\alpha_1, \alpha_2, \alpha_3\}$ *in Ex. 5.*

Sol. The linear combination of $f(x) = \sum_{i=1}^{3} \langle f, \alpha_i \rangle \alpha_i$

$$\langle f, \alpha_1 \rangle = \int_{-1}^{1} (1 + 2t + 3t^2)\frac{1}{\sqrt{2}}\, dt = \frac{2}{\sqrt{2}}\left\{\int_{0}^{1} 1\, dt + 3\int_{0}^{1} t^2 dt\right\} = \sqrt{2}\{1 + 1\} = 2\sqrt{2}$$

$$\langle f, \alpha_2 \rangle = \int_{-1}^{1} (1 + 2t + 3t^2)\sqrt{3/2}\, t\, dt = 2(\sqrt{3/2})\left\{\int_{0}^{1} 2t^2\, dt\right\} = 2\sqrt{\frac{3}{2}}.\frac{2}{3} = \frac{2\sqrt{6}}{3}$$

$$\langle f, \alpha_3 \rangle = \int_{-1}^{1} (1 + 2t + 3t^2)\sqrt{5/8}\,(3t^2 - 1)\, dt = 2(\sqrt{5/8})\left\{\int_{0}^{1} (-1 + 9t^4)\, dt\right\} = \frac{2\sqrt{10}}{5}.$$

$$\therefore\ f(x) = 2\sqrt{2}\left(\frac{1}{\sqrt{2}}\right) + \frac{2\sqrt{6}}{3}\left(\sqrt{\frac{3}{2}}x\right) + \frac{2\sqrt{10}}{5}\left(\sqrt{\frac{5}{8}}(3x^2 - 1)\right)$$

Ex. 7. *Let* $S = \{(1, 0, i), (1, 2, 1)\}$ *be a subset of the inner product space* $C^3(C)$ *with respect to standard inner product. Find* $S^{\perp}$. **(*A. U. 2012*)**

Sol. Let $S = \{\beta_1, \beta_2\}$ where $\beta_1 = (1, 0, i), \beta_2 = (1, 2, 1)$

By def. $S^{\perp} = \{\alpha \in C^3(C) \mid < \alpha, \beta_1 > = 0, < \alpha, \beta_2 > = 0\}$

Let $\alpha = (x_1, x_2, x_3)$ where $x_1, x_2, x_3 \in C$

$< \alpha, \beta_1 > = 0 \Rightarrow \bar{1} x_1 + \bar{0} x_2 + \bar{i} x_3 = 0 \Rightarrow x_1 - x_3 i = 0$ ... (1)

$< \alpha, \beta_2 > = 0 \Rightarrow \bar{1} x_1 + \bar{2} x_2 + \bar{i} x_3 = 0 \Rightarrow x_1 + 2x_2 + x_3 = 0$ ... (2)

(1), (2) $\Rightarrow x_3 i + 2x_2 + x_3 = 0 \Rightarrow x_3 (i+1) + 2x_2 = 0$

$\Rightarrow x_2 = -\frac{1}{2}(i+1) x_3$.

$\therefore \alpha = \left(i x_3, -\frac{1}{2}(1+i) x_3, x_3 \right) = x_3 \left((i) - \frac{1}{2}(1+i), 1 \right)$ and hence

$$S^{\perp} = \left\{ x_3 \left(i, -\frac{1}{2}(1+i), 1 \right) \middle| x_3 \in C \right\}$$

Ex. 8. *In the inner product space $C^3(C)$ with standard inner product find the orthonormal basis of $w = L(\{x\})$ where $x = (i, 0, 1)$* **(A. U. 2012)**

Sol. $x = (i, 0, 1) \Rightarrow \| x \|^2 = i(-i) + 0.0 + 1.1 = 2$

$w = L(\{x\}) = a(i, 0, 1)$ where $a \in C$

$\therefore$ Orthonormal basis of $w = \left(\frac{i}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right)$

EXERCISE 8 (*b*)

1. Normalize the vectors (*a*) $\{(1,1,0), (1,-1,1), (-1,1,2)\}$ (*b*) $\{(1,0,1), (1,0,-1), (0,3,0)\}$ in $R^3(R)$.

2. Find an orthonormal basis of $R^2(R)$ from the basis (*a*) $\{(1,-3), (2,2)\}$ (*b*) $\{(1,0), (3,-5)\}$

3. Applying Gram - Schmidt process obtain an orthonormal basis of $R^3(R)$ from the basis

   (*a*) $\{(1,0,1), (1,0,-1), (0,3,4)\}$ ***(S. K. D. 08, N. U. 11, 12, O. U. 12, K. U. 12)***

   (*b*) $\{(2,0,1), (3,-1,5), (0,4,2)\}$ ***(N.U. 03, 08, K. U. 11, O. U. 11)***

   (*c*) $\{(1,1,0), (-1,1,0), (1,2,1)\}$ ***(K.U. 04)*** (*d*) $\{(1,-1,2), (0,2,1), (1,2,0)\}$ ***(A. U. 11, 14)***

4. Applying Gram - Schmidt process compute the orthogonal set and then normalize to obtain an orthonormal set of $\{(1,0,1,0), (1,1,1,1), (0,1,2,1)\}$ in $R^4(R)$.

5. Find an orthonormal basis of the subspace $W = L(\{(1,0,i), (2,1,1+i)\})$ of $C^3(C)$.

6. If $\langle f, g \rangle = \int_0^1 f(t)\, g(t)\, dt$ is the inner product in $P_2(R)$ with a basis $S = \{1, x, x^2\}$ find the Fourier coefficients of $h(x) = 1 + x$ relative to an orthonormal basis.

7. If $V = L(S)$ with inner product $\langle f, g\rangle = \int_0^{\pi} f(t)\, g(t)\, dt$ and $S = \{\sin t, \cos t, 1, t\}$ find an orthonormal basis and compute the Fourier coefficients of $h(t) = 2t + 1$.

ANSWERS

1. (*a*) $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right), \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(\frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right)$

(*b*) $\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}\right), (0,1,0)$ **2.** (*a*) $\left(\frac{1}{\sqrt{10}}, \frac{-3}{\sqrt{10}}\right), \left(\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}}\right)$ (*b*) $(1,0), (0,-1)$

3. (*a*) $\left\{\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right), (0,1,0)\right\}$

(*b*) $\left\{\left(\frac{2}{\sqrt{5}}, 0, \frac{1}{\sqrt{5}}\right), \left(\frac{-7}{\sqrt{270}}, \frac{-5}{\sqrt{270}}, \frac{14}{\sqrt{270}}\right), \left(\frac{-1}{\sqrt{54}}, \frac{7}{\sqrt{54}}, \frac{2}{\sqrt{54}}\right)\right\}$

(*c*) $\left\{\frac{1}{\sqrt{2}}(1,1,0), \frac{1}{\sqrt{2}}(-1,1,0), (0,0,1)\right\}$

(*d*) $\left\{\frac{1}{\sqrt{6}}(1,-1,2), \frac{1}{\sqrt{5}}(1,0,2), \frac{30}{\sqrt{374}}\left(\frac{7}{6}, \frac{7}{30}, \frac{-1}{3}\right)\right\}$

4. $\left\{\frac{1}{\sqrt{2}}(1,0,1,0), \frac{1}{\sqrt{2}}(0,1,0,1), \frac{1}{\sqrt{2}}(-1,0,1,0)\right\}$ **5.** $\left\{\left(\frac{1}{\sqrt{2}}, 0, \frac{i}{\sqrt{2}}\right), \left(\frac{1+i}{2\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1-i}{2\sqrt{2}}\right)\right\}$

6. $3/2, 1/2\sqrt{3}, 0$

7. Orthogonal basis $= \left\{\sin t, \cos t, 1 - \frac{4}{\pi}\sin t, t + \frac{4}{\pi}\cos t - \frac{\pi}{2}\right\}$

Problems for Practicals

EIGEN VALUS AND EIGEN VECTORS

1. Find all eigen values and eigen vectors of the following matrices :

$(i)\ \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ $(ii)\ \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

2. Find an invertible matrix P such that $P^{-1}AP = D$ where D is the diagonal matrix where

(1) $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$ (2) $A = \begin{bmatrix} 2 & 0 & -1 \\ 4 & 1 & -4 \\ 2 & 0 & -1 \end{bmatrix}$

3. Let T be the linear operator defined on $P_2(R)$ by $T(f(x)) = f(x) + (x+1)f'(x)$ find the matrix of T with respect to standard basis β. Find all eigen valus of $[T]_\beta$.

4. Prove that $A = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$ is diagonalizable.

INNER PRODUCT SPACES & ORTHOGONALITY

1. Define orthogonal vectors in an inner product space. Prove that orthogonality is a symmetric relation. Prove that $(1,1,0), (1,-1,1), (-1,1,2)$ in $R^3(R)$ are mutually orthogonal.
2. Define orthonormal vectors in an inner product space. Compute the orthonormal set of $\{(1,1,0),(1,-1,1),(-1,1,2)\}$ in $R^3(R)$.
3. Define orthogonal and orthonormal set of vectors in an inner product space. Give examples of orthonormal basis of $R^2(R)$ and $R^3(R)$ with explanation.
4. If α, β are two vectors in a real inner product space then prove that $\|\alpha\| = \|\beta\| \Leftrightarrow \alpha - \beta, \alpha + \beta$ are orthogonal.
5. Find a vector of unit length which is orthogonal to $\alpha = (2,-1,6)$ in $R^3(R)$ with standard inner product.
6. Let $V(C)$ be a vector space of continuous complex valued functions on $[0,2\pi]$ with inner product $\langle f,g\rangle = \frac{1}{2\pi}\int_0^{2\pi} f(t)\,\overline{g(t)}\,dt \quad \forall f,g \in V$.

Prove that $f_m(t) = e^{imt}, f_n(t) = e^{int}$ where $m, n \in N$ and $0 \le t \le 2\pi$ are orthogonal. Are f_m, f_n orthonormal.

7. Explain the concept of Gram - Schmidt orthogonalisation process.

8. If $\{(2,1,3),(1,2,3),(1,1,1)\}$ is a basis of R^3, compute the orthonormal basis of R^3 with standard inner product.

9. If $B = \{(1,0,1,0),(1,1,1,1),(0,1,2,1)\}$ is a linearly independent subset of R^4 obtain the orthonormal set.

10. If $\{1, x, x^2\}$ is the standard ordered basis of $P_2(R)$ the vector space of all polynomials of atmost 2nd degree and $\langle f, g \rangle = \int_{-1}^{1} f(t)\, g(t)\, dt$ is the innerproduct find the orthonormal basis.

11. If $M_2(R)$ is the space of 2×2 symmetric matrices over real numbers and $\langle A, B \rangle =$ trace (AB) is the inner product prove that $\left\{\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}, \frac{1}{\sqrt{2}}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\right\}$ is orthonormal set.

12. Find orthogonal basis of $S = \{(1,1,1),(0,1,1),(0,0,1)\}$ in $R^3(R)$ with standard inner product.

13. Let V = P(R) with the inner product $\langle f(x), g(x) \rangle = \int_{-1}^{1} f(t)\, g(t)\, dt$ and consider the subspace $P_2(R)$ with the standard ordered basis β. Use the Gram - Schmidt process to replace β by an orthogonal basis $\{\upsilon_1, \upsilon_2, \upsilon_3\}$ for $P_2(R)$ and then use this orthogonal basis to obtain an orthonormal basis for $P_2(R)$.

14. In each part, apply the Gram - Schmidt process to the given subset S of the Inner product space V to obtain an orthogonal basis for span(S). Then normalize the vectors in this basis to obtain an orthonormal basis β for span(S), and compute the Fourier coefficients of the given vector relative to β.

(*i*) $V = R^3, S = \{(1,0,1),(0,1,1),(1,3,3)\}$ and $x = (1,1,2)$

(*ii*) $V = R^3, S = \{(1,1,1),(0,1,1),(0,0,1)\}$ and $x = (1,0,1)$

(*iii*) $V = P_2(R)$ with the inner product $\langle f(x), g(x) \rangle = \int_{0}^{1} f(t)\, g(t)\, dt$ and $S = \{1, x, x^2\}$, and $h(x) = 1 + x$

(*iv*) V = span (S), where $S = \{(1,i,0),(1-i,2,4i)\}$ and $x = (3+i,4i,-4)$

(*v*) $V = R^4, S = \{(2,-1,-1,3),(-2,1,-5,5),(-1,3,7,11)\}$ and $x = (-11,8,-4,18)$

OBJECTIVE TYPE QUESTIONS

MUTIPLE CHOICE QUESTIONS :

1. In the inner product space $V(F)$ for $\alpha, \beta \in V, (\overline{\alpha,\beta}) =$

(*a*) (α,β) (*b*) (β,α) (*c*) $(\bar{\alpha},\beta)$ (*d*) none

2. In the inner product space $V(F)$ where F is the field of real numbers, for $\alpha \in V, (\alpha,\alpha)$ is

(*a*) ≥ 0 (*b*) $\geq \bar{o}$ (*c*) > 0 (*d*) $= 0$

3. In the inner product space $V(F)$, for $\bar{o}, \alpha \in V$. $(\bar{o},\alpha) =$

(*a*) α (*b*) $\bar{o}$ (*c*) o (*d*) none

4 In the inner product space $V(F)$ for $a \in F, \alpha, \beta, \gamma \in V, (\alpha, a\beta + a\gamma) =$

(*a*) $a(\alpha,\beta) + a(\alpha,\gamma)$ (*b*) $\bar{a}(\beta,\alpha) + \bar{a}(\gamma,\alpha)$ (*c*) $\bar{a}(\alpha,\beta) + \bar{a}(\alpha,\gamma)$ (*d*) $\bar{a}(\overline{\alpha,\beta}) + \bar{a}(\overline{\alpha,\gamma})$

5. In the inner product space $V(F)$ for $a,b \in F; \alpha,\beta \in V, (a\alpha, b\beta) =$

(*a*) $ab(\alpha,\beta)$ (*b*) $a\bar{b}(\alpha,\beta)$ (*c*) $\bar{a}\,b(\alpha,\beta)$ (*d*) $\bar{a}\,\bar{b}(\alpha,\beta)$

6. In the inner product space $V(F)$, the norm of $\alpha \in V$ is

(*a*) non-negative real number (*b*) positive real number
(*c*) negative real number (*d*) 0

7. In the inner product space R^2 the unit vectors are

(*a*) 1 (*b*) (1,0), (0,1) (*c*) (1,0,0), (0,1,0), (0,0,1) (*d*) none

8. If α, β are two vectors of an inner product space such that $|(\alpha,\beta)| = \|\alpha\| \, \|\beta\|$ then α, β are (*a*) L. D. (*b*) L. I. (*c*) orthonormal (*d*) orthogonal

9. A unit vector orthogonal to (1,2) in R^2 is

(*a*) (1,0) (*b*) (2,−1) (*c*) $(2/\sqrt{5}, -1/\sqrt{5})$ (*d*) (0,0)

10. If $\bar{u}, \bar{v}$ are two vectors in a real inner product space such that $\|\bar{u}\| = \|\bar{v}\|$ then $\bar{u} + \bar{v}$ is orthogonal to (*a*) $\bar{u} + \bar{v}$ (*b*) $\bar{u} - \bar{v}$ (*c*) $\bar{v} - \bar{u}$ (*d*) none

11. If $S = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ is any finite orthonormal set in a finite dimensional inner product space and β is a vector then $\sum_{i=1}^{n} (\beta, \alpha_i)\,\alpha_i =$

(*a*) $\bar{o}$ (*b*) 0 (*c*) β (*d*) none

12. In an innerproduct space V (F)

(*a*) $|(\alpha,\beta)| \leq \|\alpha\| \|\beta\|$ (*b*) $\|(\alpha,\beta)\| \leq \|\alpha\| - \|\beta\|$ (*c*) $\|(\alpha,\beta)\| < \|\alpha\| \, \|\beta\|$ (*d*) none

13. If α, β are orthogonal unit vectors then $\|\alpha - \beta\| =$

(*a*) 0 (*b*) 1 (*c*) $\sqrt{2}$ (*d*) 2

14. If α, β are orthogonal vectors in a real inner product space then $\|\alpha+\beta\|^2 =$

(*a*) $\|\alpha\|+\|\beta\|^2$ (*b*) $\|\alpha\|^2+2\|\alpha\|\|\beta\|+\|\beta\|^2$ (*c*) $\|\alpha-\beta\|^2$ (*d*) none

15. In an inner product space if α, β are linearly dependent vectors then $|(\alpha,\beta)|$

(*a*) $<\|\alpha\|\ \|\beta\|$ (*b*) $>\|\alpha\|\ \|\beta\|$ (*c*) $=\|\alpha\|\ \|\beta\|$ (*d*) none

16. If $\alpha=(-1,0,1), \beta=(2,0,-2)$ then $\|\alpha+\beta\| =$

(*a*) 0 (*b*) 1 (*c*) 2 (*d*) $\sqrt{2}$

17. In the vector space $V_3(R)$ with standard inner product if θ is the angle between two non-zero vectors then

(*a*) $|\cos\theta|>1$ (*b*) $|\cos\theta|<1$ (*c*) $|\cos\theta|\le 1$ (*d*) none

18. If S is an orthonormal set of vectors in an inner product space V then for $\alpha \in S, (\alpha,\alpha) =$

(*a*) 0 (*b*) 1 (*c*) (0,0) (*d*) $\bar{o}$

19. Vector of unit length orthogonalto $\alpha=(2,-1,6)$ in $V_3(R)$ with respect to standard inner product is

(*a*) $(2/3,-2/3,-1/3)$ (*b*) $(2,-2,-1)$ (*c*) $(1,1,1)$ (*d*) $(0,0,0)$

FILL IN THE BLANKS :

20. Two vectors α, β of any inner product space are said to be orthogonal if .....

21. Any orthogonal set of non-zero vectors in an inner product space V is linearly .....

22. Schwarz's inequality in an inner product space is .....

23. In an inner product space with standard inner product if

$\alpha=(a_1,a_2,a_3), \beta=(b_1,b_2,b_3)$ then $(\alpha,\beta)=$.....

24. In an inner product space V if $\alpha \in V$ is unit vector then .....

25. In an inner product space $V(F)$ for $a \in F$, $\alpha \in V; \|a\alpha\| =$ .....

26. If $\alpha=(2,1,2)$ is a vector in $V_3(R)$ with standard inner product then unit vector = ...

27. If $(\alpha,\beta)=0$ for all $\beta \in V$ then $\alpha =$.....

28. If $(\alpha,\gamma)=(\beta,\gamma)$ for every $\gamma \in V$ then .....

29. If $\alpha,\beta \in V$ then $\text{Re}\,(\alpha,\beta) =$.....

30. If $\alpha,\beta \in V(R)$ where R is the field of real numbers then $\text{Re}\,(\alpha,\beta) =$.....

31. If α, β are vectors in an unitary space then $(\alpha,\beta) =$.....

32. In an inner product space if $\|\alpha+\beta\|=\|\alpha\|+\|\beta\|$ then the vectors α, β are .....

33. In an inner product space if two vectors α, β are orthonormal then .....

34. In an inner product space the only vector which is orthogonal to itself is .....

35. In an inner product space a set S of vectors is said to be an orthonormal set if .....

36. Any orthononnal set of vectors in an inner product space is linearly .....

37. If $B = \{\alpha_1, \alpha_2, ..., \alpha_n\}$ is any finite orthonormal set in an inner product space V and $\beta \in V$ then the Bessel's inequality is .....

38. If β is a linear combination of $\alpha_1, \alpha_2, ..., \alpha_n$, then $\sum_{i=1}^{n} |(\beta, \alpha_i)|^2 =$ .....

39. If α is a vector of n - dimensional unitary space V and α is orthogonal to n linearly independent vectors then $\alpha =$ .....

40. If $(\alpha, \beta) \in V(R)$ and $\|\alpha\| = \|\beta\|$ then $\alpha - \beta$ and $\alpha + \beta$ are.....

41. In an unitary space if $x, y \in V$ are orthogonal then $\|\alpha x + \beta y\|^2 =$ ..... $\forall \alpha, \beta \in C$

ANSWERS

EIGEN VALUES, EIGEN VECTORS

1. (*i*) 0,3,15; Vectors $\begin{bmatrix} 1/2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1/2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$ (*ii*) 2,2,8; vectors $\begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

INNER PRODUCT SPACES & ORTHOGONALITY

1. *b* **2.** *a* **3.** *c* **4.** *c* **5.** *b* **6.** *a* **7.** *b* **8.** *a* **9.** *c* **10.** *b*

11. *c* **12.** *a* **13.** *c* **14.** *a* **15.** *c* **16.** *d* **17.** *c* **18.** *b* **19.** *a*

20. $(\alpha, \beta) = 0$ **21.** independent **22.** $|(\alpha, \beta)| \le \|\alpha\| \, \|\beta\| \; \forall \alpha, \beta \in V$

23. $a_1\bar{b}_1 + a_2\bar{b}_2 + a_3\bar{b}_3$ **24.** $\|\alpha\| = 1$ **25.** $|a| \, \|\alpha\|$

26. $\left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$ **27.** $\alpha = \bar{o}$ **28.** $\alpha = \beta$

29. $\frac{1}{4}\|\alpha + \beta\|^2 - \frac{1}{4}\|\alpha - \beta\|^2$ **30.** (α, β) **31.** $\text{Re}(\alpha, \beta) + i\,\text{Re}(\alpha, i\beta)$

32. linearly dependent **33.** $(\alpha, \beta) = 0$ and $\|\alpha\| = \|\beta\| = 1$ **34.** zero vector

35. (*i*) $\alpha \in S \Rightarrow \|\alpha\| = 1$ and (*ii*) $\alpha, \beta \in S, \alpha \ne \beta \Rightarrow (\alpha, \beta) = 0$

36. independent **37.** $\sum_{i=1}^{n} |(\beta, \alpha_i)^2| \le \|\beta\|^2$ **38.** $\|\beta\|^2$

39. $\bar{o}$ **40.** orthogonal **41.** $\|\alpha x\|^2 + \|\beta y\|^2$

KEY
TO
A TEXT BOOK OF
B.Sc. MATHEMATICS (A.P.) III
5th Sememster, Paper-6
(Linear Algebra)

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UNIT - I - Vector Spaces - I

1. VECTOR SPACES

EXERCISE 1 (*a*)

1. Let $\alpha=(x_1,x_2,x_3),\beta=(y_1,y_2,y_3),\gamma=(z_1,z_2,z_3)$

(*i*) By definition $\alpha+\beta$ is a triad of real numbers. Hence the set of triads is closed.

(*ii*) $(\alpha+\beta)+\gamma=(x_1+y_1,x_2+y_2,x_3+y_3)+(z_1,z_2,z_3)$

$=\{(x_1+y_1)+z_1,(x_2+y_2)+z_2,(x_3+y_3)+z_3\}$

$=\{x_1+(y_1+z_1),x_2+(y_2+z_2),x_3+(y_3+z_3)\}$ since + is associative in $\mathbb{R}$.

$=(x_1,x_2,x_3)+(y_1+z_1,y_2+z_2,y_3+z_3)$

$=\alpha+(\beta+\gamma)$ $\quad\therefore$ Vector addition is associative.

(*iii*) $\alpha+\bar{O}=(x_1,x_2,x_3)+(0,0,0)=(x_1+0,x_2+0,x_3+0)=(x_1,x_2,x_3)=\alpha$

Similarly $\bar{O}+\alpha=\alpha\Rightarrow\alpha+\bar{O}=\bar{O}+\alpha=\alpha$ $\quad\therefore\ \bar{O}=(0,0,0)$ is the identity.

(*iv*) $\alpha+(-\alpha)=(x_1,x_2,x_3)+(-x_1,-x_2,-x_3)=(x_1-x_1,x_2-x_2,x_3-x_3)=(0,0,0)=\bar{O}$

$-\alpha$ is the additive inverse of α

(*v*) $\alpha+\beta=(x_1+y_1,x_2+y_2,x_3+y_3)$

$=(x_1+y_1,x_2+y_2,x_3+y_3)=(y_1+x_1,y_2+x_2,y_3+x_3)=\beta+\alpha$

(*vi*) By definition $c\alpha$ is a triad.

(*vii*) Let $x,y\in\mathbb{R}$. $\quad x(y\alpha)=x(yx_1,yx_2,yx_3)=\{x(yx_1),x(yx_2),x(yx_3)\}$

$=\{(xy)x_1,(xy)x_2,(xy)x_3\}=xy(x_1,x_2,x_3)=(xy)\alpha$

(*viii*) $x\in\mathbb{R}$ and $x(\alpha+\beta)=x(x_1+y_1,x_2+y_2,x_3+y_3)$

$=\{x(x_1+y_1),x(x_2+y_2),x(x_3+y_3)\}=(xx_1+xy_1,xx_2+xy_2,xx_3+xy_3)$

$=(xx_1,xx_2,xx_3)+(xy_1,xy_2,xy_3)=x\alpha+x\beta$

(*ix*) $x,y\in\mathbb{R}$ and $(x+y)\alpha=(x+y)(x_1,x_2,x_3)$

$=\{(x+y)x_1,(x+y)x_2,(x+y)x_3\}=\{xx_1+yx_1,xx_2+yx_2,xx_3+yx_3\}$

$=(xx_1,xx_2,xx_3)+(yx_1+yx_2,yx_3)=x\alpha+y\alpha$

(*x*) $1.\alpha=1(x_1,x_2,x_3)=(x_1,x_2,x_3)=\alpha$

Since all postulates are verfied the set of all triads is a vector space.

2. Let P be the set of all polynomials over the field $\mathbb{R}$. Clearly P(R) is non-empty.

Let $f(x) = a_0 + a_1x + a_2x^2 + ...,\ g(x) = b_0 + b_1x + b_2x^2 + ... \in P(R)$ and

$c \in \mathbb{R},\ f(x) + g(x) = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + ... \in P(R)$

$cf(x) = ca_0 + ca_1x + ca_2x^2 + ... \in P(R)$. Let $h(x) = c_0 + c_1x + c_2x^2 + ... \in P(R)$

(*i*) Since $f(x) + g(x) \in P(R)$. Hence P(R) is closed.

(*ii*) $[f(x) + g(x)] + h(x) = [(a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + ...] + (c_0 + c_1x + c_2x^2 + ...)$

$= [(a_0 + b_0) + c_0] + [(a_1 + b_1) + c_1]\,x + [(a_2 + b_2) + c_2]\,x^2 + ...$

$= [a_0 + (b_0 + c_0)] + [a_1 + (b_1 + c_1)]\,x + [a_2 + (b_2 + c_2)]\,x^2 + ...$

$= (a_0 + a_1x + a_2x^2 + ...) + [(b_0 + c_0) + (b_1 + c_1)x + (b_2 + c_2)x^2 + ...]$

$= f(x) + [g(x) + h(x)]$ $\quad \therefore$ + is associative in P(R).

(*iii*) Clearly $O = O(x) = 0 + 0x + 0x^2 + ... \in P(R)$

$f(x) + O(x) = (a_0 + 0) + (a_1 + 0)x + (a_2 + 0)x^2 + ... = a_0 + a_1x + a_2x^2 + ... = f(x)$

Similarly $O(x) + f(x) = f(x)$

$\therefore f(x) + O(x) = O(x) + f(x) = f(x)$ $\quad \therefore O(x)$ is the additive identity in P(R)

(*iv*) Let $f(x) = a_0 + a_1x + a_2x^2 + ... \in P(R)$

Now $(-f)(x) = (-a_0) + (-a_1)x + (-a_2)x^2 + ... \in P(R)$

$\therefore f(x) + (-f)(x) = [a_0 + (-a_0)] + [a_1 + (-a_1)]\,x + [a_2 + (-a_2)]x^2 + ...$

$= 0 + 0x + 0x^2 + ... = 0(x)$. Similarly $(-f)(x) + f(x) = O(x)$

$\therefore f(x) + (-f)(x) = (-f)(x) + f(x) = O(x)$

$\therefore (-f)(x)$ in the additive inverse of $f(x)$ in P(R)

(*v*) $f(x) + g(x) = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + ...$

$= (b_0 + a_0) + (b_1 + a_1)x + (b_2 + a_2)x^2 + ... = g(x) + f(x)$

$\therefore$ + is commutative in P(R). $\quad \therefore (P(R), +)$ is an abelian group.

(*vi*) For $c \in \mathbb{R},\ cf(x) = ca_0 + ca_1x + ca_2x^2 + ... \in P(R)$

(*vii*) Let $a, b \in \mathbb{R},\ a(bf(x)) = a(ba_0 + ba_1x + ba_2x^2 + ...)$

$= aba_0 + aba_1x + aba_2x^2 + ... = ab(a_0 + a_1x + a_2x^2 + ...) = ab\,f(x)$

(*viii*) Let $a \in \mathbb{R}$ and $f(x), g(x) \in \mathrm{P(R)} \Rightarrow$

$$a\,(f(x)+g(x) = a\,[(a_0+b_0)+(a_1+b_1)\,x+(a_2+b_2)\,x^2+...]$$

$$= \{a\,(a_0+b_0)+a\,(a_1+b_1)\,x+a\,(a_2+b_2)\,x^2+....\}$$

$$= \{(aa_0+ab_0)+(aa_1+ab_1)\,x+(aa_2+ab_2)\,x^2+....\}$$

$$= \{(aa_0+aa_1x+aa_2x^2+...)+(ab_0+ab_1x+ab_2x^2+....)\}$$

$$= \{a(a_0+a_1x+a_2x^2+....)+a\,(b_0+b_1x+b_2x^2+....)\} = af(x)+ag(x)$$

(*ix*) $a,b \in \mathbb{R}$ and $f(x) \in \mathrm{P(R)} \Rightarrow\ (a+b)f(x) = (a+b)(a_0+a_1x+a_2x^2+....)$

$$= \{(a+b)\,a_0+(a+b)\,a_1x+(a+b)\,a_2x^2+....\}$$

$$= \{(aa_0+ba_0)+(aa_1x+ba_1x)+(aa_2x^2+ba_2x^2)+.....\}$$

$$= \{(aa_0+aa_1x+aa_2x^2+...)+(ba_0+ba_1x+ba_2x^2+....)\}$$

$$= \{a(a_0+a_1x+a_2x^2+....)+b\,(a_0+a_1x+a_2x^2+....)\} = af(x)+bf(x)$$

(*x*) $1.f(x) = 1.(a_0+a_1x+a_2x^2+....) = a_0+a_1x+a_2x^2+..... = f(x)$

Since all postulates are verfied P(R) is a vector space.

3. $(\mathrm{F},+,\bullet)$ in a field. Here addition is treated as internal composition and multiplication is treated as external composition.

To prove F (F) in a vector space we require to prove all the postulates of the vector space. Since $(\mathrm{F},+,\bullet)$ is a field. $\therefore$ $(\mathrm{F},+)$ is abelian.

$\therefore$ Clousure, associative, identity, inverse and commutativity are satisfied.

The scalar multiplication is the same as multiplication of field because the set F and field are same, so $a \in \mathrm{F}, x \in \mathrm{F} \Rightarrow ax \in \mathrm{F}$

$(a+b)\,x = ax+bx$, $\quad a\,(x+y) = ax+ay$

$1.x = x.1 = x$ $\quad \forall\ 1,a,b \in \mathrm{F}$, $\forall\ x,y \in \mathrm{F}$. Hence F(F) is a vector space.

4. Let $\alpha,\beta,\gamma \in \mathbb{R}^n$, where $\alpha = (x_1,x_2,.....,x_n), \beta = (y_1,y_2,.....,y_n)$ and

$\gamma = (z_1,z_2,.....,z_n)$ where $x_i,y_i,z_i \in \mathbb{R}, i = 1$ to n

(*i*) If $\alpha,\beta \in \mathbb{R}^n, \alpha+\beta = (x_1+y_1, x_2+y_2, x_n+y_n) \in \mathbb{R}^n$

(*ii*) $(\alpha+\beta)+\gamma = (x_1+y_1,x_2+y_2,.....,x_n+y_n)+(z_1,z_2,.....,z_n)$

$$= \{(x_1+y_1)+z_1,(x_2+y_2)+z_2,.....,(x_n+y_n)+z_n\}$$

$= \{x_1 + (y_1 + z_1), x_2 + (y_2 + z_2), ..., x_n + (y_n + z_n)\}$

$= (x_1, x_2, ..., x_n) + (y_1 + z_1, y_2 + z_2, ..., y_n + z_n) = \alpha + (\beta + \gamma)$

$\therefore$ Vector addition is associative in $\mathbb{R}^n$

(iii) $\alpha + \bar{O} = (x_1, x_2, ..., x_n) + (0, 0, ..., 0) = (x_1 + 0, x_2 + 0, ..., x_n + 0) = (x_1, x_2, ..., x_n) = \alpha$

Similarly, $\bar{O} + \alpha = \alpha$. $\quad \therefore \alpha + \bar{O} = \bar{O} + \alpha = \alpha$

$\therefore$ The n-tuple $\bar{O} = (0, 0, ..., 0)$ is the identity in $\mathbb{R}^n$

(iv) $\alpha + (-\alpha) = (x_1, x_2, ..., x_n) + (-x_1, -x_2, ..., -x_n)$

$= (x_1 - x_1, x_2 - x_2, ..., x_n - x_n) = (0, 0, ..., 0) = \bar{O}$. $\quad \therefore \alpha + (-\alpha) = \bar{O}$

$\Rightarrow (-\alpha) \in \mathbb{R}^n$ is the additive inverse of α in $\mathbb{R}^n$.

(v) $\alpha + \beta = (x_1 + y_1, x_2 + y_2, ..., x_n + y_n) = (y_1 + x_1, y_2 + x_2, ..., y_n + x_n) = \beta + \alpha$

(vi) By definition, $a \in \mathbb{R}$, $a\alpha = (ax_1, ax_2, ..., ax_n) \in \mathbb{R}^n(\mathbb{R})$

(vii) Let $a, b \in \mathbb{R}$, $a(b\alpha) = a(bx_1, bx_2, ..., bx_n)$

$= \{a(bx_1), a(bx_2), ..., a(bx_n)\} = \{(ab)x_1, (ab)x_2, ..., (ab)x_n\}$

$= (ab)(x_1, x_2, ..., x_n) = (ab)\alpha$

(viii) $a \in \mathbb{R}$ and $\alpha, \beta \in \mathbb{R}^n \Rightarrow a(\alpha + \beta) = a(x_1 + y_1, x_2 + y_2, ..., x_n + y_n)$

$= \{a(x_1 + y_1), a(x_2 + y_2), ..., a(x_n + y_n)\} = \{ax_1 + ay_1, ax_2 + ay_2, ..., ax_n + ay_n\}$

$= (ax_1, ax_2, ..., ax_n) + (ay_1, ay_2, ..., ay_n)$

$= a(x_1, x_2, ..., x_n) + a(y_1, y_2, ..., y_n) = a\alpha + a\beta$

(ix) $a, b \in \mathbb{R}$ and $\alpha \in \mathbb{R}^n \Rightarrow (a + b)\alpha = (a + b)(x_1, x_2, ..., x_n)$

$= \{(a + b)x_1, (a + b)x_2, ..., (a + b)x_n\} = \{ax_1 + bx_1, ax_2 + bx_2, ..., ax_n + bx_n\}$

$= (ax_1, ax_2, ..., ax_n) + (bx_1, bx_2, ..., bx_n)$

$= a(x_1, x_2, ..., x_n) + b(x_1, x_2, ..., x_n) = a\alpha + b\alpha$

(x) $1 . \alpha = 1(x_1, x_2, ..., x_n) = (x_1, x_2, ..., x_n) = \alpha$

Since all the postulates are verified $\mathbb{R}^n(\mathbb{R})$ is a vector space.

5. Here F is the set of vectors.

Addition of vectors is the addition composition in the field F. Since F is a field,

$\therefore$ (F, +) is an abelian group.

Further the elements of the subfield K constitute the set of scalars.

The composition of scalar multiplication is the multiplication composition in the field F.

F is a field, $\therefore a\alpha \in F \; \forall a \in K, \forall \alpha \in F$, because botha a and α are elements of F.

If l is the unity element of F, then 1 is also the unity element of the subfield K.

We make the following observations.

(*i*) $a(\alpha+\beta) = a\alpha + a\beta \; \forall \; a \in K$ and $\alpha, \beta \in F$ (*ii*) $(a+b)\alpha = a\alpha + b\alpha \; \forall \; a, b, \in K \; \forall \alpha \in F$.

(*iii*) $(ab)\alpha = a(b\alpha) \; \forall a, b \in K$ and $\forall \alpha \in F$ (*iv*) $1\alpha = \alpha \; \forall \alpha \in K$.

Hence F(K) is a vector space.

6. V(R) in a vector space. $W = \{(x, y) \mid x, y \in V\}$

(I) To prove (W,+) is abelian, let $x = (x_1, y_1), y = (x_2, y_2)$.

$x + y = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$

***a)* Clousure property :** $x + y = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \in W$

Since $x_1 + x_2 \in V, y_1 + y_2 \in V$ $\therefore x + y \in W$

***b)* Associative property :** $x, y, z \in W$

$x = (x_1, y_1), y = (x_2, y_2), z = (x_3, y_3)$ where $x_1, x_2, x_3, y_1, y_2, y_3 \in V$

$(x+y)+z = (x_1 + x_2, y_1 + y_2) + (x_3, y_3) = ((x_1 + x_2) + x_3, (y_1 + y_2) + y_3)$

$= (x_1 + (x_2 + x_3), y_1 + (y_2 + y_3)) = x + (y + z)$

$\therefore \; \forall \; x, y, z \in W$. $(x+y)+z = x+(y+z)$

***c)* Identity property :** $\bar{0} = (0,0) \in W$

$x + \bar{0} = (x_1, y_1) + (0,0) = (x_1, y_1) = x$. $\therefore \; \forall x \in W$. $x + \bar{0} = \bar{0} + x = x$

***d)* Inverse property :** $x_1, y_1 \in V \Rightarrow -x_1, -y_1 \in V$

$-x = (-x_1, -y_1)$. $\therefore \; \forall \; x \in W$ $x + (-x) = (-x) + x = \bar{0}$

***e)* Commutativity :** $x, y \in W$

$(x+y) = (x_1 + x_2, y_1 + y_2) = (x_2 + x_1, y_2 + y_1) = y + x$. $\therefore \; \forall \; x, y \in W$ $x + y = y + x$

(II) $\alpha = a_1 + ia_2 \in C$, $x \in W$. $\alpha x = (a_1 + a_2 i)(x_1, y_1) = (a_1 x_1 - a_2 y_1, a_2 x_1 + a_1 y_1)$

Since $a_1 x_1 - a_2 y_1, a_2 x_1 + a_1 y_1 \in V$. $\therefore \; \alpha x \in W$

Scalar multiplication is well defined.

(III) (*i*) $\alpha \in C$, $x, y \in W$, $\beta = (b_1 + b_2 i) \in C$

$\alpha(x+y) = \alpha(x_1 + x_2, y_1 + y_2) = (a_1(x_1 + x_2) - a_2(y_1 + y_2), a_2(x_1 + x_2) + a_1(y_1 + y_2)$

$= (a_1 x_1 + a_1 x_2 - a_2 y_1 - a_2 y_2, a_2 x_1 + a_2 x_2 + a_1 y_1 + a_1 y_2)$

$\alpha x + \alpha y = \alpha (x_1, y_1) + \alpha (x_2, y_2) = (a_1 x_1 - a_2 y_1, a_2 x_1 + a_1 y_1) + (a_1 x_2 - a_2 y_2, a_2 x_2 + a_1 y_2)$

$= (a_1 x_1 + a_1 x_2 - a_2 y_1 - a_2 y_2, a_2 x_1 + a_2 x_2 + a_1 y_1 + a_1 y_2)$

$\therefore \alpha (x + y) = (\alpha x + \alpha y) \qquad \forall \alpha \in C, x, y \in W$

(ii) $(\alpha + \beta) x = ((a_1 + b_1) + i (a_2 + b_1)) (x_1, y_1)$

$= ((a_1 + b_1) x_1 - (a_2 + b_2) y_1, (a_2 + b_2) x_1 + (a_1 + b_1) y_1)$

$= (a_1 x_1 + b_1 x_1 - a_2 y_1 - b_2 y_1, a_2 x_1 + b_2 x_1 + a_1 y_1 + b_1 y_1)$

$\alpha x + \beta x = (a_1 + a_2 i) (x_1, y_1) + (b_1 + b_2 i) (x_1, y_1)$

$= (a_1 x_1 - a_2 y_1, a_2 x_1 + a_1 y_1) + (b_1 x_1 - b_2 y_1, b_2 x_1 + b_2 x_1 + b_1 y_1)$

$= (a_1 x_1 - a_2 y_1 + b_1 x_1 - b_2 y_1,\ a_2 x_1 + a_1 y_1 + b_2 x_1 + b_1 y_1)$. $\qquad \therefore (\alpha + \beta) x = \alpha x + \beta x$

(iii) $\alpha (\beta x) = \alpha [(b_1 + b_2 i) (x_1, y_1)] = \alpha (b_1 x_1 - b_2 y_1, b_2 x_1 + b_1 y_1)$

$= (a_1 + a_2 i) (b_1 x_1 - b_2 y_1,\ b_2 x_1 + b_1 y_1)$

$= (a_1 (b_1 x_1 - b_2 y_1) - a_2 (b_2 x_1 + b_1 y_1), a_2 (b_1 x_1 - b_2 y_1) + a_1 (b_2 x_1 + b_1 y_1)$

$= (a_1 b_1 x_1 - a_1 b_2 y_1 - a_2 b_2 x_1 - a_2 b_1 y_1,\ a_2 b_1 x_1 - a_2 b_2 y_1 + a_1 b_2 x_1 + a_1 b_1 y_1)$

$(\alpha\beta) x = [(a_1 + a_2 i) (b_1 + b_2 i)] (x_1, y_1)$

$= [(a_1 b_1 - a_2 b_2) + i (a_1 b_2 + a_2 b_1)] (x_1, y_1)$

$= ((a_1 b_1 - a_2 b_2) x_1 - (a_1 b_2 + a_2 b_1) y_1, (a_1 b_2 + a_2 b_1) x_1 + (a_1 b_1 - a_2 b_2) y_1)$

$= (a_1 b_1 x_1 - a_2 b_2 x_1 - a_1 b_2 y_1 - a_2 b_1 y_1,\ a_1 b_2 x_1 + a_2 b_1 x_1 + a_1 b_1 y_1 - a_2 b_2 y_1)$.

$\therefore \alpha (\beta x) = (\alpha\beta) x$

(iv) $1 . x = (1 + 0i) (x_1, y_1) = (1 . x_1 - 0 . y_1, 0 . x_1 + 1 . y_1) = (x_1, y_1) = x$

$\therefore 1 . x = x . 1 = x$. $\qquad \therefore$ W(C) is a vector space.

7. Addition, multiplications are defined as follows

$(f + g) x = f(x) + g(x) \ \forall\ f, g \in V$, $\qquad (af)(x) = af(x) \ \forall\ a \in R_1 f \in V$

We have $V = \{f \mid f(x) \in R \text{ for } x \in [0,1]\}$

I. To prove $(V, +)$ is abelian

(*a*) Clousure property :

Since the sum of two continuous is also a continuous function and the sum of two real valued is also a real valued.

$\therefore (f + g)(x) = f(x) + g(x)$ asserts that $(f + g)$ is also a real valued continuous function.

$\therefore f + g \in V \ \forall\ f, g \in V$

(*b*) Associativity :

$[(f+g)+h](x) = (f+g)(x)+h(x) = \{f(x)+g(x)\}+h(x)$

$= f(x)+\{g(x)+h(x)\} = [f+(g+h)](x) \quad \forall\, x \in [0,1]$

(*c*) Identity : Let us define O define as $O(x) = O \in R \;\forall\, x \in [0,1]$ then O is a real - valued function $\in V$ so $\exists\; O \in V$ such that

$(O+f)(x) = O(x)+f(x) = O+f(x) = f(x) \;\forall\, x \in [0,1]$

Thus $O+f = f+O = f \;\; \forall\, f \in V$. Here 'O' is the identity.

(*d*) Inverse : For each $f \in V$, let us define '$-f$' as $(-f)(x) = -f(x)$

then '$-f$' is also a real valued continuous function $\in V$.

$\forall\, f \in V, \exists -f \in V$, such that

$[f+(-f)](x) = f(x)+(-f)(x) = O = O(x) \;\forall\, x \in [0,1]$

$\therefore\; f+(-f) = (-f)+f = O$. Here '$-f$' is the additive inverse of 'f'.

(*e*) Commutativity : $(f+g)(x) = f(x)+g(x) = g(x)+f(x) = (g+f)(x)$

$\therefore \forall\, f, g \in V \qquad f+g = g+f$

II. Scalar multiplication : $\forall\, a \in R, f \in V$

$(af)(x) = af(x) \in R, \quad af \in V$

III. (*i*) $[a(f+g)](x) = a[(f+g)](x) \;\forall\, x \in [0,1] = a[f(x)+g(x)] = af(x)+ag(x)$

$= (af)(x)+(ag)(x) = (af+ag)(x) \;\forall\, x \in [0,1]$

$\therefore\; a(f+g) = (af+ag)$

(*ii*) $a, b \in R, \; f \in V$

$[(a+b)f](x) = (a+b)f(x) \;\forall\, x \in [0,1] = af(x)+bf(x) = (af)(x)+(bf)(x)$

$= (af+bf)(x) \;\forall\, x \in [0,1]$

(*iii*) $[(ab)f](x) = (ab)f(x) = a[bf(x)] = a[(bf)(x)] \qquad \forall\, x \in [0,1]$

(*iv*) $(1.f)(x) = 1.f(x) = f(x) \;\; \forall\, x \in [0,1]$. $\therefore$ V(R) in a vector space.

8. Let $\alpha = (x_1, y_1), \beta = (x_2, y_2)$. $a(\alpha+\beta) = a(3y_1+3y_2, -x_1-x_2)$ (by given definition)

$= (-3ax_1+3ax_2, -3ay_1-3ay_2)$.

But, $a\alpha+a\beta = a(x_1, y_1)+a(x_2, y_2) = (ax_1, ay_1)+(ax_2, ay_2) = (3ay_1+3ay_2, -ax_1-ax_2)$

$\therefore a(\alpha+\beta) \neq a\alpha+a\beta$. Hence V is not a vector space over the field F.

9. Additive identity does not exist.

10. C is the set of all complex numbers and R is the set of all real numbers.

Now $(R,+,\bullet)$ is a field. Here addition of complex numbers is vector addition.

Clearly $(C,+)$ is an abelian group.

Here multiplication of complex numbers is scalar multiplication.

Let $\alpha,\beta \in C$; $a,b \in R$. $a,b \in \mathbb{R} \Rightarrow a,b \in C$

Since $(C,+,\bullet)$ is a field, we have the following

$(i)\ a(\alpha+\beta)=a\alpha+b\beta$ $(ii)\ (a+b)\alpha=a\alpha+b\alpha$ $(iii)\ (ab)\alpha=a(b\alpha)$ $(iv)\ 1.\alpha=\alpha$

$\therefore$ $C(\mathbb{R})$ is a vector space. For $\alpha \in \mathbb{R}$ and $a=a_1 i+a_2 \in C$, then $a\alpha=(a_1 i+a_2)\alpha \notin \mathbb{R}$

Hence $\mathbb{R}(C)$ is not a vector space.

11. Let $\alpha=(x,y)$ and $a,b \in F$. Now $(a+b)\alpha=(a+b)(x,y)=((a+b)x,y)=(ax+bx,y)$

and $a\alpha+b\alpha=a(x,y)+b(x,y)=(ax,y)+(bx,y)=(ax+bx,2y)$

$\therefore$ $(a+b)\alpha \neq a\alpha+b\alpha$. Hence V(F) is not a vector space.

EXERCISE 1 (*b*)

1. (*i*) Let $\alpha,\beta \in W$ where $\alpha=(x_1,2y_1,3z_1)$ and

$\beta=(x_2,2y_2,3z_2)$ for some $x_1,y_1,z_1,x_2,y_2,z_2 \in R$.

Let $a,b \in R$. $\therefore$ $a\alpha+b\beta=a(x_1,2y_1,3z_1)+b(x_2,2y_2,3z_2)$

$=(ax_1,2ay_1,3az_1)+(bx_2,2by_2,3bz_2)=(ax_1+bx_2,2ay_1+2by_2,3az_1+3bz_2)$

clearly $ax_1+bx_2,2ay_1+2by_2,3az_1+3bz_2 \in R \Rightarrow a\alpha+b\beta \in W$ for all $a,b \in R$ and $\alpha,\beta \in W$.

Hence W is a subspace of $R^3(R)$

(*ii*) Let $\alpha,\beta \in W$ where $\alpha=(x_1,x_1,x_1)$ and $\beta=(x_2,x_2,x_2)$ for some $x_1,x_2 \in R$.

Let $a,b \in R$. $\therefore$ $a\alpha+b\beta=a(x_1,x_1,x_1)+b(x_2,x_2,x_2)$

$=(ax_1,ax_1,ax_1)+(bx_2,bx_2,bx_2)=(ax_1+bx_2,ax_1+bx_2,ax_1+bx_2)$

clearly $ax_1+bx_2 \in R \Rightarrow a\alpha+b\beta \in W$ for all $a,b \in R$ and $\alpha,\beta \in W$.

Hence W is a subspace of $R^3(R)$.

2. By definition $W \neq \phi$. Let $\alpha,\beta \in W$ where $\alpha=(x_1,y_1,z_1)$,

$\beta=(x_2,y_2,z_2)$ for for some $x_1,x_2,y_1,y_2,z_1,z_2 \in R$.

Then by definition $x_1-3y_1+4z_1=0$ .... (1) and $x_2-3y_2+4z_2=0$ ... (2)

If $a,b \in F$, then we have $a\alpha+b\beta=a(x_1,y_1,z_1)+b(x_2,y_2,z_2)$

$= (ax_1, ay_1, az_1) + (bx_2, by_2, bz_2) = (ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2)$

Now $(ax_1 + bx_2) - 3(ay_1 + by_2) + 4(az_1 + bz_2)$

$= (ax_1 - 3ay_1 + 4az_1) + (bx_2 - 3by_2 + 4bz_2)$

$= a(x_1 - 3y_1 + 4z_1) + b(x_2 - 3y_2 + 4z_2) = a.0 + b.0 = 0$ ($\because$ from (1) & (2))

$\therefore\ a\alpha + b\beta \in W$. Hence W is a subspace of $R^3(R)$.

3. Let $\alpha, \beta \in W$ where $\alpha = (x_1 + 2y_1, y_1, -x_1 + 3y_1)$,

$\beta = (x_2 + 2y_2, y_2, -x_2 + 3y_2)$ for some $x_1, y_1, x_2, y_2 \in R$.

Let $a, b \in R$. $\alpha, \beta \in W$

$\therefore\ a\alpha + b\beta = a(x_1 + 2y_1, y_1, -x_1 + 3y_1) + b(x_2 + 2y_2, y_2, -x_2 + 3y_2)$

$= (ax_1 + 2ay_1, ay_1, -ax_1 + 3ay_1) + (bx_2 + 2by_2, by_2, -bx_2 + 3by_2)$

$= (ax_1 + 2ay_1 + bx_2 + 2by_2, ay_1 + by_2, -ax_1 + 3ay_1 - bx_2 + 3by_2)$

$= ((ax_1 + bx_2) + 2(ay_1 + by_2), ay_1 + by_2, -(ax_1 + bx_2) + 3(ay_1 + by_2)) \in W$

Hence W is a subspace of $V_3(R)$.

4. Let $\alpha, \beta \in W$ where $\alpha = (x_1, y_1, z_1)$ and $\beta = (x_2, y_2, z_2)$

for some $x_1, x_2, y_1, y_2, z_1, z_2 \in R$. Then by definition

$x_1 + y_1 + 2z_1 = 0$ ... (1), $x_2 + y_2 + 2z_2 = 0$ .... (2)

If $a, b \in R$, then we have $a\alpha + b\beta = a(x_1, y_1, z_1) + b(x_2, y_2, z_2)$

$= (ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2)$

Now $(ax_1 + bx_2) + (ay_1 + by_2) + 2(az_1 + bz_2) = (ax_1 + ay_1 + 2az_1) + (bx_2 + by_2 + 2bz_2)$

$= a(x_1 + y_1 + 2az_1) + b(x_2 + y_2 + 2z_2)$

$= a(x_1 + y_1 + 2z_1) + b(x_2 + y_2 + 2z_2) = a.0 + b.0 = 0$ ($\because$ from (1) & (2))

$\therefore\ a\alpha + b\beta = (ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2) = a.0 + b.0 = 0 \in W$

Hence W is a subspace of $R^3(R)$.

5. Use the hint and solve the problem

6. $W = \{(a, b, c) \mid a^2 + b^2 + c^2 \le 1\}$. $x = (1, 0, 0), y = (0, 1, 0) \in W$

$x + y = (1, 1, 0) \notin W$ because $1^2 + 1^2 + 0 \not\le 1$.

$\therefore$ Addition in W not satisfied.

$\therefore$ W is not a subspace of R^3

7. Let $A, B \in W$ $\therefore AT = TA$ and $BT = TB$.

For $a,b \in F$, $(aA + bB)T = (aA)T + (bB)T$

$= a(AT) + b(BT) = a(TA) + b(TB) = (aT)A + (bT)B$

$= (Ta)A + (Tb)B = T(aA) + T(bB) = T(aA + bB)$

$\therefore aA + bB \in W$ for all $a,b \in F$ and $A,B \in W$. Hence W is not a subspace of V.

8. Use the hint and solve the problem

9. (*i*) Let $f,g \in W_1$. $\therefore f(0) = 0$ and $g(0) = 0$ Let $a,b \in F$.

Now $(af + bg)(0) = (af)(0) + (bg)(0) = af(0) + bg(0) = a.0 + b.0 = 0$

$\therefore af + bg \in W_1$ for all $a,b \in F$ and $f,g \in W_1$

Hence W_1 is a subspace of V.

(*ii*) Let $f,g \in W_2$. $\therefore f(1) = f(-1)$ and $g(1) = g(-1)$

Let $a,b \in F$. Now $(af + bg)(1) = (af)(1) + (bg)(1) = af(1) + bg(1)$

$= af(-1) + bg(-1) = (af)(-1) + (bg)(-1) = (af + bg)(-1)$

$\therefore af + bg \in W_2$ for all $a,b \in F$ and $f,g \in W_2$. Hence W_2 is a subspace of V.

(*iii*) Let $f,g \in W_3$. $\therefore f(x) = 0, g(x) = 0$ if $x < 0$. Let $a,b \in F$.

Now $(af + bg)(x) = (af)(x) + (bg)(x) = af(x) + bg(x) = a.0 + b.0 = 0$ if $x < 0$

$\therefore af + bg \in W_3$ for all $a,b \in F$ and $f,g \in W_3$. Hence W_3 is a subspace of V.

(*iv*) Let $f,g \in W_4$. $\therefore f(x) = f(-x)$ and $g(x) = g(-x)$

Let $a,b \in F$. $(af + bg)(x) = (af)(x) + (bg)(x) = af(x) + bg(x)$

$= af(-x) + bg(-x) = (af)(-x) + (bg)(-x) = (af + bg)(-x)$

$\therefore af + bg \in W_4$ for all $a,b \in F$ and $f,g \in W_4$. Hence W_4 is a subspace of V.

10. (*i*) Let $\alpha,\beta \in W$. Where $\alpha = (\alpha_1,\alpha_2,\alpha_3), \beta = (\beta_1,\beta_2,\beta_3)$

and $\alpha_1,\alpha_2,\alpha_3,\beta_1,\beta_2,\beta_3$ are real.

Now $a,b \in C$, $a\alpha + b\beta = a(\alpha_1,\alpha_2,\alpha_3) + b(\beta_1,\beta_2,\beta_3)$

$= (a\alpha_1, a\alpha_2, a\alpha_3) + (b\beta_1, b\beta_2, b\beta_3) = (a\alpha_1 + b\beta_1, a\alpha_2 + b\beta_2, a\alpha_3 + b\beta_3)$

where $a\alpha_1 + b\beta_1$ may not be real.

For e.g. $a = 2i, b = 3i$, then $a\alpha_1 + b\beta_1 = 2i\alpha_1 + 3i\beta_1$ is not real.

$\therefore a\alpha + b\beta \notin W$. Hence W is not a subspace of $C^3(C)$.

(*ii*)Let $\alpha,\beta \in W$ where $\alpha = (\alpha_1,\alpha_2,\alpha_3), \beta = (\beta_1,\beta_2,\beta_3)$ such that either $\alpha_1 = 0$ or $\alpha_2 = 0$ and $\beta_1 = 0$ or $\beta_2 = 0$.

For $a,b \in C$, $a\alpha + b\beta = a(\alpha_1,\alpha_2,\alpha_3) + b(\beta_1,\beta_2,\beta_3)$

$= (a\alpha_1, a\alpha_2, a\alpha_3) + (b\beta_1, b\beta_2, b\beta_3) = (a\alpha_1 + b\beta_2, a\alpha_2 + b\beta_2, a\alpha_3 + b\beta_3)$

If $\beta_2 \neq 0$, then $a\alpha_1 + b\beta_2, a\alpha_2 + b\beta_2$ may not be equal to zero. $\therefore a\alpha + b\beta \notin W$.

Hence W is not a subspace of V.

11. (*i*) $W = \{f \mid f \in F[x] \text{ and } \deg f \leq 10\}$. W is a subspace of $F[x]$.

Reason : Consider $f,g \in W$, then $\deg f \leq 10, \deg g \leq 10$

For all $a,b \in F$. $(af + bg)(x)$ is a polynomial of degree ≤ 10

$\Rightarrow (af + bg)(x) \in W \Rightarrow W$ is a subspace of $F[x]$.

(*ii*) $W = \{f \mid f \in F[x]$ and α,β are the roots in $F\}$,

$W = \{f \mid f \in F[x] \text{ and } (x-\alpha)(x-\beta) = f(x)\}$

Let $f(x) = (x-\alpha)(x-\beta)$ which is a polynomial in 'x' of degree 2

and $f(\alpha) = 0 = f(\beta)$ so $f(x) \in W$. $\therefore W \neq \phi$

Let $f,g \in W$. Consider $f(x) = (x-\alpha)(x-\beta)$, $g(x) = (x-\alpha)(x-\beta) \in W$

$(af + bg)(x) = af(x) + bg(x) = (a+b)(x-\alpha)(x-\beta) \in W$

$\therefore$ W is a subspace of $F[x]$

(*iii*) W is a subspace $W = \{f \mid f \in F[x]$ and $f(x) = (x-0)\, g_1(x)\}$

Reason : Let $f,g \in W$. $f(x) = (x-0)g_1(x)$ ($\because f(x)$ divisible by x)

$g(x) = (x-0)g_2(x)$ where $g_1(x), g_2(x) \in F[x]$

$(af + bg)(x) = af(x) + bg(x) = axg_1(x) + bxg_2(x) = x(ag_1(x) + bg_2(x)) \in W$

Because $(af + bg)(x)$ is divisible by x). $\therefore$ W is a subspace.

(*iv*) $W = \{f \mid f \in F[x]$ and $2f(0) = f(1)$ or $f(0) = \frac{1}{2}f(1)\}$

Reason : let $f,g \in W$. $\therefore f(0) = \frac{1}{2}f(1)$, $g(0) = \frac{1}{2}g(1)$

$(af + bg)(0) = af(0) + bg(0) = a\frac{1}{2}f(1) + b\frac{1}{2}g(1)$

$= \frac{1}{2}(af + bg)(1) \in W$ $\therefore$ W is a subspace.

12. Let $\alpha, \beta \in W$ where $\alpha = \lambda(1,1,1)$ and $\beta = k(1,1,1)$ for $\lambda, k \in R$

Now for $a, b \in R$, $a\alpha + b\beta = a \cdot \lambda(1,1,1) + b \cdot k(1,1,1)$

$= a\lambda(1,1,1) + bk(1,1,1) = (a\lambda + bk)(1,1,1) \in W$ since $a\lambda + bk \in R$

$\therefore$ $a\alpha + b\beta \in W$ for all $a, b \in R$ and $\alpha, \beta \in W$

Hence W is a subspace of R^3

EXERCISE 1 (*c*)

1. (*i*) Let $S = \{(1,0,0),(1,1,0),(1,1,1)\}$

Let $(a,b,c) \in R^3(R)$ then $(a,b,c) = x(1,0,0) + y(1,1,0) + z(1,1,1)$ ... (1)

$\Rightarrow (a,b,c) = (x+y+z, y+z, z) \Rightarrow x+y+z = a, y+z = b, z = c \Rightarrow x = a-b, y = b-c, z = c$.

$\therefore (1) \Rightarrow (a,b,c) = (a-b)(1,0,0) + (b-c)(1,1,0) + c(1,1,1)$

$\Rightarrow (a,b,c) \in L(S) \Rightarrow R^3(R) \subseteq L(S)$.

But $L(S) \subseteq R^3(R)$. Hence $L(S) = R^3(R)$.

(*ii*) Let $S = \{(1,2,3),(0,1,2),(0,0,1)\}$

Let $(a,b,c) \in R^3(R)$ then $(a,b,c) = x(1,2,3) + y(0,1,2) + z(0,0,1)$ ... (1)

$\Rightarrow (a,b,c) = (x, 2x+y, 3x+2y+z) \Rightarrow x = a, 2x+y = b, 3x+2y+z = c$

$\Rightarrow x = a, y = -2a+b, z = -3a+2b+c$.

$\therefore$ (1) $\Rightarrow (a,b,c) = a(1,2,3) + (-2a+b)(0,1,2) + (-3a+2b+c)(0,0,1)$

$\Rightarrow (a,b,c) \in L(S) \Rightarrow R^3(R) \subseteq L(S)$. But $L(S) \subseteq R^3(R)$. Hence $L(S) = R^3(R)$.

(*iii*) Let $S = \{(1,2,1),(2,1,0),(1,-1,2)\}$

Let $(a,b,c) \in R^3(R)$ then $(a,b,c) = x(1,2,1) + y(2,1,0) + z(1,-1,2)$ ... (1)

$\Rightarrow (a,b,c) = (x+2y+z, 2x+y-z, x+2z) \Rightarrow x+2y+z = a, 2x+y-z = b, x+2z = c$

Solving x, y, z in terms of a, b, c we get

$$\Rightarrow x = \frac{1}{9}(-2a+4b+3c), y = \frac{1}{9}(5a-b-3c), z = \frac{1}{9}(a-2b+3c)$$

$$\therefore (1) \Rightarrow (a,b,c) = \frac{1}{9}(-2a+4b+3c)(1,2,1) + \frac{1}{9}(5a-b-3c)(2,1,0) + \frac{1}{9}(a-2b+3c)(1,-1,2)$$

$\Rightarrow (a,b,c) \in L(S) \Rightarrow R^3(R) \subseteq L(S)$.

But $L(S) \subseteq R^3(R)$. Hence $L(S) = R^3(R)$.

2. Let $S=\{\alpha,\beta\}$ and $T=\{\gamma,\delta\}$. $L(T)=$ l.c. of the vectors of $T=\{a\gamma+b\delta/a,b\in R\}$.

Let $\alpha=x\gamma+y\delta\Rightarrow(1,2,-1)=x(4,1,3)+y(-3,1,2)\Rightarrow 4x-3y=1,\ x+y=2,\ 3x+2y=-1$

$\exists$ no x,y which satisfy all the three equations.

$\therefore\ \alpha\notin L(T)$. Similarly $\beta\notin L(T)$. $\therefore\ L(S)\not\subseteq L(T)$. Hence $L(S)\neq L(T)$

3. Let W be the subspace spanned by S.

Then each linear combination say $\alpha=a_1\alpha_1+a_2\alpha_2+.....+a_m\alpha_m$ of vectors of S

i.e. $\alpha_1,\alpha_2,.....\alpha_m\in S$ is clearly in W. Thus W contains the set L (S)

i.e. all linear combinations of vectors in S.

The set L (S), on the other hand, contains S and L (S) is non empty.

If α,β belong to L (S) then α is a linear combination of 'S' elements

i.e. $\alpha=a_1\alpha_1+a_2\alpha_2+....+a_m\alpha_m$ of vectors $\alpha_i\in S$ and β is a linear combination.

$\beta=b_1\beta_1+b_2\beta_2+.....+b_n\beta_n$ of vectors β_j in S.

For each scalar C, $C\alpha+\beta=\sum_{i=1}^{m}(ca_i)\,\alpha_i+\sum_{j=1}^{n}(b_j+\beta j)$

Hence $C\alpha+\beta$ belongs to L (S). Thus L (S) is a subspace of V.

Now we have shown that L (S) is a subspace of V which contains S, and also that any subspace which contains S contains L (S).

It follows that L (S) in the intersection of all subspaces containing S.

i.e. that L (S) is the subspace spanned by the set S.

EXERCISE 1 (*d*)

1. (*a*) Since $(6,-9)=3(2,-3)$ one vector is a scalar multiple of other.

$\Rightarrow$ The set $\{(2,-3),(6,-9)\}$ in $R^2(R)$ is L. D.

(*b*) Let $a,b\in R$, then $a(4,3,-2)+b(2,-6,7)=\bar{O}\Rightarrow(4a+2b,3a-6b,-2a+7b)=(0,0,0)$

$\Rightarrow 4a+2b=0,\ 3a-6b=0,\ -2a+7b=0\Rightarrow a=0,\ b=0,\ c=0$

$\therefore$ The given vectors are linearly independent.

(*c*) Since $B=\begin{bmatrix}3&-6&12\\9&3&-3\end{bmatrix}=3\begin{bmatrix}1&-2&4\\3&1&-1\end{bmatrix}=3A$

One vector is scalar multiple of other $\Rightarrow$ The set $\{A,B\}$ in $R^2(R)$ is L. D.

(*d*) Let $a,b\in R$, then $ap_1(x)+bp_2(x)=\bar{O}$

$\Rightarrow a(1-2x+3x^2)+b(2-3x)=\bar{O}\Rightarrow 3ax^2+(-2a-3b)x+(a+2b)=\bar{O}$

$\Rightarrow 3a=0,-2a-3b=0,a+2b=0 \Rightarrow a=0,b=0$. $\therefore$ The set $\{p_1(x),p_2(x)\}$ is L.I.

2. (*a*) Let $a,b,c\in R$ then $a(1,-2,1)+b(2,1,-1)+c(7,-4,1)=\bar{O}$

$\Rightarrow (a+2b+7c,-2a+b-4c,a-b+c)=(0,0,0)$

$\Rightarrow a+2b+7c=0,-2a+b-4c=0,a-b+c=0 \Rightarrow a=-3c,b=-2c$

$\Rightarrow$ The set $\{(1,-2,1),(2,1,-1),(7,-4,1)\}$ is L. D.

(*b*)Let $a,b,c,d\in R$ then $a(2,0,5)+b(3,-5,8)+c(4,-2,1)+d(0,0,1)=\bar{O}$

$\Rightarrow 2a+3b+4c=0,-5b-2c=0,5a+8b+c+d=0 \Rightarrow b=(-2/5)C$

$d=5c, a=\dfrac{7c}{5}$ $\Rightarrow$ The set $\{(2,0,5),(3,-5,8),(4,-2,1),(0,0,1)\}$ is L. D.

(*c*) Let $a,b,c\in R$ then, $a(-1,2,1)+b(3,0,-1)+c(-5,4,3)=\bar{O}$

$\Rightarrow (-a+3b-5c,2a+4c,a-b+3c)=(0,0,0)$

$\Rightarrow -a+3b-5c=0,\ 2a+4c=0,\ a-b+3c=0$

$\Rightarrow a=0,b=0,c=0$. $\Rightarrow$ The given set of vectors in L.I.

3. (*a*) Let $a,b,c\in Q$ then $a(2,1,1,1)+b(1,3,1,-2)+c(1,2,-1,3)=\bar{O}$

$\Rightarrow (2a+b+c,a+3b+2c,a+b-c,a-2b+3c)=(0,0,0,0)$

$\Rightarrow 2a+b+c=0,a+3b+2c=0,a+b-c=0,a-2b+3c=0$

$\Rightarrow a=0,b=0,c=0$. $\therefore$ The given vectors are L.I.

(*b*)Let $a,b,c,d\in Q$, then $a(0,1,0,1)+b(1,2,3,-1)+c(1,0,1,0)+d(0,3,2,0)=\bar{O}$

$\Rightarrow (b+c,a+2b+3d,3b+c+2d,a-b)=(0,0,0,0)$

$b+c=0,\ a+2b+3d=0,\ 3b+c+2d=0,\ a-b=0$

$\Rightarrow b+c=0,a+2b+3d=0,3b+c+2d=0,a-b=0$

$\Rightarrow a=b=d=-c$ $\therefore$ The given vectors are L.D.

(*c*) Let $a,b,c,d\in Q$, then $a(1,2,-1,1)+b(0,1,-1,2)+c(2,1,0,3)+d(1,1,0,0)=\bar{O}$

$\Rightarrow a+2c+d=0,2a+b+c+d=0,-a-b=0,a+2b+3c=0$

$\Rightarrow a=3c,b=-3c,d=-4c$. $\therefore$ The given vectors are L.D.

(*d*)Let $a,b,c,d\in Q$, then $a(1,1,2,4)+b(2,-1,-5,2)+c(1,-1,-4,0)+d(2,1,1,6)=\bar{O}$

$\Rightarrow (a+2b+c+2d,a-b-c+d,2a-5b-4c+d,4a+2b+6c)=(0,0,0,0)$

$\Rightarrow a+2b+c+2d=0,a-b-c+d=0,2a-5b-4c+d=0,4a+2b+6c=0$

$\Rightarrow d=c=-a=-b$. $\therefore$ The given vectors are L.D.

4. Let $a,b,c \in R$, then $a.1+b.x+c(x-x^2)=\bar{O}$

$\Rightarrow a+bx+cx-cx^2=\bar{O}$ $\Rightarrow -c.x^2+(b+c)x+a=0.x^2+0.x+0$

$\Rightarrow a=0, b=0, c=0$. $\therefore$ the given set is L.I.

5. Given $\{\alpha+a\beta+b\gamma,\beta,\gamma\}$ is L. D. set

$\therefore \exists\, x,y,z$ the scalares are not all zero's such that

$x(\alpha+a\beta+b\gamma)+y(\beta)+z\gamma=\bar{O} \Rightarrow x\alpha+(xa+y)\beta+(xb+z)\gamma=O$

In the above linear combination all the scalar are not zeros. $\therefore \{\alpha,\beta,\gamma\}$ is L. D. set.

6. If $a_1,a_2,.....a_n \in F$, then $a_1\alpha_1+a_2\alpha_2+.....+a_n\alpha_n=\bar{O}$

$\Rightarrow a_1=a_2=.....=a_n=0$ ($\because$ S is L.I.)

Again if, $a_1,a_2,.....a_n,b \in F$, then $a_1\alpha_1+a_2\alpha_2+....+a_n\alpha_n+b\beta=\bar{O}$

$\Rightarrow 0.\alpha_1+0.\alpha_2+....+0.\alpha_n+b\beta=0$ ($\because$ S is L.I.)

$\Rightarrow b\beta=0 \Rightarrow b=0$ $\therefore$ All $a_1,a_2,.....,a_n,b=0$

$\Rightarrow$ The set $\{\alpha_1,\alpha_2,.....\alpha_n,\beta\}$ is L. I.

7. The set $\{(\beta,1,0),(1,\beta,1),(0,1,\beta)\}$ is L. D. $\Rightarrow \begin{vmatrix} \beta & 1 & 0 \\ 1 & \beta & 1 \\ 0 & 1 & \beta \end{vmatrix}=0 \Rightarrow \beta=\pm\sqrt{2}$.

8. If $a_0,a_1,a_2,.....,a_n \in R$, then $a_0f(x)+a_1f'(x)+a_2f''(x)+.....+a_nf^{(n)}(x)=\bar{O}$

$\Rightarrow a_0f(x)+a_1f'(x)+a_2f''(x)+.....+a_nf^{(n)}(x)$

$=0.f(x)+0.f'(x)+0.f''(x)+.....+0.f^{(n)}(x) \Rightarrow a_0=0,a_1=0,a_2=0,.....a_n=0$.

$\therefore$ The given system is L. I.

2. BASIS AND DIMENSION

EXERCISE 2 (*a*)

1. Write the elements of the set $\{(1,0,0),(1,1,0),(1,1,1)\}$ as rows of a matrix

we get $S=\begin{bmatrix}1&0&0\\1&1&0\\1&1&1\end{bmatrix}$, Applying $R_2-R_1, R_3-R_1 :, S\sim\begin{bmatrix}1&0&0\\0&1&0\\0&1&1\end{bmatrix}$

Again R_3-R_2, $S\sim\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

In the last matrix contains three non-zero rows, hence the given vectors are L.I. vectors.

$\therefore$ 'S' is a basis set of $C^3(C)$.

2. (*i*) Given : $\alpha=(4,5,6), x=(1,1,1), y=(-1,1,1), z=(1,0,-1)$

If $\alpha=ax+by+cz$, then a,b,c (scalars) are called coordinates of 'α' w.r.to $\{x,y,z\}$.

$\therefore\ (4,5,6)=a(1,1,1)+b(-1,1,1)+c(1,0,-1)$

$\therefore\ a-b+c=4$ .... (1) $a+b=5$ ... (2) $a+b-c=6$ .... (3)

Adding (1), (3) we get $a=5$. Substituting 'a' in eqn. (2) we get $b=0$

Putting a,b in (1) we get $c=-1$. $\therefore$ The coordinates are $(5,0,-1)$.

(*ii*) $\alpha=(1,0,-1), x=(0,1,-1), y=(1,1,0), z=(1,0,2)$

$(1,0,-1)=a(0,1,-1)+b(1,1,0)+c(1,0,2)$

$\therefore\ b+c=1$ ... (1) $a+b=0$ .... (2) $-a+2c=-1$ .... (3)

Putting $b=-a$ in (3), we get $b+2c=-1$ .... (4)

Solving (1), (4) : we get $c=-2$. From (1) $b=3$; From (2) $a=-3$.

$\therefore$ The coordinates are $(-3,3,2)$

(*iii*) $\alpha=(2,1,3), x=(1,1,1), y=(-1,1,0), z=(1,0,-1)$

$(2,1,3)=a(1,1,1)+b(-1,1,0)+c(1,0,-1)$

$\therefore\ a-b+c=2$ .... (1) $a+b=1$ ... (2) $a-c=3$ ... (3)

Adding (1), (2) eqn.s we get $2a+c=3$ .... (4)

Solving (3), (4) eqn.s we get $a=2$. From (2) substituting $a=2$ we get $b=-1$.

From (1) substituting $a=2, b=-1$ we get $c=-1$

$\therefore$ The coordinates are $(2,-1,-1)$.

3. Given : $\{(2,1,0),(0,1,2),(-7,2,5),(8,0,0)\}$

writing the given set of vectors as rows of matrix we have

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 2 \\ -7 & 2 & 5 \\ 8 & 0 & 0 \end{bmatrix} \text{ or } \begin{bmatrix} 8 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \\ -7 & 2 & 5 \end{bmatrix}$$

By using row operations reducing the matrix into echelon form, we get $\begin{bmatrix} 8 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 8 \\ 0 & 0 & 0 \end{bmatrix}$

But in the matrix last row contains all the zeros, hence the given vectors are L. D. vectors. Therefore given set cannot form basis of $R^3(R)$

4. Consider $a,b,c \in R$ such that $a(\alpha_1+\alpha_2)+b(\alpha_2+\alpha_3)+c(\alpha_3+\alpha_1)=0$

$\therefore (a+c)\alpha_1+(a+b)\alpha_2+(b+c)\alpha_3=0$

But $\{\alpha_1,\alpha_2,\alpha_3\}$ is basis set . $\therefore$ $\{\alpha_1,\alpha_2,\alpha_3\}$ is L.I. set.

$\therefore$ All the scalars $a,b,c \in R$ are zeros $\Rightarrow a+c=0, a+b=0, b+c=0 \Rightarrow a=b=c=0$

$\therefore$ $\{\alpha_1+\alpha_2,\alpha_2+\alpha_3,\alpha_3+\alpha_1\}$ is L.I. set.

Now we prove that $\{\alpha_1+\alpha_2,\alpha_2+\alpha_3,\alpha_3+\alpha_1\}$ generates entire $R^3(R)$

Then we can write $\alpha = x(\alpha_1+\alpha_2)+y(\alpha_2+\alpha_3)+z(\alpha_1+\alpha_3)$, where $\alpha \in R^3$.

or $\alpha=\alpha_1(x+z)+\alpha_2(x+y)+\alpha_3(y+z)$ ... (1)

But $\alpha \in R^3(R)$ and $\{\alpha_1,\alpha_2,\alpha_3\}$ is a basis of $R^3(R)$

$\therefore$ We can express 'α' as the l.c. of $\{\alpha_1,\alpha_2,\alpha_3\}$

$\alpha=b_1\alpha_1+b_2\alpha_2+b_3\alpha_3$ for some scalars b_1,b_2,b_3 .... (2)

Comparing (1), (2) : we get $b_1=x+z, b_2=x+y, b_3=y+z$. $\therefore$ $b_2-b_3=x-z$.

From these equations $x=\frac{b_1+b_2-b_3}{2}, z=\frac{b_1-b_2+b_3}{2}, y=b_2-x=\frac{b_2-b_1+b_3}{2}$

$\therefore$ b_1,b_2,b_3 are not all zeros $\Rightarrow$ x,y,z are all not zeros

$\therefore$ $u=x(\alpha_1+\alpha_2)+y(\alpha_2+\alpha_3)+z(\alpha_3+\alpha_1)$

$$u=\left(\frac{b_1+b_2-b_3}{2}\right)(\alpha_1+\alpha_2)+\left(\frac{b_1-b_2+b_3}{2}\right)(\alpha_2+\alpha_3)+\left(\frac{b_2-b_1+b_3}{2}\right)(\alpha_3+\alpha_1)$$

$$= \text{l. c. of } \{\alpha_1+\alpha_2,\alpha_2+\alpha_3,\alpha_3+\alpha_1\}$$

$\therefore \{\alpha_1+\alpha_2,\alpha_2+\alpha_3,\alpha_3+\alpha_1\}$ generates $R^3(R)$

$\therefore \{\alpha_1+\alpha_2,\alpha_2+\alpha_3,\alpha_3+\alpha_1\}$ is a basis of $R^3(R)$

5. (*i*) Writing the given elements in row matrix we get $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & -1 & 2 \end{bmatrix}$

Reducing the matrix to echelon form we get : $\begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -2 \\ 0 & -3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -2 \\ 0 & 0 & 3 \end{bmatrix}$

In the last matrix contains all the 3 rows are non-zeros, hence these vectors form L.I. set.

Let $(a,b,c) \in V_3(F)$ and $x,y,z \in F$ such that

$$(a,b,c) = x(1,2,1) + y(2,1,0) + z(1,-1,2)$$

$\therefore\ x+2y+z=a$ .... (1) $2x+y-z=b$ ... (2) $x+2z=c$ .... (3)

Solving (1), (2), (3) for getting x,y,z in terms of a,b,c we have

$$x=\frac{-2a+4b-3c}{9}, y=\frac{5a-b+9c}{9}, z=\frac{a-2b+3c}{9}$$

$\therefore$ $\{(1,2,1),(2,1,0),(1,-1,2)\}$ set forms a basis of $V_3(F)$.

(*ii*) Given $\{(2,1,4),(1,-1,2),(3,1,-2)\}$.

Writing the given set elements in row matrix we get $\begin{bmatrix} 2 & 1 & 4 \\ 1 & -1 & 2 \\ 3 & 1 & -2 \end{bmatrix}$.

Reducing the matrix into echelon form we get $\begin{bmatrix} 2 & 1 & 4 \\ 0 & -3 & 0 \\ 0 & -1 & -16 \end{bmatrix} \sim \begin{bmatrix} 2 & 1 & 4 \\ 0 & -3 & 0 \\ 0 & 0 & -48 \end{bmatrix}$

In the last matrix contains all the 3 rows are non-zeros, hence these vectors form L.I. set.

Let $(a,b,c) \in R^3(R)$ and $x,y,z \in R$ such that

$$(a,b,c) = x(2,1,4) + y(1,-1,2) + z(3,1,-2)$$

$\therefore\ 2x+y+3z=a$ .... (1) $x-y+z=b$ ... (2) $4x+2y-2z=c$ .... (3)

Solving (1), (2), (3) , for getting x,y,z in terms of a,b,c we have

$$x=\frac{2b+c}{6}, y=\frac{6a-16b+c}{24}, z=\frac{2a-c}{8}$$

$\therefore$ $\{(2,1,4),(1,-1,2),(3,1,-2)\}$ set forms a basis of $R^3(R)$.

6. Writing the elements of the set $\{(1,i,0),(2i,1,1),(0,1+i,1-i)\}$ as rows of a matrix we get

$$S=\begin{bmatrix} 2i & 1 & 1 \\ 0 & 1+i & 1-i \\ 1 & i & 0 \end{bmatrix}$$

Applying $2iR_3 - R_1$, $S \sim \begin{bmatrix} 2i & 1 & 1 \\ 0 & 1+i & 1-i \\ 0 & -3 & -1 \end{bmatrix}$

Again $R_3(1+i)+3R_2$, $S \sim \begin{bmatrix} 2i & 1 & 1 \\ 0 & 1+i & 1-i \\ 0 & 0 & 2-4i \end{bmatrix}$

In the last matrix contains three non-zero rows, hence the given vectors are L.I. vectors. Therefore the set S is basis set.

Let $(x,y,z) \in V_3(C)$ and $a,b,c \in C$

$\therefore\ (x,y,z) = a\,(1,i,0) + b\,(2i,1,1) + c\,(0,1+i,1-i)$

$\Rightarrow a+2ib = x,\ ai+b+c\,(1+i) = y,\ b+c\,(1-i) = z$

Solving these equations we get $a = \dfrac{-2xi-2y\,(1+i)+2z\,(1-i)}{2\,(1-2i)}$,

$b = \dfrac{-z\,(1-i)-(xi-y)\,(1-i)}{2\,(1-2i)}$, $c = \dfrac{xi-y+3z}{2\,(1-2i)}$

Now $(x,y,z) = \dfrac{-2xi-2y\,(1+i)+2z\,(1-i)}{2\,(1-2i)}(1,i,0) +$

$+\dfrac{[-z(1+i)-(xi-y)\,(1-i)]}{2\,(1-2i)}(2i,1,1) + \dfrac{xi-y+3z}{2\,(1-2i)}(0,1+i,1-i)$ ... (1)

Using linear combinations for (1), we find the coordinates of $(1,0,0),(0,1,0),(0,0,1)$

Coordinates of $(1,0,0)$ *i.e.* substituting $x=1, y=0, z=0$ in (1) we get

$$(1,0,0) = \left(\frac{-2i}{2\,(1-2i)}, \frac{1-i}{2\,(1-2i)}, \frac{i}{2\,(1-2i)}\right)$$

Similarly $(0,1,0) = \dfrac{1}{2\,(1-2i)}(-2\,(1+i), 1\,(1-i), -1)$, $(0,0,1) = \dfrac{1}{2\,(1-2i)}(3, -(1+i), 3)$

7. We solve this problem by proceeding Problem 6 method discussed above.

8. $\{(1,1,2),(1,2,5),(5,3,4)\}$. Writing the given set of vectors in row matrix we have $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 5 \\ 5 & 3 & 4 \end{bmatrix}$

Reducing the matrix to echelon form we get $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & -2 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$.

In the last matrix contains zeros in the 3rd row.

Hence the given set is L. D set. $\therefore$ Given set cannot form basis set of $R^3(R)$.

9. First we find the coordinates of $C^3(C)$ w. r. to the basis $\{(1,0,0),(1,1,0),(1,1,1)\}$

Let $(a,b,c) \in C^3(C)$ and $x,y,z \in C$, such that $(a,b,c) = x(1,0,0) + y(1,1,0) + z(1,1,1)$.... (1)

$\therefore \quad x+y+z=a;\ y+z=b;\ z=c$

Solving above equations for x,y,z and expressing in a,b,c we get

$x=a-b,\ y=b-c,\ z=c$.

Putting x,y,z in (1) we get $(a,b,c) = (a-b)(1,0,0) + (b-c)(1,1,0) + c(1,1,1)$ .... (2)

(*i*) coordinates of $(2i, 3+4i, 5)$

Putting $a=2i,\ b=3+4i,\ c=5$ in equation (2)

we get $(2i,3+4i,5) = (2i-3-4i)(1,0,0) + (3+4i-5)(1,1,0) + 5(1,1,1)$

$\therefore\ (2i,3+4i,5) = (-3-2i)(1,0,0) + (-2+4i)(1,1,0) + 5(1,1,1)$

$\therefore$ Coordinates of $(2i,3+4i,5)$ w.r.to the given basis are $(-3-2i,-2+4i,5)$

(*ii*) Given $(6i,7,8i)$. Putting $a=6i,\ b=7,\ c=8i$ in (2) we get

$(6i,7,8i) = (6i-7)(1,0,0) + (7-8i)(1,1,0) + 8i(1,1,1)$

$\therefore$ Coordinates are $(6i-7, 7-8i, 8i)$

(*iii*) Given : $(3+4i, 6i, 3+7i)$. Putting $a=3+4i,\ b=6i,\ c=3+7i$ in (2) we get

$(3+4i, 6i, 3+7i) = (3+4i-6i)(1,0,0) + (6i-3-7i)(1,1,0) + (3+7i)(1,1,1)$

$(3+4i, 6i, 3+7i) = (3-2i)(1,0,0) + (-3-i)(1,1,0) + (3+7i)(1,1,1)$

$\therefore$ Coordinates are $(3-2i, -3-i, 3+7i)$

EXERCISE 2 (*b*)

1. Given P, Q are the two subspaces of R^4. $P = \{(a,b,c,d)/b+c+d=0\} \subset R^4$

$(a,b,c,d) = (a,-c-d,c,d) = a(1,0,0,0) + c(0,-1,1,0) + d(0,-1,0,1)\}$ $(\because b=-c-d)$

$\therefore$ Subspace 'P' in generated by L. I. set, say $S_1 = \{(1,0,0,0),(0,-1,1,0),(0,-1,0,1)\}$

$\dim P = 3 =$ No. of elements spanned by P.

$Q = \{(a,b,c,d) : a+b=0, c=2d\} \subset R^4$. Let $(a,b,c,d) \in R^4;\ b=-a,\ c=2d$

$(a,b,c,d) = (a,-a,2d,d)$; $\quad (a,b,c,d) = a(1,-1,0,0) + d(0,0,2,1)$

Subspace 'Q' is generated by L. I. set, say $S_2 = \{(1,-1,0,0),(0,0,2,1)\}$. $\therefore\ \dim Q = 2$

From Rank - Nullity theorem, $\dim R^4 = \dim P + \dim Q - \dim(P \cap Q)$

$4 = 3+2-\dim(P \cap Q) \Rightarrow \dim(P \cap Q) = 1$

Another method to find $\dim(P \cap Q)$:

$P \cap Q = \{(a,b,c,d) : b+c+d=0, a+b=0, c=2d\}$

$a+b=0 \Rightarrow b=-a$ $(\because b=-a \Rightarrow -a+c+d=0 \Rightarrow a=c+d)$

$b+c+d=0 \Rightarrow -a+c+d=0 \Rightarrow a=c+d \Rightarrow a=2d+d \Rightarrow a=3d$. $\therefore b=-3d; c=2d$

$(a,b,c,d)=(3d,-3d,2d,d)=d\,(3,-3,2,1)$

$P \cap Q$ in generated by $\{(3,-3,2,1)\}$. $\therefore \dim P \cap Q = 1$

2. Let $S_1=\{(1,1,0,-1),(1,2,3,0),(2,3,3,-1)\}$

$S_2=\{(1,2,2,-2),(2,3,2,-3),(1,3,4,-3)\}$ generates the subspaces W_1, W_2 respectively of R^4.

(*i*) Writing the vectors as rows of a matrix and reducing to echelon form

$$\begin{bmatrix}1&1&0&-1\\1&2&3&0\\2&3&3&-1\end{bmatrix}\sim\begin{bmatrix}1&1&0&-1\\0&1&3&1\\0&1&3&1\end{bmatrix}\sim\begin{bmatrix}1&1&0&-1\\0&1&3&1\\0&0&0&0\end{bmatrix}$$

In the last matrix contains zero vector which is L. D. vector.

Hence $\{(1,1,0,-1),(0,1,3,1)\}$ generates the subspace W_1.

Hence $\dim W_1 = 2$.

(*ii*)Similarly writing S_2 elements in rows of a matrix and reducing to echelon form

$$\begin{bmatrix}1&2&2&-2\\2&3&2&-3\\1&3&4&-3\end{bmatrix}\sim\begin{bmatrix}1&2&2&-2\\0&-1&-2&1\\0&1&2&-1\end{bmatrix}\sim\begin{bmatrix}1&2&2&-2\\0&-1&-2&1\\0&0&0&0\end{bmatrix}$$

Hence $\{(1,2,2,-2),(0,-1,-2,1)\}$ generates the subspace W_2. Hence $\dim W_2 = 2$.

(*iii*) We know that $W_1 + W_2 = L(W_1 \cup W_2)$ *i.e.* $W_1 + W_2$ generated by $W_1 \cup W_2$

Writing all the elements of S_1 and S_2 in the form of row matrix and reducing it into echolon form we have

$$\begin{bmatrix}1&1&0&-1\\1&2&3&0\\2&3&3&-1\\1&2&2&-2\\2&3&2&-3\\1&3&4&-3\end{bmatrix}\sim\begin{bmatrix}1&1&0&-1\\0&1&3&1\\0&1&3&1\\0&1&2&-1\\0&1&2&-1\\0&2&4&-2\end{bmatrix}\sim\begin{bmatrix}1&1&0&-1\\0&1&3&1\\0&0&0&0\\0&1&2&-1\\0&0&0&0\\0&0&0&0\end{bmatrix}\sim\begin{bmatrix}1&1&0&-1\\0&1&3&1\\0&1&2&-1\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{bmatrix}$$

$\therefore \dim (W_1+W_2)=3$

(*iv*) By using Rank - Nullity theorem $\dim (W_1 + W_2) = \dim W_1 + \dim W_2 - \dim (W_1 \cap W_2)$

$3 = 2+2-\dim (W_1 \cap W_2) \Rightarrow \dim (W_1 \cap W_2) = 1$

3. LINEAR TRANSFORMATIONS

EXERCISE 3 (*a*)

1. (*a*) $T: V_1 \to V_3$, $T(x) = (x, 2x, 3x)$. Let $\alpha = x$, $\beta = y \in V_1$, $a, b \in R$

We prove that $T(a\alpha + b\beta) = aT(\alpha) + bT(\beta)$

L.H.S. $T(a\alpha + b\beta) = T(ax + by) = (ax + by, 2(ax + by), 3(ax + by))$

$= (ax + by, 2ax + 2by, 3ax + 3by)$

$= (ax, 2ax, 3ax) + (by, 2by, 3by)$

$= a(x, 2x, 3x) + b(y, 2y, 3y) = aT(x) + bT(y)$

$= aT(\alpha) + bT(\beta) = R.H.S.$

$\therefore$ $T: V_1 \to V_3$ is Linear Transformation.

(*b*) $T: R^2 \to R^2$ defined by $T(a, b) = (2a + 3b, 3a - 4b)$

Let $\alpha = (a, b)$, $\beta = (x, y) \in R^2$, $p, q \in R$.

$T(p\alpha + q\beta) = T(pa + qx, pb + qy) = (2(pa + qx) + 3(pb + qy), 3(pa + qx) - 4(pb + qy))$

$= (2pa + 2ax + 3pb + 3qy, 3pa + 3qx - 4pb - 4qy)$

$= (2pa + 3pb, 3pa - 4pb) + (2qx + 3qy, 3qx - 4qy)$

$= p(2a + 3b, 3a - 4b) + q(2x + 3y, 3x - 4y) = pT(a, b) + qT(x, y)$

$T(p\alpha + q\beta) = pT(\alpha) + qT(\beta)$.

$\therefore$ $T: R^3 \to R^3$ is Linear Transformation.

(*c*) $x, y \in P$. $a, b \in R$. $T(x) = x^2 + x$, $T(y) = y^2 + y$

$T(ax + by) = (ax + by)^2 + (ax + by) = a^2x^2 + b^2y^2 + 2abxy + (ax + by)$

$= aT(x) + bT(y) = a(x^2 + x) + b(y^2 + y) = ax^2 + by^2 + ax + by$

$\therefore$ $T(ax + by) \neq aT(x) + bT(y)$.

$\therefore$ $T: P \to P$ is not Linear Transformation.

(*d*) $T: P \to P$ is a mapping defined by $T[p(x)] = p(0) = xp'(0) + \frac{x^2}{2!}p''(0)$

Consider $p(x), q(x) \in P$ and $a, b \in F$.

We prove that $T[ap + bq] = aT(p) + bT(q)$

L.H.S. $= T[ap + bq] = T\left[ap(0) + axp'(0) + a\frac{x^2}{2!}p''(0) + bq(0) + bxq'(0) + b\frac{x^2}{2!}q''(0)\right]$

$= T\left[ap(0) + bq(0) + axp'(0) + bxq'(0) + a\frac{x^2}{2!}p''(0) + b\frac{x^2}{2!}q''(0)\right]$

$$= T\left[(ap+bq)(0) + x(ap'+bq')(0) + \frac{x^2}{2!}(ap''+bq'')(0)\right]$$

$$= T[ap+bq](0) + xT(ap'+bq')(0) + \frac{x^2}{2!}T(ap''+bq'')(0).$$

But $T(ap+bq)(0) = (ap+bq)(0) + 0(ap'+bq')(0) + 0(ap''+bq'')(0) = (ap+bq)(0)$.

Similarly $T(ap'+bq')(0) = (ap'+bq')(0)$.

$T(ap''+bq'')(0) = (ap''+bq'')(0)$

$\therefore$ L.H.S. $= (ap+bq)(0) + x(ap'+bq')(0) + \frac{x^2}{2!}(ap''+bq'')(0)$

$$\text{R.H.S.} = aT(p) + bT(q) = a\left[p(0) + xp'(0) + \frac{x^2}{2!}p''(0)\right] + b\left[q(0) + xq'(0) + \frac{x^2}{2!}q''(0)\right]$$

$$= ap(0) + axp'(0) + a\frac{x^2}{2!}p''(0) + bq(0) + bxq'(0) + b\frac{x^2}{2!}q''(0)$$

$$= ap(0) + bq(0) + axp'(0) + bxq'(0) + a\frac{x^2}{2!}p''(0) + b\frac{x^2}{2!}q''(0)$$

$$= (ap+bq)(0) + x(ap'+bq')(0) + \frac{x^2}{2!}(ap''(0)+bq''(0)) = \text{L. H. S.}$$

$\therefore$ $T: P \to P$ is Linear Transformation.

(*e*) $T: R^3 \to R^3$; $T(x, y, z) = (x+1, y, z)$

Let $\alpha = (x_1, y_1, z_1)$, $\beta = (x_2, y_2, z_2) \in R^3$; $a, b \in R$

$T(a\alpha + b\beta) = T(ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2) = (ax_1 + bx_2 + 1, ay_1 + by_2, az_1 + bz_2)$

$aT(\alpha) + bT(\beta) = a(x_1 + 1, y_1, z_1) + b(x_2 + 1, y_2, z_2)$

$= (ax_1 + bx_2 + a + b, ay_1 + by_2, az_1 + bz_2)$.

$T(a\alpha + b\beta) \neq aT(\alpha) + bT(\beta)$

$\therefore$ $T: R^3 \to R^3$ is not Linear Transformation.

(*f*) Let $\alpha = (x_1, y_1)$, $\beta = (x_2, y_2) \in R^3$; $a, b \in R$

$T(a\alpha + b\beta) = T(ax_1 + bx_2, ay_1 + by_2) = ((ax_1 + bx_2)^3, (ay_1 + by_2)^3)$

$= (a^3x_1^3 + b^3x_2^3 + 3abx_1x_2(ax_1 + bx_2), a^3y_1^3 + b^3y_2^3 + 3aby_1y_2(ay_1 + by_2))$

$aT(\alpha) + bT(\beta) = a(x_1^3, y_1^3) + b(x_2^3, y_2^3) = (ax_1^3 + bx_2^3, ay_1^3 + by_2^3)$

$\therefore$ $T(a\alpha + b\beta) \neq aT(\alpha) + bT(\beta)$.

$\therefore$ $T: R^2 \to R^2$ is not Linear Transformation.

2. Given V(F) be a vector space over a field F contains $m \times n$ matrices
P, Q are fixed matrices of order $m \times m$ and $n \times n$ respectively over a field F.
$T : V \to V$ is a mapping defined by $T(A) = PAQ \ \forall \ A \in V_{m\times n}(F)$.

Let $A, B \in V_{m\times n}(F)$ and $P_{m\times m}$, $Q_{n\times n}$ fixed matrices and $a, b \in F$.

$T[aA + bB] = P(aA + bB)Q = (aPA + bPB)Q = aPAQ + bPBQ = aT(A) + bT(B)$

$\therefore$ $T : V \to V$ is Linear Transformation.

3. Given $T(1, 0) = (1, 1)$ and $T(0, 1) = (-1, 2)$.

First we prove that the set $S = \{(1, 0), (0, 1)\} \subset R^2$ is a basis set.

(*i*) To prove S is L.I set. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and its det $\neq 0$

$\therefore$ $\{(1, 0), (0, 1)\}$ is L.I. set or $x(1, 0) + y(0, 1) = \bar{0} = (0, 0)$

$x = 0$; $y = 0$. All the scalars are zeros. $\therefore$ $\{(1, 0), (0, 1)\}$ is L.I. set.

(*ii*) To prove $L(S) = R_2$. Let $\alpha = (a, b) \in R^2$ and $x, y \in R$

$\alpha = (a, b) = x(1, 0) + y(0, 1) \Rightarrow x = a, y = b$

$\therefore$ $(a, b) = a(1, 0) + b(0, 1)$ $\therefore$ S generates $R_2(R)$. $\therefore$ S is a basis of R_2.

(*iii*) To determine Linear Transformation.

$T = (x, y) = xT(1, 0) + yT(0, 1) = x(1, 1) + y(-1, 2)$

$T = (x, y) = (x - y, x + 2y)$ is the required Linear Transformation.

4. We prove that $S = \{(1, 2), (2, 1)\}$ is a basis set of V_2.

(*i*) To prove S is L.I. set. $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and its determinant $= -3 \neq 0$. $\therefore$ S is L.I. set.

(*ii*) To prove $L(S) = V_2$. Let $\alpha = (a, b) \in V_2$ and $x, y \in R$

$(a, b) = x(1, 2) + y(2, 1)$

$x + 2y = a$ .....(1) $2x + y = b$ .....(2)

Solving (1), (2) and express x, y interms of a, b we have

$x = \dfrac{2b - a}{3}$, $y = \dfrac{2a - b}{3}$. $\therefore$ $(a, b) = \dfrac{(2b - a)}{3}(1, 2) + \dfrac{(2a - b)}{3}(2, 1)$

$\therefore$ $L(S) = V_2$. $\therefore$ 'S' is a basis of V_2. $(x, y) \in V_2$

$T(x, y) = \left(\dfrac{2y - x}{3}\right) T(1, 2) + \dfrac{(2x - y)}{3} T(2, 1)$

$= \dfrac{(2y - x)}{3}(3, 0) + \dfrac{(2x - y)}{3}(1, 2) = (2y - x, 0) + \left(\dfrac{2x - y}{3}, \dfrac{2(2x - y)}{3}\right)$

$$= \left(2y - x + \frac{2x - y}{3},\ \frac{2(2x - y)}{3}\right) = \left(\frac{6y - 3x + 2x - y}{3},\ \frac{4x - 2y}{3}\right)$$

$$T(x, y) = \left(\frac{5y - x}{3},\ \frac{4x - 2y}{3}\right).$$

5. First we prove that $S = \{(0, 1, 2), (1, 1, 1)\}$ is a basis set

(*i*) For $a, b \in R$ scalars : $a(0, 1, 2) + b(1, 1, 1) = (0, 0, 0)$,

$\Rightarrow a = 0;\ b = 0$. $\therefore$ S is L.I. set.

(*ii*) To prove $L(S) = V_3$, let $(x, y, z) \in V_3$ for $a, b \in R$

$(x, y, z) = a(0, 1, 2) + b(1, 1, 1) \Rightarrow b = x$

$a + b = y \Rightarrow a = y - x \Rightarrow (x, y, z) = (y - x)(0, 1, 2) + x(1, 1, 1)$

$\therefore$ $L(S) = V_3$. $\therefore$ S is a basis of V_3.

$T(x, y, z) = (y - x)T(0, 1, 2) + xT(1, 1, 1) = (y - x)(3, 1, 2) + x(2, 2, 2)$

$= (3y - 3x + 2x,\ y - x + 2x,\ 2y - 2x + 2x)$. $T(x, y, z) = (3y - x,\ y + x,\ 2y)$.

6. First we prove that $S = \{(1, 2), (0, 1)\}$ is a basis set.

(*i*) For $a, b \in R$ scalars $\Rightarrow$ $a(1, 2) + b(0, 1) = (0, 0)$

$a = 0$, $2a + b = 0 \Rightarrow b = 0$. $\therefore$ S is L.I. set.

(*ii*) To prove $L(S) = V_2(R)$: Let $(x, y) \in V_2(R)$ for $a, b \in R$

$(x, y) = a(1, 2) + b(0, 1) \Rightarrow a = x$. $2a + b = y \Rightarrow b = y - 2x$

$\therefore$ $(x, y) = x(1, 2) + (y - 2x)(0, 1)$

$\therefore$ $L(S) = V_2(R)$. $\therefore$ S is a basis of $V_2(R)$.

$T(x, y) = xT(1, 2) + (y - 2x)T(0, 1) = x(3, -1, 5) + (y - 2x)(2, 1, -1)$

$= (3x + 2y - 4x,\ -x + y - 2x,\ 5x - y + 2x)$. $T(x, y) = (2y - x,\ y - 3x,\ 7x - y)$.

7. First we prove that $S = \{(2, -5), (3, 4)\}$ is a basis set of $R^2(R)$.

(*i*) For $a, b \in R$ scalars. $a(2, -5) + b(3, 4) = (0, 0)$

$\Rightarrow$ $2a + 3b = 0$ .....(1) $-5a + 4b = 0$ .....(2)

Solving (1), (2) for a, b we get : $a = 0;\ b = 0$

$\therefore$ S is L.I. set.

(*ii*) Let $(x, y) \in V_2(R)$ for $a, b \in R \Rightarrow (x, y) = a(2, -5) + b(3, 4)$

$2a + 3b = x$ .....(1) $-5a + 4b = y$ .....(2)

Solving (3), (4) and getting a, b interms of x, y we have $a = \frac{4x - 3y}{23}$; $b = \frac{5x + 2y}{23}$

$\therefore$ $(x, y) = \frac{(4x - 3y)}{23}(2, -5) + \frac{(5x + 2y)}{23}(3, 4)$. $\therefore$ $L(S) = V_2(R)$.

$\therefore$ S is a basis of $V_2(R)$.

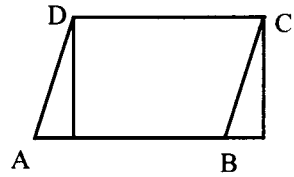
$$T(x, y) = \left(\frac{4x-3y}{23}\right) T(2, -5) + \left(\frac{5x+2y}{23}\right) T(3, 4)$$

$$= \left(\frac{4x-3y}{23}\right)(-1, 2, 3) + \left(\frac{5x+2y}{23}\right)(0, 1, 5)$$

$$T(x, y) = \left(\frac{3y-4x}{23}, \frac{13x-4y}{23}, \frac{37x+y}{23}\right).$$

8. We know that $S = \{(1, 0), (0, 1)\}$ is a standard basis set of R^2.

$\therefore$ We find Linear Transformation $T: R^2 \to R^2$.



$(x, y) = x(1, 0) + y(0, 1)$

$T = (x, y) = xT(1, 0) + yT(0, 1) = x(1, 1) + y(-1, 2)$

$T = (x, y) = (x - y, x + 2y)$

Let $T(0, 0) = (0, 0) = A$, $T(1, 0) = (1, 1) = B$

$T(1, 1) = (0, 3) = C$, $T(0, 1) = (-1, 2) = D$

Now $AB = \sqrt{2}$, $BC = \sqrt{5}$, $CD = \sqrt{2}$, $DA = \sqrt{5}$

Mid point $AC = \left(0, \frac{3}{2}\right)$; Mid point $BD = \left(0, \frac{3}{2}\right)$. $\therefore$ ABCD is a parallelogram.

EXERCISE 3 (*b*)

1. $T: R^3 \to R^2$; $H: R^2 \to R^3$.

$T(x, y, z) = (x - 3y - 2z, y - 4z)$, $H(x, y) = (2x, 4x - y, 2x + 3y)$

HT Product is possible because range of T *i.e.* R^2 is the domain of H.

$\therefore$ $HT(x, y, z) = H[T(x, y, z)]$ ($\because$ $HT(\alpha) = H[T(\alpha)]$)

$= H[x - 3y - 2z, y - 4z]$

$= [2(x - 3y - 2z), 4(x - 3y - 2z) - y + 4z, 2(x - 3y - 2z) + 3(y - 4z)]$

$HT(x, y, z) = (2x - 6y - 4z, 4x - 13y - 4z, 2x - 3y - 16z)$

Product TH is possible because range of H i.e. R^3 is the domain of T.

$TH(\alpha) = T[H(\alpha)]$. $TH(x, y) = T[H(x, y)] = T[2x, 4x - y, 2x + 3y)]$

$= (2x - 3(4x - y) - 2(2x + 3y), 4x - y - 4(2x + 3y))$

$TH(x, y) = (-14x - 9y, -4x - 13y)$

But $TH \neq HT$. Products of two L.T.'s is not commutative.

2. Given $H(x, y) = (0, x)$; $T(x, y) = (x, 0)$

TH is possible because Range of H is the domain of T.

$TH(x, y) = T[H(x, y)] = T[0, x]$

$TH(x, y) = (0, 0)$ $\therefore$ $TH = 0$.

Product HT is possible because range of T is the domain of H.

$\therefore$ $HT(x, y) = H[T(x, y)] = H[x, 0)$

$HT(x, y) = (0, x)$ $\therefore$ $HT \neq TH$

$T^2(x, y) = T[T(x, y)] = T[x, 0] = (x, 0) = T(x, y)$ $\therefore$ $T^2 = T$.

3. $T : R^3 \to R^3$ is a linear operation defined such that $T(x, y, z) = (0, x, y)$

$\therefore$ $T \neq O$, $T^2 \neq O$. Now we prove that $T^3 = O$.

$T^2(x, y, z) = T[T(T(x, y, z))] = T[T(0, x, y)] = T(0, 0, y) = (0, 0, 0)$. $\therefore$ $T^3 = 0$.

4. $DS[f(x)] = D[Sf(x)] = D\left[\int_0^x f(x)\,dx\right] = \frac{d}{dx}\left[\int_0^x f(x)\,dx\right] = f(x) = I[f(x)]$

$\therefore$ $DS[f(x)] = I[f(x)] \Rightarrow DS = I$ where I is Identity function.

$SD[f(x)] = S[Df(x)] = S\left[\frac{df}{dx}\right] = \int_0^x \frac{df}{dx}\,dx = f(x) - f(0) \neq I[f(x)]$. $\therefore$ $SD \neq I$.

5. Given $T : V_3(R) \to V_3(R)$ such that $T(a, b, c) = (3a, a-b, 2a+b+c)$

$L.H.S. = (T^2 - I)(T - 3I) = (T + I)(T - I)(T - 3I)$

$(T - 3I)(a, b, c) = T(a, b, c) - 3I(a, b, c)$

$= (3a, a-b, 2a+b+c) - 3(a, b, c) = (0, a-4b, 2a+b-2c)$

$(T - I)(a, b, c) = T(a, b, c) - I(a, b, c)$

$(T - I)(0, a-4b, 2a+b-2c) = T(0, a-4b, 2a+b-2c) - I(0, a-4b, 2a+b-2c)$

$= (0, -2a+8b, a-4b)$

$(T + I)(0, -2a+8b, a-4b) = T(0, -2a+8b, a-4b) + (0, -2a+8b, a-4b)$

$= (0, 2a-8b, -2a+8b+a-4b) + (0, -2a+8b, a-4b) = (0, 0, 0)$

$\therefore$ $L.H.S. = (T^2 - I)(T - 3I) = (0, 0, 0) = O = R.H.S.$

EXERCISE 3 (*c*)

1. Given $T(\alpha_1) = (1, 1, 1)$; $T(\alpha_2) = (1, -1, 1)$; $T(\alpha_3) = (1, 0, 0)$; $T(\alpha_4) = (1, 0, 1)$

$\therefore$ $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ is a standard basis of V_4.

$T(1, 0, 0, 0) = (1, 1, 1)$; $T(0, 1, 0, 0) = (1, -1, 1)$;

$T(0, 0, 1, 0) = (1, 0, 0)$; $T(0, 0, 0, 1) = (1, 0, 1)$.

$\therefore$ $(a, b, c, d) = a(1, 0, 0, 0) + b(0, 1, 0, 0) + c(0, 0, 1, 0) + d(0, 0, 0, 1)$

$T(a, b, c, d) = aT(1, 0, 0, 0) + bT(0, 1, 0, 0) + cT(0, 0, 1, 0) + dT(0, 0, 0, 1)$

$= a(1, 1, 1) + b(1, -1, 1) + c(1, 0, 0) + d(1, 0, 1)$

$T(a, b, c, d) = (a+b+c+d, a-b, a+b+d)$

Now $\nu(T) = \dim N(T)$. We find basis for Null space of T.

Let $\alpha \in N(T) \Rightarrow T(\alpha) = \hat{0}$ where $\hat{0} = (0, 0, 0) \in V_3(R)$

$\alpha = (a, b, c, d) \in V_4(R)$, $T(a, b, c, d) = (0, 0, 0)$

$\Rightarrow (a+b+c+d, a-b, a+b+d) = (0, 0, 0)$

$\Rightarrow a+b+c+d = 0,\ a = b,\ a+b+d = 0$

We have to solve these for a, b, c, d

Now co-efficient matrix $= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

Applying $R_2 - R_1,\ R_3 - R_1 \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -2 & -1 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$

$\therefore$ The equivalent system of equations are $a+b+c+d = 0;\ -2b-c-d = 0;\ -c = 0$.

Solving again we get $(a, b, c, d) = (b, b, 0, -2b) = b(1, 1, 0, -2)$

The number of free variables is '1' namely b.

Hence Nullity $(T) = \nu(T) = 1$. Now we find $\rho(T) = \dim R(T)$.

Writing all the T-images into a matrix we get $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

[$\because$ For confirmation of basis of $R(T)$]

Applying $R_2 - R_1,\ R_3 - R_1,\ R_4 - R_1 \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & 0 \\ 0 & -1 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

Taking $2R_4 - R_2 \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$. $\therefore$ $R(T)$ is spanned by non-zero vectors of $V_3(R)$.

These are $\{(1, 1, 1), (0, -2, 0), (0, -1, -1)\}$. $\therefore$ $\dim R(T) = \rho(T) = 3$

Verification : $\dim R(T) + \dim N(T) = \dim V_4(R)$

L.H.S. $= 3+1 = 4$ [$\because$ $\dim R(T) = 3,\ \dim N(T) = 1$]

R.H.S. $= \dim V_4 = 4 =$ L.H.S.

2. $T: R^3 \to R^3$, Range space spanned by $S = \{(1, 0, -1), (1, 2, 2)\}$.

Let us include a vector $(0, 0, 0) \in R^3$ in this spanning set which will not effect the spanning property. Let $S = \{(1, 0, -1), (1, 2, 2), (0, 0, 0)\}$

Let $B = \{\alpha_1, \alpha_2, \alpha_3\}$ be the standard basis of R^3.

We know there exists a transformation $T: R^3 \to R^3$

Such that $T(\alpha_1) = (1, 0, -1)$; $T(\alpha_2) = (1, 2, 2)$; $T(\alpha_3) = (0, 0, 0)$

Now if $\alpha \in R^3 \Rightarrow \alpha = (a, b, c)$ and $\alpha = a\alpha_1 + b\alpha_2 + c\alpha_3$

$\therefore$ $T(a, b, c) = aT(\alpha_1) + bT(\alpha_2) + cT(\alpha_3) = a(1, 0, -1) + b(1, 2, 2) + c(0, 0, 0)$

$\therefore$ $T(a, b, c) = (a + b, 2b, -a + 2b)$ is the required transformation.

3. Given $T: R^3 \to R^2$ linear transformation defined by $T(x, y, z) = (x + y, y + z)$

(*i*) Basis and dimension of Range space :

Let us consider standard basis of R^3 *i.e.*, $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$

Now $T(1, 0, 0) = (1, 0)$; $T(0, 1, 0) = (1, 1)$; $T(0, 0, 1) = (0, 1)$ (using $T(x, y, z)$)

Now we verify the T-images for the basis for which we write the matrix say S_1.

$$S_1 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}. \text{ Applying } R_2 - R_1. \; S_1 \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}. \text{ Applying } R_3 - R_2. \; S_1 \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

The non-zero rows of vectors $\{(1, 0), (0, 1)\}$ form basis for Range space.

$\therefore$ $\dim R(T) = 2 = \rho(T)$.

(*ii*) Basis and dimension of Null space : $\alpha \in R^3 \Rightarrow T(\alpha) = \bar{0} = (0, 0)$

$T(x, y, z) = (0, 0) \Rightarrow (x + y, y + z) = (0, 0) \Rightarrow x + y = 0, \; y + z = 0$

Solving we get $x = -y = z$. Let $x = k$ so that $(x, y, z) = (k, -k, k) = k(1, -1, 1)$

$\{(1, -1, 1)\}$ constitutes a basis of $N(T)$. $\therefore$ $\dim N(T) = 1$

Also $\dim R(T) + \dim N(T) = 2 + 1 = \dim R^3 = 3$. Rank-Nullity theorem verified.

4. Given linear transformation $T: R^3 \to R^3$ defined by

$T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$

(*i*) Basis and dimension of Range space :

Consider standard basis of R^3 *i.e.*, $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. Now using $T(x, y, z)$

$T(1, 0, 0) = (1, 0, 1)$; $T(0, 1, 0) = (2, 1, 1)$; $T(0, 0, 1) = (-1, 1, -2)$

We verify basis for which we write the matrix say $S_1 = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ -1 & 1 & -2 \end{bmatrix}$.

Applying $R_2 - 2R_1,\ R_3 + R_1$. $S_1 \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix}$. Again $R_3 - R_2$ $S_1 \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$

$\therefore$ The non-zero rows of vectors $\{(1, 0, 1), (0, 1, -1)\}$ forms basis for Range space.

$\therefore \ \dim R(T) = \rho(T) = 2$.

(*ii*) Basis and dimension of Null space : $\alpha = (a, b, c) \in R^3 \Rightarrow T(\alpha) = \hat{0} = (0, 0, 0)$

$T(a, b, c) = (a + 2b - c, b + c, a + b - 2c) = (0, 0, 0) \Rightarrow a + 2b - c = 0,\ b + c = 0,\ a + b - 2c = 0$

We have to solve these for a, b, c.

Co-efficient matrix $= \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & -2 \end{bmatrix}$

Applying $R_3 - R_1 \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}$. Again $R_3 + R_2 \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$.

$\therefore$ The equavalent system of equations are $a + 2b - c = 0,\ b + c = 0$.

Solving we get : $\therefore \ (a, b, c) = (3c, -c, c) = c(3, -1, 1)$

$\therefore \ \{(3, -1, 1)\}$ constitute a basis of $N(T)$. $\therefore \ \dim N(T) = \nu(T) = 1$

Verification of Rank-Nullity theorem $\nu(T) = \rho(T) = \dim R^3$

L.H.S. $= 1 + 2 = 3$. R.H.S. $= \dim R^3 = 3 =$ L.H.S

5. $T : R^3 \to R^3$ defined by $T(x, y, z) = (x - y + 2z, 2x + y - z, -x - 2y)$

$\alpha \in R^3 \Rightarrow T(\alpha) = \hat{0} = (0, 0, 0)$

$T(x, y, z) = (0, 0, 0) \Rightarrow (x - y + 2z, 2x + y - z, -x - 2y) = (0, 0, 0)$

$\Rightarrow x - y + 2z = 0,\ 2x + y - z = 0,\ -x - 2y = 0$

We have to solve these for x, y, z. Co-efficient matrix $= \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & -1 \\ -1 & -2 & 0 \end{bmatrix}$.

Applying $R_2 - 2R_1,\ R_3 + R_1 \sim \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & -5 \\ 0 & -3 & 2 \end{bmatrix}$. Again $R_3 + R_2 \sim \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & -5 \\ 0 & 0 & -3 \end{bmatrix}$.

$\therefore$ The equavalent system of equations are :

$x - y + 2z = 0,\ 3y - 5z = 0,\ -3z = 0 \Rightarrow \ x = 0,\ y = 0,\ z = 0$

$\therefore$ $(x, y, z) = (0, 0, 0) = \bar{0} \in R^3(R)$ is L.D. vector. $\therefore$ $\dim N(T) = 0 = \nu(T)$.

6. $T : V_3(R) \to V_2(R)$ defined by $T(a, b, c) = (a, b)$ $\forall$ $(a, b, c) \in V_3(R)$.

Let $\alpha, \beta \in V_3(R)$, $\alpha = (a_1, b_1, c_1)$, $\beta = (a_2, b_2, c_2)$ and $p, q \in R$

We prove that $T(p\alpha + q\beta) = pT(\alpha) + qT(\beta)$

L.H.S. $T(p\alpha + q\beta) = T[pa_1 + qa_2, pb_1 + qb_2, pc_1 + qc_2]$

$= (pa_1 + qa_2, pb_1 + qb_2) = (pa_1, pb_1) + (qa_2, qb_2)$

$= pT(a_1, b_1) + qT(a_2, b_2) = pT(\alpha) + qT(\beta) = R.H.S.$

$\therefore$ $T : V_3(R) \to V_2(R)$ is Linear Transformation.

To find $N(T)$: $N(T) = \{\alpha \in V_3(R) / T(\alpha) = \hat{0}\}$ whose $\hat{0} = (0, 0) \in V_2(R)$.

$T(a, b, c) = (0, 0) \Rightarrow (a, b) = (0, 0) \Rightarrow a = 0, b = 0$

$\therefore$ $N(T)$ or Kernel of $T = \{(0, 0, c) / c \in R\} \subset V_3(R)$.

7. Given $\dim V = n$ and $T : V \to V$ is linear transformation such that Range space and Null space of T are identical. $\therefore$ $\dim R(T) = \dim N(T)$ .....(1)

From Rank-Nullity theorem,

$\dim V = \dim R(T) = \dim N(T) \Rightarrow n = \dim R(T) + \dim R(T)$ ($\because$ from (1))

$\Rightarrow n = 2 \dim R(T)$ $\Rightarrow n$ is even.

UNIT - IV - Matrix

6. CHARACTERISTIC VALUES AND CHARACTERISTIC VECTORS, CAYLEY - HAMILTON THEOREM

EXERCISE - 6 (*a*)

1. Let $A=\begin{bmatrix} a & h & g \\ 0 & b & 0 \\ 0 & c & c \end{bmatrix}$. $\therefore$ Characteristic Polynomial of A is

$$|A-\lambda I| = \begin{vmatrix} a-\lambda & h & g \\ 0 & b-\lambda & 0 \\ 0 & c & c-\lambda \end{vmatrix} = (a-\lambda)[(b-\lambda)(c-\lambda)]$$

$\therefore$ The characteristic equation of A is $|A-\lambda I|=0 \Rightarrow \lambda = a$ or $\lambda = b$ or $\lambda = c$.
Thus eigen values of the given matrix are *a, b, c*.

2. Let $A=\begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & -1 \\ 2 & -1 & 0 \end{bmatrix}$

$\therefore$ Characteristic Polynomial of A is $|A-\lambda I| = \begin{vmatrix} 0-\lambda & 1 & 2 \\ 1 & 0-\lambda & -1 \\ 2 & -1 & 0-\lambda \end{vmatrix}$

$$= (-\lambda)[(-\lambda)(-\lambda)-1]-1[-\lambda+2]+2(-1+2\lambda)$$

$$= -\lambda^3+\lambda+\lambda-2-2+4\lambda = -\lambda^3+6\lambda-4$$

$\therefore$ The characteristic equation of A is $|A-\lambda I|=0 \Rightarrow \lambda^3-6\lambda+4=0$

$$\Rightarrow (\lambda-2)(\lambda^2+2\lambda-2)=0$$

$$\therefore \lambda = 2,\ -1+\sqrt{3}, -1-\sqrt{3}$$

$\therefore$ The characteristic values of A are $2,\ -1+\sqrt{3}, -1-\sqrt{3}$.

3. Given matrix $A=\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$. $\therefore$ The characteristic equation of A is

$$|A-\lambda I|=0 \Rightarrow \begin{vmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{vmatrix} = 0 \Rightarrow (5-\lambda)(2-\lambda)-4=0$$

$$\Rightarrow \lambda^2-7\lambda+6=0 \Rightarrow (\lambda-6)(\lambda-1)=0 \qquad \Rightarrow \lambda=6 \text{ or } \lambda=1.$$

Thus eigen values of A are $6,1$.

(*i*) To find eigen vectors corresponding to $\lambda = 6$

Eigen vectors corresponding to the eigen value 6 of A are given by

$$(A-\lambda I)X = 0 \Rightarrow \begin{bmatrix} 5-6 & 4 \\ 1 & 2-6 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$R_2 - R_1$ gives $\begin{bmatrix} -1 & 4 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ *i.e.* $-x+4y=0$

Let $x = k\ (\neq 0)$, $y = 4k$. Thus $X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} k \\ 4k \end{bmatrix} = k\begin{bmatrix} 1 \\ 4 \end{bmatrix}$

Thus the eigen vectors corresponding to the eigen value 6 are $k\begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

(*ii*) To find the eigen vector corresponding to $\lambda = 1$.

Eigen vectors corresponding to the eigen value 1 of A are given by

$$(A-\lambda I)X = 0 \Rightarrow \begin{bmatrix} 5-1 & 4 \\ 1 & 2-1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$4R_2 - R_1$ gives $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad \therefore\ x+y=0$

Let $x = k\ (\neq 0)$, $y = -k$.

$$X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} k \\ -k \end{bmatrix} = k\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Thus the eigen vectors corresponding to $\lambda = 1$ are $k\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

4. (*i*) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{bmatrix}$

$\therefore$ The characteristic equation of the matrix A is $|A-\lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 & 3 \\ 0 & -4-\lambda & 2 \\ 0 & 0 & 7-\lambda \end{vmatrix} = 0$$

$$\Rightarrow 1(1-\lambda)[(-4-\lambda)(7-\lambda)-0]-2[0-0]+3(0)=0$$

$$\Rightarrow (1-\lambda)(-4-\lambda)(7-\lambda)=0$$

$\therefore\ \lambda = 1, -4, 7$

$\therefore$ The characteristic roots are $\lambda = 1$, $\lambda = -4$, $\lambda = 7$.

(*ii*) Let $A=\begin{bmatrix}5 & 6 & 8\\ 0 & 7 & 2\\ 0 & 0 & 4\end{bmatrix}$

$\therefore$ The characteristic equation of the matrix A is $|A-\lambda I|=0 \Rightarrow \begin{vmatrix}5-\lambda & 6 & 8\\ 0 & 7-\lambda & 2\\ 0 & 0 & 4-\lambda\end{vmatrix}=0$

$\Rightarrow (5-\lambda)(7-\lambda)(4-\lambda)=0.$ $\Rightarrow \lambda=5,7,4$

$\therefore$ The characteristic roots of A are 5, 7, 4.

To find the characteristic vectors

Case (*i*): Characteristic vectors corresponding to $\lambda=4$ are given by

$$(A-\lambda I)X=0 \text{ where } X=\begin{bmatrix}x\\ y\\ z\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}5-4 & 6 & 8\\ 0 & 7-4 & 2\\ 0 & 0 & 4-4\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\end{bmatrix} \Rightarrow \begin{bmatrix}1 & 6 & 8\\ 0 & 3 & 2\\ 0 & 0 & 0\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

$\therefore\ x+6y+8z=0\ ,\ 3y+2z=0$

Let $z=k\ (\neq 0)$, then $y=(-2/3)k$.

$\therefore\ x=-6y-8z=4k-8k=-4k$.

$$\text{Thus } X=\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}-4k\\ (-2/3)k\\ k\end{bmatrix}=\frac{-k}{3}\begin{bmatrix}12\\ 2\\ -3\end{bmatrix}$$

Thus the characteristic vectors corresponding to $\lambda=4$ are $k\begin{bmatrix}12\\ 2\\ -3\end{bmatrix}$.

Case (*ii*) Characteristic vectors corresponding to $\lambda=7$ are given by

$$(A-\lambda I)X=0 \text{ where } X=\begin{bmatrix}x\\ y\\ z\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}5-7 & 6 & 8\\ 0 & 7-7 & 2\\ 0 & 0 & 4-7\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 & 6 & 8 \\ 0 & 0 & 2 \\ 0 & 0 & -3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow 2R_3 + 3R_2 \begin{bmatrix} -2 & 6 & 8 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\therefore\ 2z = 0 \Rightarrow z = 0,\ -2x + 6y + 8z = 0 \Rightarrow -2x + 6y = 0 \Rightarrow -x + 3y = 0 \Rightarrow x = 3y.$

Taking $y = k\ (\neq 0)$, we have $x = 3k$.

Thus eigen vector is $\begin{bmatrix} 3k \\ k \\ 0 \end{bmatrix} = k\begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$

Thus the characterstic vectors corresponding to $\lambda = 7$ are $k\begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$.

Case (*iii*): Characteristic vectors corresponding to $\lambda = 5$ are given by

$$(A - 5I)X = 0 \text{ where } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 6 & 8 \\ 0 & 2 & -2 \\ 0 & 0 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\therefore\ z = 0,\ 2y + 2z = 0 \Rightarrow y = 0.$

Let $x = k\ (\neq 0)$.

Thus eigen vector is $\begin{bmatrix} k \\ 0 \\ 0 \end{bmatrix} = k\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$\therefore$ The characterstic vectors corresponding to $\lambda = 5$ are $k\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

5. (*a*) Let $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$. The characteristic equation of A is

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 3-\lambda & 10 & 5 \\ -2 & -3-\lambda & -4 \\ 3 & 5 & 7-\lambda \end{vmatrix} = 0$$

$\Rightarrow (3-\lambda)[(-3-\lambda)(7-\lambda) + 20] - 10(2\lambda - 2) + 5(3\lambda - 1) = 0$

$\Rightarrow (3-\lambda)(\lambda^2 - 4\lambda - 1) - 20\lambda + 20 + 15\lambda - 5 = 0$

$\Rightarrow \lambda^3 - 7\lambda^2 + 16\lambda - 12 = 0.$ The roots are $\lambda = 2, 2, 3$.

$\therefore$ 2, 2, 3 are the characteristic values of A.

To find the characterstic vector corresponding to characteristic value 2 .

Let $X=\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ so that X will be given by a non - zero solution of the equation

$$(A-2I)X=0 \; i.e., \; \begin{bmatrix} 1 & 10 & 5 \\ -2 & -5 & -4 \\ 3 & 5 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{matrix} R_2+2R_1 \\ R_3-3R_1 \end{matrix}\begin{bmatrix} 1 & 10 & 5 \\ 0 & 15 & 6 \\ 0 & -25 & -10 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\frac{R_2}{3}, \frac{R_3}{-5}\begin{bmatrix} 1 & 10 & 5 \\ 0 & 5 & 2 \\ 0 & 5 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3-R_2\begin{bmatrix} 1 & 10 & 5 \\ 0 & 5 & 2 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We get $x+10y+5z=0$, $5y+2z=0$

Take $z=k \Rightarrow y=\frac{-2k}{5}$

Then $x=-105y-5z=-4k-5k=-9k$

The corresponding eigen vector is $\begin{bmatrix} -9k \\ \frac{-2k}{5} \\ k \end{bmatrix}=\frac{-k}{5}\begin{bmatrix} 45 \\ 2 \\ -5 \end{bmatrix}$. Let $X_1=\begin{bmatrix} 45 \\ 2 \\ -5 \end{bmatrix}$

Every non-zero multiple of the column matrix X_1 is a characteristic vector of **A** corresponding to characteristic value 2.

Let $\lambda=3$. $\begin{bmatrix} 0 & 10 & 5 \\ -2 & -6 & -4 \\ 3 & 5 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\frac{R_1}{5}, \frac{R_2}{-2}\begin{bmatrix} 0 & 2 & 1 \\ 1 & 3 & 2 \\ 3 & 5 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3 - 3R_2 \begin{bmatrix} 0 & 2 & 1 \\ 1 & 3 & 2 \\ 0 & 4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3 + 2R_2 \begin{bmatrix} 0 & 2 & 1 \\ 1 & 3 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$2y + z = 0$. Let $z = k\ (\neq 0) \Rightarrow y = \frac{-k}{2}$.

$x + 3y + 2z = 0 \Rightarrow x = \frac{-3k}{2} + 2k = \frac{k}{2}$.

The corresponding eigen vector is $\begin{bmatrix} \frac{k}{2} \\ \frac{-k}{2} \\ k \end{bmatrix} = \frac{k}{2}\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Let $X_2 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$.

Every non-zero multiple of the column matrix X_2 is a characteristic vector of **A** corresponding to the characteristic value 3.

The subspaces (characteristic spaces) generated by X_1 and X_2 are the subspaces corresponding to the characteristic values 2, 3.

6. (*i*) Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$. The characteristic equation of A is $|A - \lambda I| = 0 \Rightarrow \begin{bmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{bmatrix} = 0$

$\Rightarrow (1-\lambda)(2-\lambda) = 6$

$\Rightarrow \lambda^2 - 3\lambda - 4 = 0 \Rightarrow (\lambda - 4)(\lambda + 1) = 0$

$\Rightarrow \lambda = -1, 4$ are the eigen values.

eigen vectors are relating to $\lambda = 4$ given by $(A - 4I)X = 0$ where $X = \begin{bmatrix} x \\ y \end{bmatrix}$.

$$\Rightarrow \begin{bmatrix} 1-4 & 2 \\ 3 & 2-4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow R_1 + R_2 \begin{bmatrix} -3 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow -3x + 2y = 0$$

Let $y = k\ (\neq 0)$. Then $x = \frac{2}{3}k$.

$\therefore X = \begin{bmatrix} \frac{2}{3}k \\ k \end{bmatrix} = \frac{k}{3}\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ are the characterstic vectors corresponding to the characteristic value 4 of A.

Eigen vectors relating to $\lambda = -1$ are given by $(A + 1\,I)\,X = 0$ where $X = \begin{bmatrix} x \\ y \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 1+1 & 2 \\ 3 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

$\therefore x + y = 0$

Let $y = k(\neq 0)$. Then $x = -k$.

$\therefore$ The characterstic vectors corresponding to the characteristic value -1 of A are $k\begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

(*ii*) Try your self

(*iii*) $A = \begin{bmatrix} i & 1 \\ 2 & -i \end{bmatrix}$ defined over C.

$\therefore$ The characteristic equation of A is $\det(A - \lambda I) = 0$

i.e., $\begin{vmatrix} i-\lambda & 1 \\ 2 & -i-\lambda \end{vmatrix} = 0$ *i.e.*, $-(i^2 - \lambda^2) - 2 = 0$ *i.e.*, $1 + \lambda^2 - 2 = 0 \Rightarrow \lambda = 1, -1$

$\therefore 1, -1$ are the two characteristic values of A.

Characteristic vectors relating to $\lambda = 1$ are given by $(A - 1\,I)\,X = 0$ where $X = \begin{bmatrix} x \\ y \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} i-1 & 1 \\ 2 & -i-1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow (i-1)x + y = 0,\ 2x - iy - y = 0$$

$$\Rightarrow (i^2 - 1)x + (i+1)y = 0,\ 2x - (i+1)y = 0 \Rightarrow y = \left(\frac{2}{i+1}\right)x$$

Let $x = k$ (non-zero complex number). $\therefore y = \left(\frac{2}{i+1}\right)k = \left(\frac{2(i-1)}{i^2-1}\right)k = (1-i)k$

$\therefore$ Characteristic vectors corresponding to the characteristic root 1 are $\begin{bmatrix} k \\ (1-i)k \end{bmatrix}$

i.e., $k\begin{bmatrix} 1 \\ 1-i \end{bmatrix}$

Characteristic vectors relating to $\lambda = -1$ are given by $(A + 1\,I)\,X = 0$ where $X = \begin{bmatrix} x \\ y \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} i+1 & 1 \\ 2 & -i+1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow (i+1)x + y = 0,\ 2x + (-i+1)y = 0$$

$$\Rightarrow \frac{2(i+1)}{i+1}x + \frac{2}{i+1}y = 0,\ 2x + (-i+1)y = 0$$

$$\Rightarrow 2x + \frac{2(1-i)}{1-i^2}y = 0,\ 2x + (-i+1)y = 0$$

$$\Rightarrow 2x + (1-i)y = 0 \Rightarrow y = \frac{2}{i-1}x.$$

Let $x = k$ (non-zero complex number). Then $y = (-i-1)\,k$.

$\therefore$ Characteristic vectors corresponding to the characteristic root -1 are $\begin{bmatrix} k \\ (-i-1)k \end{bmatrix}$

i.e., $k \begin{bmatrix} 1 \\ -i-1 \end{bmatrix}$

(*iv*) $A = \begin{bmatrix} 1 & 1 \\ 0 & i \end{bmatrix}$ defined over C.

$\therefore$ The characteristic equation of A is $\det(A - xI) = O$

i.e., $\begin{vmatrix} 1-\lambda & 1 \\ 0 & i-\lambda \end{vmatrix} = 0$ *i.e.*, $(1-\lambda)(i-\lambda) = 0$ *i.e.*, $\lambda = 1, i$

$\therefore$ 1 and i are two characteristic values of A.

Characteristic vectors relating to $\lambda = 1$ are given by $(A - 1\,I)X = 0$ where $X = \begin{bmatrix} x \\ y \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 1-1 & 1 \\ 0 & i-1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow 0.x + 1.y = 0,\ 0.x + (i-1)y = 0 \Rightarrow y = 0$$

Let $x = k_1$ (non-zero complex number) $\quad \therefore X = \begin{bmatrix} k_1 \\ 0 \end{bmatrix}$

$\therefore$ Characteristic vectors corresponding to $\lambda = 1$ are $\begin{bmatrix} k_1 \\ 0 \end{bmatrix}$

Characteristic vectors relating to $\lambda = i$ are given by $(A - iI)X = O$ where $X = \begin{bmatrix} x \\ y \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 1-i & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow (1-i)x + y = 0$$

Let $x = k_2$ (non-zero complex number). $\therefore\ y = (i-1)\,k_2$

$\therefore$ Characteristic vectors corresponding to $\lambda = i$ are $\begin{bmatrix} k_2 \\ (i-1)k_2 \end{bmatrix}$

7 (*i*) $T : R^2 \to R^2$. $B = \{(1,0),(0,1)\}$ is the standard basis of R^2.

$T(a,b) = (-2a+3b, -10a+9b)$

$T(1,0) = (-2+0, -10+0) = (-2,-10) = -2(1,0) - 10(0,1)$

$T(0,1) = (3,9) = 3(1,0) + 9(0,1)$

Thus the corresponding matrix $= A = [T]_B = \begin{bmatrix} -2 & 3 \\ -10 & 9 \end{bmatrix}$.

Now $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} -2-\lambda & 3 \\ -10 & 9-\lambda \end{vmatrix} = 0$

$\Rightarrow (-2-\lambda)(9-\lambda) + 30 = 0 \Rightarrow \lambda^2 - 7\lambda + 12 = 0$

$\Rightarrow (\lambda-3)(\lambda-4) = 0 \Rightarrow \lambda = 3$. The eigen values of T are $3, 4$.

(*ii*) $T : R^3 \to R^3$. $B = \{(1,0,0),(0,1,0),(0,0,1)\}$ is the standard basis of R^3.

$T(a,b,c) = (7a-4b+10c,\ 4a-3b+8c,\ -2a+b-2c)$

$T(1,0,0) = (7,4,-2) = 7(1,0,0) + 4(0,1,0) - 2(0,0,1)$

$T(0,1,0) = (-4,-3,1) = -4(1,0,0) - 3(0,1,0) + 1(0,0,1)$

$T(0,0,1) = (10,8,-2) = 10(1,0,0) + 8(0,1,0) - 2(0,0,1)$

Thus the corresponding matrix $X = A = [T]_B = A = \begin{bmatrix} 7 & -4 & 10 \\ 4 & -3 & 8 \\ -2 & 1 & -2 \end{bmatrix}$

Now $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 7-\lambda & -4 & 10 \\ 4 & -3-\lambda & 8 \\ -2 & 1 & -2-\lambda \end{vmatrix} = 0$

$\Rightarrow (7-\lambda)[(-3-\lambda)(-2-\lambda) - 8] + 4(-8-4\lambda+16) + 10(4-6-2\lambda) = 0$

$\Rightarrow (7-\lambda)[6+3\lambda+2\lambda+\lambda^2-8] + 4(8-4\lambda) + 10(-2-2\lambda) = 0$

$\Rightarrow (7-\lambda)(\lambda^2+5\lambda-2) + 32 - 16\lambda - 20 - 20\lambda = 0$

$\Rightarrow 7\lambda^2 + 35\lambda - 14 - \lambda^3 - 5\lambda^2 + 2\lambda + 32 - 16\lambda - 20 - 20\lambda = 0$

$\Rightarrow -\lambda^3 + 2\lambda^2 + \lambda - 2 = 0$

By solving we get $\lambda = -1, 1, 2$

Thus the eigen values of T are $-1, 1, 2$.

(*iii*) $T : P_2(R) \to P_2(R)$, where T is the linear operator

$T[f(x)] = x \cdot f'(x) + f(x) \cdot x + f(3)$

Let $\beta = \{1, x, x^2\}$ is the standard basis for $P_2(R)$

$f(x) = 1, T(1) = 0 + 1 \cdot x + 1 = 1 + x = 1 + 1 \cdot x + 0 \cdot x^2$

$f(x) = x, T(x) = x \cdot 1 + 2 \cdot x + 3 = 3 + 3x = 1(3) + x(3) + 0 \cdot x^2$

$f(x) = x^2, T(x^2) = x \cdot (2x) + 4 \cdot x + 9 = 1(9) + 4 \cdot x + 2(x^2)$

Thus $A = [T]_\beta = \begin{bmatrix} 1 & 3 & 9 \\ 1 & 3 & 4 \\ 0 & 0 & 2 \end{bmatrix}$

Now $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 3 & 9 \\ 1 & 3-\lambda & 4 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 0$

$\Rightarrow (1-\lambda)[(3-\lambda)(2-\lambda) - 3(0) + 9(0)] = 0$

$\Rightarrow \lambda = 1, 2, 3$. $\therefore$ The eigen values of T are $1, 2, 3$.

EXERCISE 6 (*b*)

1. Given $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$

The characteristic equations of A is $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix} = 0$

$\Rightarrow (2-\lambda)(\lambda^2 - 5\lambda + 4) - 2(1-\lambda) + 1(\lambda - 1) = 0$

$\Rightarrow -\lambda^3 + 7\lambda^2 - 14\lambda + 8 + 3\lambda - 3 = 0 \quad \Rightarrow \lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$

By Cayley - Hamilton theorem, A satisfies its charectaristic equation.

To verify that $A^3 - 7A^2 + 11A - 5I = 0$

$$A^2 = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 4+2+1 & 2+6+4 & 2+2+2 \\ 2+3+1 & 1+9+2 & 1+3+2 \\ 2+2+2 & 2+6+4 & 1+2+4 \end{bmatrix} = \begin{bmatrix} 7 & 12 & 6 \\ 6 & 13 & 6 \\ 6 & 12 & 7 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 7 & 12 & 6 \\ 6 & 13 & 6 \\ 6 & 12 & 7 \end{bmatrix}\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 32 & 62 & 31 \\ 31 & 63 & 31 \\ 31 & 62 & 36 \end{bmatrix}$$

$$A^3 - 7A^2 + 11A - 5I$$

$$= \begin{bmatrix} 32 & 62 & 31 \\ 31 & 63 & 31 \\ 31 & 62 & 36 \end{bmatrix} - \begin{bmatrix} 49 & 84 & 42 \\ 42 & 91 & 42 \\ 42 & 84 & 42 \end{bmatrix} + \begin{bmatrix} 22 & 22 & 11 \\ 11 & 33 & 11 \\ 18 & 12 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ A satisfies Cayley - Hamilton Theroem.

2. Given $A = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$

The characteristic equation of A is $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 1 & 2 \\ 3 & 1-\lambda & 1 \\ 2 & 3 & 1-\lambda \end{vmatrix} = 0$

$$\Rightarrow (1-\lambda)[(1-\lambda)^2 - 3] - 1(3 - 3\lambda - 2) + 2(9 - 2 + 2\lambda) = 0$$

$$\Rightarrow -\lambda^3 + 3\lambda^2 - 2 - 1 + 3\lambda + 14 + 4\lambda = 0$$

$$\Rightarrow -\lambda^3 + 3\lambda^2 + 7\lambda + 11 = 0 \Rightarrow \lambda^3 - 3\lambda^2 - 7\lambda - 11 = 0 \quad \ldots\ (1)$$

By Cayley - Hamilton theorem, A satisfies equation (1)

To verify that $A^3 - 3A^2 - 7A - 11I = 0$.

$$A^2 = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 8 & 5 \\ 8 & 7 & 8 \\ 13 & 8 & 8 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 8 & 8 & 5 \\ 8 & 7 & 8 \\ 13 & 8 & 8 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 42 & 31 & 29 \\ 45 & 39 & 31 \\ 53 & 45 & 42 \end{bmatrix}$$

$$A^3 - 3A^2 - 7A - 11I = \begin{bmatrix} 42 & 31 & 29 \\ 45 & 39 & 31 \\ 53 & 45 & 42 \end{bmatrix} - \begin{bmatrix} 24 & 24 & 15 \\ 24 & 21 & 24 \\ 39 & 24 & 24 \end{bmatrix} - \begin{bmatrix} 7 & 7 & 14 \\ 21 & 7 & 7 \\ 14 & 21 & 7 \end{bmatrix} - \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ A satisfies Cayley - Hamilton Theorem.

Here $|A| \neq 0$. $\quad A^3 - 3A^2 - 7A - 11I = 0 \Rightarrow A^2 - 3A - 7I - 11A^{-1} = 0$

$$\Rightarrow 11A^{-1} = A^2 - 3A - 7I \Rightarrow A^{-1} = \frac{1}{11}(A^2 - 3A - 7I)$$

$$= \frac{1}{11}\left[\begin{bmatrix} 8 & 8 & 5 \\ 8 & 7 & 8 \\ 13 & 8 & 8 \end{bmatrix} - \begin{bmatrix} 3 & 3 & 6 \\ 9 & 3 & 3 \\ 6 & 9 & 3 \end{bmatrix} - \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}\right]. \qquad \therefore A^{-1} = \frac{1}{11}\begin{bmatrix} -2 & 5 & -1 \\ -1 & -3 & 5 \\ 7 & -1 & -2 \end{bmatrix}$$

3. Given $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$.

The charectaristic equation of A is $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$

$$\Rightarrow (2-\lambda)(\lambda^2 - 4\lambda + 3) + (\lambda - 1) + (\lambda - 1)$$

$$\Rightarrow -\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0 \qquad \Rightarrow \lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0 \quad \ldots. (1)$$

By Cayley - Hamilton theorem, A satisfies equation (1)

$$(1) \Rightarrow A^3 - 6A^2 + 9A - 4I = 0$$

$$A^2 = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 5 \end{bmatrix}; A^3 = \begin{bmatrix} 22 & -21 & 21 \\ 1 & 18 & -13 \\ 20 & -20 & 20 \end{bmatrix}; 9A = \begin{bmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{bmatrix}; \quad \therefore 4I = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Here $|A| \neq 0$. $\quad A^3 - 6A^2 + 9A - 4I = 0 \Rightarrow A^2 - 6A + 9I - 4A^{-1} = 0$

$$4A^{-1} = A^2 - 6A + 9I \Rightarrow A^{-1} = \frac{1}{4}(A^2 - 6A + 9I) \Rightarrow A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

4. Given $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}$

The charectristic equation of A is $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & 0 & 2 \\ 0 & 1-\lambda & 2 \\ 1 & 2 & -\lambda \end{vmatrix} = 0$

$$\Rightarrow (1-\lambda)(\lambda^2 - \lambda - 4) + (2-\lambda) = 0 \Rightarrow \lambda^3 - 2\lambda^2 - 5\lambda + 6 = 0 \qquad \ldots. (1)$$

By Cayley Hamilton theorem, 'A' satisfies equation (1).

$(1) \Rightarrow A^3 - 2A^2 - 5A + 6I = 0$

$$A^2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 2 \\ 2 & 5 & 2 \\ 1 & 2 & 6 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 3 & 4 & 2 \\ 2 & 5 & 2 \\ 1 & 2 & 6 \end{bmatrix}\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 8 & 14 \\ 4 & 9 & 14 \\ 7 & 14 & 6 \end{bmatrix}$$

$$A^3 - 2A^2 - 5A + 6I$$

$$= \begin{bmatrix} 5 & 8 & 14 \\ 4 & 9 & 14 \\ 7 & 14 & 6 \end{bmatrix} - \begin{bmatrix} 6 & 8 & 4 \\ 4 & 10 & 4 \\ 2 & 4 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 10 \\ 0 & 5 & 10 \\ 5 & 10 & 0 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Here $|A| \neq 0$.

$\therefore A^3 - 2A^2 - 5A + 6I = 0 \Rightarrow A^2 - 2A - 5I + 6A^{-1} = 0$.

$$\Rightarrow 6A^{-1} = [5I + 2A - A^2] = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 4 \\ 0 & 2 & 4 \\ 2 & 4 & 0 \end{bmatrix} - \begin{bmatrix} 3 & 4 & 2 \\ 2 & 5 & 2 \\ 1 & 2 & 6 \end{bmatrix}$$

$$\therefore 6A^{-1} = \begin{bmatrix} 4 & -4 & 2 \\ -2 & 2 & 2 \\ 1 & 2 & -1 \end{bmatrix} \Rightarrow A^{-1} = \frac{1}{6}\begin{bmatrix} 4 & -4 & 2 \\ -2 & 2 & 2 \\ 1 & 2 & -1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 2/3 & -2/3 & 1/3 \\ -1/3 & 1/3 & 1/3 \\ 1/6 & 1/3 & -1/6 \end{bmatrix}.$$

Check : $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix}$ $\quad A^{-1} = \frac{1}{|A|}(adj\ A)$; $\quad |A| = -6$

$$adj\ A = \begin{bmatrix} -4 & 2 & -1 \\ 4 & 2 & -2 \\ 2 & -2 & 1 \end{bmatrix}^T = \begin{bmatrix} -4 & 4 & -2 \\ 2 & -2 & -2 \\ -1 & -2 & 1 \end{bmatrix} \quad A^{-1} = \frac{-1}{6}\begin{bmatrix} -4 & 4 & -2 \\ 2 & -2 & -2 \\ -1 & -2 & 1 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 4 & -4 & 2 \\ -2 & 2 & 2 \\ 1 & 2 & -1 \end{bmatrix}$$

5. Given charecteristic equation of a matrix A is $a_1 + a_2A + a_3A^2 = 0$ ..... (1)

Multiplying with A^{-1} on both sides of (1)

$A^{-1}(a_1 + a_2A + a_3A^2) = 0 \quad \Rightarrow a_1A^{-1} + a_2A^{-1}.A + a_3A^{-1}.A^2 = 0$

$\Rightarrow a_1A^{-1} + a_2I + a_3A^{-1}.A.A = 0$

$\Rightarrow a_1A^{-1} + a_2I + a_3I.A = 0 \quad \Rightarrow a_1A^{-1} + a_2I + a_3A = 0$

$\Rightarrow a_1A^{-1} = -a_2I - a_3A \qquad \therefore A^{-1} = \frac{-a_2}{a_1}I - \frac{a_3}{a_1}A.$

7. INNER PRODUCT SPACES

EXERCISE 7 (*a*)

1. (*i*) $<\alpha,\ \beta\pm\gamma> = <\overline{\beta\pm\gamma,\ \alpha}> = <\overline{\beta,\ \alpha}> \pm <\overline{\gamma,\ \alpha}>$

$= <\alpha,\beta> \pm <\alpha,\gamma>$ (Conjugativity & Distributivity)

(*ii*) $<\alpha-\beta,\alpha-\beta> = <\alpha+(-\beta)\ ,\delta>$ where $\delta=\alpha-\beta$.

$= <\alpha,\delta> + <-\beta,\delta>$ (Distributivity)

$= <\alpha,\ \alpha+(-\beta)> + <-\beta,\ \alpha+(-\beta)> = <\overline{\alpha+(-\beta),\ \alpha}> + <\overline{\alpha+(-\beta),\ -\beta}>$

$<\overline{\alpha,\alpha}> + <\overline{-\beta,\alpha}> + <\overline{\alpha,-\beta}> + <\overline{-\beta,-\beta}>$

$= <\alpha,\alpha> + <-\beta,\alpha> + <\alpha,-\beta> + <-\beta,-\beta>$

$= <\alpha,\alpha> + (-1)<\beta,\alpha> + \overline{(-1)}<\alpha,\beta> + (-1)\ \overline{(-1)}<\beta,\beta>$

$= <\alpha,\alpha> - <\beta,\alpha> - <\alpha,\beta> + <\beta,\beta>$.

(*iii*) $<a\alpha-b\beta,a\alpha-b\beta> = <a\alpha+(-b)\beta,\delta>$ where $\delta=a\alpha-b\beta$

$= a<\alpha,\delta> + (-b)<\beta,\delta> = a<\alpha,a\alpha+(-b)\beta> - b<\beta,a\alpha+(-b)\beta>$

$= a<\overline{a\alpha+(-b)\beta,\ \alpha}> - b<\overline{a\alpha+(-b)\beta,\ \beta}>$

$= a\{\bar{a}<\overline{\alpha,\alpha}> + (\overline{-b})<\overline{\beta,\alpha}>\} - b\{\bar{a}\ <\overline{\alpha,\beta}> + (\overline{-b})<\overline{\beta,\beta}>\}$

$= a\,\bar{a}<\alpha,\alpha> - a\bar{b}<\alpha,\beta> - b\bar{a}<\beta,\alpha> + b\bar{b}<\beta,\beta>$.

2. $\alpha=(1,2),\ \beta=(-1,1)\in R^2$. If $\alpha=(a_1,a_2),\ \beta=(b_1,b_2)$

then standard inner product $=\alpha.\beta=a_1b_1+a_2b_2$. Let $\gamma=(x_1,x_2)$ where $x_1,x_2\in R$.

$<\alpha,\gamma> = -1 \Rightarrow 1x_1+2x_2=-1$ ....(1) and $<\beta,\gamma>=3 \quad \Rightarrow -1x_1+1x_2=3$ ...(2)

Solving (1) and (2): $x_1=-7/3,\ x_2=2/3$. $\therefore\quad \gamma=(-7/3,\ 2/3)$.

3. $\alpha=(2,1+i,i),\ \beta=(2-i,2,1+2i)\in C^3$. If $\alpha=(a_1,a_2,a_3)$ and $\beta=(b_1,b_2,b_3)$

then inner product of $\alpha,\beta=\alpha.\beta=a_1\overline{b_1}+a_2\overline{b_2}+a_3\overline{b_3}=2\ (\overline{2-i})+(1+i)\ (\overline{2})+i\ (\overline{1+2i})$

$=2\ (2+i)+2\ (1+i)+i\ (1-2i)=8+5i$.

4. Let $A=\begin{bmatrix}a_{11} & a_{12}\\ a_{21} & a_{22}\end{bmatrix}, B=\begin{bmatrix}b_{11} & b_{12}\\ b_{21} & b_{22}\end{bmatrix}\in M_2(R)$ where $a_{11},a_{12},a_{21},a_{22},b_{11},b_{12},b_{21},b_{22}\in R$

Then $A+B=\begin{bmatrix}a_{11}+b_{11} & a_{12}+b_{12}\\ a_{21}+b_{21} & a_{22}+b_{22}\end{bmatrix}$. $<A,B>=tr(A+B)=a_{11}+b_{11}+a_{22}+b_{22}\in R$.

(*i*) $<B,A> = tr\ (B+A)=tr\ (A+B)=<A,B> \quad \Rightarrow <A,B> = <\overline{B,A}>$

(*ii*) $< A, A > = tr\ (A + A) = 2\ (a_{1\ 1} + a_{2\ 2})$. If $A = O$ then $< A, A > = 2\ (0+0) = 0$

and if $A = \begin{bmatrix} 0 & a_{12} \\ a_{21} & 0 \end{bmatrix} \neq O$, then $< A, A > = tr\ (A + A) = 2\ (0+0) = 0$.

$\therefore$ positivity is not true. Hence $< A, B > = tr\ (A + B)$ does not define inner product.

5. $\alpha = (x_1, x_2, \ldots, x_n),\ \beta = (y_1, y_2, \ldots, y_n),\ \gamma = (z_1, z_2, \ldots z_n) \in R^n$

where x's, y's and z's are real numbers.

(*i*) $< \alpha, \beta > = x_1 y_1 + x_2 y_2 + \ldots + x_n y_n = y_1 x_1 + y_2 x_2 + \ldots y_n x_n$

$= < \beta, \alpha > = < \overline{\beta, \alpha} > \Rightarrow$ Conjugate Symmetry.

(*ii*) $\alpha = (0,0,\ldots,0) \Rightarrow < \alpha, \alpha > = 0.0 + 0.0 + \ldots + 0.0 = 0$

$\alpha \neq (0, 0, \ldots, 0) \Rightarrow \alpha \Rightarrow (x_1, x_2, \ldots, x_n)$ so that atleast one of $x_1, x_2, \ldots, x_n$ is not zero.

$\therefore\ < \alpha,\ \alpha > = x_1 . x_1 + x_2 . x_2 + \ldots + x_n . x_n = x_1^2 + x_2^2 + \ldots + x_n^2 \neq 0 \Rightarrow$ Positivity.

(*iii*) For $a, b \in R$,

$< a\alpha + b\beta,\ \gamma > = (ax_1 + by_1)\ z_1 + (ax_2 + by_2)\ z_2 + \quad \ldots + (ax_n + by_n)\ z_n$

$= a\ (x_1 z_1 + x_2 z_2 + \ldots + x_n z_n) + b\ (y_1 z_1 + y_2 z_2 + \ldots + y_n z_n)$

$= a < \alpha, \gamma > + b < \beta, \gamma > \Rightarrow$ Distributivity. $\therefore\ R^n$ is inner product space.

6. Let $\alpha = (x_1, x_2),\ \beta = (y_1, y_2),\ \gamma = (z_1, z_2) \in R^2$ where x's, y's and z's are real numbers.

(*i*) $< \alpha, \beta > = x_1 + x_2 + y_1 + y_2 = y_1 + y_2 + x_1 + x_2 = < \beta, \alpha > = < \overline{\beta, \alpha} >$

(*ii*) $\alpha = O = (0,0) \Rightarrow < \alpha, \alpha > = 0 + 0 = 0$. $\alpha \neq (0,0) \Rightarrow \alpha = (-2, 0)$ say

$\Rightarrow < \alpha, \alpha > = -2 + 0 = -2 \not> 0 \Rightarrow$ Positivity is not true.

(*iii*) We can also observe that $< a\alpha + b\beta, \gamma > = a\ x_1 + b\ y_1 +$

$$ax_2 + by_2 + z_1 + z_2 \neq a\ (x_1 + x_2 + z_1 + z_2) + b\ (y_1 + y_2 + z_1 + z_2)$$

$= a < \alpha, \gamma > + b < \beta, \gamma > \Rightarrow$ Distributivity is not true.

Hence $< \alpha, \beta > = x_1 + x_2 + y_1 + y_2$ is not an inner product.

7. Let $\alpha = (x_1, x_2),\ \beta = (y_1, y_2),\ \gamma = (z_1, z_2) \in R^2$ where x's, y's and z's are real numbers.

(*i*) $< \alpha, \beta > = x_1 y_1 - x_2 y_1 - x_1 y_2 + 4x_2 y_2 = y_1 x_1 - y_2 x_1 - y_1 x_2 + 4y_2 x_2$

$= < \beta, \alpha > = < \overline{\beta, \alpha} > \Rightarrow$ Conjugativity is true.

(*ii*) $\alpha = (0,0) \Rightarrow < \alpha, \alpha > = 0$ and $\alpha \neq (0,0)$

$\Rightarrow < \alpha, \alpha > = x_1^2 - x_2 x_1 - x_1 x_2 + 4x_2^2 = x_1^2 - 2x_1 x_2 + 4x_2^2$

$= (x_1 - x_2)^2 + 3x_2^2 > 0$ ($\because$ atleast one of x_1, x_2 is not zero),

$\Rightarrow$ Positivity is true.

(*iii*) $a\alpha+b\beta=(ax_1+by_1,\ ax_2+by_2)$

$<a\alpha+b\beta,\gamma>=(ax_1+by_1)z_1-(ax_2+by_2)z_1-(ax_1+by_1)z_2$

$+4\ (ax_2+by_2)z_2=a\ (x_1z_1-x_2z_1-x_1z_2+4x_2z_2)$

$+b\ (y_1z_1-y_2z_1-y_1z_2+4y_2z_2)=a<\alpha,\gamma>+b<\beta,\gamma>$

$\Rightarrow$ Distributivity is true.

$\therefore\ <\alpha,\beta>=x_1y_1-x_2y_1-x_1y_2+4x_2y_2$ defines inner product.

8. Let $\bar{x}=(x_1,x_2,x_3),\ \bar{y}=(y_1,y_2,y_3),\ \bar{z}=(z_1,z_2,z_3)\in R^3(R)$

so that x's, y's and z's are real numbers.

(*i*) $<\bar{x},\bar{y}>=x_1y_1+x_2y_2-x_3y_3=y_1x_1+y_2x_2-y_3x_3=<\bar{y},\bar{x}>=\overline{<\bar{y},\bar{x}>}$

(*ii*) $\bar{x}=(0,0,0)\Rightarrow<\bar{x},\bar{x}>=0$

For $\bar{x}=(0,1,4)\neq(0,0,0),\ <\bar{x},\bar{x}>=0.0+1.1-4.4$

$=1-16=-15\not>0\Rightarrow$ Positivity is not true.

$\therefore\ <\bar{x},\bar{y}>=x_1y_1+x_2y_2-x_3y_3$ does not define inner product.

9. f,g are inner products on $V(F)\Rightarrow$ For $\alpha,\beta,\gamma\in V$ and $a,b\in F$

we have $f(\alpha,\beta)=\overline{f(\beta,\alpha)}, f(\alpha,\alpha)>0$ when $\alpha\neq\bar{O}$ and $f(a\alpha+b\beta,\gamma)=af(\alpha,\gamma)+bf(\beta,\gamma)$.

$g(\alpha,\beta)=g(\beta,\alpha), g(\alpha,\alpha)>0$ when $\alpha\neq\bar{O}$ and $g(a\alpha+b\beta,\gamma)=ag(\alpha,\gamma)+bg(\beta,\gamma)$

(*i*) We define $(f+g):V\times V\to F$ as $(f+g)(\alpha,\beta)=f(\alpha,\beta)+g(\alpha,\beta)\ \forall(\alpha,\beta)\in V\times V$

Conjugate symmetry.

$(f+g)(\alpha,\beta)=f(\alpha,\beta)+g(\alpha,\beta)=\overline{f(\beta,\alpha)}+\overline{g(\beta,\alpha)}=\overline{f(\beta,\alpha)+g(\beta,\alpha)}=\overline{(f+g)(\beta,\alpha)}$

Non - negativity.

$(f+g)(\alpha,\alpha)=f(\alpha,\alpha)+g(\alpha,\alpha)>0$ for $\alpha\neq\bar{O}$

Linearity.

$(f+g)(a\alpha+b\beta,\gamma)=f(a\alpha+b\beta,\gamma)+g(a\alpha+b\beta,\gamma)$

$=\{af(\alpha,\gamma)+bf(\beta,\gamma)\}+\{ag(\alpha,\gamma)+bg(\beta,\gamma)\}=a(f+g)(\alpha,\gamma)+b(f+g)(\beta,\gamma)$

(*ii*) By case (*i*), f is inner product $\Rightarrow f+f=2f$ is an inner product.

By induction we can prove that nf is an inner product.

(*iii*) Define $(f-g)(\alpha,\beta)=f(\alpha,\beta)-g(\alpha,\beta)$.

We have $f(\alpha,\alpha)>0$ and $g(\alpha,\alpha)>0$ for $\alpha\neq0$.

But $(f-g)(\alpha,\alpha)=f(\alpha,\alpha)-g(\alpha,\alpha)$ may be negative when $f(\alpha,\alpha)<g(\alpha,\alpha)$.

EXERCISE 7 (*b*)

1. (*i*) $||\alpha|| = \sqrt{1^2+(-2)^2+5^2} = \sqrt{30}$. Unit vector of $\alpha = \frac{1}{||\alpha||}\alpha = \frac{1}{\sqrt{30}}(1, -2, 5)$

(*ii*) $<\alpha, \alpha> = (3+i)(3-i)+4\ i\ (-4i)+(-4)(-4) = 42$.

$\therefore\ ||\alpha|| = \sqrt{42}$ and unit vector $= \frac{1}{\sqrt{42}}(3+i, 4i, -4)$.

2. $<\alpha, \beta> = (1-i)(\overline{2-5i})+(2+3i)(\overline{3-i}) = (1-i)(2+5i)+(2+3i)(3+i)$

$= (2+5-2i+5i)+(6-3+2i+9i) = 10+14i$

$||\alpha|| = \sqrt{<\alpha, \alpha>} = \sqrt{(1-i)(1+i)+(2+3i)(2-3i)} = \sqrt{(1^2+1^2)+(2^2+3^2)} = \sqrt{27}$

$||\beta|| = \sqrt{<\beta, \beta>} = \sqrt{(2-5i)(2+5i)+(3-i)(3+i)} = \sqrt{(2^2+5^2)+(3^2+1^2)} = \sqrt{39}$.

3. Given that $||\alpha|| = ||\beta|| \Rightarrow\ <\alpha, \alpha> = <\beta, \beta>$

$<\alpha+\beta, \alpha-\beta> = <\alpha, \alpha-\beta> + <\beta, \alpha+\beta> = <\overline{\alpha-\beta, \alpha}> + <\overline{\alpha-\beta, \beta}>$

$= <\overline{\alpha, \alpha}> - <\overline{\beta, \alpha}> + <\overline{\alpha, \beta}> - <\overline{\beta, \beta}>$

(α, β are vectors in Euclidean space $\Rightarrow <\alpha, \alpha>, <\alpha, \beta>, \ldots$

are real numbers and $<\overline{\alpha, \beta}> = <\alpha, \beta>$)

$= <\alpha, \alpha> - <\alpha, \beta> + <\alpha, \beta> - <\beta, \beta> = 0$.

4. α, β are vectors in unitary space $\Rightarrow <\alpha, \alpha>, <\alpha, \beta>, \ldots$ are complex numbers.

$<\alpha+i\beta, \alpha+i\beta> = <\alpha, \alpha> + \bar{i}<\alpha, \beta> + i<\beta, \alpha> + i\bar{i}<\beta, \beta>$

$= ||\alpha||^2 - i<\alpha, \beta> + i<\beta, \alpha> + ||\beta||^2 \qquad (\because\ i\bar{i} = i(-i) = 1)$

$\therefore\ ||\alpha+i\beta|| = \sqrt{||\alpha||^2 - i<\alpha, \beta> + i<\beta, \alpha> + ||\beta||^2}$

$<\alpha-i\beta, \alpha-i\beta> = <\alpha, \alpha> - \bar{i}<\alpha, \beta> - i<\beta, \alpha> + i\bar{i}<\beta, \beta>$

$= ||\alpha||^2 + i<\alpha, \beta> - i<\beta, \alpha> + ||\beta||^2$

$\therefore\ ||\alpha-i\beta|| = \sqrt{||\alpha||^2 + i<\alpha, \beta> - i<\beta, \alpha> + ||\beta||^2}$.

5. $||\alpha+\beta||^2 = <\alpha+\beta, \alpha+\beta> = <\alpha, \alpha> + <\alpha, \beta> + <\beta, \alpha> + <\beta, \beta>$

$||\alpha-\beta||^2 = <\alpha-\beta, \alpha-\beta> = <\alpha, \alpha> - <\alpha, \beta> - <\beta, \alpha> + <\beta, \beta>$

$||\alpha+\beta||^2 - ||\alpha-\beta||^2 = 2<\alpha, \beta> + 2<\beta, \alpha> = 2\{<\alpha, \beta> + \overline{<\alpha, \beta>}\}$

(But $Z + \bar{Z} = 2\mathrm{Re}Z$ and hence $<\alpha, \beta> + \overline{<\alpha, \beta>} = 2\mathrm{Re}<\alpha, \beta>$

$= 2\{2\mathrm{Re}<\alpha, \beta>\} = 4\mathrm{Re}<\alpha, \beta>$.

6. $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \Rightarrow \sqrt{<A, A>} = ||A|| = \sqrt{1.1+0.0+1.1+1.1} = \sqrt{3}$

$B = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \Rightarrow \sqrt{<B, B>} = ||B|| = \sqrt{0.0+2.2+0.0+0.0} = \sqrt{4}$

$<A, B> = 1.0+0.2+1.0+1.0 = 0$

$$\text{Cos}\theta = \frac{< A,\ B >}{||A||\ \ ||B||} = 0 \qquad \Rightarrow \theta = \frac{\pi}{2}.$$

7. $\alpha = (4, 3, 1, -2) \Rightarrow ||\alpha|| = \sqrt{< \alpha,\ \alpha >} = \sqrt{4^2 + 3^2 + 1^2 + (-2)^2} = \sqrt{30}$

$\beta = (-2, 1, 2, 3) \Rightarrow ||\beta|| = \sqrt{< \beta,\ \beta >} = \sqrt{(-2)^2 + 1^2 + 2^2 + 3^2} = \sqrt{18}$

$< \alpha,\ \beta > = 4\,(-2) + 3.1 + 1.2 + (-2)\,3 = -9$

$$\cos\theta = \frac{< \alpha,\ \beta >}{||\alpha||\ \ ||\beta||} = \frac{-9}{\sqrt{30}\sqrt{18}} \qquad \Rightarrow \theta = \cos^{-1}\sqrt{\frac{3}{20}}.$$

8. $||\alpha+\beta||^2 = < \alpha+\beta,\ \alpha+\beta > = < \alpha,\ \alpha > + < \alpha,\ \beta > + < \beta,\ \alpha > + < \beta,\ \beta >$

$$= ||\alpha||^2 + < \alpha,\ \beta > + \overline{< \alpha,\ \beta >} + ||\beta||^2$$

$$= ||\alpha||^2 + 2\text{Re} < \alpha,\ \beta > + ||\beta||^2$$

$(||\alpha|| + ||\beta||)^2 = ||\alpha||^2 + 2||\alpha||\ \ ||\beta|| + ||\beta||^2$

Given that $||\alpha+\beta|| = ||\alpha|| + ||\beta|| \Rightarrow ||\alpha+\beta||^2 = (||\alpha|| + ||\beta||)^2$

$\Rightarrow ||\alpha||^2 + 2\text{Re} < \alpha,\ \beta > + ||\beta||^2 = ||\alpha||^2 + 2||\alpha||\ \ ||\beta|| + ||\beta||^2$

$\Rightarrow 2\text{Re} < \alpha,\ \beta > = 2||\alpha||\ \ ||\beta|| \Rightarrow \text{Re} < \alpha,\ \beta > = ||\alpha||\ \ ||\beta||$

$\Rightarrow |< \alpha,\ \beta >| = ||\alpha||\ \ ||\beta||$ ($\because$ $\alpha, \beta \in$ real inner product space)

$\Rightarrow$ The vectors α, β are linearly dependent.

(Particular case of Schwarz's inequality).

8. ORTHOGONALITY

EXERCISE 8 (*a*)

1. Let V (F) be an inner product space and $\alpha, \beta \in V$ are orthogonal

$\Rightarrow <\alpha, \beta>=0 \quad \Rightarrow <\overline{\alpha, \beta}>=\overline{0} \Rightarrow <\beta, \alpha>=0 \Rightarrow \beta, \alpha \in V$ are orthogonal.

$\therefore$ In V (F), orthogonality is a symmetric relation.

2. $u+v$, $u-v$ are orthogonal $\Rightarrow <u+v, u-v>=0$

$\Rightarrow <u, u>-<u, v>+<v, u>-<v, v>=0$

$\Rightarrow ||u||^2 -<u, v>+\overline{<u, v>}-||v||^2=0$

$\Rightarrow ||u||^2 -||v||^2 -<u, v>+<u, v>=0$ (z is real $\Rightarrow (\bar{z}=z)$)

$\Rightarrow ||u||^2-||v||^2=0 \Rightarrow ||u||^2 = ||v||^2 \Rightarrow ||u|| = ||v||$.

3. u, v are orthogonal vectors $\Rightarrow ||u|| = ||v|| = 1$ and $<u, v>=0$.
See Ex.3.

4. Let $\alpha=(2, -1, 6)$. Let $\beta=(x_1, x_2, x_3)$ be orthogonal to α.

$\therefore \quad <\alpha, \beta>=0 \Rightarrow 2x_1-x_2+6x_3=0$.

By inspection, $x_1=4$, $x_2=2$, $x_3=-1$ is a solution.

A vector β orthogonal to $\alpha=(4, 2, -1)$.

$||\beta||=\sqrt{4^2+2^2+(-1)^2}=\sqrt{21}$. Unit vector $=\frac{1}{\sqrt{21}}(4, 2, -1)$.

5. Let $\alpha=(3/5, 0, 4/5)$, $\beta=(-4/5, 0, 3/5)$, $\gamma=(0, 1, 0)$

$||\alpha||=\sqrt{\frac{9}{25}+0+\frac{16}{25}}=1$; $||\beta||=\sqrt{\frac{16}{25}+0+\frac{9}{25}}=1$; $||\gamma||=\sqrt{0+1+0}=1$

$\Rightarrow \alpha, \beta, \gamma$ are unit vectors.

$<\alpha, \beta>=(3/5)(-4/5)+0.0+(4/5)(3/5)=0$;

$<\beta, \gamma>=(-4/5).0+0.1+(3/5).0=0$ and

$(\alpha, \gamma)=(3/5).0+0.1+(4/5).0=0 \Rightarrow \alpha, \beta, \gamma$ are mutually orthogonal.

$\therefore$ $\{\alpha, \beta, \gamma\}$ is an orthonormal set of $R^3(R)$.

6. Let $\alpha=(1, 0, 1, 0)$, $\beta=(0, 1, 0, 1)$, $\gamma=(-1, 0, 1, 0)$.

$||\alpha||=\sqrt{1+0+1+0}=\sqrt{2}$; $||\beta||=\sqrt{2}$ and $||\gamma||=\sqrt{2}$

$\therefore$ Orthonormal set $=\{1/\sqrt{2}\alpha, 1/\sqrt{2}\beta, 1/\sqrt{2}\gamma\}$.

8. Let $\alpha=(a_1, a_2, a_3)$, $\beta=(b_1, b_2, b_3)$ be two orthogonal vectors so that each of which is orthogonal with $\gamma=(4, 2, 3)$.

$\therefore \quad <\alpha, \beta> = 0 \Rightarrow a_1b_1 + a_2b_2 + a_3b_3 = 0 \qquad \ldots(1)$

$<\alpha, \gamma> = 0 \Rightarrow 4a_1 + 2a_2 + 3a_3 = 0 \quad \ldots(2) \quad <\beta, \gamma> = 0 \Rightarrow 4b_1 + 2b_2 + 3b_3 = 0 \quad \ldots(3)$

$(2) \Rightarrow a_3 = -\frac{1}{3}(4a_1 + 2a_2)$. $(3) \Rightarrow b_3 = -\frac{1}{3}(4b_1 + 2b_2)$

Substituting in (1): $a_1b_1 + a_2b_2 + \frac{1}{9}(4a_1 + 2a_2)(4b_1 + 2b_2) = 0$

$\Rightarrow 25a_1b_1 + 13a_2b_2 + 8a_1b_2 + 8a_2b_1 = 0 \Rightarrow 25\left(\frac{a_1}{a_2}\right)\left(\frac{b_1}{b_2}\right) + 13 + 8\left(\frac{a_1}{a_2}\right) + 8\left(\frac{b_1}{b_2}\right) = 0$

$\Rightarrow 25xy + 13 + 8x + 8y = 0$ where $x = \frac{a_1}{a_2}, \quad y = \frac{b_1}{b_2}$

Giving $x = 1$ we have $25y + 13 + 8 + 8y = 0 \Rightarrow 33y + 21 = 0 \Rightarrow y = -7/11$

$x = 1 \Rightarrow \frac{a_1}{a_2} = 1 \Rightarrow \frac{a_1}{1} = \frac{a_2}{1}$; $y = \frac{-7}{11} \Rightarrow \frac{b_1}{b_2} = \frac{-7}{11} \Rightarrow \frac{b_1}{-7} = \frac{b_2}{11}$

Substituting proportional values in (2): $4 + 2 + 3a_3 = 0 \Rightarrow a_3 = -2$

Substituting proportional values in (3): $-28 + 22 + 3b_3 = 0 \Rightarrow b_3 = 2$.

$\therefore \quad \alpha = (1, 1, -2)$ and $\beta = (-7, 11, 2)$.

9. Let $\alpha = (1/\sqrt{2}, 0, 1/\sqrt{2}), \quad \beta = (1/\sqrt{2}, 0, -1/\sqrt{2}), \quad \gamma = (0, 1, 0)$

Let $X = (2, 4, 6)$ be the given vector.

$X = a\ \alpha + b\ \beta + c\ \gamma$ where $a, b, c \in R$ is the linear combination.

$\therefore \quad a = <X, \alpha> = 2.1/\sqrt{2} + 4.0 + 6.1/\sqrt{2} = 8/\sqrt{2} = 4\sqrt{2}$

$b = <X, \beta> = 2.1/\sqrt{2} + 4.0 + 6.-1/\sqrt{2} = -4/\sqrt{2} = -2\sqrt{2}$

$c = <X, \gamma> = 2.0 + 4.1 + 6.0 = 4$

$\therefore \quad (2, 4, 6) = 4\sqrt{2}(1/\sqrt{2}, 0, 1/\sqrt{2}) - 2\sqrt{2}(1/\sqrt{2}, 0, -1/\sqrt{2}) + 4\ (0, 1, 0)$.

EXERCISE 8 (*b*)

1. (a) Let $\beta_1 = (1, 1, 0), \quad \beta_2 = (1, -1, 1), \quad \beta_3 = (-1, 1, 2) \in R^3(R)$

The set $\{\beta_1, \beta_2, \beta_3\}$ is mutually orthogonal.

Normalizing the vector means obtaining unit vector

$||\beta_1|| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2} \Rightarrow$ Unit vector of $\beta_1 = 1/\sqrt{2}(1, 1, 0)$

$||\beta_2|| = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3} \Rightarrow$ Unit vector of $\beta_2 = 1/\sqrt{3}(1, -1, 1)$

$||\beta_3|| = \sqrt{(-1)^2 + 1^2 + 2^2} = \sqrt{6} \Rightarrow$ Unit vector of $\beta_3 = 1/\sqrt{6}(-1, 1, 2)$.

(b) Let $\beta_1 = (1, 0, 1), \ \beta_2 = (1, 0, -1), \ \beta_3 = (0, 3, 0) \in R^3(R)$

The set $\{\beta_1, \beta_2, \beta_3\}$ is mutually orthogonal.

$||\beta_1|| = \sqrt{1^2+0^2+1^2} = \sqrt{2} \Rightarrow$ Unit vector of $\beta_1 = 1/\sqrt{2}(1, 0, 1)$

$||\beta_2|| = \sqrt{1^2+0^2+(-1)^2} = \sqrt{2} \Rightarrow$ Unit vector of $\beta_2 = 1/\sqrt{2}(1, 0, -1)$

$||\beta_3|| = \sqrt{0^2+3^2+0^2} = 3 \Rightarrow$ Unit vector of $\beta_3 = 1/3(0, 3, 0)$

2. (*a*) Let $\beta_1 = (1, -3),\ \beta_2 = (2, 2) \in R^2(R)$

By Gram - Schmidt orthogonalization process, an orthogonal basis $= \{\alpha_1, \alpha_2\}$ is given by

$$\alpha_1 = \frac{1}{||\beta_1||}\beta_1, \quad \alpha_2 = \frac{1}{||\gamma_2||}\gamma_2 \quad \text{where } \gamma_2 = \beta_2 - <\beta_2, \alpha_1>\alpha_1.$$

$||\beta_1|| = \sqrt{1^2+(-3)^2} = \sqrt{10}$ and $\alpha_1 = \frac{1}{\sqrt{10}}(1, -3)$.

$<\beta_2, \alpha_1> = 2(1/\sqrt{10}) + 2(-3/\sqrt{10}) = -4/\sqrt{10}$

$\gamma_2 = \beta_2 - <\beta_2, \alpha_1>\alpha_1 = (2, 2) - (-4/\sqrt{10})(1/\sqrt{10})(1, -3)$

$$= (2, 2) + \left(\frac{4}{10}, \frac{-12}{10}\right) = \left(2+\frac{4}{10}, 2-\frac{12}{10}\right) = \left(\frac{24}{10}, \frac{8}{10}\right) = \frac{8}{10}(3, 1)$$

$$||\gamma_2|| = \sqrt{\left(\frac{8}{10}\right)^2 (3^2+1^2)} = \frac{8}{\sqrt{10}} \qquad \therefore \quad \alpha_2 = \frac{1}{||\gamma_2||}\gamma_2 = \frac{1}{\sqrt{10}}(3, 1)$$

$$\therefore \text{ Orthonormal basis } = \{\alpha_1, \alpha_2\} = \left\{\left(\frac{1}{\sqrt{10}}, -\frac{3}{\sqrt{10}}\right), \left(\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}}\right)\right\}$$

(*b*) Let $\beta_1 = (1, 0),\ \beta = (3, -5) \in R^2(R)$

We now first find orthogonal basis $= \{\gamma_1, \gamma_2\}$ where $\gamma_1 = \beta_1$ and

$$\gamma_2 = \beta_2 - \frac{1}{||\gamma_1||^2} <\beta_2, \gamma_1>. \qquad \therefore \quad \gamma_1 = (1, 0) \text{ and } ||\gamma_1|| = \sqrt{1^2+0^2} = 1.$$

$<\beta_2, \gamma_1> = 2.1 + (-5).0 = 3$ and $\gamma_2 = (3, -5) - \frac{1}{1^2}(3)(1, 0) = (3-3, -5-0) = (0, -5)$.

$\therefore$ Orthogonal basis $= \{\gamma_1, \gamma_2\} = \{(1, 0), (0, -5)\}$.

$||\gamma_1|| = \sqrt{1^2+0^2} = 1$ and $||\gamma_2|| = \sqrt{0^2+(-5)^2} = 5$.

$$\therefore \text{ Orthonormal basis } = \{\gamma_1, \gamma_2\} = \left\{\frac{1}{||\gamma_1||}\gamma_1, \frac{1}{||\gamma_2||}\gamma_2\right\} = \{(1, 0), (0, -1)\}.$$

3. (*a*) Let $\beta_1 = (1, 0, 1),\ \beta_2 = (1, 0, -1),\ \beta_3 = (0, 3, 4) \in R^3(R)$

An orthogonal basis $= \{\gamma_1, \gamma_2, \gamma_3\}$ is given by

$$\gamma_1 = \beta_1,\ \gamma_2 = \beta_2 - \frac{<\beta_2, \gamma_1>}{||\gamma_1||^2}\gamma_1,\ \gamma_3 = \beta_3 - \frac{<\beta_3, \gamma_1>}{||\gamma_1||^2}\gamma_1 - \frac{<\beta_3, \gamma_2>}{||\gamma_2||^2}\gamma_2$$

$\gamma_1 = (1, 0, 1)$ and $||\gamma_1|| = \sqrt{1+0+1} = \sqrt{2}$

$<\beta_2, \gamma_1> = 1.1 + 0.0 + (-1).1 = 0$

$\gamma_2 = (1, 0, -1) - 0\ \gamma_1 = (1, 0, -1).\ ||\gamma_2|| = \sqrt{1+0+1} = \sqrt{2}$

$<\beta_3, \gamma_1> = 0.1 + 3.0 + 4.1 = 4;\quad <\beta_3, \gamma_2> = 0.1 + 3.0 + 4(-1) = -4$

$$\gamma_3 = (0, 3, 4) - \frac{4}{2}(1, 0, 1) - \frac{(-4)}{2}(1, 0, -1) = (0, 3, 4) - (2, 0, 2) + (2, 0, -2)$$

$= (0 - 2 + 2, 3 - 0 + 0, 4 - 2 - 2) = (0, 3, 0)$ and $||\gamma_3|| = 3$.

Orthogonal basis $= \{\alpha_1, \alpha_2, \alpha_3\} = \left\{\frac{1}{||\gamma_1||}\gamma_1, \frac{1}{||\gamma_2||}\gamma_2, \frac{1}{||\gamma_3||}\gamma_3\right\}$

$$= \left\{\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right), (0, 1, 0)\right\}$$

(*b*) Let $\beta_1 = (2, 0, 1),\ \beta_2 = (3, -1, 5),\ \beta_3 = (0, 4, 2) \in R^3(R)$

Let $\{\gamma_1, \gamma_2, \gamma_3\}$ be an orthogonal basis.

$\gamma_1 = \beta_1 = (2, 0, 1)$ and $||\gamma_1|| = \sqrt{4+0+1} = \sqrt{5}$.

$<\beta_2, \gamma_1> = 3.2 + (-1).0 + 5.1 = 11$

$$\gamma_2 = (3, -1, 5) - \frac{11}{5}(2, 0, 1) = \left(3 - \frac{22}{5}, -1 - 0, 5 - \frac{11}{5}\right)$$

$$= \left(\frac{-7}{5}, -1, \frac{14}{5}\right) \text{ and } ||\gamma_2|| = \sqrt{\frac{49}{25} + 1 + \frac{196}{25}} = \sqrt{\frac{270}{25}}$$

$$<\beta_3, \gamma_1> = 0.1 + 4.0 + 2.1 = 2;\quad <\beta_3, \gamma_2> = 0.\left(\frac{-7}{5}\right) + 4(-1) + 2\left(\frac{14}{5}\right) = 0 - 4 + \frac{28}{5} = \frac{8}{5}$$

$$\gamma_3 = (0, 4, 2) - \frac{2}{5}(2, 0, 1) - \frac{8}{5}.\frac{25}{270}\left(\frac{-7}{5}, -1, \frac{14}{5}\right)$$

$$= (0, 4, 2) - \left(\frac{4}{5}, 0, \frac{2}{5}\right) - \left(\frac{-56}{270}, \frac{-40}{270}, \frac{112}{270}\right)$$

$$= \left(0 - \frac{4}{5} + \frac{56}{270}, 4 - 0 + \frac{40}{270}, 2 - \frac{2}{5} - \frac{112}{270}\right) = \left(\frac{-16}{27}, \frac{112}{27}, \frac{32}{27}\right)$$

$= \frac{16}{27}(-1, 7, 2)$ and $\|\gamma_3\| = \sqrt{\frac{16^2}{27^2}(1+49+4)} = \frac{16}{27}\sqrt{54}$.

Orthogonal basis $= \{\gamma_1, \gamma_2, \gamma_3\}$ and orthonormal basis

$$= \left\{\frac{\gamma_1}{\|\gamma_1\|}, \frac{\gamma_2}{\|\gamma_2\|}, \frac{\gamma_3}{\|\gamma_3\|}\right\} = \left\{\frac{1}{\sqrt{5}}(2, 0, 1), \frac{1}{\sqrt{270}}(-7, -5, 14), \frac{1}{\sqrt{54}}(-1, 7, 2)\right\}$$

(*c*) Let $\beta_1 = (1, 1, 0)$, $\beta_2 = (-1, 1, 0)$, $\beta_3 = (1, 2, 1) \in R^3(R)$

Let $\{\gamma_1, \gamma_2, \gamma_3\}$ be an orthogonal basis. Then $\gamma_1 = \beta_1 = (1, 1, 0)$. $\|\gamma_1\| = \sqrt{2}$

$$<\beta_2, \gamma_1> = (-1).1 + 1.1 + 0.0 = 0$$

$\gamma_2 = (-1, 1, 0) - 0(1, 1, 0) = (-1, 1, 0)$ and $\|\gamma_2\| = \sqrt{2}$

$<\beta_3, \gamma_1> = 1.1 + 2.1 + 1.0 = 3$; $<\beta_3, \gamma_2> = 1.(-1) + 2.1 + 1.0 = 1$

$$\gamma_3 = (1, 2, 1) - \frac{3}{2}(1, 1, 0) - \frac{1}{2}(-1, 1, 0)$$

$$= \left(1 - \frac{3}{2} + \frac{1}{2}, 2 - \frac{3}{2} - \frac{1}{2}, 1 - 0 - 0\right) = (0, 0, 1) \text{ and } \|\gamma_3\| = 1$$

$\therefore$ Orthogonal basis $= \{(1, 1, 0), (-1, 1, 0), (0, 0, 1)\}$

Orthonormal basis $= \left\{\frac{1}{\sqrt{2}}(1, 1, 0), \frac{1}{\sqrt{2}}(-1, 1, 0), (0, 0, 1)\right\}$.

(*d*) Let $\beta_1 = (1, -1, 2), \beta_2 = (0, 2, 1), \beta_3 = (1, 2, 0) \in R^3(R)$

Let $\{\gamma_1, \gamma_2, \gamma_3\}$ be an orthogonal basis.

$\gamma_1 = \beta_1 = (1, -1, 2)$ and $\|\gamma_1\| = \sqrt{1+1+4} = \sqrt{6}$. $<\beta_2, \gamma_1> = 0 - 2 + 2 = 0$

$\gamma_2 = (0, 2, 1) - 0\gamma_1 = (0, 2, 1)$ and $\|\gamma_2\| = \sqrt{5}$

$<\beta_3, \gamma_1> = 1 - 2 + 0 = -1$; $<\beta_3, \gamma_2> = 0 + 4 + 0 = 4$

$$\gamma_3 = (1, 2, 0) - \frac{(-1)}{6}(1, -1, 2) - \frac{4}{5}(0, 2, 1) = \left(\frac{7}{6}, \frac{7}{30}, \frac{-1}{3}\right) \text{ etc.}$$

4. Let $\beta_1 = (1, 0, 1, 0)$, $\beta_2 = (1, 1, 1, 1)$, $\beta_3 = (0, 1, 2, 1) \in R^4(R)$

Let $\{\gamma_1, \gamma_2, \gamma_3\}$ be an orthogonal basis. Then $\gamma_1 = \beta_1 = (1, 0, 1, 0)$ and $\|\gamma_1\| = \sqrt{2}$.

$<\beta_2, \gamma_1> = 1.1 + 1.0 + 1.1 + 1.0 = 2$

$\gamma_2 = (1, 1, 1, 1) - \frac{2}{2}(1, 0, 1, 0) = (0, 1, 0, 1)$ and $\|\gamma_2\| = \sqrt{2}$

$< \beta_3, \gamma_1 > = 0.1+1.0+2.1+1.0 = 2; \quad < \beta_3, \gamma_2 > = 0.0+1.1+2.0+1.1 = 2$

$\gamma_3 = (0, 1, 2, 1) - \frac{2}{2}(1, 0, 1, 0) - \frac{2}{2}(0, 1, 0, 1)$

$= (0-1-0, 1-0-1, 2-1-0, 1-0-1) = (-1, 0, 1, 0)$

$\therefore$ Orthogonal basis $= \{(1, 0, 1, 0), (0, 1, 0, 1), (-1, 0, 1, 0)\}$

Orthonormal basis $= \left\{\frac{1}{\sqrt{2}}(1, 0, 1, 0), \frac{1}{\sqrt{2}}(0, 1, 0, 1), \frac{1}{\sqrt{2}}(-1, 0, 1, 0)\right\}$.

5. Let $\beta_1 = (1, 0, i)$, $\beta_2 = (2, 1, 1+i) \in W$.

Let $\{\gamma_1, \gamma_2\}$ be an orthogonal basis.

Then $\gamma_1 = \beta_1 = (1, 0, i)$ and $\| \gamma_1 \| = \sqrt{1.1+0.0+i(-i)} = \sqrt{2}$

$< \beta_2, \gamma_1 > = 2.1+1.0+(1+i)(-i) = 2-i+1 = 3-i$

$\gamma_2 = (2, 1, 1+i) - \frac{(3-i)}{2}(1, 0, i) = \left(2-\frac{3-i}{2}, 1-0, 1+i-\frac{3i+1}{2}\right)$

$= \left(\frac{1+i}{2}, 1, \frac{1-i}{2}\right)$ and $\| \gamma_2 \| = \sqrt{\left(\frac{1+i}{2}\right)\left(\frac{1-i}{2}\right)+1.1+\left(\frac{1+i}{2}\right)\left(\frac{1-i}{2}\right)} = \sqrt{\frac{1}{2}+1+\frac{1}{2}} = \sqrt{2}$.

Orthogonal basis $= \left\{(1, 0, i), \left(\frac{1+i}{2}, 1, \frac{1-i}{2}\right)\right\}$.

Orthonormal basis $= \left\{\frac{1}{\sqrt{2}}(1, 0, i), \frac{1}{\sqrt{2}}\left(\frac{1+i}{2}, 1, \frac{1-i}{2}\right)\right\}$.

6. From worked out example (2); Orthonormal basis of $S = \{1, x, x^2\}$ is

$S' = \{\alpha_1, \alpha_2, \alpha_3\} = \{1, \sqrt{12}(x-1/2), 3\sqrt{20}(x^2-x+1/6)\}$

Given $h(x) = 1+x$. Fourier coefficients of $h(x)$ relative to S' are

$< h(x), \alpha_1 >$, $< h(x), \alpha_2 >$ and $< h(x), \alpha_3 >$.

$< h(x), 1 > = \int_0^1 (1+t)dt = \left[t+\frac{t^2}{2}\right]_0^1 = \frac{3}{2}$

$< h(x), \sqrt{12}(x-1/2) > = \int_0^1 (1+t)\frac{\sqrt{12}}{2}(2t-1)dt = \frac{\sqrt{12}}{2}\int_0^1 (-1+t+2t^2)dt$

$= \sqrt{3}\left[-t+\frac{t^2}{2}+\frac{2t^3}{3}\right]_0^1 = \sqrt{3}\left(-1+\frac{1}{2}+\frac{2}{3}\right) = \frac{\sqrt{3}}{6} = \frac{1}{2\sqrt{3}}$

$$< h(x), 3\sqrt{20}(x^2 - x + 1/6) > = \int_0^1 (1+t)\frac{3\sqrt{20}}{6}(6t^2 - 6t + 1)dt$$

$$= \frac{\sqrt{20}}{2}\int_0^1 (6t^3 - 5t + 1)dt = \sqrt{5}\left[\frac{6t^4}{4} - \frac{5t^2}{2} + t\right]_0^1 = 0 .$$

7. Let $S = \{\beta_1, \beta_2, \beta_3, \beta_4\} = (\sin t, \cos t, 1, t)$. Let $\{\gamma_1, \gamma_2, \gamma_3, \gamma_4\}$ be an orthogonal basis.

$$\therefore \quad \gamma_1 = \beta_1 = \sin t \text{ and } ||\gamma_1||^2 = \int_0^\pi \sin^2 t\, dt = 2\int_0^{\pi/2} \sin^2 t\, dt = 2.\frac{1}{2}.\frac{\pi}{2} = \frac{\pi}{2}.$$

$$< \beta_2, \gamma_1 > = \int_0^\pi \cos t . \sin t\, dt = \left[\frac{(\sin t)^2}{2}\right]_0^\pi = 0$$

$$\gamma_2 = \cos t - 0 \sin t = \cos t \text{ and } ||\gamma_2||^2 = \int_0^\pi \cos^2 t\, dt = \pi/2$$

$$< \beta_3, \gamma_1 > = \int_0^\pi 1 . \sin t\, dt = [-\cos t]_0^\pi = 2$$

$$< \beta_3, \gamma_2 > = \int_0^\pi 1 . \cos t\, dt = [\sin t]_0^\pi = 0 .$$

$$\gamma_3 = 1 - \frac{2}{\pi/2}\sin t - 0 . \cos t = 1 - \frac{4}{\pi}\sin t$$

$$||\gamma_3||^2 = \int_0^\pi \left(1 - \frac{4}{\pi}\sin t\right)^2 dt = \int_0^\pi \left(1 - \frac{8}{\pi}\sin t + \frac{16}{\pi^2}\sin^2 t\right)dt$$

$$= \pi - \frac{8}{\pi}(2) + \frac{16}{\pi^2}.\frac{\pi}{2} = \pi - \frac{8}{\pi} = \frac{\pi^2 - 8}{\pi}$$

$$< \beta_4, \gamma_1 > = \int_0^\pi t \sin t\ dt = [-t\cos t + \sin t]_0^\pi = \pi$$

$$< \beta_4, \gamma_2 > = \int_0^\pi t \cos t\ dt = [t\sin t + \cos t]_0^\pi = -2$$

$$< \beta_4, \gamma_3 > = \int_0^\pi t\left(1 - \frac{4}{\pi}\sin t\right)dt = \int_0^\pi t\ dt - \frac{4}{\pi}\int_0^\pi t \sin t\ dt \quad = \frac{\pi^2}{2} - \frac{4}{\pi}(\pi) = \frac{\pi^2 - 8}{2}$$

$$\gamma_4 = t - \frac{\pi}{\pi/2}.\sin t - \frac{(-2)}{\pi/2}\cos t - \frac{\pi^2 - 8}{2} \times \frac{\pi}{\pi^2 - 8}\left(1 - \frac{4}{\pi}\sin t\right)$$

$$\therefore \quad \gamma_4 = t - 2\sin t + \frac{4}{\pi}\cos t - \frac{\pi}{2} + 2\sin t = t + \frac{4}{\pi}\cos t - \frac{\pi}{2}$$

$$||\gamma_4||^2 = \int_0^\pi \left(t + \frac{4}{\pi}\cos t - \frac{\pi}{2}\right)^2 dt = \int_0^\pi t^2\ dt + \frac{16}{\pi^2}\int_0^\pi \cos^2 t\ dt + \frac{\pi^2}{4}\int_0^\pi 1\ dt$$

$$+ \frac{8}{\pi}\int_0^\pi t\cos t\ dt - \pi\int_0^\pi t\ dt - 4\int_0^\pi \cos t\ dt$$

$$= \frac{1}{3}\pi^3 + \frac{16}{\pi^2}.2.\frac{1}{2}.\frac{\pi}{2} + \frac{\pi^2}{4}.\pi + \frac{8}{\pi}(-2) - \pi.\frac{\pi^2}{2} - 4.0$$

$$= \pi^3\left(\frac{1}{3} + \frac{1}{4} - \frac{1}{2}\right) + \frac{8}{\pi} - \frac{16}{\pi} = \frac{\pi^3}{12} - \frac{8}{\pi} = \frac{\pi^4 - 96}{12\pi}$$

Orthogonal basis $= \{\gamma_1, \gamma_2, \gamma_3, \gamma_4\}$ and orthonormal basis

$$= \left\{\frac{\gamma_1}{||\gamma_1||}, \frac{\gamma_2}{||\gamma_2||}, \frac{\gamma_3}{||\gamma_3||}, \frac{\gamma_4}{||\gamma_4||}\right\} = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\} = B.$$

Fourier coefficients of $h(t) = 2t + 1$ relative to the other orthonormal basis B are given by $< h(t), \alpha_1 >$, $< h(t), \alpha_2 >$, $< h(t), \alpha_3 >$ and $< h(t), \alpha_4 >$.

$$< h(t), \alpha_1 > = \int_0^{\pi} (2t+1)\sqrt{\frac{2}{\pi}} \sin t \, dt = 2\sqrt{\frac{2}{\pi}} \int_0^{\pi} t \sin t \, dt + \sqrt{\frac{2}{\pi}} \int_0^{\pi} \sin t \, dt$$

$$= 2\sqrt{\frac{2}{\pi}}.\pi + \sqrt{\frac{2}{\pi}}.2 = 2\sqrt{\frac{2}{\pi}}(\pi + 1)$$

$$< h(t), \alpha_2 > = \int_0^{\pi} (2t+1)\sqrt{\frac{2}{\pi}} \cos t \, dt = 2\sqrt{\frac{2}{\pi}}(-2) + \sqrt{\frac{2}{\pi}}.0 = -4\sqrt{\frac{2}{\pi}}$$

$$< h(t), \alpha_3 > = \int_0^{\pi} (2t+1)\sqrt{\frac{\pi}{\pi^2 - 8}}\left(1 - \frac{4}{\pi}\sin t\right) dt$$

$$= \sqrt{\frac{\pi}{\pi^2 - 8}}\left\{\pi^2 - \frac{8}{\pi}.\pi + \pi - \frac{4}{\pi}.2\right\} = \sqrt{\frac{\pi}{\pi^2 - 8}}\left(\pi^2 - 8 + \pi - \frac{8}{\pi}\right)$$

$$< h(t), \alpha_4 > = \int_0^{\pi} (2t+1)\sqrt{\frac{12\pi}{\pi^4 - 96}}\left(t + \frac{4}{\pi}\cos t - \frac{\pi}{2}\right) dt$$

$$= \sqrt{\frac{12\pi}{\pi^4 - 96}}\left\{\frac{2}{3}\pi^3 + \frac{8}{\pi}(-2) - \pi.\frac{\pi^2}{2} + \frac{\pi^2}{2} + \frac{4}{\pi}.0 - \frac{\pi}{2}.\pi\right\}$$

$$= \sqrt{\frac{12\pi}{\pi^4 - 96}}\left(\frac{\pi^3}{6} - \frac{16}{\pi}\right).$$